Part III

Option Valuation

The valuation engine, built twice. Chapter 13 builds it discretely, where the argument is visible; Chapters 14 and 15 build it in continuous time and arrive at the same answer. What follows extends the result to other underlyings, to the hedging problem, to the smile the model cannot produce, and to the numerical methods used when no formula exists. Chapter 14 carries the dagger: a reader who wants the Black–Scholes–Merton formula and its use rather than its derivation can pass from Chapter 13 straight to Chapter 15.

13 Binomial Trees
replication and risk-neutral valuation, in the case where you can see them

14 Wiener Processes and Itô's Lemma
the continuous-time machinery behind the formula

15 The Black–Scholes–Merton Model
the equation, the formula, and the assumptions that carry them

16 Options on Indices, Currencies, and Futures
one formula, adapted to three underlyings

17 Employee Stock Options and Equity Compensation
grant-date fair value, vesting, and dilution

18 The Greek Letters
the sensitivities, and what a hedged book costs to run

19 Volatility Smiles and Surfaces
the evidence that one volatility is not enough

20 Numerical Procedures
trees, finite differences, and Monte Carlo, including early exercise

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Chapter 13
Binomial Trees

Chapters 10 to 12 described what options are, what relationships they must satisfy, and how they are combined. None of them priced one. Chapter 11 established bounds, and bounds are not a price.

This chapter supplies the missing element, and it does so with almost no mathematics.

The tree here is a teaching device before it is a computational method. It is used because it is the simplest possible market in which the entire argument of option pricing can be seen at once (one period, two states, two traded instruments), and because in that setting every step can be checked by hand. How to build trees efficiently, how many steps they need, and how to make them converge quickly are questions about computation, and they belong to Chapter 20.

What the device makes visible is the point of it: **replication**, **no arbitrage**, and **risk-neutral valuation**: the argument by which a derivative acquires a price at all. That argument is the foundation of everything in Part III, and it is genuinely difficult to motivate once the machinery of continuous time is in the way. Here it fits on a page.

Sections 13.1 to 13.4 are the argument in its smallest setting: one step, risk-neutral valuation, state prices, and why two states suffice. Sections 13.5 to 13.7 extend it to many steps, to the general form, and to American exercise. Sections 13.8 to 13.10 are the choices that connect the tree to a real underlying (the up and down moves, convergence, and dividends), and Sections 13.11 and 13.12 close with the identity of pricing and hedging and with what the tree assumes.

Later chapters depend on this one. Chapters 15 and 16 take risk-neutral valuation as given; Chapter 18 builds on the identity of pricing and hedging established in Section 13.11; Chapter 19 works in the incomplete markets that Section 13.4 introduces.

13.1 A ONE-STEP BINOMIAL MODEL

Suppose a stock trades at \$20 and that in three months it will be worth either \$22 or \$18. Nothing else can happen. We want the value f of a European call struck at \$21, which pays \$1 in the up state and nothing in the down state.

Figure 13.1 carries both on one tree (the stock's two possible prices above, the call's two possible values below), and the rest of this chapter shows how the stock's two prices force the call's value today.

The method is to construct a portfolio whose value is the same in both states. Hold Δ shares and write one call. In the up state it is worth $22\Delta - 1$; in the down state, 18Δ. These are equal when

$$22\Delta - 1 = 18\Delta \quad \Longrightarrow \quad \Delta = 0.25$$

At $\Delta = 0.25$ the portfolio is worth $18 \times 0.25 = \$4.50$ in *either* state. It is riskless. A riskless portfolio must earn the risk-free rate, or there is an arbitrage, so its value today is $4.50\, e^{-rT}$. With $r = 4\%$ and $T = 0.25$,

$$4.50\, e^{-0.04\times 0.25} = 4.50\, e^{-0.01} = \$4.455$$

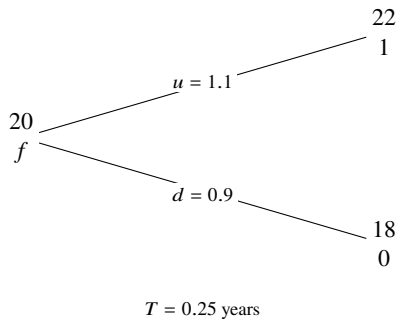


Figure 13.1: Stock price above, call value below. $K = 21$.

The portfolio today costs $20\Delta - f = 5 - f$, so

$$5 - f = 4.455 \quad \Longrightarrow \quad f = \$0.545$$

Read the same construction the other way and it is a recipe. Holding 0.25 shares and owing \$4.455 is worth $0.25 \times 22 - 4.50 = \$1$ in the up state and $0.25 \times 18 - 4.50 = \$0$ in the down state: it *replicates* the call. Its cost is $5 - 4.455 = \$0.545$, and that is the price: the call is worth what it costs to manufacture. Example 13.1 shows what happens if it trades at any other price.

Example 13.1 Checking the arbitrage

Suppose the call traded at \$0.60 rather than \$0.545. Write the call, buy 0.25 shares:

$$\text{cash out} = 0.25 \times 20 - 0.60 = \$4.40$$

Borrow the \$4.40 at 4%. In three months repay $4.40\, e^{0.01} = \$4.444$, and the portfolio is worth \$4.50 whichever way the stock moves. The profit is \$0.056 in every state, from nothing invested—and \$0.056 is $(0.60 - 0.545)e^{0.01}$, the mispricing carried forward, as in Example 11.1.

If instead the call traded at \$0.50, run the trade in reverse: buy the call, short 0.25 shares, invest the proceeds.

Notice what never entered the calculation: no one was asked how likely the up state is, or what return investors require for bearing risk. The argument works for a trader who thinks the stock will certainly rise and for one who thinks it will certainly fall. That is the feature to hold on to, because Section 13.2 shows it is not an accident of this example.

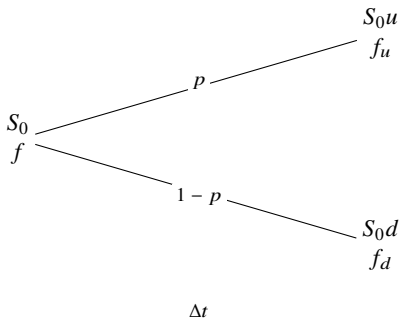


Figure 13.2: The general one-step tree. Stock price above, derivative value below; p and $1 - p$ are the risk-neutral probabilities of (13.2).

The general one-step formula. Write the stock as moving from S_0 to S_0u or S_0d over a step of length Δt, with the derivative paying f_u or f_d, as in Figure 13.2. The same construction

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gives the number of shares held per option, the hedge ratio known as the option's delta,

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d} \tag{13.1}$$

and, after substituting and rearranging,

$$\boxed{f = e^{-r\,\Delta t}\,[p f_u + (1-p) f_d]} \qquad \text{where} \qquad p = \frac{e^{r\,\Delta t} - d}{u - d} \tag{13.2}$$

For no arbitrage we need $d < e^{r\Delta t} < u$; otherwise the stock always beats or always loses to cash. That condition also guarantees $0 < p < 1$.

Check the example:

$$p = \frac{e^{0.04\times 0.25} - 0.9}{1.1 - 0.9} = 0.5503,$$
$$f = e^{-0.04\times 0.25}\,[0.5503 \times 1 + 0.4497 \times 0] = \$0.545,$$

as before. Here and throughout, intermediate values are carried unrounded and printed to four places, so the last digit of a product can differ from the product of the printed factors.

13.2 RISK-NEUTRAL VALUATION

Equation (13.2) has the form of a discounted expected payoff, with p playing the role of a probability. What p is needs care, because most of the confusion in derivative pricing starts here.

What p is. Compute the expected stock price using p:

$$\mathbb{E}[S_T] = pS_0 u + (1-p)S_0 d = S_0\,[pu + (1-p)d] = S_0 e^{r\Delta t}$$

(the last step is algebra: substitute p and simplify). So p is the probability under which the stock is expected to earn the risk-free rate. It is called the **risk-neutral probability**, and the corresponding expectation is taken in the **risk-neutral world**.

What p is not. It is not the real-world probability of the stock rising, and it is not anyone's forecast. In the example $p = 0.5503$; the true probability might be 0.7 or 0.3, and the option price would be \$0.545 either way. This is exact, not an approximation: the true probability is irrelevant.

Why this works. The reason is not that anyone is risk-neutral. It is that the option can be *replicated*: Section 13.1 built a portfolio of stock and cash that reproduces the option's payoff exactly. Two things with identical payoffs must have identical prices, whatever anyone's attitude to risk. Attitudes to risk do set the interest rate and the price of the underlying asset, but once those two prices are given they have no further say in the option's. Preferences and real-world probabilities drop out of the problem, and the only assignment of probabilities that survives (the only *measure*, in the language of Part III) is the one under which no risk premium is required, which is p.

The result generalizes into the principle underlying all of Part III:

> **Risk-neutral valuation.** The value of a derivative is the expected value of its payoff in a world where all assets earn the risk-free rate, discounted at the risk-free rate.

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The risk-neutral probability is a tool for calculating derivative prices, not a description of the world. We do not believe investors are risk-neutral. We use the risk-neutral measure because it gives the right answer to a question about relative prices, and it gives it far more easily than the alternative. Chapter 15 shows the same principle at work in continuous time; Chapter 28 shows what happens when one changes to a different measure entirely. Example 13.2 shows what the alternative involves.

Example 13.2 The real-world calculation, for comparison

Suppose the true probability of the up state is 0.7. Then the expected stock price is $0.7\times22+0.3\times18 = \$20.80$, an expected return of

$$\frac{20.80}{20} - 1 = 4\% \text{ over three months}$$

The expected call payoff is $0.7 \times 1 + 0.3 \times 0 = \$0.70$. To discount it we need the return investors require *on the option*, which is not 4%: the option is a leveraged position in the stock and is far riskier.

That discount rate cannot be looked up. It depends on how leveraged the option is, and the leverage depends on the option's price, which is what we were trying to find. The real-world approach is circular unless a preference model is supplied from outside.

Run backward from the known answer, and the circle closes. Discounting \$0.70 back to the known price \$0.545 implies a required return of $0.70/0.5448 - 1 = 28.5\%$ per quarter, an enormous risk premium, correctly reflecting how leveraged the option is. Risk-neutral valuation sidesteps this entirely. The required return on the option, which no one can observe, never has to be found: the replication argument makes the risk premium cancel out, and what remains is the risk-free rate, which everyone can observe.

13.3 STATE PRICES AND THE ABSENCE OF ARBITRAGE

Equation (13.2) can be read a second way, and the second reading connects this chapter to Chapters 11 and 12 and states the principle in its most general form.

Group the discount factor with the probabilities:

$$f = \underbrace{e^{-r\Delta t}p}_{\psi_u} f_u + \underbrace{e^{-r\Delta t}(1-p)}_{\psi_d} f_d \tag{13.3}$$

The two quantities ψ_u and ψ_d are **state prices**. Each is the price, today, of a security that pays \$1 if a particular state occurs and nothing otherwise. Such a security is called an **Arrow–Debreu security**, and once the price of each is known, *every* derivative on this underlying can be priced by multiplying its payoffs by the corresponding state prices and adding.

In the running example,

$$\psi_u = e^{-0.04\times0.25} \times 0.5503 = 0.5448,$$
$$\psi_d = e^{-0.04\times0.25} \times 0.4497 = 0.4453,$$

and Example 13.3 confirms both by replication.

Example 13.3 Building and using state prices

Verify that $\psi_u = 0.5448$ really is the price of \$1 in the up state. Replicate that payoff with Δ shares and B in cash:

$$22\Delta + Be^{0.04\times0.25} = 1, \qquad 18\Delta + Be^{0.04\times0.25} = 0$$

Subtracting gives $\Delta = (1-0)/(22-18) = 0.25$, which is (13.1) with the payoffs 1 and 0, and then

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$Be^{0.04\times0.25} = -4.50$, so $B = -\$4.4552$. The cost today is

$$0.25 \times 20 - 4.4552 = \$0.5448$$

which is ψ_u, as claimed. In this example the security paying \$1 up and \$0 down *is* the call struck at 21, which is why $f = \psi_u$. Similarly, replicating \$1 in the down state takes $\Delta = -0.25$ and $B = \$5.4453$, so $\psi_d = -0.25 \times 20 + 5.4453 = \$0.4453$.

Now price something else with no further work. A claim paying \$3 in the up state and \$7 in the down state is worth

$$3 \times 0.5448 + 7 \times 0.4453 = 1.6343 + 3.1169 = \$4.7512$$

Checking by replication, (13.1) gives $\Delta = (3-7)/(22-18) = -1$, then $B = 25e^{-0.04\times0.25} = \$24.7512$, and the cost is $-20 + 24.7512 = \$4.7512$.

Two state prices value every derivative on this stock: the replication is done once, and every payoff afterward is a weighted sum.

Note also that $\psi_u + \psi_d = e^{-r\Delta t} = 0.99005$. A claim paying \$1 in *both* states is a riskless bond, so the state prices must sum to the discount factor, which is where the discounting in (13.2) comes from, and why p and $1-p$ behave like probabilities: they are state prices *normalized* by the bond price.

The arbitrage-free condition. State prices make precise what "no arbitrage" requires. A security paying \$1 in one state and nothing otherwise cannot have a negative price (the buyer would be paid to take a payoff that is never negative), and it cannot be free, since it sometimes pays. So both state prices must be strictly positive, which given $\psi_u + \psi_d = e^{-r\Delta t}$ is exactly the condition

$$0 < p < 1 \quad \Longleftrightarrow \quad d < e^{r\Delta t} < u$$

met earlier in Section 13.1.

This is a special case of a general result:

> **The market admits no arbitrage if and only if a set of strictly positive state prices exists**; equivalently, if and only if there is a risk-neutral measure, a probability measure that gives every state a positive probability and under which every asset's expected return is the risk-free rate.

Read in that direction, risk-neutral valuation stops looking like a trick. The risk-neutral measure is not an assumption added to the model; it is another name for the absence of arbitrage. A market with no arbitrage always has such a measure, and a market with an arbitrage never does.

The connection to Chapter 12. Section 12.8 built a digital option as the limit of a tight call spread. Push that construction one derivative further: a butterfly of vanishing width centered at K pays only if the underlying finishes at K, and its price, divided by the square of its width, tends to $\partial^2 c/\partial K^2$. A butterfly of that kind *is* an Arrow–Debreu security, and its price *is* a state price. The risk-neutral density of the terminal price, which Chapter 19 reads off a continuum of traded option prices, and the state prices of this section are the same object seen from two directions, one observed in the market, the other constructed from a replication argument. Chapter 19 uses the first; most of the rest of Part III uses the second.

13.4 WHY TWO STATES? COMPLETENESS

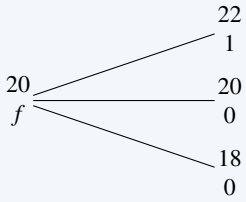
An obvious objection has been left standing since Section 13.1: a stock does not have two possible prices in three months. Section 13.9 answers part of that by letting the steps shrink, but the deeper answer is not about realism at all, and it is the most important idea in this chapter.

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The two-state assumption is what makes the market **complete**: every payoff can be replicated by trading the instruments available. There are two traded instruments (the stock and the bond) and two states, so the replicating equations are two equations in two unknowns and have a unique solution. *Every* payoff can therefore be manufactured, so every payoff has a price. Add a third state and that stops being true, as Example 13.4 shows.

Example 13.4 Three states, and the price is no longer unique

Suppose the stock at \$20 can move to \$22, \$20, or \$18. Everything else is as before: $r = 4\%$, $\Delta t = 0.25$, and a call struck at \$21 paying \$1, \$0, \$0.



Try to replicate it with Δ shares and B in cash:

$$22\Delta + Be^{0.01} = 1, \qquad 20\Delta + Be^{0.01} = 0, \qquad 18\Delta + Be^{0.01} = 0$$

The last two give $2\Delta = 0$, so $\Delta = 0$ and then $B = 0$, which contradicts the first. The call cannot be replicated, and the argument that produced a price collapses.

What survives? The state prices must still be positive and must still price the two traded instruments correctly:

$$22\psi_1 + 20\psi_2 + 18\psi_3 = 20, \qquad \psi_1 + \psi_2 + \psi_3 = e^{-0.01} = 0.99005$$

Two equations in three unknowns have a whole family of solutions rather than one. Eliminating ψ_2 gives $\psi_1 = \psi_3 + 0.0995$, and requiring $\psi_2 > 0$ and $\psi_3 > 0$ confines ψ_3 to the open interval $(0,\ 0.445274)$.

The call's value is ψ_1, so it can be anything in

$$\$0.0995 \ < \ f \ < \ \$0.5448$$

Every price in that interval is consistent with no arbitrage. The theory no longer delivers a price—it delivers bounds, and the upper one is more than five times the lower.

Note what sits at the two ends. As ψ_3 approaches 0.445274, ψ_2 goes to zero, the middle state is priced at nothing, and the value approaches \$0.5448, the binomial answer of Section 13.1. As ψ_3 approaches zero instead, the down state is priced at nothing and the value approaches \$0.0995, the binomial answer for a tree whose two moves are to \$22 and \$20. Each bound is a two-state model, the limiting case of the three-state one.

This example is worth dwelling on because it identifies precisely what the binomial assumption is doing. It is not a claim about how stocks behave. It is the assumption that makes the number of states equal the number of traded instruments, and that equality, not realism, is what converts Chapter 11's bounds into a price.

The general statement is the second half of the result quoted above:

> **The market admits no arbitrage and is complete if and only if there is exactly one risk-neutral measure**; equivalently, if and only if there is exactly one set of strictly positive state prices. Where it is unique, every derivative has one arbitrage-free price. Where it is not, derivatives have a range of arbitrage-free prices, and selecting one requires something beyond arbitrage: a model, or a view about preferences.

This is not a technicality confined to contrived examples. Real markets are incomplete for exactly this reason: prices jump, volatility moves unpredictably, and the number of things that can happen exceeds the number of instruments available to hedge them. The consequence

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appears throughout the rest of the book. It is why Chapter 19's volatility smile (options at different strikes implying different volatilities for the same underlying) cannot be argued away, why Chapter 34's exotic payoffs carry model risk that vanilla options do not, and why two desks using different models can quote different prices for the same structure without either committing an arbitrage.

Completeness is the assumption that lets this chapter say "the price." Everywhere it fails, the accurate phrase is "a price, given a model."

13.5 TWO-STEP TREES AND BACKWARD INDUCTION

With more than one period, apply (13.2) repeatedly, working backward from expiration one step at a time, the procedure called **backward induction**. Figure 13.3 extends the example to two three-month steps.

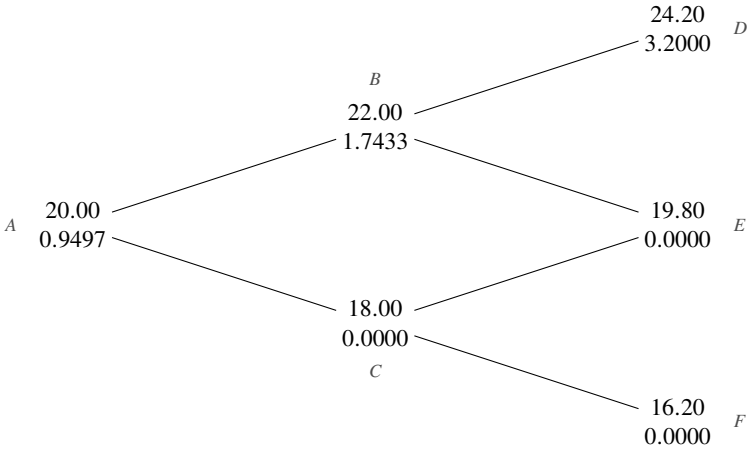


Figure 13.3: Stock price above, European call value below. $K = 21$, $r = 4\%$.

At the terminal nodes D, E, and F the call is worth $\max(S_T - 21, 0)$: \$3.20, \$0, \$0. Because u, d, and r are the same at every step, $p = 0.5503$ is the same at every node too. Working back with it and $e^{-r\Delta t} = 0.99005$:

$$\begin{aligned}
\text{node B}: &\quad e^{-0.01}\,[0.5503 \times 3.20 + 0.4497 \times 0] = \$1.7433 \\
\text{node C}: &\quad e^{-0.01}\,[0.5503 \times 0 + 0.4497 \times 0] = \$0 \\
\text{node A}: &\quad e^{-0.01}\,[0.5503 \times 1.7433 + 0.4497 \times 0] = \$0.9497
\end{aligned}$$

Two features of the procedure matter more than the arithmetic.

The tree recombines. An up-then-down move and a down-then-up move both give $S_0ud = 19.80$, so the number of nodes grows linearly rather than exponentially: $n + 1$ nodes after n steps rather than 2^n. This is what makes trees computationally practical, and Section 13.10 shows what happens when it fails.

The replication is dynamic. The riskless portfolio of Section 13.1 must be *rebalanced* at each node, because Δ from (13.1) differs at each one. At node B, $\Delta = (3.20 - 0)/(24.20 - 19.80) = 0.7273$; at node C, $\Delta = 0$. Pricing the option assumes the hedger can and does trade at every step. Section 13.12 returns to what that assumption costs.

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Example 13.5 Following the hedge through the tree

Take the tree of Figure 13.3 and follow a trader who writes the call and hedges it.

From (13.1) the deltas are $\Delta_A = (1.7433 - 0)/(22 - 18) = 0.4358$ at node A, $\Delta_B = (3.20 - 0)/(24.20 - 19.80) = 0.7273$ at node B, and $\Delta_C = 0$ at node C. The arithmetic below carries the deltas unrounded.

Today (node A). Sell the call for \$0.9497. Buy 0.4358 shares at \$20, costing \$8.7164. Borrow the difference:

$$8.7164 - 0.9497 = \$7.7667$$

Three months on. The debt has grown to $7.7667\, e^{0.01} = \$7.8448$. If the stock is at 22 (node B), the shares are worth $0.4358 \times 22 = \$9.5881$, so the position is worth

$$9.5881 - 7.8448 = \$1.7433$$

which is exactly the call's value at node B. If instead the stock is at 18 (node C), the shares are worth \$7.8448 and the position is worth precisely nothing, again the call's value there.

Rebalancing at node B. Delta must rise from 0.4358 to 0.7273, so buy 0.2915 more shares at \$22, costing \$6.4119, financed by borrowing. Total debt becomes $7.8448 + 6.4119 = \$14.2567$.

At expiration. The debt grows to $14.2567\, e^{0.01} = \$14.4000$. If the stock reaches 24.20, the shares are worth $0.7273 \times 24.20 = \$17.6000$ and the position is worth

$$17.6000 - 14.4000 = \$3.2000$$

which is exactly the call's payoff. If the stock falls to 19.80, the shares are worth \$14.4000, the position is worth zero, and the call expires worthless.

The hedge closes to the penny in every state, having been financed entirely by the \$0.9497 received at the outset. That is what it means for \$0.9497 to be the price—it is the cost of manufacturing the option's payoff.

Two things in the walkthrough deserve emphasis. The trader bought more shares at 22 and would have sold shares at 18: the hedge *buys high and sells low* by construction, and the accumulated cost of doing so is precisely the option premium. And nothing was ever at risk (no step required a view), which is the replication argument made operational rather than asserted.

13.6 THE MODEL IN GENERAL FORM

The examples have been worked one node at a time. Set out formally, the model is compact, and its structure becomes visible in a way node-by-node arithmetic hides.

The lattice. Divide the derivative's life, T years, into n steps of length $\Delta t = T/n$. The underlying starts at S_0 and at each step is multiplied by u or by d. After i steps it has taken j up moves and $i - j$ down moves, so it stands at

$$S_{i,j} = S_0\, u^j d^{\,i-j}, \qquad j = 0, 1, \ldots, i \tag{13.4}$$

Because u and d multiply, only the count j matters, not the order of the moves. That is the recombining property, and it is why step i has $i + 1$ nodes rather than 2^i.

The measure. Under the risk-neutral measure Q, each step is an up move with probability p from (13.2), independently of the others. The number of up moves in n steps is therefore *binomially distributed*, which is the model's name:

$$Q(j \text{ up moves}) = \binom{n}{j} p^j (1 - p)^{n-j}$$

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The pricing formula. For a European claim paying $h(S_T)$ at maturity, backward induction through the whole tree collapses to a single sum:

$$f_0 = e^{-rT} \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j}\, h\big(S_0 u^j d^{n-j}\big) \tag{13.5}$$

and for a European call, $h(S) = \max(S - K, 0)$, so the terms with $S_0 u^j d^{n-j} \le K$ vanish and the sum runs only over the nodes finishing in the money.

Equation (13.5) is the two-step calculation of Section 13.5 written out in one line. Applying it there with $n = 2$ reproduces \$0.9497 directly, without the intermediate node values, though those values are exactly what an American option needs, which is why Section 13.7 cannot use this shortcut.

The martingale property. There is a more revealing way to state what Q does. Under it, the *discounted* price of any traded asset is a **martingale**: its expected future value, discounted, equals its value today:

$$e^{-r\Delta t}\, \mathbb{E}^Q[S_{i+1} \mid S_i] = S_i, \qquad e^{-r\Delta t}\, \mathbb{E}^Q[f_{i+1} \mid S_i] = f_i \tag{13.6}$$

The first equality is the calculation of Section 13.2 restated; the second is equation (13.2) restated. That both hold under the same measure is the entire content of risk-neutral valuation, and it is the form in which the principle carries over to continuous time in Chapter 15 and to other numeraires (other choices of the asset in which prices are measured) in Chapter 28.

The continuous limit. With the parameter choice (13.8), Section 13.9 shows that as $n \to \infty$ the terminal distribution becomes lognormal (the logarithm of the price is normally distributed) and the sum (13.5) for a European call converges to the Black–Scholes–Merton formula, which Chapter 15 derives directly and which is the continuous-time counterpart of everything in this chapter.

13.7 AMERICAN OPTIONS

Trees remain in use because they handle early exercise, which Chapter 11 showed matters for puts and for calls on dividend-paying stocks, and which, outside a few special cases, no closed-form formula can address.

The modification is small. At each node, compute the value of continuing (equation (13.2) applied to the two successor nodes) and compare it with the value of exercising immediately. The node's value is the greater:

$$f = \max\Big(\underbrace{e^{-r\Delta t}\,[p f_u + (1-p) f_d]}_{\text{continue}},\ \underbrace{\text{intrinsic value}}_{\text{exercise}}\Big) \tag{13.7}$$

Example 13.6 applies the comparison to a two-step put. The same comparison handles any decision embedded in a security, including an issuer's right to call (redeem early) a convertible bond: each right is a max or a min at the nodes where it can be exercised, and because the rights interact node by node, the bond and its embedded options cannot be valued separately and added.

Example 13.6 An American put, and what early exercise is worth

A stock trades at \$50, with $u = 1.2$ and $d = 0.8$ over each of two one-year steps. Consider an American put struck at \$52, with $r = 5\%$.

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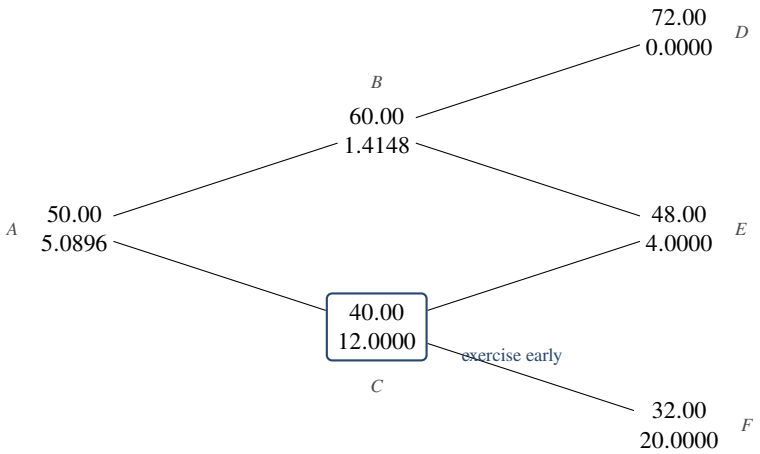


Figure 13.4: Stock price above, American put value below. $K = 52$, $r = 5\%$.

First, $e^{0.05\times 1} = 1.051271$ and

$$p = \frac{1.051271 - 0.8}{1.2 - 0.8} = \frac{0.251271}{0.4} = 0.6282$$

Figure 13.4 sets out the tree. At the terminal nodes D, E, and F the put is worth \$0, \$4, and \$20.

Node B ($S = 60$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 0 + 0.3718 \times 4] = 0.951229 \times 1.4873 = \$1.4148$$

Exercising yields $\max(52 - 60, 0) = 0$. Hold. Value \$1.4148.

Node C ($S = 40$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 4 + 0.3718 \times 20] = 0.951229 \times 9.9492 = \$9.4639$$

Exercising yields $52 - 40 = \$12$. **Exercise.** Value \$12.

Node A (today). Continuation value is

$$e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 12] = 0.951229 \times 5.3506 = \$5.0896$$

Exercising yields $52 - 50 = \$2$. Hold. Value \$5.0896.

Now price the European put for comparison. The only difference is at node C, where the European put is worth its continuation value of \$9.4639 rather than \$12:

$$p_{\text{Eur}} = e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 9.4639] = \$4.1927$$

The early-exercise premium is $5.0896 - 4.1927 = \$0.897$, more than 21% of the European value. Section 11.9 argued that an American put is worth more than its European counterpart; here is the gap, computed.

The tree also *locates* the exercise decision, which is its practical advantage. Exercise is optimal at node C and not at nodes A or B, so the critical price S^* of Figure 11.5 lies between 40 and 60 at that date, and below 50 today. Figure 13.5 traces it date by date on a 1,000-step tree with $\sigma = \ln(1.2/0.8)/2 = 20.3\%$, the volatility of the two-step tree; the curve $S^*(t)$ it draws is the put's **early-exercise boundary**. The boundary starts just above 40, rises to about 42 by the one-year date the two-step tree bracketed, and climbs the rest of the way to the strike only in the final weeks.

13.8 CHOOSING u AND d

So far u and d have been given. In practice they are chosen so that the tree reproduces the volatility of the underlying, which is the only parameter of the model that must be estimated.

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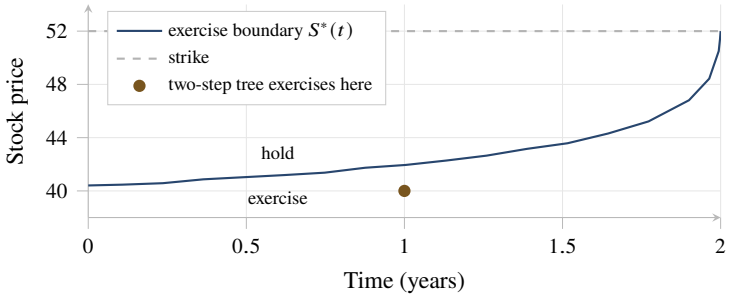


Figure 13.5: Early-exercise boundary of the Example 13.6 put on a 1,000-step tree, $\sigma = 20.3\%$.

The standard choice, due to Cox, Ross, and Rubinstein, is

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}} \tag{13.8}$$

where σ is the stock's **volatility**, the annualized standard deviation of its continuously compounded return, and Δt is the length of a step in years.

These match the variance of the stock's return over each step to $\sigma^2\Delta t$, to first order in Δt, which is the same variance the continuous-time model of Chapter 15 assumes. Setting $d = 1/u$ also centers the tree, so that an up move followed by a down move returns exactly to the starting price.

This leaves σ itself, the one input that cannot be read off a screen. It is the same number in the real world and in the risk-neutral one: moving from the true probability to p changes the tree's drift but, to first order in Δt, not its variance, so a volatility measured from historical prices can go into the tree unchanged. Estimating it, from history or by inverting a traded option's price, is taken up once, in Chapter 15; here σ is given, and Example 13.7 builds a tree from it.

Example 13.7 Building a tree from a volatility

A stock has volatility 30% a year. Build a tree with monthly steps, $\Delta t = 1/12$:

$$u = e^{0.30\sqrt{1/12}} = e^{0.086603} = 1.09046, \qquad d = 1/1.09046 = 0.91704$$

With $r = 4\%$,

$$p = \frac{e^{0.04/12} - 0.91704}{1.09046 - 0.91704} = \frac{1.003339 - 0.91704}{0.17342} = 0.49762$$

One sanity check is worth performing every time: $d < e^{r\Delta t} < u$, equivalently $0 < p < 1$. Here $0.917 < 1.0033 < 1.090$. It fails when the step is too long for the volatility and rate, and the symptom is a negative probability.

Note also how little p moved from one-half. Expanding $e^{r\Delta t}$, u, and d to order Δt gives

$$p \approx \frac{1}{2} + \frac{1}{2}\left(\frac{r - \sigma^2/2}{\sigma}\right)\sqrt{\Delta t} = \frac{1}{2} + \frac{1}{2} \times \frac{0.04 - 0.3^2/2}{0.30} \times 0.2887 = 0.4976,$$

the exact value to four places. So the risk-neutral probability is close to a coin flip and the drift enters only as a small correction. The volatility sets u and d, and with them how far the tree spreads at each step.

13.9 CONVERGENCE

A two-step tree is a crude description of a stock. The model is justified because refinement removes the crudeness: as the number of steps rises, the tree value settles down to a limit, the

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Black–Scholes–Merton value that Chapter 15 derives. Here the point is only that the limit exists, so that the arbitrary-looking choice of two branches does not determine the answer.

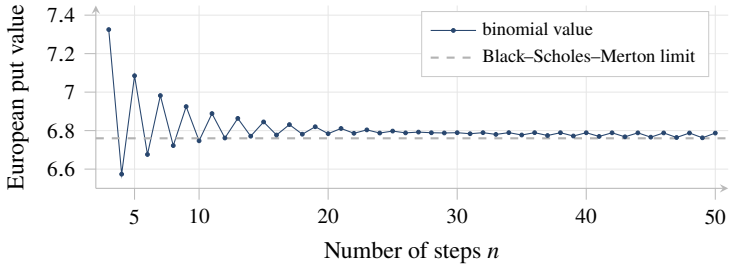


Figure 13.6: European put value against number of steps, $n \geq 3$. $S_0 = 50$, $K = 52$, $\sigma = 30\%$, $r = 5\%$, $T = 2$.

Figure 13.6 shows the characteristic behavior, and two features of it matter in practice.

Convergence is slow. The error decays roughly as $1/n$, so halving it requires doubling the number of steps, and since a tree of n steps has about $n^2/2$ nodes, that quadruples the work.

Convergence oscillates. Successive values zigzag, because where the strike falls between adjacent terminal nodes changes with the parity of n; in the figure the even-n values sit near 6.79 while the odd-n values drift down toward the limit, and both are still above it at $n = 50$. Reading convergence off a single value of n is therefore unreliable, and at small n adding one step can move the price by more than a one-dollar move in the stock would.

Both are numerical rather than conceptual difficulties, and Chapter 20 addresses them. What matters here is what convergence establishes about the *argument*: the price does not depend on the artificial two-branch assumption, because refining that assumption away leaves the answer essentially unchanged.

Why use a tree at all, then? Not for European values, which have a formula. The tree's worth is that it puts the pricing principle on display (replication, no arbitrage, risk-neutral valuation) and that the principle survives when a decision has to be made at an intermediate date, which is Section 13.7's situation and what a closed-form formula cannot express.

Why the limit is lognormal

It is worth seeing, without any stochastic calculus, why the tree converges to the model of Chapter 15 rather than to something else.

Over n steps the stock price is $S_0 u^j d^{n-j}$, where j is the number of up moves. Taking logarithms,

$$\ln S_T = \ln S_0 + j \ln u + (n - j) \ln d$$

so $\ln S_T$ is a *linear function of* j, and j is a sum of n independent binary draws. The central limit theorem then does the rest: as n grows, j becomes normally distributed, so $\ln S_T$ becomes normal, so S_T becomes **lognormal**.

Two things follow. The lognormal distribution that Chapter 15 assumes is not an extra assumption imposed from outside; it is what any model built from small independent multiplicative shocks produces. And the parameter choice (13.8) was designed to make the variance come out at $\sigma^2 T$ in the limit, so the limiting distribution has exactly the volatility that was put in.

The same reasoning identifies what would have to change to get a different limit. Shocks that are not independent, whose size depends on the price level or varies at random, or that are occasionally very large break the conditions under which the limit is lognormal; the last three are precisely the modifications that Chapters 19 and 21 make to reproduce the volatility smile.

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13.10 DIVIDENDS AND OTHER UNDERLYINGS

A known dividend yield. If the stock pays a continuous yield q, the stock itself grows at $r - q$ in the risk-neutral world, so only p changes:

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d} \tag{13.9}$$

Everything else is untouched. The same substitution handles a stock index (with q the dividend yield), a currency (with $q = r_f$, the foreign rate, following Section 5.7), and a futures contract.

Entering a futures contract costs nothing, so in the risk-neutral world a futures price must have zero expected growth. Setting $q = r$ in (13.9) gives

$$p = \frac{1 - d}{u - d} \tag{13.10}$$

Table 13.1 collects the cases, and Example 13.8 applies the last of them.

Table 13.1: The probability p for different underlyings.

Underlying	Risk-neutral probability p
Non-dividend-paying stock	$\left(e^{r\Delta t} - d\right)/(u - d)$
Stock or index with yield q	$\left(e^{(r-q)\Delta t} - d\right)/(u - d)$
Currency, foreign rate r_f	$\left(e^{(r-r_f)\Delta t} - d\right)/(u - d)$
Futures contract	$(1 - d)/(u - d)$

Example 13.8 An American call on futures, exercised early

A futures price is 100, with $\sigma = 25\%$, $r = 4\%$, and two three-month steps. Value an American call struck at 100.

From (13.8), $u = e^{0.25\sqrt{0.25}} = 1.133148$ and $d = 0.882497$. From (13.10),

$$p = \frac{1 - 0.882497}{1.133148 - 0.882497} = 0.4688$$

Terminal futures prices are 128.40, 100.00, and 77.88, giving payoffs 28.4025, 0, and 0.

Upper node ($F = 113.3148$). Continuation is $e^{-0.01}\,[0.4688 \times 28.4025] = 13.1824$. Exercising yields $113.3148 - 100 = 13.3148$. **Exercise.**

Lower node. Both successors pay nothing; value 0.

Today. Continuation is $e^{-0.01}\,[0.4688 \times 13.3148] = 6.1798$, against an exercise value of 0. Hold.

The American call is worth 6.1798; the European is worth 6.1183.

Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early—and here is one exercised early, with no dividend anywhere in the problem.

There is no contradiction, only a limit to that result's scope. The proof relied on the underlying being an asset that must be paid for, so that deferring payment of the strike earns interest. Any positive yield weakens the argument, because waiting now costs the income the underlying pays. A futures contract costs nothing to enter, which is exactly what (13.10) encodes by setting $q = r$: the yield equals the interest rate, and deferring the strike earns nothing net. Early exercise of a deep-in-the-money American call is therefore a live decision on futures options, on currencies with a high foreign interest rate, and on high-yielding indices.

A known cash dividend. A fixed dollar dividend is harder, and the difficulty is worth noting because it is the first place the method's convenience breaks. If the stock drops by D on an ex-dividend date, the tree *stops recombining*: an up-then-down path and a down-then-up path

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no longer meet, because the dividend is subtracted at different price levels. The node count no longer grows linearly; with a dividend at every step it reaches 2^n. Nothing about the pricing principle changes (the replication argument at each node is untouched), but the computation becomes impractical, and Chapter 20 deals with it.

13.11 PRICING AND HEDGING ARE ONE CALCULATION

One consequence of the argument deserves to be stated on its own, because it is easy to pass over and it governs everything in Part III.

The hedge ratio Δ of (13.1) is not a by-product of the pricing calculation. It *is* the pricing calculation: the price was obtained by finding the Δ that makes the portfolio riskless, and the option's value fell out of that construction. There is no separate step in which one first prices the option and then asks how to hedge it.

This is why the sensitivities of an option are not, in this theory, sensitivity analysis. They are the recipe for manufacturing it, and Example 13.5 carried that recipe out node by node. A trader who cannot hedge a position cannot claim to know what it is worth, because the price was defined as the cost of the hedge. Chapter 18 develops the full set of sensitivities on that understanding.

13.12 WHAT THE TREE ASSUMES

The argument of Section 13.1 is valid given its assumptions. They are the assumptions of the continuous-time model the tree converges to, and Section 15.1 lists them and says where each fails: volatility is neither known nor constant, prices jump rather than moving by u or d, and borrowing is not at r. One consequence belongs to the tree. Rebalancing at every node is what makes the hedge riskless, so with transaction costs the imperfection grows with the number of rebalances, and the limit that makes the model converge is also the limit that makes hedging infinitely expensive. Chapter 18 confronts this directly.

None of this makes the model useless. What survives is the *principle* that a derivative replicable from traded assets has a price independent of anyone's beliefs, and that principle is what the rest of Part III builds on.

SUMMARY

A binomial tree lets the underlying move to one of two values over a short interval. Holding $\Delta = (f_u - f_d)/(S_0u - S_0d)$ shares against a written derivative is riskless, so the portfolio must earn the risk-free rate, and that fixes the price. Pricing and delta hedging are one calculation, not two.

The same price is a discounted expectation, $f = e^{-r\Delta t}[p f_u + (1-p) f_d]$ with $p = (e^{r\Delta t} - d)/(u-d)$. The real-world probability of an up move never appears: p is the probability under which the stock earns the risk-free rate, and the device works because the derivative can be replicated, so preferences cancel out. With the discount factor absorbed, the weights are state prices, which are positive exactly when there is no arbitrage; the risk-neutral measure is another name for the absence of arbitrage.

Two states and two traded assets make the market complete, and that is what turns Chapter 11's bounds into a single price. With a third state the option cannot be replicated and the theory delivers only a range; wherever completeness fails, the accurate phrase is "a price, given a model."

Multi-period trees are solved backward. They recombine, so nodes grow linearly, and an American option takes the greater of continuation and exercise value at each node, which prices the early-exercise premium and locates the boundary. With $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$ only volatility must be estimated, and as the steps shrink the terminal distribution becomes lognormal

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and the value converges, oscillating, to the Black–Scholes–Merton formula. Dividend yields, currencies, and futures change only p; for futures $p = (1 - d)/(u - d)$, which is why an American call on futures can be worth exercising early with no dividend anywhere.

The tree assumes costless rebalancing, constant known volatility, no jumps, and borrowing at the risk-free rate, and all four fail in practice. What survives is the principle: a derivative that can be replicated from traded assets has a price independent of anyone's beliefs.

FURTHER READING

The original papers are the place to start: Cox, Ross, and Rubinstein (1979) for the model itself, and Rendleman and Bartter (1979), published independently in the same year. Cox and Ross (1976) is where risk-neutral valuation first appears as a general principle. For convergence behavior, Broadie and Detemple (1996), a comparison of American option valuation methods, documents the odd–even oscillation and sets out which refinements are worth their cost.

PRACTICE QUESTIONS

13.1 A stock at \$40 will move to \$44 or \$36 in six months. The rate is 5%. Value a European call struck at \$40 by constructing the riskless portfolio, then verify the answer using (13.2).

13.2 In the previous question, compute the risk-neutral probability and verify that the expected stock price under it equals $S_0 e^{rT}$.

13.3 Explain why the real-world probability of an up move does not appear in the option price. Would the answer change if it were 0.9 rather than 0.5?

13.4 A stock at \$100 has $u = 1.15$, $d = 0.87$, and two six-month steps, with $r = 6\%$. Value a European put struck at \$100.

13.5 Repeat the previous question for an American put and compute the early-exercise premium. At which nodes is exercise optimal?

13.6 A stock has volatility 25% a year. Compute u, d, and p for a tree with three-month steps when $r = 3\%$. Verify that $d < e^{r\Delta t} < u$.

13.7 Explain why the binomial value oscillates as n increases, and why the existence of a limit matters for the argument of Section 13.1.

13.8 Show that for a futures contract $p = (1 - d)/(u - d)$, and explain in words why the interest rate disappears.

13.9 A stock at \$50 can move to \$56, \$50, or \$44 over one period, with $r = 4\%$ and $\Delta t = 0.5$. Show that a call struck at \$52 cannot be replicated, and find the range of arbitrage-free prices by imposing positive state prices.

13.10 Explain why a one-period model with two states can be completed by the stock and the bond, and what fails when a third state is added. State what completeness has to do with the option having a single arbitrage-free price rather than a range of them.

13.11 Explain why pricing an option and hedging it are one calculation in a binomial tree rather than two. State where the hedge ratio appears in the pricing arithmetic, and what that implies about a model that prices well but hedges badly.

13.12 State four assumptions the binomial tree makes, and for each name a market condition that violates it. Identify which of the four the convergence result of Section 13.9 is most sensitive to, and why.

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FURTHER QUESTIONS

13.13 Example 13.2 computed a 28.5% quarterly required return on the call by working backward from its known price. Repeat this for the put in the same one-step model, and explain why the two required returns differ so sharply in sign and magnitude. What does this tell you about using risk-adjusted discount rates for options?

13.14 A known cash dividend breaks the recombining property. Demonstrate this explicitly with a two-step tree, a \$50 stock, $u = 1.2$, $d = 0.8$, and a \$2 dividend paid between the steps. Then explain why the pricing principle is unaffected even though the computation is not.

13.15 Section 13.12 notes that the limit making the tree converge is also the limit making hedging infinitely expensive under transaction costs. Formalize this: with a proportional cost k per trade, estimate how total hedging cost scales with n, and identify the trade-off between model error and trading cost. (Chapter 18's gamma sets the size of each rebalancing trade.)

13.16 Using a three-step tree, trace the early-exercise boundary for an American put and compare it with the critical price described in Section 11.9. How does the boundary move as you approach expiration, and why must it converge to the strike?

13.17 An analyst prices a convertible bond by valuing the bond and the conversion option separately and adding them. Explain why this is wrong when the bond is callable, and identify which feature of backward induction the additive approach discards.

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Chapter 14
Wiener Processes and Itô's Lemma †

Chapter 13 priced options without any continuous-time mathematics. It could do so because the binomial world has finitely many states, and finitely many states can be handled with algebra. Chapter 15 does the same job in continuous time, where the states form a continuum and algebra is no longer enough. This chapter supplies the language for that. A reader who wants the Black–Scholes–Merton formula and its use rather than its derivation can pass from Chapter 13 straight to Chapter 15 without loss of continuity. Chapters 15, 16, 19, and 22 do cite four results here worth taking even on a skip: the lognormal property (Section 14.9: the median grows at $\mu - \sigma^2/2$, more slowly than the mean, a gap behind common errors in performance reporting), the $\sigma/16$ rule (Section 14.6), the volatility of a ratio (Section 14.11) and the model's failures (Section 14.12). Chapters 20, 21, 28, and 30 assume the whole chapter.

Sections 14.1 to 14.4 build the process for a stock price, from the Markov property through Wiener processes to geometric Brownian motion; Sections 14.5 and 14.6 say what its two parameters mean. Sections 14.7 and 14.8 are Itô's lemma and why it delivers a hedge; Sections 14.9 and 14.10 are the lognormal property and the risk-neutral process. Sections 14.11 and 14.12 extend the model to two sources of uncertainty and state what it assumes, and Section 14.13 introduces fractional Brownian motion, which Chapter 21's rough volatility will need.

14.1 THE MARKOV PROPERTY

A **Markov process** is one in which only the present value of the variable is relevant for predicting the future. The path taken to get there carries no information.

Stock prices are usually assumed to be Markov. The motivation is **weak-form market efficiency**, the hypothesis that past prices cannot be used to predict future ones: if the history of prices contained information about future prices beyond what today's price contains, that information could be traded on, and trading on it would remove it. The assumption is not exactly true: under volatility clustering, which Chapter 38 models when it forecasts volatility, the size of tomorrow's move is partly predictable even though its direction is not. The Markov assumption is nevertheless close enough to be the standard starting point.

The model adds one assumption this chapter relies on throughout: changes over non-overlapping periods are independent, so their variances add. The Markov property alone does not deliver this: the mean-reverting interest rates of Chapter 31 are Markov, yet their changes are negatively correlated, and their variance levels off instead of growing linearly.

Example 14.1 Why variances, not standard deviations, add

Suppose a stock at \$40 has a price change with standard deviation \$10 over one year. What is the uncertainty over two years?

The change over two years is the change in year one plus the change in year two. These are assumed independent, so their *variances* add:

$$\text{Var}(2\text{ years}) = 100 + 100 = 200, \qquad \text{standard deviation} = \sqrt{200} = \$14.14$$

Uncertainty, measured by standard deviation, grows with $\sqrt{T}$, not with T; it is the variance that grows with T. Over six months the standard deviation is $10\sqrt{0.5} = \$7.07$; over three months, $10\sqrt{0.25} = \$5.00$.

This fact is why Section 11.1 found option value growing roughly as $\sqrt{T}$. It is why volatility is quoted per annum and rescaled by $\sqrt{\Delta t}$. And it is why the binomial parameters of Section 13.8 were $u = e^{\sigma\sqrt{\Delta t}}$ rather than $e^{\sigma \Delta t}$. The square root was not a modeling choice but a consequence of independence.

14.2 WIENER PROCESSES

The building block is a process W with two properties. Over a short interval Δt,

$$\Delta W = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim N(0,1) \tag{14.1}$$

and the values of ΔW for any two non-overlapping intervals are independent. A process with these properties is a **Wiener process**, also called **Brownian motion**.

Two consequences follow immediately. The mean of ΔW is zero and its variance is Δt, so the variance per unit time is exactly 1, which is what makes W a natural unit of randomness. And over a long interval T, made of $T/\Delta t$ independent steps,

$$W(T) - W(0) \sim N(0, T)$$

by the same variance-addition argument as Example 14.1.

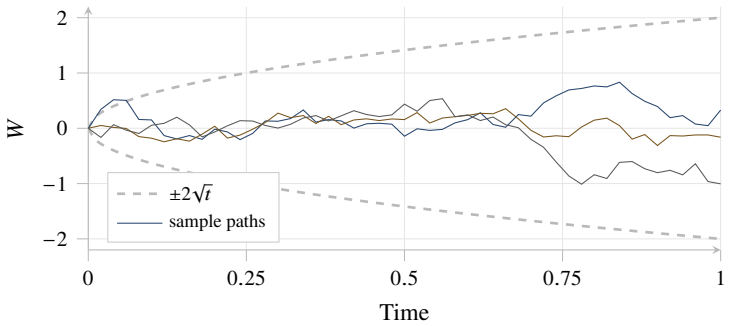


Figure 14.1: Three illustrative simulated Wiener paths and the $\pm 2\sqrt{t}$ envelope.

Two features of the paths in Figure 14.1 recur throughout Part III.

The paths are continuous but extremely rough. As Δt shrinks, the standard deviation of ΔW shrinks only as $\sqrt{\Delta t}$, so that of $\Delta W/\Delta t$ is $1/\sqrt{\Delta t}$ and grows without bound. A Wiener path has, with probability one, no derivative anywhere. This is why the calculus of Section 14.7 has to be different from the ordinary kind: one cannot simply write dW/dt.

And the envelope widens as $\sqrt{t}$, not linearly. The typical distance from zero grows as time passes, but at a decreasing rate.

14.3 GENERALIZED WIENER PROCESSES

A Wiener process has zero **drift**, the expected change per unit time, and a unit **variance rate**, the variance per unit time. Both can be generalized. A variable x follows a **generalized Wiener process** if

$$dx = a\,dt + b\,dW \tag{14.2}$$

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with a and b constants. The $a\,dt$ term is a deterministic trend (x drifts at rate a per unit time), and the $b\,dW$ term, the **diffusion**, adds noise with variance rate b^2.

Over an interval T,

$$x(T) - x(0) \sim N\left(aT,\ b^2T\right)$$

The mean grows linearly in T and the standard deviation as $\sqrt{T}$, so the trend eventually dominates the noise, but only eventually, and the crossover can be a long way off, as Example 14.2 shows.

Example 14.2 A cash position

A company's cash balance, in millions, follows a generalized Wiener process with drift 20 per year and variance rate 900 per year. It currently holds \$50 million.

After one year the balance is distributed $N(50 + 20,\ 900)$, that is, mean \$70 million with standard deviation \$30 million. After six months it is $N(60,\ 450)$: mean \$60 million, standard deviation $\sqrt{450}$ = \$21.2 million.

Writing $N(x)$ for the standard normal cumulative distribution function, the probability that the balance is negative at the end of one year is

$$\Pr(x < 0) = N\left(\frac{0 - 70}{30}\right) = N(-2.333) = 0.98\%$$

and at the end of six months, $N(-60/21.2) = N(-2.83) = 0.23\%$. The balance is expected to *grow*, yet the probability is *higher* at one year than at six months, because uncertainty accumulates faster than the drift does over short horizons.

Running out of cash, though, means touching zero at any time, not being below it on a given date. That probability is larger, 2.7% within the year (by the reflection argument Chapter 34 uses for barrier options), and it never falls: over an unlimited horizon it is $e^{-2\times20\times50/900} = 10.8\%$. This is the same structure as the funding failures of Chapter 8: a position with a positive expected trend can still fail on the way, and the distribution on any one date understates how often.

An **Itô process** generalizes (14.2) once more by letting the coefficients depend on the current value and on time:

$$dx = a(x,t)\,dt + b(x,t)\,dW \tag{14.3}$$

This is the form most of the rest of the book uses. It is shorthand for an integral equation,

$$x(T) - x(0) = \int_0^T a(x,t)\,dt + \int_0^T b(x,t)\,dW$$

and the second integral is not an ordinary one, because W has no derivative to integrate against. It is an **Itô integral**, the limit of the sums

$$\sum_{i=0}^{n-1} b\big(x(t_i), t_i\big)\,\Delta W_i, \quad \Delta W_i = W(t_{i+1}) - W(t_i), \quad 0 = t_0 < \cdots < t_n = T,$$

as the longest interval shrinks to zero, in which b is evaluated at the *start* of each interval, before the increment ΔW_i is known. In finance this is not a convention but a fact about trading: a hedge ratio is chosen before the price move it is exposed to, so a trading gain is a sum of exactly this kind. The choice makes the integral a fair game (a martingale, the term Section 13.6 defined), with expected value zero whatever b does, and it is the source of everything that separates Itô calculus from the ordinary kind, beginning with the extra term in Section 14.7. Evaluating b at the midpoint of each interval instead gives the Stratonovich integral, which obeys the ordinary chain rule but is not a fair game and does not describe what a trader can earn. The derivation in [TN14] works through the **quadratic variation** of W, the limit of the sum of its squared increments, which over $[0, T]$ is exactly T; Section 14.7 shows why.

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14.4 THE PROCESS FOR A STOCK PRICE

What process should a stock price follow? A generalized Wiener process is the obvious candidate, and it is wrong for two reasons that determine the model that replaces it.

Constant absolute drift is not how returns work. A \$10 stock and a \$1,000 stock do not have the same expected dollar gain. Investors require a *percentage* return, so the drift should be proportional to the price.

Constant absolute volatility is not how risk works either. A \$1 move means something very different at \$10 than at \$1,000. And a constant b permits the price to go negative, which for a limited-liability share it cannot.

Both objections are met by making the coefficients proportional to S:

$$\boxed{dS = \mu S\,dt + \sigma S\,dW} \tag{14.4}$$

This is **geometric Brownian motion**, and it is the model underlying everything from Chapter 15 onward. Dividing through, the *proportional* return over a short Δt is close to normal:

$$\frac{dS}{S} = \mu\,dt + \sigma\,dW, \qquad \frac{\Delta S}{S} \sim N\left(\mu\,\Delta t,\ \sigma^2\Delta t\right).$$

Here μ is the expected return per year and σ is the volatility, the same σ that Section 13.8 fed into the binomial tree.

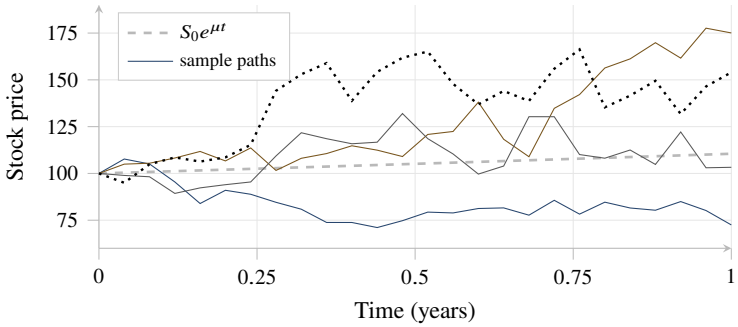


Figure 14.2: Four illustrative simulated paths. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$.

Figure 14.2 shows four paths from this process. Note that none of them tracks the dashed line $S_0e^{\mu t}$ at all closely, and that half of them finish below it despite μ being positive. Section 14.9 explains why that is exactly what should be expected.

Example 14.3 Reading the process over a short interval

A stock trades at \$50 with $\mu = 16\%$ and $\sigma = 30\%$. Over one week, $\Delta t = 1/52 = 0.01923$:

$$\frac{\Delta S}{S} \sim N\left(0.16 \times 0.01923,\ 0.30^2 \times 0.01923\right) = N(0.003077,\ 0.001731)$$

so the weekly return has mean 0.31% and standard deviation $\sqrt{0.001731} = 4.16\%$. In dollar terms,

$$\Delta S \sim N(0.154,\ 2.08^2)$$

The noise term is about thirteen and a half times the drift term over one week, and since the drift scales with Δt while the noise scales with $\sqrt{\Delta t}$, the ratio gets worse as the interval shortens: over one day it is about thirty to one.

This is why estimating μ from historical returns is hopeless in a way that estimating σ is not: to measure a drift buried under thirty times its own size in noise, one needs decades of data, over which the drift will not have stayed constant.

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The discrete-time version

Equation (14.4) is a statement about infinitesimals. To simulate it, or to connect it to Chapter 13's tree, it must be discretized, and there are two ways of doing so that are not equivalent.

The direct reading of (14.4) gives

$$\Delta S \approx \mu S\,\Delta t + \sigma S\,\varepsilon\sqrt{\Delta t} \tag{14.5}$$

which is accurate only for small Δt and can produce a negative price if ε is sufficiently negative. The alternative uses the solution of the equation rather than the equation itself, which Section 14.9 will derive:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\varepsilon\sqrt{\Delta t}\right] \tag{14.6}$$

This is *exact* for any Δt, and it cannot go negative. Figure 14.2 was generated with (14.6).

The comparison is worth keeping in mind because the same choice arises whenever a process is discretized, and Chapter 20 returns to it. Note also that (14.6) is recognizably Chapter 13's tree with the two-point distribution of ε replaced by a normal one: $u = e^{\sigma\sqrt{\Delta t}}$ was the same expression with $\varepsilon = \pm 1$ and the drift moved out of the step and into p.

Where the model came from. Louis Bachelier's thesis of 1900 used the arithmetic process of Section 14.3, which this section has just rejected for stocks. Norbert Wiener gave the process its rigorous construction in 1923. Kiyoshi Itô defined the integral of Section 14.3 in 1944 and set out the calculus built on it, including the lemma of Section 14.7, in 1951. Section 15.12 traces the economics, from Bachelier to Black, Scholes, and Merton.

14.5 EXPECTED RETURN

The parameter μ was introduced as "the expected return," and that phrase turns out to name two different numbers. The distinction causes more confusion than any other point in this chapter, and it is not confined to theory.

From (14.4), the expected *proportional* return over a short interval satisfies

$$\mathbb{E}\left[\frac{\Delta S}{S}\right] \approx \mu\,\Delta t$$

so μ is the expected return in the sense of an arithmetic average of short-period returns. But from (14.10), which Section 14.9 derives, the expected *continuously compounded* return over $[0, T]$ is

$$\mathbb{E}\left[\frac{1}{T}\ln\frac{S_T}{S_0}\right] = \mu - \frac{\sigma^2}{2}$$

which is the expected rate earned by an investor who holds the asset throughout. Example 14.4 shows the gap in five years of returns.

Example 14.4 Arithmetic and geometric averages

An asset returns +30%, −10%, +20%, −5%, and +15% over five years. The arithmetic average is

$$\frac{30 - 10 + 20 - 5 + 15}{5} = 10.0\%$$

But an investor who held throughout finished with

$$1.30 \times 0.90 \times 1.20 \times 0.95 \times 1.15 = 1.53387$$

times their money, an annual compounded return of $1.53387^{1/5} - 1 = 8.93\%$.

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In log returns the five years average 8.56% ($e^{0.0856} - 1 = 8.93\%$), 1.44 percentage points below the arithmetic average, and half the sample variance of the five simple returns is $0.0288/2 = 1.44$ percentage points, a match to the second decimal that in general holds only approximately. This is the $\sigma^2/2$ that separates (14.13) from (14.14), seen in five numbers.

The arithmetic average is not wrong; it answers a different question. It is the right input for an expected one-period return, and therefore the right thing to compare with μ. The compounded figure is the right answer to "what did an investor make," and it corresponds to $\mu - \sigma^2/2$.

The practical consequence is a reporting convention that repays skepticism. A track record summarized by the arithmetic average of its annual returns, common in strategy marketing and backtests, exceeds what an investor who held throughout received, and the excess grows with volatility. The standardized figure a US mutual fund must publish is not of this kind: the SEC's Form N-1A defines its average annual total return as the compounded rate.

14.6 VOLATILITY

As Section 13.8 defined it, σ is the annualized standard deviation of the continuously compounded return. Over a period Δt it is $\sigma\sqrt{\Delta t}$, by the variance-addition property of Section 14.1.

One rule of thumb is used constantly on trading desks. With about 252 trading days in a year and $\sqrt{252} \approx 15.9$, the daily standard deviation is approximately

$$\sigma_{\text{daily}} \approx \frac{\sigma}{16}$$

so a stock with 32% annual volatility has a daily standard deviation of about 2%, and a 6% move is a three-standard-deviation event. It converts an abstract parameter into something a trader can check against the screen.

How σ is estimated (from historical returns, or by inverting an option price) is taken up in Chapter 15, where it is needed for the first time.

14.7 ITÔ'S LEMMA

Assume that a derivative's value depends only on the current stock price and time, so that it can be written $f(S, t)$. That is an assumption, not a general truth: it holds for standard calls and puts, and fails for contracts whose payoff depends on the path, such as Chapter 34's Asian and lookback options, whose value depends on where the stock has been and not only on where it is now. If S follows (14.4), what process does $f(S, t)$ follow? Ordinary calculus gives the wrong answer, and the reason it does is the single technical fact this chapter exists to deliver.

Consider a Taylor expansion of f:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \cdots$$

In ordinary calculus the ΔS^2 term is discarded, because ΔS is proportional to Δt and Δt^2 is negligible. Here it is not. From (14.4), $\Delta S \approx \mu S\,\Delta t + \sigma S\varepsilon\sqrt{\Delta t}$, so

$$\Delta S^2 = \sigma^2 S^2 \varepsilon^2\,\Delta t + \text{terms of order } \Delta t^{3/2} \text{ and smaller}$$

whose leading term is of order Δt, *not* Δt^2. It survives. A single $\varepsilon^2\Delta t$ is still random: its mean is Δt and its standard deviation $\sqrt{2}\,\Delta t$, of the same size. But over an interval T, made of $T/\Delta t$ steps, the squared increments add to a total with mean T and variance $2T\Delta t$, which vanishes as the steps shrink. The sum, the quadratic variation of Section 14.3, becomes non-random and equal to T, as Figure 14.3 shows on one path. In the notation that results,

$$dW^2 = dt \tag{14.7}$$

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a rule about sums over many steps, not about any single increment. [TN14], reproduced in Appendix A, makes this precise.

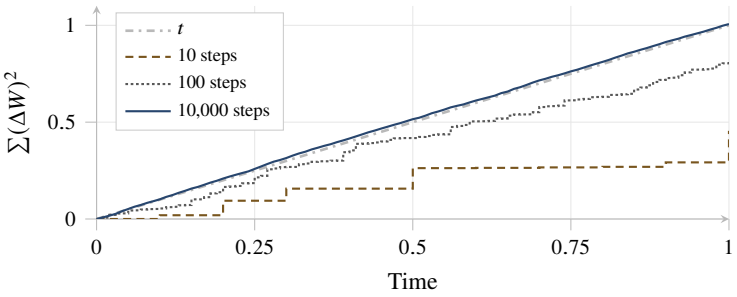


Figure 14.3: Running sum of squared increments along one simulated Wiener path, sampled on grids of 10, 100 and 10,000 steps.

Substituting dS and $dS^2 = \sigma^2 S^2\,dt$ into the expansion gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW} \tag{14.8}$$

Everything except the third term in the drift is what ordinary calculus would give. That third term is the entire difference, and it is the reason a derivative's expected return is not simply the chain rule applied to the stock's. It is proportional to the second derivative, the convexity, which is why it will reappear in Chapter 18 as gamma and in Chapter 30 as the source of convexity adjustments.

Example 14.5 Itô's lemma applied to a forward price

The forward price of a stock that pays no dividend, for delivery at T is $F = Se^{r(T-t)}$, from Section 5.2. What process does F follow?

The partial derivatives are

$$\frac{\partial F}{\partial S} = e^{r(T-t)}, \qquad \frac{\partial F}{\partial t} = -rSe^{r(T-t)}, \qquad \frac{\partial^2 F}{\partial S^2} = 0$$

Substituting into (14.8), and noting that the second-derivative term vanishes:

$$dF = \left(e^{r(T-t)}\mu S - rSe^{r(T-t)}\right)dt + e^{r(T-t)}\sigma S\,dW = (\mu - r)F\,dt + \sigma F\,dW$$

So the forward price also follows geometric Brownian motion, with *the same volatility* σ as the spot but a drift of $\mu - r$ rather than μ.

The equality of volatilities is why a volatility quoted on the spot can be used to price an option on the forward, which Chapter 16 relies on throughout. And the drift being the *excess* return $\mu - r$ is the continuous-time statement of Section 13.10's observation that a futures price has zero growth in the risk-neutral world: setting $\mu = r$, as risk-neutral valuation does, makes the forward's drift exactly zero.

Common applications

Table 14.1 collects the cases used elsewhere in the book. Each is (14.8) with the relevant derivatives substituted, and each is worth being able to reproduce.

14.8 WHY THE LEMMA DELIVERS A HEDGE

The last row of Table 14.1 is the bridge between Chapter 13 and Chapter 15.

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Table 14.1: Processes followed by functions of a stock price obeying (14.4).

Function	Process	Where used
$\ln S$	$\left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW$	The lognormal property, Section 14.9
S^n	$S^n\left[\left(n\mu + \frac{n(n-1)}{2}\sigma^2\right) dt + n\sigma\, dW\right]$	Power payoffs; $n = 2$ in Practice Question 5
$Se^{r(T-t)}$	$(\mu - r)F\, dt + \sigma F\, dW$	Forward prices, Example 14.5
general $f(S,t)$	drift as in (14.8), diffusion $\dfrac{\partial f}{\partial S}\sigma S\, dW$	Every derivative, Section 14.8

For *any* derivative $f(S,t)$, Itô's lemma gives

$$df = (\cdots)\, dt + \frac{\partial f}{\partial S}\, \sigma S\, dW \tag{14.9}$$

Whatever the drift term turns out to be, the *random* part of the derivative's movement is the stock's own shock dW, scaled by $\sigma S\, \partial f/\partial S$. And the random part of the stock's movement, from (14.4), is $\sigma S\, dW$: the same dW.

Consider a portfolio that is short one derivative and long $\partial f/\partial S$ shares. Its change over an infinitesimal time interval is

$$-df + \frac{\partial f}{\partial S}\, dS = -(\cdots)\, dt - \frac{\partial f}{\partial S}\sigma S\, dW + \frac{\partial f}{\partial S}\, (\mu S\, dt + \sigma S\, dW)$$

and the two dW terms cancel exactly. The portfolio has no random component at all. It is riskless, so it must earn r, and imposing that condition is precisely what Chapter 15 does to obtain the Black–Scholes–Merton equation.

Three observations follow, and they connect this chapter back to the last one.

It is Chapter 13's argument in continuous time. Section 13.1 found the Δ that made a portfolio worth the same in both states. Here $\partial f/\partial S$ makes a portfolio's change contain no dW. The discrete hedge ratio

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

is the difference quotient whose limit is $\partial f/\partial S$.

μ cancels too. The derivative's drift, the $(\cdots)$ above, contains the term $\mu S\, \partial f/\partial S$; the short derivative subtracts it and the shares add it back, so μ drops out of the portfolio's change along with dW, leaving only the time and convexity terms for Chapter 15 to set equal to the riskless return. This is the continuous-time version of Chapter 13's observation that the real-world probability never entered, and it is why an option can be priced without forecasting the stock.

The hedge is instantaneous, not permanent. The cancellation holds over a single dt. Since $\partial f/\partial S$ changes as S and t change, the position must be adjusted continuously: the dynamic replication of Section 13.5, now requiring infinitely frequent trading. That requirement is harmless in the model and expensive in reality, which is the subject of Chapter 18 and the reason the assumptions of Section 14.12 matter.

14.9 THE LOGNORMAL PROPERTY

The most useful application of Itô's lemma is to $G = \ln S$. Here

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

and (14.8) gives

$$d\ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW \tag{14.10}$$

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The right-hand side has constant coefficients, so $\ln S$ follows a *generalized* Wiener process, and Section 14.3 already told us the answer for those:

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right) \tag{14.11}$$

Because its logarithm is normal, S_T is lognormal. This is the result Chapter 13 anticipated by a different route: Section 13.9 showed that a product of many small independent multiplicative shocks becomes lognormal by the central limit theorem. Equation (14.11) is that conclusion in continuous time, and the agreement is not a coincidence: the binomial model was built to converge to this one.

Moments of the lognormal

Two results are used constantly from Chapter 15 onward. If $\ln X \sim N(m, s^2)$, then

$$\mathbb{E}[X] = e^{m+s^2/2}, \qquad \mathrm{Var}(X) = e^{2m+s^2}\left(e^{s^2} - 1\right) \tag{14.12}$$

Applying the first to (14.11), with $m = \ln S_0 + (\mu - \sigma^2/2)T$ and $s^2 = \sigma^2 T$:

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

The two rates are consistent: μ is the expected growth rate of the price, and Itô's lemma makes the logarithm, and with it the median, grow $\sigma^2/2$ more slowly.

Where the $\sigma^2/2$ comes from, and why it matters

The term $-\sigma^2/2$ in (14.10) is the Itô correction term of (14.8), and it is the most consequential piece of notation in this chapter.

From (14.11), two different quantities both have a claim to be called "the" growth rate:

$$\mathbb{E}[S_T] = S_0 e^{\mu T} \tag{14.13}$$

$$\mathrm{median}(S_T) = S_0 e^{(\mu - \sigma^2/2)T} \tag{14.14}$$

The expected value grows at μ. The median (and the continuously compounded return actually realized on a typical path) grows only at $\mu - \sigma^2/2$. The gap is entirely due to volatility, and it is sometimes called **volatility drag**.

Example 14.6 Expected value against typical outcome

A stock trades at \$100 with $\mu = 10\%$ and $\sigma = 40\%$. After one year:

$$\mathbb{E}[S_T] = 100e^{0.10} = \$110.52, \qquad \text{median} = 100e^{0.10-0.08} = 100e^{0.02} = \$102.02$$

The expected price is \$110.52 and yet half of all outcomes fall below \$102.02. Worse, ask how often the expected value is reached at all. Since $\ln S_T \sim N(\ln 100 + 0.02,\ 0.40^2)$,

$$\Pr(S_T > 110.52) = 1 - N\left(\frac{\ln 1.1052 - 0.02}{0.40}\right) = 1 - N(0.20) = 42.1\%$$

The stock finishes below its own expected value 58% of the time. Nothing is wrong with the arithmetic: the mean is dragged up by a long right tail that occurs rarely.

A 95% interval for the terminal price makes the dispersion concrete. With probability 0.95,

$$\ln 100 + 0.02 - 1.96 \times 0.40 \le \ln S_T \le \ln 100 + 0.02 + 1.96 \times 0.40,$$

that is, $\$46.58 \le S_T \le \$223.45$ at 40% volatility over a single year.

Two common errors follow directly from ignoring the distinction, besides the fund-reporting one of Example 14.4. Quoting an "expected return" of 10% invites investors to expect \$110.52,

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when the typical outcome is \$102.02. And a structured product promising participation in an index's *average* growth is offering something systematically better than participation in its *realized* growth, a difference of $\sigma^2/2$ per year, which at 40% volatility is eight percentage points.

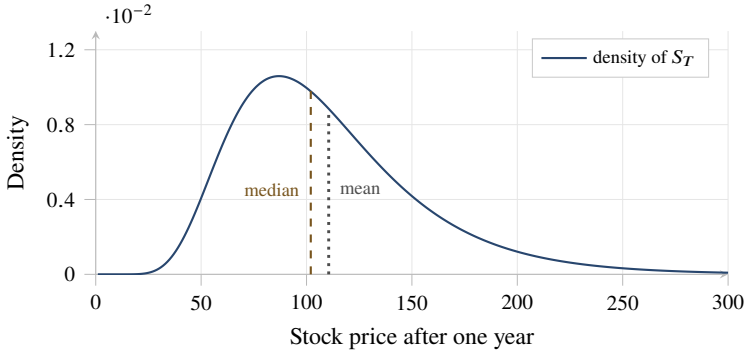


Figure 14.4: Density of S_T. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$, $T = 1$.

Figure 14.4 shows why. The distribution is bounded below by zero and has a long right tail; the mean sits to the right of the median, dragged there by outcomes that are large but rare. The mode is lower still, at $S_0 e^{(\mu - 3\sigma^2/2)T} = \$86.94$.

Which rate does an option price use? Neither: Section 14.10 replaces μ by r, and the $-\sigma^2/2$, a property of the lognormal distribution rather than of the drift, survives.

Case Study 14.1 Leveraged exchange-traded funds

A widely sold class of funds promises a fixed multiple (usually two or three times) of the *daily* return of an index. Investors often assume this delivers the same multiple over longer horizons. It does not, and Itô's lemma says exactly by how much it fails.

If the index follows (14.4), a fund whose value V delivers k times its instantaneous return has $dV/V = k\,dS/S = k\mu\,dt + k\sigma\,dW$, a geometric Brownian motion with drift $k\mu$ and volatility $k\sigma$, before the financing cost of the borrowed $k-1$ units of exposure, which lowers the drift by a further $(k-1)r$. From (14.14), its median grows at

$$k\mu - \frac{(k\sigma)^2}{2} = k\mu - \frac{k^2\sigma^2}{2}$$

while k times the index's own median growth would be $k\left(\mu - \sigma^2/2\right)$. The shortfall is

$$\frac{k(k-1)}{2}\,\sigma^2 \text{ per year}$$

which for $k = 2$ is σ^2 and for $k = 3$ is $3\sigma^2$.

Numbers make the size clear. Take an index with $\mu = 8\%$ and $\sigma = 25\%$. Over one year the index's median return is $e^{0.08 - 0.03125} - 1 = 5.0\%$, so an investor might expect roughly 10% from a two-times fund. The fund's median return is

$$e^{0.16 - 0.125} - 1 = 3.6\%$$

At $\sigma = 40\%$ the arithmetic turns brutal: the index's median return is $e^{0.08 - 0.08} - 1 = 0\%$, and the two-times fund's median is $e^{0.16 - 0.32} - 1 = -14.8\%$. In the median case the index goes nowhere and the leveraged fund loses fifteen percent, before financing, fees, and the cost of daily rebalancing.

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Nothing here is a defect in the product, which does precisely what its prospectus says: it delivers k times the return *each day*. The gap is a mathematical consequence of compounding a scaled-up volatility, and it is the same $\sigma^2/2$ that separates the mean from the median in Example 14.6. These funds are sound instruments for a position held for days and poor ones for a position held for years, and the boundary between the two is set by σ.

14.10 THE RISK-NEUTRAL PROCESS

Chapter 13 established that a derivative may be valued as though every asset earned the risk-free rate. In the language of this chapter that statement becomes a single substitution: under the risk-neutral measure Q (Section 13.6), with W^Q a Wiener process under Q,

$$dS = rS\,dt + \sigma S\,dW^Q \tag{14.15}$$

and therefore, repeating Section 14.9 with μ replaced by r,

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right), \qquad \mathbb{E}^Q[S_T] = S_0 e^{rT} \tag{14.16}$$

Equation (14.16) is the entire input to Chapter 15. Everything the Black–Scholes–Merton formula knows about the stock is contained in it.

Only the drift changes. The volatility σ is identical under both measures, as Section 13.8 found for the tree. The formal statement is Girsanov's theorem (a change to an equivalent measure adds a drift to a Wiener process and changes nothing else), which [TN14] sketches; the intuition is that changing the measure re-weights which paths are likely but cannot change how rough they are.

The correction term survives. Substituting r for μ does not remove the $-\sigma^2/2$, because that term came from Itô's lemma rather than from the drift. This is why the Black–Scholes–Merton formula contains $r - \sigma^2/2$; the correction is the Itô term of the logarithm, not a property of the expected return.

The connection back to Chapter 13 is exact. There, p made the discounted stock price a martingale (Section 13.6), the condition that the stock's expected growth be r. Here, $\mathbb{E}^Q[S_T] = S_0 e^{rT}$ is the same condition, written for a continuum of states.

14.11 TWO SOURCES OF UNCERTAINTY

Many instruments depend on more than one uncertain quantity: a quanto (a foreign asset's payoff converted at an exchange rate fixed in advance, Chapter 16) on both an asset and an exchange rate, a spread option (an option on the difference between two prices) on two prices. The apparatus needs one addition.

Let W_1 and W_2 be Wiener processes whose increments have correlation ρ, so that, by the argument behind (14.7),

$$dW_1\,dW_2 = \rho\,dt \tag{14.17}$$

alongside $dW_1^2 = dW_2^2 = dt$ from (14.7). For a function $f(x, y, t)$ of two Itô processes $dx = a_1 dt + b_1 dW_1$ and $dy = a_2 dt + b_2 dW_2$, Itô's lemma becomes

$$\begin{aligned} df = &\left(\frac{\partial f}{\partial x}a_1 + \frac{\partial f}{\partial y}a_2 + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}b_1^2 + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}b_2^2 + \frac{\partial^2 f}{\partial x\,\partial y}b_1 b_2 \rho\right)dt \\ &+ \frac{\partial f}{\partial x}b_1\,dW_1 + \frac{\partial f}{\partial y}b_2\,dW_2 \end{aligned} \tag{14.18}$$

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The only new term, the last in the drift, exists for the same reason as the two second-derivative terms: the cross-product $\Delta x\,\Delta y$ is of order Δt rather than Δt^2. It is also the source of the correlation term in Chapter 30's quanto adjustment. Example 14.7 applies it to a price ratio.

Example 14.7 The volatility of a ratio

Two assets follow geometric Brownian motions with expected returns μ_1 and μ_2, volatilities σ_1 and σ_2, and correlation ρ. What process does the ratio $G = S_1/S_2$ follow?

Applying (14.18) and collecting terms gives

$$\frac{dG}{G} = \left(\mu_1 - \mu_2 + \sigma_2^2 - \rho\sigma_1\sigma_2\right) dt + \sigma_1\, dW_1 - \sigma_2\, dW_2$$

so the ratio is itself a geometric Brownian motion, with volatility

$$\sigma_G = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{14.19}$$

Put numbers on it. With $\sigma_1 = 20\%$, $\sigma_2 = 12\%$, and $\rho = 0.4$:

$$\sigma_G = \sqrt{0.04 + 0.0144 - 2 \times 0.4 \times 0.20 \times 0.12} = \sqrt{0.0352} = 18.8\%$$

The ratio is *less* volatile than S_1 alone, because the two assets partly move together and the common movement cancels in the ratio. Reverse the correlation to $\rho = -0.4$ and $\sigma_G = \sqrt{0.0544 + 0.0192} = \sqrt{0.0736} = 27.1\%$, more volatile than either input.

Equation (14.19) is used more often than almost any other result in this chapter:

- It gives the volatility of a cross exchange rate from two rates against a common currency.
- It gives the volatility of a stock measured in a foreign currency, which Chapter 16 needs for composite options (options on a foreign stock whose payoff is converted at the expiration exchange rate). A product S_1S_2 is the ratio of S_1 to $1/S_2$, whose correlation with S_1 is $-\rho$, so the minus sign in (14.19) becomes a plus.

It is the formula for the standard deviation of a difference, which is what a ratio becomes after taking logarithms. That is the quickest way to remember it.

14.12 WHAT THE MODEL ASSUMES

Geometric Brownian motion is a modeling choice, not a fact. Section 15.1 collects the assumptions of the pricing model built on it and says where each fails; three ways real prices depart from it are worth having here. Volatility is not constant but clusters (calm follows calm and violent follows violent), with consequences Chapter 19 documents in the volatility surface. Prices are not continuous: the negative oil settlement of Section 5.9, the LME nickel suspension of Chapter 2 and any earnings announcement produce moves no continuous path can accommodate, and a model with jumps assigns much more probability to extreme outcomes than (14.11) does. And returns are independent across time in direction but not in magnitude: clustering violates the Markov property of Section 14.1 for the price alone, though not for price and volatility jointly, and Section 14.13 introduces the standard process whose increments are not independent. Of the two parameters, σ must be estimated (Chapter 15 takes up how), while Example 14.3 showed that μ effectively cannot be, which does not matter for option pricing and matters a great deal for everything else.

14.13 FRACTIONAL BROWNIAN MOTION

Rough volatility, the finding that the path of an asset's volatility is far more jagged than a Wiener path, together with the models built on that finding, appears in Sections 21.8 and 23.6 and in [TN21], so the process behind it is introduced here. It is also the standard process for which Section 14.1's variance rule fails.

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Definition. **Fractional Brownian motion** B_t^H is a continuous process, jointly normal at any set of dates, with $B_0^H = 0$, mean zero, and covariance

$$\operatorname{Cov}\left[B_s^H, B_t^H\right] = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right), \qquad 0 < H < 1 \tag{14.20}$$

Its increments are **stationary**, as a Wiener process's are: the variance of a change depends only on the length of the interval,

$$\operatorname{Var}\left[B_t^H - B_s^H\right] = |t-s|^{2H} \tag{14.21}$$

The exponent H is the **Hurst parameter**, after the hydrologist who first measured it in the flood records of the Nile. At $H = 1/2$ the covariance (14.20) is $\min(s,t)$, which is the Wiener process of Section 14.2: the variance over T is T and increments over disjoint intervals are independent.

The increments are correlated. Take two consecutive years. The change over the second year, $B_2^H - B_1^H$, has by stationarity the same variance as the change over the first, and the two-year variance is the sum of the two plus twice their covariance. With (14.21) on both sides, $2^{2H} = 1 + 1 + 2\rho$, so the correlation between successive equal increments is

$$\rho = 2^{2H-1} - 1 \tag{14.22}$$

which is -0.43 at $H = 0.1$, zero at $1/2$, and $+0.74$ at $H = 0.9$. Below one-half the process is **anti-persistent**: each move partly undoes the last. Above one-half it is **persistent**, with a long memory: a rise makes a further rise more likely. Section 14.1's argument that variances add needed exactly this cross term to vanish, and it vanishes only at one-half.

Scaling and roughness. Equation (14.21) says the process is **self-similar**: stretching time by a factor c scales the path's distribution by c^H, so an increment over a short interval Δt has standard deviation Δt^H. For the Wiener process that is $\sqrt{\Delta t}$, the scaling Section 14.2 built on. For $H < 1/2$, Δt^H is larger than $\sqrt{\Delta t}$ for small Δt, so the path moves more over short intervals and is *rougher*; for $H > 1/2$ it is smoother. Figure 14.5 shows the three cases built from the same random draws: the rough path reverses constantly, the persistent one trends, and the Wiener path sits between them.

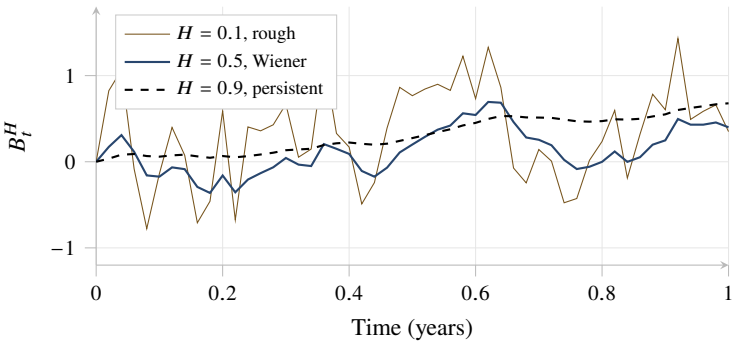


Figure 14.5: Fractional Brownian motion at $H = 0.1$, 0.5 and 0.9, built from the same random draws; each has unit variance at one year.

The roughness matters for the calculus of this chapter. Section 14.7 rested on $dW^2 = dt$: the statement that the squared increments of a Wiener path over $[0,T]$ add up to T. For fractional Brownian motion the n squared increments over a grid of step T/n add up to about

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$n\,(T/n)^{2H} = T^{2H}\,n^{1-2H}$, which grows without bound for $H < 1/2$ and shrinks to zero for $H > 1/2$. There is no $dB^2 = dt$, and Itô's lemma in the form of Section 14.7 does not apply. Nor is the process Markov: by (14.20) the next increment is correlated with every past increment, so the current value does not summarize the path and forecasting needs the whole history.

Example 14.8 Variances that do not add

Take $H = 0.1$, the value Chapter 21 estimates for log-volatility, and unit variance over one year.

Two years. By (14.21) the two-year variance is $2^{0.2} = 1.15$, against 2 for a Wiener process, and the two-year standard deviation is $2^{0.1} = 1.07$ against $\sqrt{2} = 1.41$. Doubling the horizon adds 7% to the uncertainty, not 41%, because the second year's move is correlated at −0.43 with the first and partly undoes it.

Annualizing daily data. Section 14.6 turned a daily standard deviation into an annual one by multiplying by $\sqrt{252}$. For a process with Hurst parameter H the factor is 252^H instead: $252^{0.1} = 1.74$ at $H = 0.1$, against $252^{0.5} = 15.9$. Applying the square-root rule to such a series would overstate the one-year standard deviation by $252^{0.4} = 9.1$ times. The same arithmetic at $H = 0.9$ understates it by the same factor. The square-root-of-time rule is the $H = 1/2$ case of T^H, and it is only as good as the independence it assumes.

Built from a Wiener process. One standard construction writes the process as a weighted sum of past Wiener increments,

$$B_t^H = \sqrt{2H} \int_0^t (t-s)^{H-1/2}\, dW_s \tag{14.23}$$

The kernel $(t-s)^{H-1/2}$ weights each past shock by its age. At $H = 1/2$ it is one, and the integral is W_t. Below one-half the exponent is negative and the weight on the most recent shocks rises without bound as $s \to t$, which is where the roughness comes from, while the weight on old shocks decays as a power of their age rather than dying off exponentially, which is where the memory comes from. The $\sqrt{2H}$ makes the variance at each date exactly t^{2H}, but this version, started at zero, has stationary increments only approximately; it is the Riemann–Liouville form, a close relative. Mandelbrot and Van Ness's two-sided integral has the exact covariance (14.20) and stationary increments, and nothing in this book turns on the difference. Section 21.8 uses this kernel, with the integral inside an exponential and a volatility-of-volatility in front.

What it can and cannot model. As a model for a traded price it fails, and for a reason the correlation (14.22) makes plain: a process whose next move is predictable from its last can be traded on. With $H > 1/2$ a rise predicts a rise, with $H < 1/2$ a rise predicts a fall, and in either case a strategy that trades on the prediction at ever shorter intervals earns a profit with a risk that shrinks toward zero. Trading gains cannot be defined as Itô integrals against it (in the technical term, it is not a semimartingale), no risk-neutral measure exists, and the apparatus of Chapter 15 onward has nothing to attach to. It is therefore not a candidate to replace (14.4). Mandelbrot proposed it for prices in the 1960s, with $H > 1/2$, and the proposal did not survive. The arbitrage is fragile in practice: Cheridito (2003) shows that a minimal delay between trades removes it, and Guasoni (2006) that proportional transaction costs of any size do, so the process is not absurd as a description of prices; it is merely useless as a pricing model. A "fractional Black–Scholes" formula exists and is sometimes quoted. It is not a price, its implied volatilities look nothing like a market surface, and no strategy replicates the option at that price; Section 8 of [TN21] explains all three.

Volatility is different. Spot volatility is not itself a traded asset, so no arbitrage argument rules out a rough path for it, and nothing forbids modeling it with a process that would be

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arbitrageable as a price. Gatheral, Jaisson, and Rosenbaum (2018) measured the roughness of realized volatility across many assets and found $H \approx 0.1$, far rougher than a Wiener path, and Section 21.8 shows what follows for option prices: the one number reproduces a feature of the volatility surface that two decades of classical models could not, and Example 14.8's arithmetic is the reason.

Simulating it. A Wiener path is simulated one independent increment at a time, and a fractional path cannot be, because the increments are not independent. The exact method writes down the covariance matrix (14.20) on the time grid, takes its Cholesky factor (the lower-triangular matrix L with $LL^\top$ equal to the covariance matrix), and multiplies a vector of standard normal draws by it, which is how Figure 14.5 was made, and why its three paths are the same randomness seen through three kernels. Each path costs the square of the number of steps and the factorization its cube, and Chapter 21's case study on rough volatility explains what replaced the method in production.

SUMMARY

A Markov process is one in which only the current value matters for the future. With independent changes over separate periods, variances add, so uncertainty grows with $\sqrt{T}$ rather than T. That is why volatility is scaled by $\sqrt{\Delta t}$ and why Chapter 13's tree used $u = e^{\sigma\sqrt{\Delta t}}$.

A Wiener process has increments $\Delta W = \varepsilon\sqrt{\Delta t}$ with ε standard normal and non-overlapping increments independent, so $W(T) - W(0) \sim N(0, T)$. Its paths are continuous everywhere and differentiable nowhere, which is why a different calculus is required.

A generalized Wiener process adds a constant drift and variance rate, $dx = a\,dt + b\,dW$; an Itô process lets both depend on the state and time.

Stock prices are modeled by geometric Brownian motion, $dS = \mu S\,dt + \sigma S\,dW$, because returns rather than dollar changes are what scale with the price level, and because the proportional form keeps the price positive. Over short intervals the noise term dominates the drift by an order of magnitude, which is why μ is hard to estimate; the hedge of Section 14.8 removes it from pricing.

Itô's lemma gives the process followed by a function $f(S, t)$. It differs from ordinary calculus by one term, $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2\,dt$, which survives because $dW^2 = dt$. That term is proportional to convexity and reappears as gamma in Chapter 18 and as the convexity adjustment in Chapter 30.

Applying the lemma to $\ln S$ gives $d\ln S = (\mu - \sigma^2/2)dt + \sigma\,dW$, so $\ln S_T$ is normal and S_T is lognormal. The expected price grows at μ while the median grows only at $\mu - \sigma^2/2$, so the gap widens with volatility and a volatile stock finishes below its own expected value more often than not. Confusing the two rates produces real and common errors in return reporting and product design.

Because a derivative and its stock are driven by the same dW, a position long $\partial f/\partial S$ shares and short the derivative carries no randomness, and μ cancels with it: Chapter 13's hedge in continuous time. Under the risk-neutral measure only the drift changes, from μ to r; the volatility carries over and the $-\sigma^2/2$ survives.

The model assumes constant volatility, continuous paths, independent returns, and known parameters. All four fail; knowing which one a given phenomenon violates identifies where in Parts III and IV its repair is found. Fractional Brownian motion is the standard process without independent increments: its variance over T is T^{2H}, so variances no longer add, and below $H = 1/2$ its paths are rougher than a Wiener path. It cannot model a traded price without admitting arbitrage, but it can model volatility, which Chapter 21 does with H far below $1/2$.

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FURTHER READING

The derivation of Itô's lemma is in [TN14], reproduced in Appendix A, which also makes the $dW^2 = dt$ argument precise. Beyond that, Neftci's introduction to the mathematics of financial derivatives bridges this level and a rigorous treatment, and Baxter and Rennie's account is a short exposition of why the change of measure works. Karatzas and Shreve's *Brownian Motion and Stochastic Calculus* is the standard graduate text on stochastic calculus, for a reader who wants the construction of the Itô integral itself rather than the rule that follows from it. For the empirical failures of geometric Brownian motion, Chapter 19's references are the place to start, since they organize the evidence around what the option market prices. Fractional Brownian motion entered finance through Mandelbrot and Van Ness (1968), who introduced the process and the Hurst parameter; Rogers (1997) is the short proof that it admits arbitrage as a price process, and Cheridito (2003) and Guasoni (2006) show the arbitrage vanishing under a minimal trading delay and under transaction costs respectively; the literature on the fractional Black–Scholes formula is cited in [TN21]. Gatheral, Jaisson, and Rosenbaum (2018) is the paper that found the process, with $H \approx 0.1$, in volatility.

PRACTICE QUESTIONS

14.1 A variable's change over one year has standard deviation 5. What is the standard deviation of its change over three years, and over one month? State the property you used.

14.2 A variable follows a generalized Wiener process with drift 3 per year and variance rate 16 per year, starting at 25. Give its distribution after two years and compute the probability that it is negative.

14.3 Explain why $dS = \mu\,dt + \sigma\,dW$ is unsuitable as a model for a stock price, giving two distinct reasons.

14.4 A stock at \$80 has $\mu = 12\%$ and $\sigma = 25\%$. Give the distribution of the proportional change over two weeks, and compare the sizes of the drift and noise terms.

14.5 Using Itô's lemma, find the process followed by $G = S^2$ when S follows (14.4).

14.6 A stock at \$60 has $\mu = 14\%$ and $\sigma = 35\%$. Compute the expected price and the median price after two years, and explain why they differ.

14.7 For the stock in the previous question, find the range within which the price after two years lies with 95% probability.

14.8 Explain in words why the term $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2$ appears in Itô's lemma but not in the ordinary chain rule.

14.9 State the Markov property and explain why it is an assumption about information rather than about volatility. Explain what weak-form market efficiency has to do with it, and give one observed feature of returns that sits uneasily with it.

14.10 A stock has expected return μ. Explain why the expected continuously compounded return per year is $\mu - \sigma^2/2$ rather than μ, and state which of the two a manager quoting the arithmetic average of past annual returns is reporting.

14.11 Explain why Itô's lemma delivers a hedge and not merely a change-of-variable formula. State which term in the lemma the hedge removes, which term survives it, and why the surviving term is what the option is paid for.

14.12 State the risk-neutral process for a stock price. Explain what changes and what does not when moving to it from the real-world process, and why the volatility is untouched while the drift is not.

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FURTHER QUESTIONS

14.13 Example 14.6 found that a stock finishes below its expected value 58% of the time. Derive a general expression for $\Pr(S_T > \mathbb{E}[S_T])$ in terms of σ and T alone, show that it does not depend on μ, and evaluate it for $\sigma\sqrt{T} = 0.2$, 0.6, and 1.0. What happens as volatility rises?

14.14 A fund reports the arithmetic average of its annual returns over ten years as 12%. Explain what an investor's compounded return would have been if the volatility was 30%, state which of the two figures is the relevant one for an investor, and identify the circumstances in which the gap is largest.

14.15 Apply Itô's lemma to the forward price when the underlying pays a continuous dividend yield q, and show that under risk-neutral valuation the forward's drift is again zero. Explain the connection to Section 13.10.

14.16 Section 14.12 notes that volatility clustering violates the Markov property. Construct an argument that this does *not* contradict weak-form market efficiency, and identify precisely what a trader could and could not profit from if returns were unpredictable but their magnitudes were not.

14.17 Section 14.13 states that a change in fractional Brownian motion over any interval of length T has variance T^{2H}. Extend the argument that gives the correlation $2^{2H-1} - 1$ between consecutive yearly increments to the first year's increment and the increment over year $k+1$, and show that the correlation is $\tfrac{1}{2}\left[(k+1)^{2H} - 2k^{2H} + (k-1)^{2H}\right]$. Evaluate it for $k = 1, 5, 20$ at $H = 0.9$, and explain why a decay of order k^{2H-2}, slower than exponential, is the long memory the section names.

14.18 The Itô correction term is proportional to $\partial^2 f / \partial S^2$. Using this, explain without calculation whether the correction raises or lowers the drift of a long call position, and connect your answer to the statement in Chapter 12 that long-volatility positions are hurt by the passage of time.

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Chapter 15
The Black–Scholes–Merton Model

Chapter 13 established the principle: a derivative that can be replicated has a price independent of anyone's beliefs. Chapter 14 supplied the mathematics for carrying that principle into continuous time, and ended one step short of the result. This chapter takes the step.

What emerges is a closed-form formula for European options—the most widely used equation in finance, and the one that made options a mass market rather than a specialist's trade. Its importance is not that it is right. Chapter 19 documents at length that it is not, in a specific and measurable way. Its importance is that it turned option pricing into a calculation with five inputs, of which four are read off a screen and the fifth became a traded quantity in its own right.

Sections 15.1 to 15.3 are the assumptions, the differential equation, and the formula. Sections 15.4 and 15.5 say what its terms mean and how it behaves. Sections 15.6 and 15.7 are the fifth input, volatility, estimated from history and implied from prices; Sections 15.8 to 15.10 are the formula in practice, with dividends, and for American options. Section 15.11 is what the model gets wrong and Section 15.12 where it came from.

Section 15.2 can be read for its conclusions alone; the formula in Section 15.3 is self-contained.

15.1 THE ASSUMPTIONS

The model rests on assumptions that Chapter 14 has already named. They are collected here because knowing them is how one knows when the answer will be wrong.

1. The stock follows geometric Brownian motion with constant μ and σ.
2. Short selling is permitted, with full use of the proceeds.
3. There are no transaction costs or taxes, and securities are perfectly divisible.
4. No dividends are paid over the option's life (relaxed in Section 15.9).
5. There are no arbitrage opportunities.
6. Trading is continuous.
7. The risk-free rate r is constant and the same for borrowing and lending.

Every one of these fails in reality, and the book has already said where. Assumption 1 fails through volatility clustering and jumps (Section 14.12; Chapter 19 measures what they do to option prices). Assumptions 2 and 7 fail in the way Section 5.15 documented at length: borrowing is not at r, and shorting is neither free nor always possible. Assumption 3 is what Chapter 18 is about. Assumption 6 fails whenever a market closes or a stock is suspended.

The list is not a disclaimer; it is a diagnostic tool. When a market price disagrees with the formula, the useful question is which of the seven has broken, because that identifies the repair.

15.2 THE BLACK–SCHOLES–MERTON DIFFERENTIAL EQUATION

Section 14.8 supplies the derivation. Take a stock following

$$dS = \mu S\,dt + \sigma S\,dW$$

and a derivative on it whose price at time t can be expressed as $f(S,t)$, a function of the current stock price and time alone. Itô's lemma gives

$$df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW$$

and a portfolio short one derivative and long $\partial f/\partial S$ shares has no dW term at all. Writing Π for its value,

$$\Pi = -f + \frac{\partial f}{\partial S}S, \qquad d\Pi = \left(-\frac{\partial f}{\partial t} - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt$$

The portfolio is riskless over dt, so it must earn r, or Section 13.1's arbitrage is available. Setting $d\Pi = r\Pi\,dt$ and rearranging gives the **Black–Scholes–Merton differential equation**:

$$\boxed{\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} = rf} \tag{15.1}$$

Three observations matter more than the derivation.

μ is absent. The stock's expected return entered through Itô's lemma and canceled when the hedge was imposed. This is Chapter 13's central result (the real-world probability is irrelevant) appearing in continuous time. Two investors who disagree completely about where the stock is going must still agree on the option's value.

The equation is the same for every derivative whose value depends only on S and t. Nothing in the derivation used the payoff, so forwards, calls, puts, digitals, and the barrier options of Chapter 34 all satisfy (15.1). What distinguishes them is the **boundary condition**, the value at $t = T$:

European call: $f = \max(S-K,0)$, forward: $f = S-K$, digital: $f = \mathbf{1}\{S > K\}$

The instrument is the boundary condition, not the equation. The qualification "only on S and t" matters. A payoff that depends on the path (an average, a running maximum) needs an extra state variable and the equation gains a term; an American option satisfies it only in the region where holding beats exercising, which the free boundary of Section 15.10 encloses; and a derivative whose value also moves with volatility or interest rates needs those as variables too. Each of those is a different equation, but the argument that produces it is this one.

The hedge is instantaneous. The portfolio is riskless over dt only, because $\partial f/\partial S$ changes as S and t change. Section 14.8 flagged this and Chapter 18 confronts its cost.

15.3 THE FORMULA

Solving (15.1) subject to the European call's boundary condition is a piece of applied mathematics carried out in [TN15], reproduced in Appendix A. The route taken there is the one Chapter 13 makes natural: rather than solving the equation directly, take the risk-neutral expectation of the payoff.

Under the risk-neutral measure, Section 14.10 gives

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right)$$

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and taking the expected payoff $\max(S_T - K, 0)$ under that distribution and discounting it at r produces the **Black–Scholes–Merton formulas**, which the market calls simply Black–Scholes:

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{15.2}$$

$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1) \tag{15.3}$$

where $N(\cdot)$ is the standard normal cumulative distribution function and

$$d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{15.4}$$

The formula takes five inputs: S_0, K, T, r, and σ. Four are observable; the fifth is not, and Sections 15.6 and 15.7 are about it.

Two routes to the same answer

There are two ways to get from (15.1) to (15.2), and both are worth recognizing because different parts of the book use different ones.

The **differential-equation route** solves (15.1) directly, subject to $f = \max(S - K, 0)$ at T. A change of variables turns it into the heat equation of physics, whose solution has been known since the nineteenth century. This is Black and Scholes's original route, and it is the one that generalizes to the numerical methods of Chapter 20.

The **risk-neutral route** takes the discounted expectation instead:

$$c = e^{-rT} \int_K^\infty (S_T - K)\, g(S_T)\, dS_T$$

where g is the lognormal density of (14.16). Splitting the integrand,

$$c = \underbrace{e^{-rT} \int_K^\infty S_T\, g\, dS_T}_{S_0 N(d_1)} - \underbrace{K e^{-rT} \int_K^\infty g\, dS_T}_{K e^{-rT} N(d_2)}$$

The second integral is $Q(S_T > K)$, the source of $N(d_2)$; the first is an expectation of S_T restricted to the exercise region, the source of $N(d_1)$. [TN15] evaluates both.

This route is the one to remember, because it makes the formula's two terms mean something (Section 15.4 unpacks them), and because it is the route that extends to every payoff in Chapter 34 without modification.

A check on the equation. It is worth confirming that (15.1) really does govern instruments other than options. Take a forward contract with delivery price K, worth $f = S - K e^{-r(T-t)}$ by Section 5.5. Then

$$\frac{\partial f}{\partial t} = -rK e^{-r(T-t)}, \qquad \frac{\partial f}{\partial S} = 1, \qquad \frac{\partial^2 f}{\partial S^2} = 0$$

and substituting into the left-hand side of (15.1),

$$-rK e^{-r(T-t)} + rS \times 1 + 0 = r\left(S - K e^{-r(T-t)}\right) = rf$$

The equation is satisfied identically. The forward contract's value from Chapter 5, derived there from a cash-and-carry argument with no model at all, is a solution of the Black–Scholes–Merton equation, which it must be, since both rest on the same no-arbitrage principle.

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Example 15.1 Pricing a call and a put

A non-dividend stock trades at \$58. Value six-month options struck at \$55 with $r = 4\%$ and $\sigma = 24\%$.

First $\sigma\sqrt{T} = 0.24\sqrt{0.5} = 0.16971$, and

$$d_1 = \frac{\ln(58/55) + (0.04 + 0.0288) \times 0.5}{0.16971} = \frac{0.053110 + 0.034400}{0.16971} = 0.5157$$

$$d_2 = 0.5157 - 0.1697 = 0.3460$$

From normal tables, $N(0.5157) = 0.69697$ and $N(0.3460) = 0.6353$. Then

$$c = 58 \times 0.69697 - 55e^{-0.02} \times 0.6353 = 40.42 - 34.25 = \$6.17$$

and for the put, $N(-d_1) = 0.30303$, $N(-d_2) = 0.3647$:

$$p = 55e^{-0.02} \times 0.3647 - 58 \times 0.30303 = 19.66 - 17.58 = \$2.08$$

Always check parity. From Section 11.4, $c - p$ must equal $S_0 - Ke^{-rT}$:

$$6.1731 - 2.0840 = 4.0891, \qquad 58 - 55e^{-0.02} = 58 - 53.9109 = 4.0891$$

They agree exactly, and they must: the formulas satisfy parity identically, because $N(d)+N(-d) = 1$ for every d. A discrepancy in a spreadsheet is an arithmetic error, not a modeling question, and this is the fastest test of any implementation.

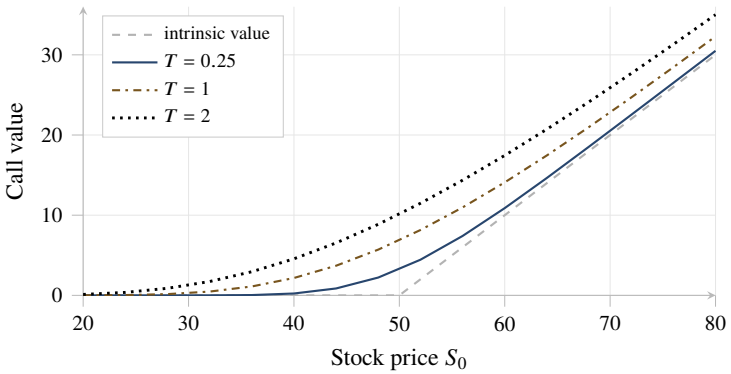


Figure 15.1: Call value against the stock price. $K = 50$, $r = 4\%$, $\sigma = 30\%$.

Figure 15.1 plots (15.2) against S_0 for three maturities. This is Figure 10.2 with a formula behind it: the curves lie above intrinsic value by the time value, the gap is widest near the money, and it shrinks toward zero as $T \to 0$. The model has not produced a new shape—it has produced a number for a shape Chapter 10 already described.

15.4 WHAT $N(d_1)$ AND $N(d_2)$ MEAN

The formula is more memorable, and more useful, once its terms are read rather than computed.

Rewrite (15.2) as

$$c = e^{-rT}\Big[\underbrace{S_0 e^{rT} N(d_1)}_{\text{expected stock received}} - \underbrace{K\,N(d_2)}_{\text{expected strike paid}}\Big]$$

Under the risk-neutral measure, the terms read as follows:

$N(d_2)$ **is the probability that the option is exercised.** It is $Q(S_T > K)$, the risk-neutral probability of finishing in the money. In Example 15.1, $N(d_2) = 0.6353$, so the market prices a 64% chance of exercise. This is not the real-world probability, for the reason Chapter 13 gave:

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Q weights states so that the stock earns r, not μ. A stock with a high expected return has a higher true probability of exercise than $N(d_2)$ suggests.

$N(d_1)$ **is the option's delta**, the hedge ratio $\partial c/\partial S$ that Sections 13.1 and 14.8 built the entire theory around. In the example it is 0.6970, so replicating the call requires holding 0.6970 shares per option written. Chapter 18 takes this as its starting point.

$S_0N(d_1)$ **is the present value of receiving the stock in the states where the option is exercised.** It is not S_0 times the exercise probability. The option is exercised only in the high-price states, those with $S_T > K$, so the stock's average value across those states exceeds its average over all states, the forward price S_0e^{rT}; $N(d_1)$ is $N(d_2)$ scaled up by that ratio, which is why $N(d_1) > N(d_2)$.

That last point resolves a persistent confusion. Only $N(d_2)$ is the exercise probability under the risk-neutral measure; $N(d_1)$ is the exercise probability under the measure that uses the stock as numeraire ([TN15]; Chapter 28).

15.5 PROPERTIES OF THE FORMULA

Any formula should be tested at the edges. Equation (15.2) passes every test, and each reproduces a result Chapter 11 obtained *without* a model.

Deep in the money. As $S_0 \to \infty$, both d_1 and $d_2 \to \infty$, so $N(d_1), N(d_2) \to 1$ and

$$c \to S_0 - Ke^{-rT}$$

The call becomes a forward: exercise is certain, so the holder is committed to buy at K.

Deep out of the money. As $S_0 \to 0$, both arguments $\to -\infty$ and $c \to 0$.

Zero volatility. As $\sigma \to 0$, the option's outcome is certain. Then $d_1, d_2 \to \pm\infty$ according to whether S_0 exceeds Ke^{-rT}, and

$$c \to \max\left(S_0 - Ke^{-rT},\, 0\right)$$

which is exactly the lower bound (11.1) of Section 11.2.

Infinite volatility. As $\sigma \to \infty$, $d_1 \to +\infty$ while $d_2 \to -\infty$, so $N(d_1) \to 1$, $N(d_2) \to 0$, and

$$c \to S_0$$

the upper bound of Section 11.2. The formula cannot be made to violate the bounds however the inputs are abused.

At expiration. As $T \to 0$, $c \to \max(S_0 - K, 0)$.

The put runs the same checks in mirror image. As $S_0 \to \infty$, both $N(-d_1)$ and $N(-d_2)$ vanish and $p \to 0$; as $S_0 \to 0$ they tend to one and $p \to Ke^{-rT} - S_0 \to Ke^{-rT}$, the put's upper bound of Section 11.2, which is also its value as $\sigma \to \infty$. As $\sigma \to 0$, $p \to \max(Ke^{-rT} - S_0, 0)$, the put's lower bound (11.2), and as $T \to 0$, $p \to \max(K - S_0, 0)$. None of these needs a separate calculation: since (15.3) follows from (15.2) by parity, each put limit is the corresponding call limit plus $Ke^{-rT} - S_0$.

The agreement with Chapter 11 is not a coincidence. Those bounds followed from no-arbitrage alone; this formula followed from no-arbitrage *plus* a distributional assumption. A model that added an assumption and thereby violated a conclusion reachable without it would be internally inconsistent. Extra assumptions may narrow a range of admissible prices; they must never move it.

Computing the normal distribution function

Equation (15.2) requires $N(x)$, which has no elementary closed form. Pricing libraries compute it as $\operatorname{erfc}(-x/\sqrt{2})/2$ from the double-precision complementary error function, to machine

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precision. By hand or in a spreadsheet a polynomial serves; a standard one, accurate to 7.5×10^{-8}, sets $\gamma = 0.2316419$ and $k = 1/(1 + \gamma x)$ for $x \geq 0$, then

$$N(x) \approx 1 - \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \left(a_1 k + a_2 k^2 + a_3 k^3 + a_4 k^4 + a_5 k^5 \right)$$

with $a_1 = 0.319381530$, $a_2 = -0.356563782$, $a_3 = 1.781477937$, $a_4 = -1.821255978$, $a_5 = 1.330274429$. For negative arguments, $N(-x) = 1 - N(x)$.

That accuracy is far beyond what pricing near the money requires (a bid–ask spread is typically five orders of magnitude wider), but not in the far wings, where $N(-d)$ is around 10^{-6}, nor when implied volatility is computed by iteration, where an error in N is divided by the option's sensitivity to volatility, which is small far from the money.

15.6 ESTIMATING VOLATILITY FROM HISTORY

Volatility must be estimated, and the two ways to do it answer different questions.

The historical estimator. Take $n + 1$ successive closing prices $S_0, S_1, \ldots, S_n$ and form the log returns $u_i = \ln(S_i/S_{i-1})$. The sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (u_i - \bar{u})^2} \tag{15.5}$$

where $\bar{u}$ is the mean of the u_i, estimates volatility per observation interval, and is annualized by multiplying by $\sqrt{252}$ for daily data, $\sqrt{52}$ for weekly, $\sqrt{12}$ for monthly.

Table 15.1: Ten daily returns.

Day	Close	u_i	Day	Close	u_i
0	50.00	—	6	51.10	+0.013793
1	50.75	+0.014889	7	51.85	+0.014570
2	50.20	−0.010897	8	51.60	−0.004833
3	51.30	+0.021676	9	52.30	+0.013475
4	51.05	−0.004885	10	52.00	−0.005753
5	50.40	−0.012814			

Example 15.2 Computing a historical volatility

From Table 15.1, $\sum u_i = 0.039221$ and $\sum u_i^2 = 0.00163889$. Then

$$s^2 = \frac{1}{9} \left[0.00163889 - \frac{(0.039221)^2}{10} \right] = 0.000165007$$

so $s = 0.012846$ per day, and

$$\sigma = 0.012846\sqrt{252} = 0.012846 \times 15.8745 = 20.4\%$$

Three cautions apply.

Ten observations is far too few. The standard error of a volatility estimate is approximately $\sigma/\sqrt{2n}$, here $20.4/\sqrt{20} = 4.6$ percentage points. A two-standard-error band runs from about 11% to 30%, which is not a usable input. Ninety to 180 observations is the usual compromise: long enough to reduce the error, short enough that volatility has not yet changed.

Trading days, not calendar days. Volatility accumulates while markets are open, not while time passes: French and Roll (1986) found the variance over a weekend far below three days' worth. So the annualization factor is $\sqrt{252}$; using 365 would overstate volatility by about 20%.

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The mean is usually dropped. Over short intervals the drift is negligible next to the noise (Example 14.3 showed it is outweighed about thirteen and a half to one over a week), so setting $\bar{u} = 0$ in (15.5), and dividing by n rather than $n - 1$ since no mean is now estimated, typically improves the estimate by removing a noisily estimated parameter. Here it gives $s = 0.012802$, changing the annualized figure by 0.1 percentage points.

What an error in volatility costs

The standard error above is easy to accept and easy to underrate; Table 15.2 prices it.

Table 15.2: Value of the call of Example 15.1 at different volatilities.

σ	Call value	Difference	Percent
14%	4.8292	−1.3440	−21.8%
19%	5.4736	−0.6995	−11.3%
24%	6.1731	—	—
29%	6.9003	+0.7272	+11.8%
34%	7.6428	+1.4697	+23.8%

The outer rows are roughly two standard errors of a ten-observation estimate on either side of 24% (the same $\pm 2\sigma/\sqrt{2n}$ band Example 15.2 computed, here about ±10 points): a 14% to 34% range for volatility corresponds to a \$4.83 to \$7.64 range for the option. The historical estimate from ten observations does not determine the price to better than about ±20%.

Two conclusions follow. Historical volatility needs far more data than a casual calculation suggests. And the relationship is close to linear over this range, which is the sensitivity Chapter 18 names vega and which Section 15.7 exploits to invert the formula.

The deeper objection to historical volatility is not statistical. It is that the estimator answers the wrong question: it measures what the stock *did*, and an option's value depends on what it *will do*.

15.7 IMPLIED VOLATILITY

The alternative inverts the problem. Take the option's market price as given and ask what volatility makes (15.2) reproduce it. That number is the **implied volatility**.

The inversion is well posed because a call's value is increasing in σ, as Section 11.1 argued without a model, and strictly so in the formula, whose vega is positive for every $\sigma > 0$; Figure 15.2 shows it for the call of Example 15.3. A strictly increasing function takes each value at most once, so an implied volatility is unique, and it exists whenever the price lies strictly between the bounds of Section 11.2. No closed form exists, so it is found numerically, which on modern hardware is instantaneous.

Figure 15.2 also shows a feature that matters in practice. Below about 5% volatility the curve is indistinguishable from the lower bound $S_0 - Ke^{-rT} = 1.149$: the call is in the money, and with so little volatility it almost never finishes out of the money, so extra volatility adds almost nothing. The function is still strictly increasing, but it is nearly flat there, so a small error in the price moves the implied volatility a long way. The put at the same strike has the same vega, so switching to it does not steepen the curve; what it changes is the price being read. Most of the in-the-money call's price is the lower bound, and only the remainder carries information about volatility, whereas the out-of-the-money put's price is all time value and usually trades on a narrower spread. That is why implied volatility at each strike is read from the out-of-the-money option, here the put, which by parity implies the same volatility as the call.

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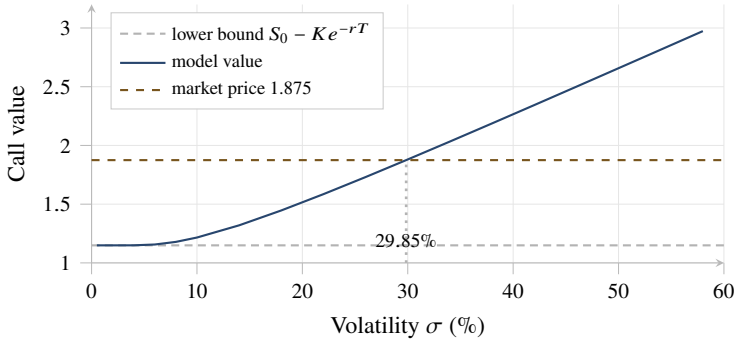


Figure 15.2: Call value against volatility, and the implied volatility. $S_0 = 21$, $K = 20$, $r = 3\%$, $T = 0.25$.

Example 15.3 Backing out an implied volatility

A three-month call on a \$21 stock, struck at \$20 with $r = 3\%$, trades at \$1.875.

Try $\sigma = 20\%$: the formula gives \$1.5148, too low. Try 30%: \$1.8807, slightly too high. Try 29%: \$1.8429. Narrowing further gives

$$\sigma_{\text{implied}} = 29.85\%$$

The market price is consistent with 29.85% volatility over the next three months. Had this stock's own history given a figure near 20%, as it did for the stock of Example 15.2, two readings would be possible: the market anticipates something the past does not contain (an earnings announcement, a pending decision), or option buyers are paying the **volatility risk premium**, the amount by which implied volatility tends to exceed the volatility later realized, which Chapter 19 documents. Usually both are true.

No one used (15.2) to compute a price here; the price was given, and the formula converted it into a volatility. That is what it is mostly used for in practice, and Section 10.6 already reported the consequence: dealers quote in volatility because volatility is the coordinate in which prices are stable.

The formula as a language. This reverses the usual account of what the formula is for. As a device for computing prices from a volatility forecast, it is unreliable, because volatility is neither constant nor reliably forecastable over an option's life. As a device for converting between prices and volatilities (a common language in which options of different strikes and maturities can be compared), it is indispensable, and its failures are informative rather than fatal.

Chapter 19 shows that implied volatility varies systematically with strike and maturity, so the model is wrong; the market uses it anyway, because a wrong but agreed transformation is still a usable quoting convention. The VIX of Chapters 19 and 22 carries the same idea to an index: one volatility number read from a strip of option prices.

15.8 USING THE FORMULA IN PRACTICE

Three conventions decide whether two people pricing the same option agree.

Which interest rate? The model says "the" risk-free rate, and Section 5.15 explained why no single such rate exists: which one a participant faces depends on who they are and what they can pledge. In practice, the rate used is the one at which the hedge can be financed over the option's life, which for a dealer's OTC trade, cleared or collateralized under a CSA, is the collateral rate: the OIS rate of Chapter 4. For listed options, desks commonly take r and q

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(with any borrow cost) jointly from the implied forward of Section 11.6 and the box-spread rate of Section 11.7. Using a government yield instead is a small error for short-dated equity options and a large one for long-dated rate derivatives, which is why Chapters 7 and 8 built the discounting apparatus they did.

Which maturity? Section 15.6 annualized volatility using *trading* days, and the two clocks must be kept apart: variance accrues on trading days, interest on calendar days. An option expiring in 30 calendar days has standard deviation $\sigma\sqrt{21/252}$ over its life but discount factor $e^{-30r/365}$, so d_1 carries rT_{cal} beside $\sigma^2 T_{\text{trd}}/2$. Mixing the conventions is a common implementation error. On average the two agree, since $252/365 \approx 21/30$, but for an option with a week or two to run the placement of weekends and holidays inside its life moves the variance by several percent. Some desks go further and weight days unequally (Chapter 23's business time: a day containing an earnings announcement carries more variance than a quiet Tuesday), which is a refinement of the same idea.

Which volatility? This is the substantive one. The formula assumes a single σ governs the stock, so any option on it should use the same number. The market disagrees: implied volatilities differ across strikes and maturities, as Section 15.11 notes and Chapter 19 documents.

The working resolution is to abandon the pretense of one volatility and use, for each option, the implied volatility of that option, interpolated from the traded surface if the exact strike and maturity are not quoted. The formula is then a quoting convention, strike by strike, rather than a model of the stock. A desk that prices a book with a single volatility is not using Black–Scholes conservatively; it is using a model the market has already rejected.

15.9 DIVIDENDS

Assumption 4 can be relaxed in two ways, according to how dividends are described.

A continuous dividend yield. If the asset pays a known yield q, its holder receives q and the option holder does not. Section 5.4 showed the effect: the asset behaves like one worth $S_0 e^{-qT}$ today. Replacing S_0 by $S_0 e^{-qT}$ throughout gives

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{15.6}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This single substitution covers stock indices (q = dividend yield), currencies ($q = r_f$, the foreign rate), and futures ($q = r$), and Chapter 16 develops each; Example 15.4 applies it to an index.

The differential equation changes as well, and the change explains the substitution. The shares in the hedge portfolio now pay dividends of

$$qS\frac{\partial f}{\partial S}\,dt$$

over each interval, which the riskless return must include, so the argument of Section 15.2 gives

$$\frac{\partial f}{\partial t} + (r - q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf \tag{15.7}$$

The asset's risk-neutral drift is $r - q$ while discounting is still at r: the option holder forgoes the yield, so the price the option sees grows more slowly than the riskless rate. Solving (15.7)

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with the call's boundary condition gives (15.6), and setting $q = r$ removes the first-derivative term altogether, which is the form Chapter 16 uses for options on futures.

Example 15.4 An index option

An index stands at 1,500 with a 2.5% dividend yield. Value a six-month call struck at 1,550 with $r = 4\%$ and $\sigma = 18\%$.

$$d_1 = \frac{\ln(1500/1550) + (0.04 - 0.025 + 0.0162) \times 0.5}{0.18\sqrt{0.5}} = -0.135056, \qquad d_2 = -0.262335$$

$$c = 1500e^{-0.0125} \times 0.446284 - 1550e^{-0.02} \times 0.396532 = 661.11 - 602.45 = 58.66$$

At the \$100 multiplier of Section 10.3, one contract costs \$5,866.

Here $N(d_2) = 0.3965$: the market prices a 40% risk-neutral chance that the index gains more than 3.3% in six months.

Known cash dividends. For an individual stock the dividend is a known amount on a known date, not a yield. The standard treatment subtracts the present value of the dividends payable during the option's life and applies (15.2) to what remains (Example 15.5).

Example 15.5 A call on a dividend-paying stock

A stock trades at \$50 and will pay \$0.80 in two months and \$0.80 in five months. Value a six-month call struck at \$50 with $r = 4\%$ and $\sigma = 30\%$.

The present value of the dividends is

$$0.80e^{-0.04(2/12)} + 0.80e^{-0.04(5/12)} = 0.7947 + 0.7868 = \$1.5815$$

so the adjusted stock price is $S^* = 50 - 1.5815 = \$48.4185$. Applying (15.2) to S^* gives

$$c = \$3.83$$

against \$4.70 for an otherwise identical option on a stock paying nothing, so the dividends have cost the call 18.5% of its value.

The volatility in this calculation is that of S^*, not of S, and the two differ because S^* is a smaller number carrying the same absolute risk. For a modest yield the difference is negligible; for a high yield over a long horizon it is not. [TN15] quantifies it.

15.10 AMERICAN OPTIONS

Equations (15.2) and (15.3) price *European* options. Chapter 11 established when that matters.

American calls on non-dividend stocks need nothing: Section 11.8 proved early exercise is never optimal, so $C = c$ and (15.2) is exact.

American puts have no closed-form solution, and this is not a gap awaiting a cleverer derivation—it is a consequence of the problem's structure. The holder's exercise decision must be made at every date before expiration, so the boundary condition of Section 15.2 is not fixed at T but applies along a **free boundary** (the critical stock price at each date below which immediate exercise is optimal), whose location is itself part of the answer. Equation (15.1) still holds; what is missing is a closed form for where it stops holding.

Three responses are used, in ascending order of cost.

Use the European value as a lower bound. Since $P \geq p$ always (Section 11.9), the formula gives a floor. How slack the floor is depends on how deep in the money the option is and on the level of rates. The two-year put of Section 13.9's convergence figure is worth 6.7601 as a European, the limit the tree settles to, and 7.4719 as an American on a fine tree, an early-exercise premium of 10.5%, far too large to ignore. For a one-month out-of-the-money put on the same

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stock it is under a percent.

Use an analytic approximation. Several exist. The best known, the Barone-Adesi–Whaley quadratic approximation, drops a small term from the early-exercise premium's equation so that it solves in closed form, then finds the boundary by iteration. Such approximations are fast and typically accurate to a few tenths of a percent, inside the bid–ask spread for most listed options.

Use a lattice. Chapter 13's backward induction handles the free boundary exactly by testing exercise at every node, and Chapter 20 makes it efficient. This, or the finite-difference grid of Chapter 20, is what most dealers' pricing systems use, and a large part of why the formula has not retired the tree.

American calls on dividend-paying stocks may, in a frictionless market, be exercised early, but only immediately before an ex-dividend date (Section 11.10). That restriction permits a serviceable approximation, worked through in Example 15.6.

Example 15.6 Black's approximation

A stock trades at \$52 and pays a \$1.50 dividend in five months. Value an American call struck at \$50 expiring in six months, with $r = 4\%$ and $\sigma = 30\%$.

First check whether early exercise is even possible, using condition (11.13):

$$D = 1.50 \quad \text{against} \quad K\left(1 - e^{-r(T-t_1)}\right) = 50\left(1 - e^{-0.04/12}\right) = \$0.17$$

where $t_1 = 5/12$ is the ex-dividend date. The condition is comfortably satisfied, so exercise just before the ex-dividend date may be optimal.

Black's approximation values the American call as the greater of two European calls: one expiring at T on the dividend-adjusted price, and one expiring just before the ex-dividend date at t_1 on the unadjusted price.

European to T. Subtracting the dividend's present value, $\text{PV}(D) = 1.50e^{-0.04(5/12)} = \$1.4752$, gives $S^* = \$50.5248$, and (15.2) with $T = 0.5$ gives \$5.00.

European to $t_1 = 5/12$. Here the option is assumed exercised before the dividend, so no adjustment applies. With $S_0 = 52$ and $T = 5/12$, (15.2) gives \$5.48.

Black's approximation takes the larger:

$$C \approx \max(5.00,\ 5.48) = \$5.48$$

so, of the two policies, exercising just before the ex-dividend date is worth more than holding to expiration.

The approximation is crude, and it always understates: it chooses between the two policies today, while the holder chooses at the dividend date after seeing the price; that option to decide later is what it omits. Its merit is that it requires two evaluations of a formula rather than a lattice, and with one dividend it is usually within a few cents of the tree value.

15.11 WHAT THE MODEL GETS WRONG

The formula was published in 1973, the options market adopted it within two or three years, and it has been known to be wrong since at least 1987. Both facts are worth taking seriously.

Volatility is not constant across strikes. If the model were right, every option on a stock would imply the same σ. They do not, and the failure can be measured on the book's own data.

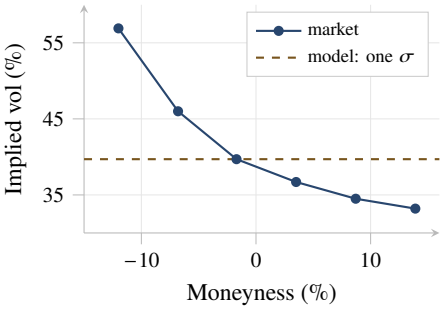
Table 15.3 inverts (15.2) across the option chain of Table 10.2, using the \$1.00 dividend that Section 11.6 extracted from the same quotes.

The implied volatility falls monotonically from 56.9% to 33.2% as the strike rises. This is a single stock, on a single day, at a single maturity, and the model requires all six numbers to be identical. They differ by a factor of 1.7. The calls are American, so part of each price, most at the lowest strikes, is early-exercise premium; inverting a European formula folds it into the implied volatility, which a first reading accepts and Chapter 19 refines.

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Table 15.3: Call implied volatilities from the chain of Table 10.2. $S_0 = 48.30$, $r = 5\%$, discounting over 45 calendar days, variance over 31 trading days; moneyness is $K/S_0 - 1$.

Strike	Moneyness	Call mid	Implied vol
42.50	−12.0%	6.650	56.9%
45.00	−6.8%	4.425	46.0%
47.50	−1.7%	2.670	39.7%
50.00	+3.5%	1.475	36.7%
52.50	+8.7%	0.710	34.5%
55.00	+13.9%	0.305	33.2%



The shape (high implied volatility at low strikes, falling as the strike rises) is the equity skew. It has been a stable feature of index and single-stock options since the crash of October 1987, and it says something specific: the market assigns more probability to large downward moves than the lognormal distribution of (14.11) allows. Chapter 19 is devoted to reading it.

Put–call parity (Section 11.4) holds identically for the Black–Scholes–Merton formulas, so a call and a put at the same strike and maturity must imply the same volatility. If European options do not, the cause is a data problem (a stale quote, a wrong dividend or borrow-fee assumption, a mismatched timestamp) and not a market phenomenon; on American-style listed options a residual gap is the early-exercise premium, which Section 11.6 already met in this chain. Checking it is the first test on any option dataset.

Volatility is not constant across time either. Implied volatility varies with maturity, and that term structure (the pattern of implied volatility across maturities) itself moves over time.

Returns are not lognormal. Real returns have fatter tails than the lognormal distribution behind (15.2), and jumps that no continuous path allows. The skew of Table 15.3 is that disagreement showing up in prices: low strikes are priced for a fatter lower tail than the model allows.

The frictionless assumptions (2, 3, and 7) fail as Section 15.1 noted. The model remains the market's language for the reason Section 15.7 gave: it is used not as a price generator but as an invertible map between prices and a single number, and for that purpose being wrong in a stable, well-understood way is tolerable. The volatility surface of Chapter 19 is not a correction to the model—it is a catalog of exactly how the model is wrong, quoted in the model's own units.

Case Study 15.1 The formula that arrived with the market

The Chicago Board Options Exchange opened on April 26, 1973. Black and Scholes's paper appeared in the *Journal of Political Economy* that May, having been rejected twice before publication; Merton's paper, developed alongside theirs, appeared in the *Bell Journal* the same spring. A market and its pricing model were born within weeks of each other.

Adoption was extraordinarily fast for an academic result. Traders on the CBOE floor were using the formula within two or three years, from about 1975 first on Black's printed sheets of model values and then on a handheld calculator sold pre-programmed with it. By the end of the decade it was the market's common language, and Scholes and Merton received the Nobel Prize in 1997. Black had died in 1995, and the prize is not awarded posthumously.

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What makes the episode instructive is that the model's assumptions were violated on the day it was published, and everyone involved knew it. Stocks pay dividends, volatility moves, markets close overnight, borrowing is not at the risk-free rate. Black said so himself, at length, in a 1988 essay titled "The Holes in Black–Scholes"—a list of the model's defects written by one of its authors, which remains accurate today.

The formula spread anyway, as a quoting convention rather than a pricing model, and no better convention has displaced it in fifty years.

15.12 WHERE THIS CAME FROM

Bachelier (1900) is the beginning, and it is remarkable twice over: he modeled the price itself, not its logarithm, as Brownian motion (the arithmetic Wiener process $dS = \sigma\, dW$ of Section 14.3) five years before Einstein described the physics, and he derived option values from it. The normal model's flaw (prices can go negative) kept it out of use for most of the twentieth century, and it returned in the 2010s when negative rates made it the natural choice for rate options, and again in April 2020, when WTI futures settled below zero and CME moved its crude-oil options to it for four months.

The intervening attempts show what was missing. Samuelson (1965), adopting the geometric Brownian motion Osborne had proposed, got the distribution right, but priced the option by discounting its expected payoff at a rate that had to be assumed. No one could say what that rate should be, and that single gap is what Black and Scholes (1973) closed: a continuously rebalanced hedge removes the risk, so no risk premium is needed and no view about the stock is required. Merton (1973) reached the same formula by a different route in the same year and extended it to dividends and to stochastic rates.

Cox and Ross (1976) then extracted the principle from the formula. If the hedge argument works, the option can be valued as though every asset drifted at the risk-free rate—not because investors are risk-neutral but because the answer does not depend on whether they are. Cox, Ross and Rubinstein (1979), with Rendleman and Bartter (1979) independently in the same year, turned that principle into Chapter 13's tree, and Harrison and Kreps (1979) with Harrison and Pliska (1981) gave the general statement Chapter 28 uses: no arbitrage holds if and only if an equivalent martingale measure exists, a risk-neutral measure of Chapter 13's kind under which every discounted asset price has zero drift.

SUMMARY

The Black–Scholes–Merton equation follows from Chapter 14's observation that a portfolio short one derivative and long $\partial f/\partial S$ shares has no random component. Setting its return to r gives $\partial f/\partial t + rS\,\partial f/\partial S + \frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2 = rf$. The stock's expected return μ cancels, which is Chapter 13's principle in continuous time, and the equation governs every derivative whose value depends only on S and t: the instrument is the boundary condition, not the equation. A forward contract satisfies it identically, which confirms that Chapter 5 and this chapter rest on the same principle.

Solving it for a European call gives $c = S_0 N(d_1) - Ke^{-rT}N(d_2)$, with the put following from parity. The formulas satisfy put–call parity identically, which is the fastest test of any implementation.

Of the two normal terms, $N(d_2)$ is the risk-neutral probability of exercise, not the real-world one. $N(d_1)$ is the delta, and $S_0 N(d_1)$ is the present value of receiving the stock in the states where the option is exercised.

Volatility is the one input that must be estimated. The historical estimator uses the sample standard deviation of log returns, annualized by $\sqrt{252}$; over a short sample its standard error

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is large, and it answers the wrong question, measuring what the stock did rather than what it will do. Implied volatility inverts the formula against a market price. The inversion is unique because option value is strictly increasing in volatility.

The formula's real role is as a language. It converts prices into a single comparable number, which is why dealers quote in volatility and why a model known to be wrong remains universal.

A continuous dividend yield is handled by replacing S_0 with $S_0 e^{-qT}$ (equivalently, the equation's drift term becomes $(r - q)S\,\partial f/\partial S$), which also covers indices, currencies, and futures. Known cash dividends are subtracted at present value.

American calls on non-dividend stocks are priced exactly by the European formula. American puts need a lattice or an analytic approximation. American calls on dividend-paying stocks can be valued by Black's approximation, the greater of two European values.

The model is wrong in documented ways: volatility varies with strike and maturity, returns have fat tails and jumps, and the frictionless-market assumptions fail. Measured on the book's own option chain, implied volatility differs widely across strikes. Chapter 19 catalogs the first of these in the model's own units.

FURTHER READING

[TN15] derives (15.2) from the risk-neutral expectation and shows the equivalence with solving the differential equation. The original papers, Black and Scholes (1973) and Merton (1973), are worth reading for how differently the two derive the same result: Black and Scholes through a hedged portfolio and the heat equation; Merton through a self-financing replicating portfolio (one that needs no cash added or withdrawn after it is set up) under weaker assumptions; the risk-neutral route this chapter prefers is Cox and Ross (1976). Black's 1988 essay "The Holes in Black–Scholes" is a short account of the model's failures by one of its authors, and its list still applies nearly four decades on.

PRACTICE QUESTIONS

15.1 A non-dividend stock trades at \$30. Value a three-month European call struck at \$29 with $r = 5\%$ and $\sigma = 25\%$. Then value the put and verify put–call parity.

15.2 Explain why the stock's expected return does not appear in (15.1), and state which chapter established the same result in a discrete setting.

15.3 In Example 15.1, state the risk-neutral probability that the option is exercised and explain why this is not the probability an investor should use when forecasting.

15.4 Compute the historical volatility from the last six observations of Table 15.1 and compare it with the estimate from all ten. What does the difference tell you about the reliability of short samples?

15.5 A call on a \$60 stock struck at \$60 with six months to run and $r = 4\%$ trades at \$5.20. Estimate the implied volatility to the nearest half percent by trial and error.

15.6 An index at 2,000 with a 2% dividend yield has three-month calls struck at 2,050 trading at 42. With $r = 3\%$, find the implied volatility.

15.7 A stock at \$40 pays a \$0.60 dividend in three months. Value a five-month European call struck at \$40 with $r = 5\%$ and $\sigma = 28\%$.

15.8 Explain why the European formula prices an American call on a non-dividend stock exactly, but not an American put.

15.9 Verify the limiting behavior of (15.2) as $\sigma \to \infty$ and confirm that it reproduces the upper bound of Section 11.2. Do the same as $\sigma \to 0$.

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15.10 Show that a forward contract satisfies (15.1), and explain what this demonstrates about the relationship between Chapter 5 and this chapter.

15.11 Using Table 15.3, state by what factor the implied volatilities differ across strikes and explain in one sentence what the model requires them to be.

FURTHER QUESTIONS

15.12 Show algebraically that (15.2) and (15.3) satisfy put–call parity for all parameter values, using $N(d) + N(-d) = 1$. Then explain why this makes parity useless as a test of whether the *model* is correct, though useful as a test of whether an *implementation* is.

15.13 Section 15.4 states that $N(d_1) > N(d_2)$ always. Prove this from (15.4), and give the economic reason in one sentence.

15.14 Example 15.3 found an implied volatility of 29.85%; suppose the same stock's own history gave 20.4%, as Example 15.2's did. Construct two distinct explanations consistent with both numbers, describe an observation that would distinguish them, and state which is more likely if the gap persists across all strikes and maturities.

15.15 Black's approximation always understates the American call value. Explain why, identify the choice available to the holder that it omits, and construct a case in which the understatement would be largest.

15.16 Example 15.5 notes that the volatility in the dividend adjustment refers to S^* rather than S. Quantify this: for a stock with a 4% annual yield and a five-year option, estimate the proportional difference between σ_{S^*} and σ_S, and state whether ignoring it overstates or understates the option value.

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Chapter 16
Options on Indices, Currencies, and Futures

Chapter 15 derived the Black–Scholes–Merton formula for options on a non-dividend-paying stock, a small corner of the option market by notional. The largest option markets in the world are written on interest rates, currencies, and futures contracts, and almost none of them involve a share that pays nothing. Currency options are also the first option market in this book that trades mainly over the counter, dealer to client, rather than on an exchange; Section 10.10 set out what that changes.

The extension needed is smaller than it looks. Section 15.9 already stated it: replacing S_0 with $S_0 e^{-qT}$ covers every case, with q interpreted differently in each. This chapter works through what q means in each market, what conventions surround the resulting prices, and one genuinely new structure, the quanto, where an option's payoff is fixed in a currency other than the one its underlying trades in.

It also introduces Black's model, the form the Black–Scholes–Merton formula takes when written on a forward or futures price. That form, not the original, is the market standard almost everywhere outside single-stock equity options, and Chapter 29's caps and swaptions are written in it or in its normal-model variant (Section 29.8).

Section 16.1 states the one substitution. Sections 16.2 to 16.4 apply it to index options and to currency options, including how the FX market actually quotes. Sections 16.5 and 16.6 are Black's model, on futures and beyond them, and Section 16.7 is the quanto. Sections 16.8 and 16.9 close with the two practical questions: where the forward comes from, and which convention each market uses.

16.1 ONE SUBSTITUTION, THREE MARKETS

The observation that unifies the chapter is that a *yield* on the underlying and a *dividend* on a stock enter the pricing problem identically. Both mean that holding the asset produces income the option holder does not receive, so the asset behaves as though it were worth $S_0 e^{-qT}$ today.

With that substitution, equations (15.2)–(15.4) become

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{16.1}$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1) \tag{16.2}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Table 16.1 gives the interpretation of q in each market. Nothing else changes.

The last row gets a section of its own: Section 16.5 shows that setting $q = r$ gives Black's model.

Table 16.1: What q means in equation (16.1).

Underlying	q	Reason
Non-dividend stock	0	Holding it produces no income
Dividend-paying stock or index	Dividend yield	Holder receives dividends; option holder does not
Currency	Foreign rate r_f	Foreign currency earns interest abroad
Futures contract	r	Costs nothing to hold, so it grows at zero under Q

16.2 OPTIONS ON STOCK INDICES

For an index, q is the dividend yield of the constituent stocks over the option's life: an average, weighted by index weight, and known reasonably well in advance because dividend policies change slowly.

Section 15.9 priced a straightforward index call. The more instructive application is the one Section 12.10 introduced without a model: using index puts to insure a portfolio.

Example 16.1 Insuring a portfolio, and why the strike is not obvious

A fund holds \$30 million tracking an index at 1,500, with a dividend yield of 2%. The manager wants a floor at \$28.5 million (a 5% loss) over six months, with $r = 4\%$ and $\sigma = 18\%$.

If the portfolio mirrors the index, the natural instrument is a put struck 5% below the current level, at 1,425. The contract multiplier is \$100, so

$$\text{contracts} = \frac{30{,}000{,}000}{1{,}500 \times 100} = 200$$

and from (16.2) with $K = 1{,}425$, the put is worth 37.27 index points, costing

$$200 \times 100 \times 37.27 = \$745{,}400$$

or 2.48% of the portfolio for six months.

A floor at 1,425 does not produce a floor at \$28.5 million. The portfolio also collects dividends over the six months (about 1% at a 2% annual yield), so at index 1,425 the fund holds roughly $30\text{m} \times 0.95 + 0.30\text{m} = \$28.8$ million before the put pays anything, and about \$28.05 million net of the premium, below the floor that was wanted. Separately, the put's payoff arrives at expiration while the insurance was wanted throughout.

A manager who wants a genuine floor on *terminal wealth* must solve for the strike that delivers it, accounting for dividends received and the premium paid. If the portfolio is not an index tracker its beta enters as well; the next paragraph works that out. The mechanical calculation is easy and the specification of what is being insured is not, which is where most portfolio-insurance mandates go wrong.

When the portfolio is not the index. A portfolio with beta β against the index does not fall 5% when the index does, so the strike that protects it has to be found through the beta. Write V for the portfolio's value (Section 3.6's V_A), and keep S for the index the put is written on. The capital asset pricing model (CAPM) relates the two total returns over the option's life,

$$\frac{V_T}{V_0} + q_V T - e^{rT} = \beta \left(\frac{S_T}{S_0} + qT - e^{rT} \right)$$

and solving for the index level gives

$$\frac{S_T}{S_0} = \frac{1}{\beta} \left(\frac{V_T}{V_0} + q_V T \right) - qT + \left(1 - \frac{1}{\beta} \right) e^{rT} \tag{16.3}$$

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where q_V and q are the dividend yields of the portfolio and the index, and S_T, the index level at which the portfolio reaches its floor, is the put strike. Take the fund of Example 16.1 with $\beta = 1.5$, dividend yields of 2% on both portfolio and index, and $r = 4\%$. A \$28.5 million floor on terminal wealth, dividends included but before the premium, is $V_T/V_0 + q_V T = 0.95$, so

$$\frac{S_T}{S_0} = \frac{0.95}{1.5} - 0.01 + \left(1 - \frac{1}{1.5}\right) e^{0.02} = 0.9634$$

and the puts are struck at $0.9634 \times 1{,}500 = 1{,}445$, not 1,425: a portfolio with beta 1.5 has lost 5%, dividends included, by the time the index level is down only 3.7%. The number of contracts scales with beta as well, $1.5 \times 200 = 300$, exactly as Section 3.6's hedge ratio did. At $\beta = 1$ equation (16.3) reduces to $S_T/S_0 = V_T/V_0 + (q_V - q)T$, the dividend point made in the example.

Example 16.1 bought a put, but through the mid-1980s a large body of institutional money instead manufactured one, a technique marketed as **portfolio insurance**: selling index futures as the market fell, in the proportion a put's delta prescribes (Section 18.11). Section 40.2 tells how that played out on October 19, 1987. The equity skew dates from that day, and Chapter 19's case study "The day the skew appeared" asks why.

Who trades index options

Index options are the largest equity option market by notional, and the participants divide cleanly.

Protection buyers are the users of Example 16.1: pension funds, insurers, and asset managers buying puts or put spreads against a portfolio. Section 12.10 established the arithmetic that governs them: continuous protection struck 5% out of the money costs roughly 5% a year, so almost everyone buys a spread or a collar instead.

Overwriters systematically sell calls against index holdings. Section 12.12's case study on derivative-income funds described the largest such program, and Example 10.7 gave the shape: an uncertain equity return converted into a capped one.

Volatility traders take positions in index volatility itself, delta-hedged as Section 12.4 described. They are the natural counterparty to both groups above, and Part IV treats them.

Dispersion traders exploit the relationship between index volatility and the volatilities of its constituents. An index is less volatile than the average of its members because they are imperfectly correlated (Section 14.11's two-asset volatility in miniature), so selling index volatility and buying single-stock volatility is a position in correlation. Chapter 22 develops it.

The structure of that list explains a persistent feature of the market. Protection buyers and overwriters both create one-directional demand, for puts and against calls respectively, and dealers absorb the imbalance. The equity skew is partly the price of that imbalance, which is a market-structure explanation to set alongside the crash-risk explanation that dates the skew to 1987. Chapter 19 weighs them.

16.3 CURRENCY OPTIONS

A foreign currency is an asset that pays interest at the foreign rate, so $q = r_f$. Substituting into (16.1) gives what is often called the Garman–Kohlhagen formula, though it is the same equation:

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2) \tag{16.4}$$

with S_0 the spot exchange rate expressed as units of domestic currency per unit of foreign.

That last phrase carries more difficulty than the formula. Every quantity in an FX option must be labeled with which currency it is measured in, and errors of this kind are a common source of loss in currency options among people who understand the mathematics perfectly.

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Example 16.2 A euro call

A US firm expects to pay €1 million in six months and wants protection against a rising euro. Spot is 1.0850 dollars per euro, US rates are 4%, euro rates 2.5%, and volatility is 8%. Value a call on the euro struck at 1.1000.

Here the domestic currency is the dollar, so $r = 4\%$, $q = r_f = 2.5\%$, $S_0 = 1.0850$, $K = 1.1000$.

$$d_1 = \frac{\ln(1.0850/1.1000) + \left(0.04 - 0.025 + 0.08^2/2\right) \times 0.5}{0.08\sqrt{0.5}} = -0.0819, \qquad d_2 = -0.1384$$

$$c = 1.0850e^{-0.0125} \times 0.467383 - 1.1000e^{-0.02} \times 0.444954 = 0.50081 - 0.47976 = 0.02105$$

On a €1 million notional the premium is $21,053.

Check the forward, since it is the reference every FX desk uses:

$$F = 1.0850\, e^{(0.04-0.025)\times 0.5} = 1.09317$$

The strike of 1.1000 is above the forward, so the call is out of the money *forward*, and indeed $d_2 < 0$, a risk-neutral exercise probability of 44.5%.

The premium is in dollars per euro of notional; Table 16.2 sets out the three other ways the same trade is quoted.

Who buys currency options, and what they buy

Currency options are bought mostly by people with a currency exposure they did not choose, and the structures reflect that.

Importers and exporters face the exposure of Example 16.2: a known foreign-currency payment or receipt at a known date. A plain option caps the cost, but the premium is a visible expense that must be defended, and Section 12.10 explained why that matters more than it should. The usual response is a **range forward**, also called a zero-cost collar or cylinder, which Example 16.3 constructs.

Example 16.3 A zero-cost range forward

The importer of Example 16.2 owes €1 million in six months and does not want to pay a premium. They buy a euro call struck at 1.1300 and sell a euro put struck so the premiums offset.

The call at 1.1300 is worth 0.01063 dollars per euro. Solving for the put strike with the same value gives $K_p = 1.0587$, and the structure costs nothing at inception.

The result is a band:

Euro at expiration	What happens	Cost of €1m
Below 1.0587	Put exercised against them	$1,058,748
1.0587 to 1.1300	Both expire; buy at spot	$1,058,748–$1,130,000
Above 1.1300	They exercise the call	$1,130,000

Compare the outright forward, which locks in $1,093,168 with certainty. The collar gives up the chance of paying less than $1,058,748 in exchange for capping the cost at $1,130,000, and costs nothing today.

The trade-off is the one Section 12.10 identified for equity index collars, with one difference. In equities the skew forces an asymmetric band: the put bought is further from spot than the call sold. Here, at a flat 8% volatility, the band is nearly symmetric around the forward (3.1% below it, 3.4% above); with a smile like Figure 16.1's, where euro puts carry the higher volatility, the put the importer sells is worth more and can be struck further out. The shape of the collar a client can be offered is a direct read on the shape of the smile, which is why Section 16.4's quoting conventions matter commercially and not just descriptively.

Cross-border investors hedge the currency exposure that comes with a foreign asset.

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Options are used rather than forwards when the size of the foreign holding is uncertain: an equity portfolio's value moves, so the required hedge moves with it, and an option's payoff scales in a way a fixed forward does not.

Carry traders buy protection on the funding currency, the low-yield currency borrowed to finance the position. A position long a high-yield currency against a low-yield one earns the differential and loses if the funding currency rallies, which tends to happen abruptly, when many such positions are unwound at once. An out-of-the-money option on the funding currency converts an open-ended tail into a premium, and the demand for it is visible in the risk reversal.

Structured product issuers embed currency options in notes sold to retail investors. The simplest is the dual-currency deposit, in which the investor sells a currency option in exchange for an enhanced deposit rate. Section 12.12's test applies: work out what option the buyer has written, and ask whether they would have written it deliberately.

Macro funds trade currency volatility and the risk reversal outright, as views on an exchange rate rather than hedges of it.

16.4 HOW THE FX MARKET ACTUALLY QUOTES

Currency options are quoted differently from every other option market, and the conventions are worth knowing because Chapter 19 uses them and they are more sensible than they look.

Which currency is the premium in? Table 16.2 sets out the four correct ways to quote the premium of Example 16.2.

Table 16.2: One premium, four quotations. €1m call struck at 1.1000; spot 1.0850.

Convention	Computed as	Value
Domestic per unit foreign	the formula's direct output	0.02105
% of foreign notional	premium / S_0	1.940%
% of domestic notional	premium / K	1.914%
Domestic cash	premium × foreign notional	\$21,053

The first and second differ by whether the premium is paid in dollars or euros, the second and third by which leg's notional is the base, and the fourth is the first multiplied by the notional. Every one is correct and they are not equal, which is why FX confirmations state the premium currency, amount, and settlement date rather than a percentage.

A related trap concerns delta. A euro call is also a dollar put, so a position has two deltas depending on which currency is treated as the asset, and the two are related through the **premium-adjusted delta**: the ordinary delta less the premium expressed in units of the foreign currency, $\Delta - c/S_0$, which is the delta seen by a holder who pays the premium in that currency. Desks additionally distinguish *spot* delta from *forward* delta, and whether the premium has been deducted. The number of ways to be right is large, and the number of ways to be consistent with a counterparty is one.

Volatility, not price. As Section 10.6 described generally, FX options trade on implied volatility. But the FX market goes further: it identifies options by *delta* rather than by strike. A "25-delta put" is the put whose delta is −0.25, whatever strike that turns out to be today.

The reason is that a strike ages badly. What matters for an option's price is how far its strike sits from the forward in standard deviations,

$$\frac{\ln(K/F_0)}{\sigma\sqrt{T}}$$

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and for a fixed strike that distance changes every day: the forward moves with spot, and even if it did not, the shrinking $\sigma\sqrt{T}$ pushes the same strike further from the money in standard-deviation terms as expiration approaches. A strike set at today's forward is at the money *forward* today (the sense Example 16.2 used, which coincides with the spot sense only when $r = r_f$) and is unlikely to be at the money in either sense next month. A 25-delta option is defined by that distance itself, so it is always the same distance out of the money. Delta is a moneyness measure that is automatically rescaled by volatility and time, and quoting in it keeps a surface comparable across days. Figure 19.4 draws exactly this, marking one strike on the smile before and after a move in spot.

Three quotes per maturity. Rather than a volatility for each strike, the FX market quotes three numbers at each maturity, the at-the-money (ATM) volatility, the risk reversal (RR), and the butterfly (BF), from which the whole smile is reconstructed, as Example 16.4 does:

$$\text{ATM} = \text{volatility of the delta-neutral straddle}$$

$$\text{RR}_{25} = \sigma_{\text{25-delta call}} - \sigma_{\text{25-delta put}} \tag{16.5}$$

$$\text{BF}_{25} = \frac{1}{2}\left(\sigma_{\text{25-delta call}} + \sigma_{\text{25-delta put}}\right) - \text{ATM} \tag{16.6}$$

The delta-neutral straddle is struck close to the forward but not at it; long-dated and some emerging-market pairs use the at-the-money-forward straddle instead. Strictly, the quoted butterfly fixes the price of a strangle valued at the single volatility ATM + BF, and (16.5) and (16.6) are the first-order reading desks use for a quick reconstruction.

The risk reversal quote measures the smile's *skew* (whether calls or puts are more expensive in volatility terms), and Section 12.6 showed it is the instrument that trades exactly that. The butterfly quote measures the smile's *curvature*, how much the wings exceed the middle, and the instrument that isolates it is a strangle bought against a straddle in vega-neutral proportions.

This is not a coincidence: the market quotes what it trades, and it trades level, skew, and curvature because those are the three things a smile can do.

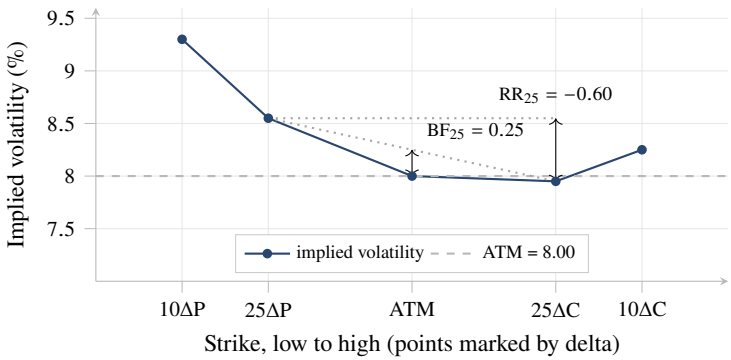


Figure 16.1: Smile built from ATM 8.00, $\text{RR}_{25} = -0.60$, $\text{BF}_{25} = 0.25$; the 10-delta points are extrapolated. Dotted: the 25-delta put level, and the chord joining the two 25-delta points. Illustrative.

Example 16.4 Reconstructing a smile from three numbers

A dealer quotes six-month EUR/USD as ATM 8.00, 25-delta risk reversal −0.60, 25-delta butterfly 0.25.

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Solving (16.5) and (16.6) for the two wing volatilities:

$$\sigma_{25C} = \text{ATM} + \text{BF} + \frac{1}{2}\text{RR} = 8.00 + 0.25 - 0.30 = 7.95\%$$

$$\sigma_{25P} = \text{ATM} + \text{BF} - \frac{1}{2}\text{RR} = 8.00 + 0.25 + 0.30 = 8.55\%$$

The three quotes fix three points on the smile, the at-the-money and the two 25-delta points of Figure 16.1; the figure's 10-delta points are extrapolated. Adding 10-delta quotes fixes five points, which is what most desks use.

Read the risk reversal. It is negative, meaning euro puts are more expensive than euro calls in volatility terms: the market is paying more for protection against a falling euro. In equity markets the skew has this sign essentially always (Section 15.11); in FX it changes sign as sentiment moves, which makes it a genuinely informative price rather than a structural constant.

16.5 OPTIONS ON FUTURES AND BLACK'S MODEL

The last row of Table 16.1 produces the most widely used version of the formula.

A futures contract costs nothing to enter, so in the risk-neutral world its price has zero expected growth; Section 13.10 derived this from the tree, and Section 14.7 from Itô's lemma. Setting $q = r$ in (16.1) makes the $S_0 e^{-qT}$ term equal $F_0 e^{-rT}$, and the formula collapses to

$$c = e^{-rT}\,[F_0 N(d_1) - K N(d_2)] \tag{16.7}$$

$$p = e^{-rT}\,[K N(-d_2) - F_0 N(-d_1)] \tag{16.8}$$

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This is **Black's model**, and Example 16.5 applies it. Three features make it the market's preferred form.

The interest rate appears only as a discount factor. It does not enter d_1 or d_2 at all. Everything about the underlying is contained in F_0, which is directly observable, so the model requires no view on carry, dividends, convenience yields, or foreign rates, all of which are already embedded in the futures price.

The forward is the natural reference. Setting $K = F_0$ makes $c = p$, which Section 11.4 derived from parity alone. Black's model reproduces it immediately, and this is why the at-the-money-forward straddle is the usual volatility reference point outside FX (Section 16.4).

It generalizes. Anything with an observable forward price can be priced this way, whether or not a futures contract exists; Section 16.6 shows how far that reaches.

Example 16.5 An option on a futures contract

A futures price is 100, with $r = 4\%$, $\sigma = 25\%$, and six months to expiration. Value calls and puts struck at 100.

$$d_1 = \frac{0 + 0.0625 \times 0.5/2}{0.25\sqrt{0.5}} = 0.0884, \qquad d_2 = -0.0884$$

$$c = e^{-0.02}\,[100 \times 0.535216 - 100 \times 0.464784] = 0.9802 \times 7.043 = 6.904$$

and the put comes to the same 6.904, as it must at $K = F_0$.

As a consistency check, note that for a non-dividend stock $F_0 = S_0 e^{rT}$, so Black's model on the futures price and the Black–Scholes–Merton formula on the spot must agree. Taking Example 15.1's option ($S_0 = 58$, $K = 55$, $r = 4\%$, $\sigma = 24\%$, $T = 0.5$) gives $F_0 = 58e^{0.02} = 59.1717$, and (16.7) returns \$6.1731, identical to four decimals. The two are the same formula written in different variables.

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How exercise settles. Exercise does not deliver the commodity, and it does not happen at an intraday price. At CME exercise notices are processed after the close: the holder of a call is given a long futures position opened at the strike, and that position is marked to the exercise day's settlement price like any other. The first mark-to-market therefore pays the holder that settlement price less the strike, and leaves a long futures position at the settlement price. The writer, on assignment, pays the cash and takes the short position.

Together the cash and the position are a futures contract bought at K: the cash is the settlement price less K, and the position, worth nothing once marked, earns only the moves after the close, so it is a new futures position rather than part of the payoff. The book therefore takes the payoff as $\max(F - K, 0)$ with F the settlement price on the day of exercise. The price at which the holder decided to exercise does not enter: the move from it to the settlement is added to a call's payoff and subtracted from a put's, which is one reason many contracts now settle straight to cash. Example 16.6 works through a call and a put.

Example 16.6 Exercising a futures call, and a futures put

A trader holds one call on the December crude oil futures contract struck at \$70.00, on 1,000 barrels. With the futures at \$73.40 during the day, when the call's intrinsic value is

$$1{,}000 \times (73.40 - 70.00) = \$3{,}400,$$

the trader submits an exercise notice. The contract settles that evening at \$73.85, and the notice is processed after the close: a long futures position opened at \$70.00 and marked to \$73.85. The move after the decision adds

$$1{,}000 \times (73.85 - 73.40) = \$450$$

for a total of \$3,850, which is $1{,}000 \times (73.85 - 70.00)$: the exercise day's settlement price less the strike, exactly as (16.7) assumes.

The put is the mirror image. A put on the same contract struck at \$76.00, exercised at the same moment, is worth

$$1{,}000 \times (76.00 - 73.40) = \$2{,}600$$

at \$73.40, but it opens a *short* futures position at \$76.00, marked to \$73.85. The same \$450 the call gained is lost, leaving \$2,150, which is $1{,}000 \times (76.00 - 73.85)$. The move from the intraday price to the settlement price helps one and hurts the other by the same amount, and both holders end with the intrinsic value at the settlement price.

Keeping the futures position leaves the cash and an outright position at \$73.85, long for the call holder and short for the put holder. That is a bet on crude, not a larger payoff, and it is the choice an assigned writer faces in reverse.

One settlement feature, not universal, does change the value. Where the exchange margins the option itself (**futures-style margining**, used at Eurex and ICE Futures Europe), the premium is not paid up front but posted and marked daily, so there is nothing to discount and Black's model loses its discount factor:

$$c = F_0 N(d_1) - K N(d_2)$$

At the rates of the 1980s the missing discount factor was a large difference; at recent rates it is small but not zero. Exercising early would also release nothing, so an American futures-style option is worth its European value.

Futures options versus spot options. Where both exist, they are not interchangeable, for four reasons, the first of which Section 10.3 gave.

Exercise delivers a margined position, not an asset to fund. That changes early exercise, so the agreement checked in Example 16.5 holds only for European options. Example 13.8

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showed why an American call on a futures contract can be exercised early with no dividend anywhere in the problem: a futures contract costs nothing to hold, which is exactly what $q = r$ encodes, so the premise of Section 11.8's argument, that deferring the strike payment earns interest, fails, and exercise releases the intrinsic value as margin that can earn interest. Two practical consequences follow. Listed commodity and rate options with the premium paid up front are usually American, so Black's model gives a lower bound rather than the value, and desks price them on lattices as Section 15.10 described. And a writer of American futures options should expect assignment whenever the option is deep in the money, not merely near expiration.

Where the futures contract expires with the option, a European futures option and a European spot option are worth the same. American ones are not: when the futures price stands above spot (contango), the American call on the futures is worth more than the American call on the spot and the American put less, and in backwardation the ranking reverses.

The underlying futures contract usually expires shortly after the option, so the option's underlying is a specific contract month rather than the spot. On a commodity with a steep curve, that difference is material.

And the futures price already contains the carry. Where the carry is uncertain (an unknown convenience yield, an unpredictable dividend), pricing off the futures removes an estimation problem that the spot-based formula would have to solve.

Who uses them. Section 10.3 named the hedgers; a refiner buying crude calls or a grower buying corn puts gets a cap or a floor where a futures hedge would be a lock. Asset managers and mortgage servicers use options on Treasury and rate futures to hedge duration and the convexity of Chapter 26's prepayment risk. By notional, the largest listed rate options are those on SOFR futures, which absorbed the Eurodollar options in 2023; since the futures price is 100 less the rate (Section 6.1), a call on the price is a put on the rate, which is why a SOFR call hedges a fall in rates. Exchanges also list weekly and daily expirations on equity index, Treasury, crude oil and gold futures, the futures-market counterpart of Chapter 23's short-dated options. Commodity trading advisors and volatility funds trade them for the volatility itself, and they are the natural counterparty to the hedgers.

Where commodity and energy options live

Black's model is not a special case in commodities—it is the whole market. Almost every listed option on oil, gas, metals, or grain is written on a *futures contract* rather than on the physical, and there are good reasons.

The spot price may not exist in a usable form. Section 2.1's contract specification showed that a commodity's price is a location, a grade, and a delivery window, not a single number, and Section 36.9 develops the point. A futures contract fixes all three, so it is the only unambiguous price to write an option on.

Delivery is avoided. Exercising a futures option produces a futures position, which can be closed. Exercising a physical option produces an obligation to take delivery of crude oil, which most option users are not equipped to do.

Wherever the underlying is awkward to hold, the market writes the option on the forward and prices it with Black's model; Section 16.6 carries this to interest rates.

16.6 BLACK'S MODEL BEYOND FUTURES

Nothing in (16.7) requires a futures contract. All it needs is a forward price that is observable and that has zero expected growth under the measure used for discounting. That condition holds far beyond the futures market, and the world's largest option market runs on it.

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A caplet is a call on a forward interest rate: Black's model applies with F_0 the forward rate and K the cap rate, scaled by the accrual fraction and discounted to the payment date at the end of the period. **A swaption** is an option on a forward swap rate, and the same form applies with the discount factor replaced by the *annuity*, the present value of one unit paid on each fixed-leg date (Chapter 7's PV01 leg). Chapter 29 derives both formulas and the normal-model form in which the market now mostly quotes them (Section 29.8), and covers who trades them: borrowers capping floating funding, issuers of callable debt, and pension funds hedging a fall in rates.

The condition that must be checked. Black's model requires the forward to have zero drift under the discounting measure, and this is where the apparatus can be misapplied. A forward rate is a martingale under the measure associated with *its own* payment date. A forward swap rate is a martingale under the measure associated with *the annuity*. Use one where the other belongs (pay a rate at the wrong date, observe it in the wrong currency, or average it over a period) and the forward acquires a drift.

That drift is exactly the **convexity, timing, and quanto adjustments** of Chapter 30, and Section 16.7 shows the simplest case next. Black's model is not wrong in those situations; it is being fed the wrong forward, and the adjustment corrects the forward rather than the formula.

16.7 QUANTOS AND COMPOSITE OPTIONS

Only one structure in this chapter needs new machinery: an option on a foreign asset whose payoff is settled in the domestic currency.

Two versions exist, and confusing them is expensive, as Example 16.8 shows.

A **composite** (or "compo") option pays the foreign-currency payoff converted at the *prevailing* exchange rate at expiration. The holder is exposed to both the asset and the currency, and the option is effectively written on the asset measured in domestic currency. Write V for the asset's foreign-currency price and S for the exchange rate in domestic units per foreign unit, as in Section 16.3. The option's volatility is that of the product VS, which by Section 14.11 is

$$\sigma_{\text{compo}} = \sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$$

where σ_V and σ_S are the volatilities of the asset and of the exchange rate, and ρ is their correlation.

A **quanto** pays the foreign-currency payoff converted at a *fixed* rate agreed in advance, typically 1.0, so that a foreign index moving 100 points pays 100 domestic currency units. The holder has no currency exposure at all, which is what makes quantos popular: an investor gets foreign equity exposure without an FX position. CME's dollar-denominated Nikkei 225 futures contract is a listed quanto; over the counter, structured-note issuers are major buyers of quanto calls, and the dealers who sell them carry the correlation risk described below.

That convenience has a price, and the price is not obvious.

The quanto adjustment. The issuer of a quanto must hedge by holding the foreign asset, and the size of that foreign holding, measured in domestic currency, changes as the exchange rate moves. If the asset and the currency are correlated, the hedge must be adjusted in a way that is systematically profitable or costly, and the option's price must reflect it.

The result, derived in Chapter 30, is that the foreign asset's forward price must be adjusted:

$$F_0^{\text{quanto}} = V_0\, e^{\left(r_f - q_f - \rho\,\sigma_V\sigma_S\right)T} \tag{16.9}$$

where V_0 is the asset's foreign-currency price today, r_f the foreign rate, and q_f its dividend yield. The option is then priced with Black's model (16.7), discounting at the domestic rate.

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The correction term $-\rho\sigma_V\sigma_S$ is the **quanto adjustment**. Its sign follows the correlation: when the foreign asset and the foreign currency move together, the adjustment reduces the forward and cheapens a call.

Example 16.7 Pricing a quanto call

A foreign index stands at 3,000 with a 2% dividend yield. Foreign rates are 1%, domestic rates 4%. Index volatility is 20%, exchange-rate volatility 10%, and their correlation is +0.3. Value a one-year quanto call struck at 3,100, paying one domestic unit per index point.

From (16.9),

$$F_0^{\text{quanto}} = 3000\, e^{(0.01-0.02-0.3\times0.20\times0.10)\times1} = 3000\, e^{-0.016} = 2{,}952.38$$

Without the correction the forward would be $3000e^{-0.01} = 2{,}970.15$, so the quanto adjustment lowers the forward by 17.77 points, or 0.60%.

Applying Black's model with $\sigma = 20\%$ and discounting at the domestic 4%:

$$d_1 = -0.1439, \qquad d_2 = -0.3439, \qquad c = 167.52$$

against 175.18 if the adjustment were omitted: an error of 4.6% in the option's value from a 0.6% error in the forward. Options magnify errors in the forward, so the adjustment cannot be waved away as second order.

Two practical points follow. The price depends on ρ, a correlation that is not directly traded and must be estimated, so quantos carry a risk that vanilla options do not. That risk is why quanto books carry **correlation reserves**: part of the book's model value held back against the chance that the correlation has been misestimated. And the sign matters: had the correlation been −0.3, the adjustment would have *raised* the forward and the call.

Example 16.8 The same view, two structures

Take the market of Example 16.7 and price the composite alongside the quanto. Assume the exchange rate starts at 1.00 domestic unit per foreign unit, so both structures start from an index level of 3,000 in domestic terms, and both are struck at 3,100 for one year.

Composite. The underlying is the index measured in domestic currency, whose volatility is

$$\sigma_{\text{compo}} = \sqrt{0.20^2 + 0.10^2 + 2\times0.3\times0.20\times0.10} = 24.90\%$$

Its forward is $3000e^{(r-q_f)T} = 3000e^{0.02} = 3{,}060.60$, and Black's model gives

$$c_{\text{compo}} = 274.69$$

Quanto. From Example 16.7, $c_{\text{quanto}} = 167.52$.

The composite costs 64% more, for two reasons: it carries a higher volatility (24.9% against 20%), and its forward is higher, 3,060.60 against 2,952.38. Most of that gap is the foreign currency's own forward premium, since the domestic rate exceeds the foreign rate by three percentage points; the remaining 0.6% is the quanto adjustment the composite escapes.

The payoffs explain the price gap better than the formulas do:

Index	FX rate	Quanto pays	Composite pays
3,400	0.90	300	0
3,400	1.00	300	300
3,400	1.10	300	640
3,100	1.10	0	310
2,800	0.90–1.10	0	0

The quanto pays on the index alone. The composite pays on the index *and* the currency, so it can pay when the index ends at the strike and the currency has risen—and pay nothing when the index has risen and the currency has fallen.

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These are different views, not different prices for the same view. An investor who wants foreign equity exposure and no currency exposure buys the quanto and pays for the correlation adjustment. An investor who is bullish on both buys the composite and pays for the extra volatility. Selling one to a client who wanted the other is an unsuitable sale that the payoff table makes obvious and the price comparison does not.

16.8 GETTING THE FORWARD RIGHT

Every formula in this chapter takes a forward as its real input. Black's model says so explicitly; equation (16.1) says so in disguise, since $S_0 e^{-qT}$ is the discounted forward $F_0 e^{-rT}$. The practical question in each market is not which formula to use but where the forward comes from.

Table 16.3 gives the answer, and one rule governs all of it.

Table 16.3: Where F_0 comes from in each market.

Market	Computed as	But in practice
Equity index	$S_0 e^{(r-q)T}$	Read from index futures
Single stock	$S_0 e^{rT}$ − FV(divs)	Implied from the option chain
Currency	$S_0 e^{(r-r_f)T}$	Read from the FX forward market
Commodity	Not computable	The futures price
Interest rate	From the curve	From the curve (Ch. 4)

The rule is to use the traded forward wherever one exists, never the computed one. Earlier chapters gave the reasons, and they are not small.

For currencies, Section 5.7 showed that covered interest parity has been persistently violated since 2008, by tens of basis points in the deepest markets in existence. A forward computed from $S_0 e^{(r-r_f)T}$ will therefore differ from the forward one can actually trade at, and the difference is the cross-currency basis of Chapter 7. Pricing an FX option off the computed forward embeds an error that counterparties can trade against.

For commodities the forward is not computable even in principle (Section 5.8); the futures price is the primary datum.

For single stocks, Section 11.6 showed the implied forward can be read directly from the option chain; Example 16.9 does the same for an index and shows what that forward contains.

Example 16.9 Extracting the dividend yield from index options

An index stands at 1,500. Six-month options struck at 1,500 trade at 68.50 (call) and 61.20 (put), with $r = 4\%$. What dividend yield is the market assuming?

Put–call parity with a yield, from Section 11.4, requires

$$c - p = S_0 e^{-qT} - K e^{-rT}$$

Substituting,

$$7.30 = 1500\, e^{-0.5q} - 1500\, e^{-0.02} = 1500\, e^{-0.5q} - 1470.30$$

so $1500 e^{-0.5q} = 1477.60$, giving

$$e^{-0.5q} = 0.985067 \quad \Longrightarrow \quad q = 3.01\%$$

Three points follow, and the third is the one that matters.

The calculation needs no volatility and no model: it is parity, which Chapter 11 established without either.

It gives the market's dividend expectation, which for an index is often better than a published forecast, because index dividends are actively traded.

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> And the number is not purely a dividend yield. It is whatever makes the forward come out right, which bundles expected dividends with the cost of financing or borrowing the index basket. What has been measured is the implied forward, expressed as a yield: the same warning Section 11.6 gave for single stocks. A desk that treats 3.01% as a dividend forecast and separately adds a financing spread has counted the same thing twice.

16.9 WHICH CONVENTION FOR WHICH MARKET

Table 16.4 collects what a reader needs to move between markets without importing the wrong assumption.

Table 16.4: Pricing form and quoting convention by market.

Market	Form used	Quoted in	Moneyness by
Single-stock equity	Black–Scholes–Merton, discrete dividends	Price	Strike
Equity index	Black–Scholes–Merton with q	Price or volatility	Strike
Currency	Black–Scholes–Merton with r_f	Volatility	Delta
Commodity, energy	Black's model on futures	Volatility	Strike or delta
Interest rate (caps, swaptions)	Black or normal model on forward rates	Volatility	Strike

Two habits follow from Table 16.4.

Ask what the forward is before asking what the option is worth. In every row, the model takes a forward as its input, and errors in the forward are magnified in the option; Example 16.7 put the magnification at roughly eight to one. Most disputes about an option's value turn out to be disputes about the forward.

Ask which currency each number is in. Table 16.2 showed one premium admitting four correct expressions. The same discipline applies to notionals, strikes, and deltas in any cross-currency structure.

Why volatilities differ so much between markets

A reader moving between the markets of Table 16.4 meets volatility numbers that differ by an order of magnitude, and the differences are structural rather than accidental.

Table 16.5: Indicative implied volatility ranges.

Market	Typical range
Major currency pairs	5–12%
Emerging-market currencies	8–25%
Equity indices	12–25% calm, 30–80% in crisis
Single stocks	20–60%
Commodities	20–60%, energy higher

The most striking entry in Table 16.5 is the first, and Section 14.11 already explains it. An exchange rate is a *ratio* of two currencies, and by equation (14.19) the volatility of a ratio is

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Two large economies share most of their shocks (global growth, commodity prices, risk appetite), so ρ is high and the common component largely cancels. What remains in an exchange rate is the difference between two economies, not the level of either, and differences are smaller than levels.

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Equities sit at the other end for the opposite reason. A share is a residual claim on a leveraged balance sheet, so the volatility of the assets is amplified by the leverage before it reaches the equity. Single stocks exceed indices because diversification removes the idiosyncratic part but not the leverage.

The practical use of Table 16.5 is as a sanity check. A quoted volatility outside its market's range is far more often a units or convention error than a discovery, and Section 14.6's rule of thumb (daily move $\approx \sigma/16$) converts any of these into something checkable against a screen in seconds.

SUMMARY

Replacing S_0 with $S_0 e^{-qT}$ covers indices, currencies, and futures, with q the dividend yield, the foreign rate, and the domestic rate. Nothing else in the formula changes.

Portfolio insurance is mechanically simple and conceptually not: a floor on the index level is not a floor on terminal wealth, because dividends are received, premium is paid, and a portfolio with beta other than one moves further than the index.

In currencies the difficulty is not the formula but the labeling: a premium can be correctly quoted four ways, and every quantity must carry its currency. The market quotes volatility rather than price and identifies options by delta rather than strike, because a strike ages and a delta does not. Each maturity is three numbers (ATM, 25-delta risk reversal, 25-delta butterfly) measuring the smile's level, skew, and curvature, which are what the corresponding instruments trade. Unlike equity skew, the FX risk reversal changes sign.

Setting $q = r$ gives Black's model, $c = e^{-rT}[F_0 N(d_1) - K N(d_2)]$: the rate is only a discount factor, and everything about the underlying sits in the observable forward. At $K = F_0$ the call and put are equal. It prices anything with an observable forward, caplets and swaptions included, provided that forward has no drift under the discounting measure; where it does, Chapter 30's adjustments correct the forward, not the formula. Every formula here takes a forward as its real input, and the traded forward is to be used wherever one exists: covered interest parity fails, commodity carry is unobservable, and an implied dividend yield bundles financing with dividends.

Exercising a futures option opens a futures position at the strike, marked to the exercise day's settlement price, so the holder collects the intrinsic value at that settlement price and is left holding the futures. Early exercise is possible without dividends, and the underlying is a specific contract month.

A composite option converts the foreign payoff at the prevailing rate and carries volatility $\sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$; a quanto converts at a fixed rate and removes the currency exposure, at the cost of a $-\rho\sigma_V\sigma_S$ correction to the foreign forward's drift. Even a small correction to the drift moves the price materially, and the dependence on an untraded correlation is why quanto books carry correlation reserves.

FURTHER READING

Merton (1973) gives the continuous-yield extension (16.1), Garman and Kohlhagen (1983) is the standard citation for the currency-option form (16.4), and Black (1976) introduces the futures form. On FX quoting conventions (premium currency, spot or forward delta, premium adjustment, and which straddle is at the money), Reiswich and Wystup (2010) set them out in one paper, and Clark's *Foreign Exchange Option Pricing* (2011) works them through as a desk uses them. Chapter 30 derives the quanto adjustment.

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PRACTICE QUESTIONS

16.1 An index stands at 800 with a dividend yield of 3%. Value a four-month European call struck at 820 with $r = 4\%$ and $\sigma = 22\%$.

16.2 Explain why $q = r$ for a futures contract, and state which chapters established this in discrete and continuous time.

16.3 Spot GBP/USD is 1.2600, US rates are 4%, UK rates 4.5%, and six-month volatility is 9%. Value a call on sterling struck at 1.2800, and express the premium both in dollars per pound and as a percentage of the sterling notional.

16.4 A dealer quotes three-month USD/JPY as ATM 9.50, $\mathrm{RR}_{25} = +1.20$, $\mathrm{BF}_{25} = 0.30$. Compute the two 25-delta wing volatilities and state which currency the market is paying more to protect against.

16.5 Using Black's model, value a six-month call on a futures price of 60 struck at 58, with $r = 3\%$ and $\sigma = 30\%$.

16.6 Show that at $K = F_0$ Black's model gives $c = p$, and identify the result from Chapter 11 that this reproduces.

16.7 Explain the difference between a quanto and a composite option, and state which one an investor wanting foreign equity exposure without currency risk should buy.

16.8 A foreign index at 2,500 has $q_f = 1.5\%$, $r_f = 2\%$, $\sigma_V = 22\%$, $\sigma_S = 9\%$, and $\rho = -0.25$. Compute the one-year quanto forward and state whether the adjustment raises or lowers it.

16.9 Every formula in this chapter takes a forward as its real input. Using Table 16.3, state the rule that governs how the forward is obtained for an equity index, a currency, and a commodity, and explain why a desk that computes the forward from spot and a carry rate can be wrong even when its rates are right.

16.10 Using Table 16.4, state which convention applies to an equity index option, a currency option, and an option on a futures contract. Explain what goes wrong when a convention is carried across from the market it belongs to, and give one case where two conventions would both be defensible.

FURTHER QUESTIONS

16.11 Example 16.1 notes that a floor on the index level is not a floor on terminal wealth. Set up the equation that determines the strike delivering a genuine \$28.5 million floor, accounting for dividends received and premium paid, and explain why the answer depends on whether the premium is funded from the portfolio or from outside it.

16.12 The FX market quotes by delta rather than strike. Construct the argument for why this makes a volatility surface more stable across days, then identify a circumstance in which it makes the surface *less* stable and state what desks do about it.

16.13 Example 16.7 found that a 0.60% adjustment to the forward changed the option value by 4.6%. Derive the general relationship between a proportional error in the forward and the resulting proportional error in an at-the-money option's value, and explain why the magnification is largest for short-dated options.

16.14 A trader observes that Black's model on a stock index futures price and the Black–Scholes–Merton formula on the index spot give different answers. List the possible causes in order of likelihood, and state what data you would examine to distinguish them.

16.15 Quanto pricing depends on a correlation that is not directly traded. Explain what this

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implies for how a quanto book should be risk-managed, what a correlation reserve is intended to cover, and why the problem is worse for long-dated quantos than for short-dated ones.

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Chapter 17
Employee Stock Options and Equity Compensation

Every option in this book so far has been a traded contract. Someone wrote it, someone bought it, and either could close the position at any time by trading it back. Chapters 13 through 16 rest on that: a price is the cost of a replicating strategy, and the replicating strategy requires that positions can be established and unwound.

None of this holds for an employee stock option. It cannot be sold. It is forfeited on departure. Its holder is unhedged, undiversified, and forbidden from shorting the stock. It runs ten years when listed stock options run at most about three. And it is issued by the company on its own shares, so exercise creates stock rather than transferring it.

The result is an option whose value depends on human behavior, and the models in this chapter say so explicitly. That is not a retreat from Chapters 13–16. It is those chapters applied consistently: when replication is impossible, the price is no longer pinned by arbitrage, and what fills the gap must be stated as an assumption rather than left implicit.

The chapter also covers the instruments that have largely displaced options in listed-company pay, restricted stock units and performance shares, and the accounting rules that drove the change. Section 10.10 promised this treatment for both warrants and employee options, and the two are the same problem: an option written by the issuer on its own equity.

Sections 17.1 and 17.2 are the contract and why it is not a traded call. Sections 17.3 to 17.5 value it: the models, the dilution that exercise causes, and the exercise decision itself. Section 17.6 is the accounting, Section 17.7 the restricted stock units and performance shares that displaced options, and Section 17.8 private-company equity. Sections 17.9 and 17.10 are warrants and the backdating episode, and Section 17.11 who uses equity pay and why.

17.1 THE CONTRACT

A grant of employee stock options specifies more terms than a listed contract does.

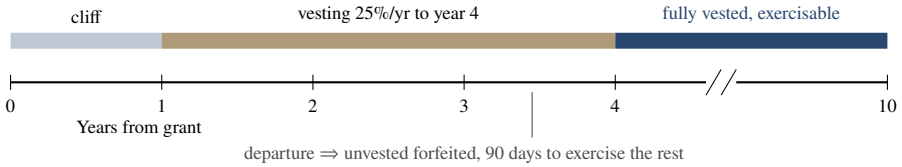


Figure 17.1: A typical grant: one-year cliff, then vesting to year four, ten-year term. The axis is broken between years 4 and 10.

Vesting. The option cannot be exercised until it vests. The common US pattern is a one-year **cliff** (nothing vests until the first anniversary, then 25% at once) followed by monthly or quarterly vesting to year four. Figure 17.1 shows the shape. Unvested options are forfeited outright on departure, which makes vesting the retention mechanism the grant exists for.

Term. Ten years from grant is standard, though four-year and seven-year terms appear. Compare this with the roughly three-year outer limit on single-stock LEAPS from Section 10.10: an employee option outlasts any standard listed option on a stock.

The post-termination exercise window. On leaving, an employee typically has 90 days to exercise vested options or lose them. This clause does most of the damage to the value, because it turns a ten-year option into one that expires 90 days after an event neither party controls.

Non-transferability. The option cannot be sold, pledged, or given away. There is no secondary market and no way to monetize it except by exercising and selling.

Strike. The strike is almost always the market price on the grant date. Sections 17.6 and 17.10 explain why: an at-the-money grant used to carry no accounting charge at all, and still carries the cleanest tax treatment.

Dilution. Exercise creates new shares. The company receives the strike price in cash and issues stock, so the share count rises. Section 17.4 works out what this does to the valuation.

17.2 WHY AN EMPLOYEE OPTION IS NOT A TRADED CALL

Four of the restrictions above break the arbitrage argument, all in the same direction.

The holder cannot hedge. Chapter 13's replication argument requires that a position be constructed and maintained. An employee cannot short the company's stock (most equity plans forbid it and insider-trading policy forbids the rest), so the option's value to the holder is not the cost of replicating it. It is whatever an undiversified, risk-averse person with a concentrated position thinks it is worth, which is *less*.

The holder is undiversified. An employee's human capital is already invested in the firm. Adding options on the same firm concentrates rather than diversifies, so the marginal option is worth less to the employee than to a diversified investor. The size of the gap depends on how risk-averse the holder is and on how much of their wealth is already tied to the firm, and the utility-based estimates run wide: Hall and Murphy (2002) put an executive's **certainty-equivalent** value (the sure sum of cash the executive would accept in place of the option) at roughly half to two-thirds of the Black–Scholes–Merton cost for a typical at-the-money grant, and Meulbroek (2001) found discounts of the same order for concentrated holdings. Whatever the number, it is a real cost of paying in options, and it appears nowhere in the accounts.

Forfeiture is a random early expiration. Departure forfeits unvested options and leaves 90 days to exercise vested ones. Modeling this requires an assumption about employee turnover, and the assumption matters: Section 17.3 shows a value moving from 16.10 to 12.59 as the assumed exit rate goes from zero to 6% a year.

Early exercise is the norm, not the exception. Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early, and employees exercise early constantly. Bettis, Bizjak and Lemmon (2005) find ten-year options exercised a median of 4.25 years before expiration, under six years into the term; Carpenter's (1998) sample of top executives gives a mean of 5.83 years after grant, at a stock price 2.75 times the strike. Five to six years at two to three times the strike is the working summary, and it is where the exercise multiple of Section 17.3 comes from.

This is not employees being irrational. The proof in Section 11.8 assumes the holder can sell the option instead. An employee cannot. For a holder choosing between exercising and holding an unsellable, unhedgeable, forfeitable claim on a single company, exercising to diversify is the sensible choice. The theorem does not fail; its hypothesis does.

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That said, exercise-and-hold is common and usually a mistake: it gives up the remaining time value and, in the US, prepays ordinary income tax on the spread to acquire exposure the option already provided (Example 17.4).

17.3 VALUING AN EMPLOYEE OPTION

Three approaches are in use. They disagree, and by how much is the main finding.

1. Black–Scholes with the contractual term. Put the ten-year term into the formula of Chapter 15 and read off the answer. This overstates the value, because it ignores forfeiture and early exercise entirely.

2. Black–Scholes with an expected term. Replace the contractual term with the expected time to exercise or forfeiture, estimated from the company's own history. This is the workhorse: it is what most disclosures use, it is auditable, and it is defensible precisely because it makes no claim to be a model of behavior. A company whose own history gives no reasonable basis for the estimate may use the simplified method of the SEC's Staff Accounting Bulletins (SABs) 107 and 110, which sets the expected term of a plain at-the-money grant midway between vesting and the contractual term: 6.5 years for Example 17.1's grant.

Its weakness is that expected life is not a parameter of the option but an *outcome*, and it depends on the stock price path. A company whose stock triples will see exercises much earlier than one whose stock stagnates, so the expected term estimated from history is conditioned on a history that will not repeat.

3. A lattice with an exercise rule. Build the tree of Chapter 13 and replace optimal exercise with a behavioral rule: the employee exercises voluntarily once the stock reaches a multiple M of the strike, and is forced to exercise (or forfeit, if unvested) on departure, which arrives at rate λ. This is the Hull–White approach, and it is the one that spells out what the other two conceal.

Example 17.1 Three answers for one grant

A company grants at-the-money options: $S_0 = K = 40$, ten-year term, three-year cliff vesting, $\sigma = 30\%$, $r = 4\%$, dividend yield $q = 1\%$.

Method	Value	% of grant price
Black–Scholes, 10-year contractual term	16.704	41.8%
Black–Scholes, 6.5-year simplified term	13.808	34.5%
Black–Scholes, 6-year expected term	13.295	33.2%
Black–Scholes, 4.5-year expected term	11.552	28.9%
Lattice, $M = 2.5$, exit rate 3%/yr	14.22	35.5%

The spread is 16.704 to 11.552, a factor of 1.45. ASC 718 rules out the first row, which ignores early exercise, but the methods it allows still span 15.56 (the lattice's value conditional on vesting, the standards' basis; Example 17.5) to 11.552, and on one million options that vest that is a difference of $4.0 million in reported compensation expense, produced entirely by choices about the model rather than about the contract.

The lattice makes the sensitivity explicit:

	exit 0%	exit 3%	exit 6%
$M = 2.0$	—	13.54	—
$M = 2.5$	16.10	14.22	12.59
$M = 3.0$	—	14.52	—

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Turnover matters more than the exercise multiple: moving the exit rate from 0% to 6% costs 3.51 of value, while moving the exercise multiple from 2.0 to 3.0 gains only 0.98, less of it from 2.5 to 3.0 than from 2.0 to 2.5, because a higher threshold matters only on the paths that reach it.

The practical implication is that the most important input is a human-resources (HR) statistic. The company's own attrition rate drives the number more than volatility does over the plausible range, and a firm with high turnover is genuinely granting less, option for option.

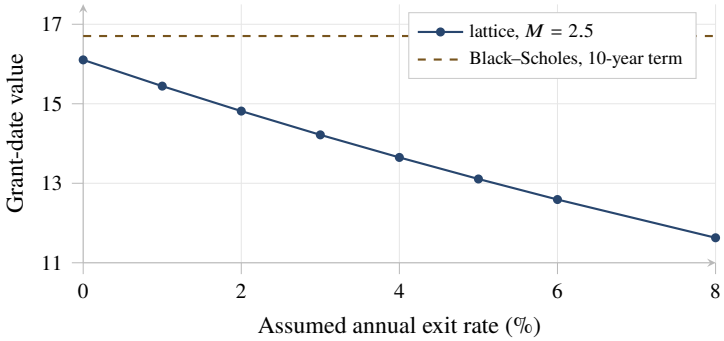


Figure 17.2: Lattice value against the assumed annual exit rate. $S_0 = K = 40$, $\sigma = 30\%$, ten-year term.

Figure 17.2 makes the point visually: the reported expense is close to linear in an assumption that no market quotes and no auditor can verify beyond checking that the company applied its own history consistently.

This is the position of the whole chapter. An employee option cannot be traded, so its reported value is a model output whose most sensitive input is behavioral.

17.4 DILUTION

Exercise creates shares. Existing holders own a smaller fraction of a company that has received the strike price in cash, and **dilution** is the size of that transfer: the fraction of the company passing to the option holders. Three measures of it appear in this section and they are not equal. The *share-count* measure is the new shares as a fraction of the enlarged total, and it is what governs a warrant's value. The *economic* measure is the grant's value as a fraction of market capitalization. The *reported* measure counts only the incremental shares the treasury stock method leaves after assumed buybacks, and Example 17.3 finds it running at about half the economic figure.

Take a company with N shares and no debt, whose assets, excluding anything received from the option holders, will be worth V_T at date T. It has issued M warrants with strike K (in this section M counts warrants; it is no longer the exercise multiple of Section 17.3). If they are exercised, the company receives MK and issues M shares, so the post-exercise share price is

$$\frac{V_T + MK}{N + M}$$

and each warrant is worth that less K. A line of algebra gives

$$\text{warrant payoff} = \frac{N}{N+M}\,\max\!\left(\frac{V_T}{N} - K,\ 0\right) \tag{17.1}$$

So a warrant is $N/(N+M)$ of a call option on V/N, the per-share value of the company *excluding* the warrant proceeds. Both parts of that sentence matter, and the second is where practitioners go wrong. Example 17.2 works through the first.

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Example 17.2 What dilution costs, and what it does not

A company has 10 million shares. It issues 2 million five-year warrants struck at \$40, with $\sigma = 30\%$, $r = 4\%$, $q = 1\%$, and a per-share value of \$40 before the warrants exist.

A five-year call on \$40 struck at \$40 is worth **12.171**. The dilution factor is $10/12 = 0.8333$, so each warrant is worth

$$\frac{10}{12} \times 12.171 = \mathbf{10.143}$$

16.7% less than the equivalent call, which is exactly the fraction of the company the new shares represent.

The total is $2\text{m} \times 10.143 = \$20.3$ million. If the warrants are handed out for nothing, the share price falls by $20.3/10 = \$2.03$ on announcement, from \$40.00 to \$37.97. The value did not vanish; it moved from the existing shareholders to the warrant holders, and the price move is the transfer.

The mistake worth avoiding. Once the warrants or options are outstanding and known, the observed share price already reflects them. Applying (17.1) to the market price then double-counts: the market price is V/N net of the warrant liability, not gross of it.

This has a direct consequence for employee options, and it is the opposite of what intuition suggests. An outstanding employee option grant requires no dilution adjustment at all. The market knows the options exist (they are disclosed), so the share price already embeds them, and the grant-date fair values of Section 17.3 are computed on the observed price with no $N/(N + M)$ factor. The adjustment applies only when valuing an instrument the market has not yet been told about, which for a public company is essentially never.

The dilution that matters for employee options is therefore a governance question about the size of the grant, not a valuation adjustment: how much of the company is being transferred, over what period, and in exchange for what.

Diluted earnings per share

Dilution is quantified and published in only one place, and it ignores the valuation above.

Diluted earnings per share (EPS) is computed by the **treasury stock method**: assume the options are exercised, assume the proceeds are used to buy back shares at the average market price, and count the difference as incremental shares.

Example 17.3 Reported dilution against economic dilution

A company has 10 million shares, net income of \$50 million, and 1 million options struck at \$30. The average share price for the period is \$50. The options were granted at the money with the stock at \$30, worth about \$10 each on the parameters below, and \$6 million of that \$10 million grant-date cost, which is expensed over the vesting period (Section 17.6), remains unrecognized on average over the period.

Basic EPS is $50/10 = \mathbf{\$5.00}$.

Assumed proceeds are $1\text{m} \times 30 = \$30$ million, plus the \$6 million of unrecognized cost (the value of the service employees still owe, which ASC Topic 260 also treats as proceeds): \$36 million, which buys back $36/50 = 0.72$ million shares. Incremental shares are $1.00 - 0.72 = 0.28$ million, so

$$\text{diluted EPS} = \frac{50}{10.28} = \mathbf{\$4.864}, \qquad \text{reported dilution } \frac{0.28}{10} = 2.8\%$$

The reported figure is the increase in the share count. Earnings per share itself falls by 2.7%: a share count 2.8% larger divides into the same earnings.

Valued with the stock at \$50, six years, $\sigma = 30\%$, $r = 4\%$ and $q = 1\%$, each option is worth \$25.68, so the grant is worth \$25.7 million against a market capitalization of \$500 million, or **5.1%**

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of the company.

The reported dilution is roughly half the economic dilution, and the reason is structural: the treasury stock method counts the option at its intrinsic value less unrecognized compensation cost, which ignores time value entirely. It is an accounting convention for allocating this period's earnings, not a measure of what has been given away.

The gap widens exactly where it matters most. For a volatile company whose options are near the money, intrinsic value is small and time value is nearly the whole of the option, so the treasury stock method reports almost no dilution on grants that transfer a substantial fraction of the equity.

An analyst who wants the economic number must compute it, and the inputs are disclosed: option counts, strikes, and grant-date fair values all appear in the equity compensation footnote. Section 17.11 returns to why this matters for reading buyback announcements.

17.5 THE EXERCISE DECISION

Section 17.2 said employees exercise early and that this is not irrational. Example 17.4 works through what the choice actually looks like, because the arithmetic is not obvious and the common advice is wrong in both directions.

Example 17.4 Three things to do with a vested option

An employee holds 10,000 vested non-qualified options (NSOs) struck at \$20: the ordinary kind, taxed as income on the spread at exercise (Section 17.8 covers the incentive kind). The stock is \$50, four years remain, $\sigma = 35\%$, $r = 4\%$, no dividend. Their marginal tax rate is 40%.

The corresponding traded call is worth **33.483** against an intrinsic value of **30.00**, so the time value is **3.483**, or 11.6% of the intrinsic value, and \$34,826 across the position.

Action	Cash today	Position afterward
Hold	\$0	options worth \$334,826 before tax
Exercise and sell	+\$180,000	nothing
Exercise and hold	−\$320,000	\$500,000 of stock

Read the third row carefully. Exercising costs \$200,000 in strike and creates \$300,000 of ordinary income, so \$120,000 of tax comes due whether or not any stock is sold. Holding would defer that tax, not avoid it, so exercise-and-hold prepays it: the employee spends \$320,000 in cash to acquire exposure the option already gave them, and gives up the \$34,826 of time value.

What the prepayment buys is that later appreciation is taxed as a capital gain rather than as ordinary income. That is the real content of "I want to start the capital gains clock," an offset worth weighing on a stock expected to rise far, though the time value, the deferral and the concentration usually outweigh it. "I believe in the company" is no reason at all: belief is an argument for keeping the exposure, and the option *is* the exposure, with leverage and without the cash outlay.

The case for exercising is the case for diversifying, and it is the first two rows that are in genuine tension. They must be compared after tax on both sides, since the options also owe the tax when eventually exercised: at a constant 40% rate the \$334,826 is worth $0.6 \times \$334{,}826 = \$200{,}896$ after tax, against \$180,000 of cash from exercising and selling. Diversifying therefore costs about \$20,900, the after-tax time value. An employee whose net worth is already concentrated in the firm may rationally pay that, and Section 17.2's undiversified-holder discount is exactly the statement that the model value is not what the position is worth to them.

Two things sharpen the decision. *Time value does not disappear as the stock rises*: at \$120 the same option's time value would still be 2.98, barely changed from the 3.483 here. It never falls to zero: for a non-dividend-paying stock it converges to the interest on the deferred strike, $K(1 - e^{-rT})$, which is 2.96 for this option and is Section 11.8's result seen from the other side. What does fall is the time value measured against what is at stake: 3.0% of the intrinsic value rather than 11.6%, so an employee exercising to diversify surrenders proportionally less at the higher

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price. *A near departure forces it*: the 90-day window converts the choice into a deadline, which is why so many exercises cluster around resignations rather than around stock prices.

17.6 ACCOUNTING AND EXPENSING

Until 2006 a US company with a calendar fiscal year could grant at-the-money options and record no expense whatsoever. Accounting Principles Board Opinion 25 (APB 25, 1972) measured the charge as the option's *intrinsic* value at grant, which for an at-the-money option is zero, and Financial Accounting Standard 123 (FAS 123, 1995), though it encouraged expensing fair value, let companies keep that method with footnote disclosure; nearly all did. Companies that would never have paid $14 million in cash bonuses granted one million options worth $14 million and reported nothing.

FAS 123R (issued in December 2004), now ASC 718, and IFRS 2, in force from 2005, ended that. In outline, the current rules are these:

Measurement at grant-date fair value. The charge is the model value on the grant date, computed as in Section 17.3 but conditional on vesting: departures before vesting reduce the number of awards expensed, not the value of each. It is never remeasured for subsequent stock price moves. An option valued at $15 at grant that expires worthless was still a $15 expense, and so was one that ends $200 in the money.

Recognition over the vesting period. The total is spread across the service period, but the pattern depends on whether the grant is treated as one award or several. **Straight-line recognition** treats it as one award and expenses equal amounts each year; **graded recognition** treats each vesting tranche as a separate award expensed over its own vesting period, so the early years carry a share of every tranche. Example 17.5 sets the two side by side.

Equity or liability classification. Everything above assumes settlement in shares. An award settled in cash is a liability. It may be a **stock appreciation right (SAR)**, which pays the spread without the holder paying the strike, or phantom stock, which pays a share's value. A liability is remeasured to fair value every reporting date until it settles, so a cash-settled SAR, unlike the equity-settled option above, rises and falls with the share price for years and ends at whatever the award pays. Companies accept that volatility to pay the award without issuing shares, though the cost to existing shareholders is the same: the cash is theirs too, so settling in it keeps the share count fixed but transfers the same value. SARs are common where local law restricts issuing shares to employees and among private companies with no market to settle into.

Forfeiture. Expense on awards forfeited for failure to vest is reversed. Since ASU 2016-09, ASC 718 lets a company either estimate forfeitures or account for them as they occur; IFRS 2 requires an estimate. A company that estimates uses the exit rate *twice*, for different purposes: once inside the model, after vesting, to shorten the option's life, and again outside it to reduce the number of awards expected to vest.

Pay versus performance. For fiscal years ending on or after December 16, 2022, US proxy statements must also report "compensation actually paid" (Item 402(v) of Regulation S-K, adopted in August 2022), which remeasures equity awards at fair value at each year end and at vesting. It is the remeasurement the expense never makes, published beside it.

Example 17.5 Straight-line against graded recognition

A grant of 100,000 options vests 25% a year over four years. Each annual tranche is a separate option with its own vesting date, so each has its own value (Example 17.1's lattice, with the cliff set at the tranche's vesting date). The values are the expected cost per option granted, including the chance of leaving before vesting:

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Tranche	Options	Value	Cost
1-year cliff	25,000	14.52	$363,000
2-year cliff	25,000	14.37	$359,250
3-year cliff	25,000	14.22	$355,500
4-year cliff	25,000	14.05	$351,250
Total	100,000		**$1,429,000**

Per option granted, longer vesting costs *less*, not more, because vesting later means more exposure to forfeiture before the option becomes exercisable. The effect is small (3.2% from the first tranche to the fourth), but it runs the opposite way from the usual "more time is worth more" intuition: the extra time is time the holder cannot use. The standards measure fair value conditional on vesting instead, and put pre-vesting departures into the number of awards. On that basis each value is divided by the probability of staying until the cliff, $e^{-0.03c}$ for a cliff of c years, and the tranches are worth 14.96, 15.26, 15.56 and 15.85. These rise with the cliff: once pre-vesting departures are set aside, what costs value is early exercise after vesting, voluntary at the multiple or forced by departure, and a later cliff postpones both. The expected totals are the same on either basis.

Now recognize that $1,429,000 over the vesting period:

	Year 1	Year 2	Year 3	Year 4
Straight-line	$357,250	$357,250	$357,250	$357,250
Graded	$748,938	$385,938	$206,312	$87,812

Graded recognition puts 52% of the total cost in the first year against straight-line's 25%. Both are permitted under ASC 718 for awards with only a service condition (vesting on continued employment alone); IFRS 2 requires the graded pattern. The total is identical and the timing is not, so two companies with identical pay can report materially different first-year expense.

The difference washes out in steady state for a company that grants every year, but *not* for one that has just started granting or made a large one-off award, precisely the cases where the number is being looked at.

Repricing and modification. When the stock falls far enough that outstanding options are hopelessly underwater, they stop retaining anyone. Companies respond by repricing (lowering the strike) or by exchanging old options for fewer new ones or for restricted stock.

Accounting treats this as a modification: the incremental charge is the fair value *added* by the change, measured immediately before and after, on top of any unrecognized original cost. So a repricing is not free, and shareholders usually must approve it. The economics are worse than the accounting suggests: an employee who learns that underwater options get reset has learned that the downside of the option was never real, which removes most of the incentive the grant was supposed to create.

Repricing and exchanging are usually compared on cost; Example 17.6 shows that this framing gets the incentive question backward.

Example 17.6 Repricing against a value-for-value exchange

A company granted 500,000 ten-year options struck at $80. The stock is now $30 with six years remaining; volatility is 45%, the rate 4%, and the stock pays no dividend. The existing options are worth $6.5944 each and a new six-year at-the-money option would be worth $14.6527.

Repricing all 500,000 to a $30 strike adds $14.6527 − $6.5944 = $8.0583 of fair value per option. The alternative is a value-for-value exchange: cancel the underwater options and grant fewer new ones at $30, in a ratio chosen so the fair value handed over matches the fair value taken

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back: \$6.5944/\$14.6527 = 0.45005 new per old, or 225,025 in all. The two compare as follows:

	Reprice	Exchange
Options held afterward	500,000	225,025
Fair value before	\$3,297,207	\$3,297,207
Fair value after	\$7,326,327	\$3,297,207
Incremental expense	**\$4,029,120**	**\$0**

The exchange delivers exactly what it took away, so nothing is recognized. That is why boards prefer it, and why proxy advisers (firms advising institutional shareholders how to vote) accept it.

Now look at what each does to the employee's exposure rather than to the reported number. Incentive intensity is the delta of the grant, how many shares of exposure the employee actually holds:

	Delta	Shares of exposure
Underwater \$80 options	0.452	225,931
Repriced \$30 options	0.779	389,507
Exchanged \$30 options	0.779	175,297

The exchange that costs nothing to report leaves the employee with 22% *less* exposure to the share price than the underwater options they surrendered. A deeply out-of-the-money option has a low value and a surprisingly high delta (nearly half a share), so trading it for a smaller number of at-the-money options is a bad trade in delta terms even when it is a fair trade in value terms.

So the two alternatives fail in opposite directions. The repricing multiplies the value of the employee's grant by 2.2 and their share exposure by 1.7, and charges shareholders \$4.0 million for doing it, having taught everyone that strikes follow the stock down. The exchange reports nothing, costs nothing, and quietly reduces the incentive it was adopted to restore: the one that is easy to disclose does less of what the grant was for.

17.7 RESTRICTED STOCK UNITS AND PERFORMANCE SHARES

Options have been displaced. Most large listed companies now grant the majority of equity compensation as restricted stock units (RSUs), and the reason is visible in one calculation.

An **RSU** is a promise to deliver one share on vesting. Having no strike, it is not an option but a prepaid forward on the stock (equivalently, a zero-strike call) with delivery contingent on service. Its grant-date fair value is the share price, less the present value of dividends forgone during vesting if the units carry no **dividend equivalents**: cash or extra units matching the dividends paid on the shares while the units are unvested.

Example 17.7 Why RSUs displaced options

Take the same company and grant terms: $S_0 = 40$, $\sigma = 30\%$, $q = 1\%$, Example 17.1's three-year cliff and 3% annual exit rate.

The two must be valued on one basis. The option's 14.22 is an expected cost per option granted and carries the chance of leaving before the cliff (Example 17.5), so the RSU must carry it too: a unit without dividend equivalents is a prepaid forward on the share that is forfeited on departure before vesting, worth $40e^{-(0.01+0.03)\times 3} = \mathbf{35.48}$. So one RSU costs the same as **2.50 options** (a ratio unchanged conditional on vesting, where both values rise by $e^{0.09}$), which is why a company moving from options to RSUs grants one unit in place of two to three options.

Now suppose the stock falls immediately after the grant, with the same strike and the same vesting still ahead:

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Stock price	RSU value	change	Option value	change
40	35.48	—	14.22	—
30	26.61	−25%	8.41	−41%
20	17.74	−50%	3.68	−74%

A halving of the stock destroys 74% of the option grant's value and exactly 50% of the RSU grant's. Figure 17.3 shows why: the RSU is a straight line (a prepaid forward is linear in the share price), while the option curve bends away from it and flattens toward zero, so the same fall in the stock takes proportionally more of the option. After 2008 and again after 2022, boards that had granted options found their entire retention apparatus nearly worthless at exactly the moment they most needed to retain people, while boards that had granted RSUs found theirs merely diminished.

That is the whole argument, and it is an argument about the *purpose* of the grant. An option pays only for outperformance; an RSU is deferred stock that pays for staying. If the objective is retention, as it is for most employees below the executive level, the RSU does the job with less accounting expense per unit of retention delivered.

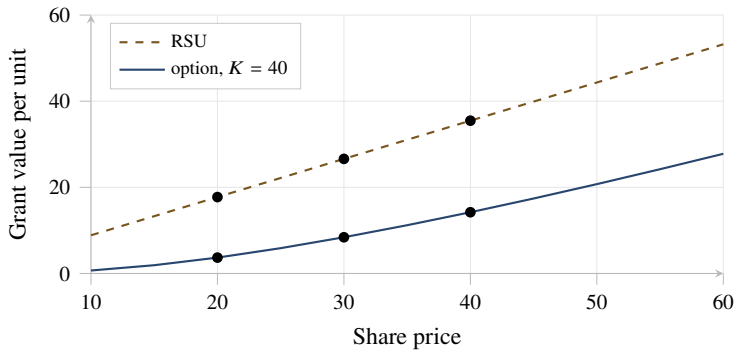


Figure 17.3: Grant value against the share price, on Example 17.7's terms. Marks show the table's three cases.

Performance share units. At executive level, grants increasingly vest on a performance condition rather than on service alone, and the accounting treats two kinds differently.

A **non-market condition** (revenue, earnings per share, a regulatory approval) is not built into the fair value. The grant-date value is computed as if the condition will be met, and the number of awards expected to vest is estimated and trued up each period. If the target is missed, the expense is reversed entirely.

A **market condition**, such as total shareholder return (TSR: price appreciation plus dividends) measured against an index or a peer group, *is* built into the fair value and is never trued up. The award is valued at grant with a Monte Carlo simulation of the company's stock against the benchmark, and that value is expensed whether or not the target is hit, provided the employee completes the service period.

The asymmetry is easily missed. A relative TSR award that pays nothing still generates expense for its full grant-date value, because the possibility of paying was what was purchased. An EPS-target award that pays nothing generates no expense at all. Two grants that look identical in the proxy statement can behave completely differently in the income statement, and Chapter 20's simulation machinery is what values the first of them.

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17.8 PRIVATE COMPANIES AND STARTUP EQUITY

Everything above assumes an observable share price. A private company has none, with consequences beyond valuation.

The 409A valuation. Section 409A of the US Internal Revenue Code requires that options be struck at or above fair market value, and for a private company that value is in practice established by an independent appraisal (a **409A valuation**, which the tax authorities presume reasonable) refreshed at least annually or after any material event. If the strike turns out to be below fair market value, the option fails Section 409A, and the statute puts the tax on the employee, not the company: the in-the-money amount is taxed as the option vests, plus a 20% additional tax and interest, on a gain the employee cannot sell. The employee cannot check the valuation, which is why companies buy the appraisal (the tax authorities can overturn it only by showing it grossly unreasonable) and why some reimburse employees when one fails.

The appraisal is a genuine exercise in valuation under uncertainty: comparable-company multiples, discounted cash flow, and an allocation of the resulting equity value across the preferred and common share classes, since the common stock employees receive sits behind the preferred holders' **liquidation preferences**, their right to be paid back first when the company is sold or wound up. That subordination is why the 409A price routinely sits far below the price paid in the latest preferred round (often under half at an early-stage company, a gap that narrows only as an exit approaches), and why an employee's strike price can be far below the company's headline valuation.

Illiquidity has no offsetting benefit. The employee holds an option on an asset with no market, so the diversification argument of Section 17.2 bites hardest here. There is not even the escape of exercising and selling; Example 17.8 shows the cost to a leaver.

Example 17.8 The exercise-and-hold trap

An engineer holds 100,000 vested incentive stock options (ISOs) struck at \$0.50. The latest 409A valuation is \$8.00. They leave the company, and the 90-day window starts.

Exercising costs $100{,}000 \times 0.50 = \$50{,}000$ in cash. That is the visible cost. The invisible one is the tax. Exercising an ISO creates no regular taxable income, but the alternative minimum tax (AMT), a parallel US tax computation, counts the spread as income:

$$(8.00 - 0.50) \times 100{,}000 = \$750{,}000$$

which at the 28% AMT rate produces a bill on the order of **\$210,000** the following April, on an unrealized gain in shares that cannot be sold.

The total cash required is roughly \$260,000, against a position that may be worth nothing. The alternative is to let the options lapse and walk away from a nominal \$750,000.

This is not a rare edge case; it is the standard situation for anyone leaving a private company with appreciated options, and it is why several structural responses exist: extended post-termination windows (some companies now allow seven or ten years, though an option exercised more than three months after departure loses ISO status), company-run tender offers that create periodic liquidity, and third-party financing against the shares. Each of these is a response to the same defect: an instrument that generates a tax liability before it generates cash.

Incentive against non-qualified options. Incentive stock options avoid ordinary income tax at exercise but trigger AMT as above, and keep the favorable treatment only if the shares are held two years from grant and one from exercise; at most \$100,000 of stock, valued at grant, may first become exercisable as ISOs in a year. Non-qualified options are taxed as ordinary income on the spread at exercise, withheld through payroll, with no AMT complication, and the company deducts that income; an ISO gives it no deduction unless the shares are sold early. The choice turns on whether the employee will hold long enough to qualify.

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Why startups grant options at all. Cash-constrained companies pay in equity because they have equity and not cash, and the option's leverage means a given accounting expense buys a larger notional bet on the company's success, which is precisely the alignment a startup wants and a mature company does not. The instrument matches the situation. It is when a mature, cash-generating company keeps using it out of habit that the mismatch of Section 17.7 appears.

17.9 WARRANTS BEYOND EMPLOYEE OPTIONS

Section 10.10 introduced warrants alongside employee options. Three families are worth distinguishing, because only two of them dilute.

Corporate warrants are issued by the company, usually attached to a bond or preferred issue as a sweetener, and exercise creates shares. Equation (17.1) applies. The attachment is the point: a warrant lets an issuer sell debt at a lower coupon by handing the buyer equity upside, the trade of Chapter 25's convertible bond with the two parts separable.

Special purpose acquisition company (SPAC) warrants were the highest-volume example of the 2020–21 boom. A SPAC is a listed shell company that raises cash from public investors to merge with a private operating company, which thereby becomes listed. It sells units (typically one share plus a fraction of a warrant), with the warrant struck around 15% above the unit price and running five years from the merger. They dilute, they are usually redeemable by the issuer once the stock trades well above the strike, and that redemption right, a call the issuer holds on the warrant, caps the holder's upside. Redemption and, before the merger, the risk that no deal closes and the warrants expire worthless are why SPAC warrants trade well below the plain Black–Scholes–Merton value of their terms, and a valuation that ignores them is badly wrong.

Covered warrants are issued by a bank, not by the company, on someone else's stock. They are the dominant retail option instrument in Hong Kong, Germany, and several Asian markets, and they are simply calls and puts in a retail wrapper: exercise settles in cash or in existing shares, so there is no dilution and (17.1) does not apply. What they carry instead is issuer credit risk (the uncollateralized counterparty exposure that Chapter 9's CVA prices) and a bid–ask spread set by a single market maker who is also the issuer.

The test is always the same question: does exercise create a share or move one? If it creates one, the dilution factor applies and the instrument is a claim on the company. If it moves one, the instrument is an ordinary option and the issuer is just a counterparty.

17.10 BACKDATING

Between roughly 1996 and 2005, a substantial number of US companies granted options and then chose the grant date afterward, selecting a day on which the stock had been low, and reported the grant as at-the-money. It was not.

The mechanism was pure accounting arbitrage. Under APB 25 an at-the-money grant carried zero expense and an in-the-money grant carried an expense equal to its intrinsic value. Backdating converted the second into the first in the records while delivering the second in substance. Example 17.9 puts numbers on both halves of that statement.

Example 17.9 The arithmetic of choosing the lowest price

Take a stock with 45% annual volatility, $r = 4\%$ and $q = 1\%$. Over a 21-trading-day month, the expected minimum is **8.3%** below the expected price on a randomly chosen day.

Grant 500,000 ten-year options honestly at the month's typical price of \$40 and the strike is \$40. Backdate to the low and the strike is \$36.67.

The option value rises from 21.478 to 22.190, only 3.3%, or \$356,000 across the grant, because a ten-year option at 45% volatility is already worth 54% of the stock price and an 8% strike

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reduction moves it little.

The intrinsic value is the real number. The backdated option is immediately $3.33 in the money, so under APB 25 it carried a required expense of

$$3.33 \times 500{,}000 = \$1.67 \text{ million}$$

which was reported as zero.

That gap is the point: little value to the executive, a great deal of concealed expense to the company. It was the misreporting, not the transfer, that made it a securities-law violation rather than a compensation decision.

Case Study 17.1 How backdating was found, and what ended it

The practice was uncovered statistically rather than by a whistleblower. Erik Lie, an academic studying option grant dates, noticed that stock returns were systematically *negative* before grant dates and *positive* after, across hundreds of companies, over years.

No plausible innocent story fits that pattern. Executives cannot reliably forecast a bottom, and boards do not schedule compensation meetings for market troughs. A pattern of buying at the low across a large sample is neither skill nor luck. It is hindsight.

What ended it was not enforcement but a change in reporting infrastructure. The Sarbanes–Oxley Act of 2002 required insiders to report option grants within **two business days** rather than, as previously, up to 45 days after the end of the fiscal year. Backdating requires a reporting lag, and the lag was removed. Heron and Lie (2007) show the anomaly shrinking sharply after 2002 and surviving mainly among grants reported late.

The episode matters for a reason beyond the fraud: it is a clean case of a disclosure rule making a behavior mechanically difficult rather than merely illegal, as Section 2.7's position limits and Section 2.11's clearing mandate did.

By March 2007 the research firm Glass Lewis counted 257 companies with internal reviews or SEC or Justice Department investigations, the SEC staff put its own count at about 140, dozens restated earnings, and a small number of executives were prosecuted. The much larger effect was on practice: grants are now made on pre-scheduled dates by formula, a lasting improvement bought with a scandal. A restatement now carries a further cost: under the SEC's Rule 10D-1, adopted in 2022 and in force through the exchanges' listing standards since October 2, 2023, a listed company must recover, regardless of fault, the incentive pay its executive officers received in the three fiscal years before a required restatement in excess of what the restated figures would have paid.

Grant timing today. The successor to backdating is **spring-loading**, granting just before good news is released. SAB 120 (November 2021) says such an award's fair value should reflect the pending news, and Item 402(x), adopted in December 2022, requires a company to state its grant-timing policy and tabulate executive option grants made from four business days before to one business day after a 10-Q, a 10-K or an 8-K disclosing material nonpublic information. The same release imposed a 90- to 120-day cooling-off period on officers' and directors' Rule 10b5-1 trading plans, which carry much of the selling of Section 17.11.

17.11 WHO USES EQUITY COMPENSATION, AND WHAT FOR

The instruments in this chapter exist to solve specific problems, and each solves a different one.

Retention is the dominant purpose below executive level, and RSUs do it best. Unvested units are forfeited on departure, so the employee carries a rolling balance of value they lose by leaving. An option grant achieves the same thing only while it is above water, which is why the

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drawdown behavior shown in Example 17.7 drove the shift to RSUs.

Alignment is the argument for options and for performance shares. An option pays only if the share price rises above the grant-date level, so the holder's payoff resembles a shareholder's, levered. Whether this genuinely changes behavior is contested; what is not contested is that it changes the *distribution* of the executive's payoff, and a convex payoff rewards volatility. An executive paid in options has a personal interest in the firm taking more risk, aligned with shareholders only when they want more risk, and never with creditors (Chapter 24).

Cash conservation is the startup's reason (Section 17.8).

Tax and accounting arbitrage was a reason and is much less of one. From 1994, Section 162(m) of the Internal Revenue Code capped a public company's deduction for each top executive's pay at $1 million but exempted performance-based pay, at-the-money options included, a push toward options that the 2017 tax act removed. The zero-expense era ended in 2006 and the reporting lag closed in 2002. What remains is genuine but modest: the corporate deduction on option exercise can exceed the recorded expense when the stock has risen, producing a windfall tax benefit.

Where this shows up in market prices. Two channels connect the chapter to the rest of the book. Large option and RSU programs generate predictable, calendar-driven *selling* (vesting dates and same-day-sale exercises, in which the shares are sold at once to cover the strike and the tax), which single-stock market makers track explicitly. And companies that grant heavily buy back stock to offset dilution, making buyback programs partly a mechanical consequence of the compensation plan rather than a signal about valuation. Reading a buyback announcement as a statement about intrinsic value, when it is offsetting a vesting schedule, is the same category of error as reading a borrow cost as a directional view.

SUMMARY

An employee stock option is non-transferable, forfeited on departure, exercisable only after vesting, and long-dated. Those restrictions break the replication argument on which Chapters 13–16 rest, so its value is not enforced by arbitrage and must be modeled with a behavioral assumption stated explicitly. Employees exercise early (typically about halfway through the term, once the stock has risen far above the strike), which does not contradict Section 11.8's theorem: its hypothesis, that the holder can sell the option instead, does not hold. Defensible methods give materially different values for the same grant, and the most sensitive input is turnover: an HR statistic, not a market observable.

Dilution is measured three ways and they disagree. An outstanding, disclosed grant needs no valuation adjustment at all, because the share price already reflects it, and applying the $N/(N+M)$ factor to the market price double-counts. The reported measure, the treasury stock method behind diluted EPS, understates economic dilution, and the gap is widest exactly where time value dominates: near-the-money options on volatile stocks.

For the holder, exercising and holding is usually the mistake: it spends the strike, prepays the tax and gives up the time value to acquire exposure the option already provided, in exchange for capital-gains treatment of later appreciation. The case for exercising is diversification, not belief in the company, and its price is the after-tax time value; belief in the company is an argument for keeping the option.

Expensing at grant-date fair value replaced intrinsic-value accounting in 2005–06, and the value is never remeasured, so an option that expires worthless was still an expense. Graded recognition moves the cost well forward of where straight-line puts it. Performance awards divide on the same principle: a market condition is priced into the grant-date value and never trued up, while a non-market condition is trued up and reversed if missed. RSUs have displaced

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options at most large listed companies, and the reason is drawdown behavior: a fall in the stock destroys a larger fraction of an option grant's value than of an RSU's, which loses value exactly in proportion to the price. Repricing underwater options is expensed as the value it adds; a value-for-value exchange reports nothing but leaves the employee with less exposure to the share price than the options surrendered, because a deep out-of-the-money option carries a high delta for its value.

Private companies strike at a 409A value below the preferred round, because common stock is subordinate, and leaving with appreciated ISOs creates an AMT liability on a gain that cannot be sold to pay it. Warrants divide by whether exercise creates a share or moves one: corporate and SPAC warrants dilute, while a covered warrant issued by a bank on someone else's stock is an ordinary option carrying issuer credit risk. Backdating exploited the old intrinsic-value rule to conceal a substantial expense. It was detected statistically and ended by Sarbanes–Oxley's two-day reporting rule, a disclosure change that made the practice mechanically difficult rather than merely illegal.

FURTHER READING

Hull and White's (2004) paper on employee stock option valuation sets out the exercise-multiple lattice of Section 17.3 and is the direct source for the approach, with Huddart and Lang (1996), Carpenter (1998) and Bettis, Bizjak and Lemmon (2005) supplying the exercise evidence it is calibrated to; it is also explicit about what the model does and does not claim. Hall and Murphy (2002) and Meulbroek (2001) are where the undiversified-holder discounts quoted in Section 17.2 come from; both make plain how much the answer depends on the risk aversion assumed. Lie (2005) and Heron and Lie (2007) are the empirical backdating papers and repay reading as an exercise in inference from a distribution rather than from a document. On the accounting, ASC 718 and IFRS 2 differ in ways that matter for the graded-recognition question of Example 17.5, and any comparison of US and European compensation disclosures should start there. The practitioner literature on 409A valuation is largely produced by appraisal firms and should be read with that in mind.

PRACTICE QUESTIONS

17.1 Explain why Section 11.8's theorem on early exercise does not apply to an employee option, being precise about which assumption fails.

17.2 A grant has a ten-year term but an expected life of five years. Explain why using the expected life in Black–Scholes is an approximation rather than a correction, and whether its error has a definite sign.

17.3 Using Example 17.1, state which input the reported expense is most sensitive to, and explain why that is awkward for an auditor.

17.4 A company with 20 million shares issues 5 million warrants, and the equivalent call is worth \$8.00. What is each warrant worth? Explain why an outstanding, disclosed employee grant instead requires no dilution adjustment, and state which of a covered and a corporate warrant needs the factor in (17.1).

17.5 A grant of 200,000 options has a grant-date fair value of \$12 each and vests 25% a year over four years. Compute the first-year expense under straight-line recognition, and compare it with graded recognition.

17.6 Explain why one RSU is worth two to three options, why the ratio would be smaller for a more volatile company, and what happens to the expense of a performance award with a market and with a non-market condition if the target is missed.

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17.7 An employee leaves a private company holding 50,000 ISOs struck at $1.00 with a 409A value of $6.00. Compute the cash needed to exercise and the AMT income created.

17.8 Explain how backdating was detected, and why the Sarbanes–Oxley two-day reporting rule stopped it where prohibition had not.

17.9 A vested option is worth $18.40 with the stock at $60 and a strike of $45. State the time value, explain what an employee gives up by exercising, and use Example 17.4 to explain why "I believe in the company" is an argument against exercising rather than for it.

17.10 A company has 40 million shares, 4 million options struck at $25, and an average share price of $40. Compute the incremental shares under the treasury stock method ignoring unrecognized cost, explain why the method understates economic dilution, and identify the type of company for which the understatement is largest.

17.11 Using Example 17.6, state the incremental expense of the repricing and of the value-for-value exchange, and explain why the exchange is cheaper to report and weaker as an incentive.

17.12 A company grants 100,000 cash-settled SARs, fully vested at grant, worth $8 each; a year later they are worth $13. Explain how the expense differs from an equity-settled option grant of the same value, and what the pay-versus-performance table of Item 402(v) would show for the year.

FURTHER QUESTIONS

17.13 Example 17.1 shows the lattice value falling from 16.10 to 12.59 as the assumed exit rate rises from 0% to 6%. Explain the mechanism by which turnover destroys option value, distinguishing its effect on vested from unvested options, and discuss whether a company with high turnover should regard its option program as cheap or as ineffective.

17.14 Section 17.2 states that an employee values the option below its model value because they are undiversified, and that the gap appears nowhere in the accounts. Construct the argument that the accounting is nonetheless correct, then the argument that it is not, and state what each side takes the expense to be measuring.

17.15 An executive is paid entirely in at-the-money options. Show that their payoff is convex in the share price, and explain what this implies about their preferred level of firm risk. Extend the argument to creditors, and explain why the same grant that aligns the executive with shareholders sets them against bondholders.

17.16 Example 17.7 shows an option grant losing 74% of its value on a halving of the stock while an RSU grant loses 50%. A board argues that this makes options the better instrument because they punish failure. Evaluate the argument, distinguishing the retention objective from the incentive objective, and state what the board's conclusion should be for a senior executive and for a mid-level engineer.

17.17 Example 17.9 finds that backdating raised the option's value by only 3.3% while concealing $1.67 million of expense. Explain why the accounting effect so greatly exceeded the value transferred, and use this to argue that the practice is better understood as a financial-reporting offense than as a compensation abuse. Then explain why the two are hard to separate in a company where executive pay is benchmarked against reported earnings.

17.18 Section 17.11 claims that buyback announcements from heavy option grantors are partly mechanical rather than informational. Design a test that would separate the two motives using publicly available data, stating what you would need and what each hypothesis predicts.

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Chapter 18
The Greek Letters

Every chapter since Chapter 13 has rested on one idea: an option's price is the cost of manufacturing its payoff. This chapter verifies it by simulation: a delta-hedging program run 20,000 times costs, on average, exactly the formula's value.

The same simulation produces a second number, and it is this chapter's subject: at five rebalances the standard deviation of the hedging cost is $1.21 on an option worth $3.19. The model assumes continuous, costless hedging; the market permits neither.

The Greeks are the vocabulary for managing that gap. They are not sensitivity analysis added afterward to a pricing model; Section 13.11 established that delta *is* the pricing calculation. This chapter takes the remaining derivatives of the same formula and shows that each answers a specific question a dealer must answer before the end of the day.

Section 18.1 states the dealer's problem. Sections 18.2 to 18.6 take the Greeks one at a time: delta, gamma, theta, vega, rho, and the rest. Sections 18.7 to 18.9 are the book: managing it, what desks actually hedge, and how often to rebalance. Sections 18.10 to 18.13 go past the Greeks to synthetic options, the feedback when hedging moves the market, and static hedging.

18.1 THE DEALER'S PROBLEM

Start where a dealer starts. A client buys 100,000 call options on a stock at $63, struck at $64, with eighteen weeks to expiration: $S_0 = 63$, $K = 64$, $T = 18/52$, $\sigma = 22\%$, $r = 4\%$, and no dividend. These are the inputs behind every number in this chapter. Chapter 15's formula values each option at $3.19438, so the dealer receives

$$100{,}000 \times 3.19438 = \$319{,}438$$

The dealer now has a liability that will be worth $\max(S_T - 64, 0) \times 100{,}000$ in eighteen weeks. What should they do?

Do nothing: a naked position. If the stock finishes at $54 the options expire worthless and the dealer keeps $319,438. If it finishes at $74 they owe $1 million, for a loss of $680,562.

Buy the stock: a covered position. Buying 100,000 shares at $63 costs $6.3 million. If the stock finishes at $74 the shares gain $1.1 million and the options cost $1 million, leaving a profit of $419,438. But if it finishes at $54, the shares have lost $900,000 and the dealer is down $580,562.

Stock at expiration	Naked	Covered
$54	+$319,438	−$580,562
$74	−$680,562	+$419,438

Figure 18.1 plots the two. *Neither is a hedge.* Each converts the dealer's position into a directional bet, merely in opposite directions, and each can lose roughly twice the premium collected. A dealer who intermediates thousands of such trades cannot run either.

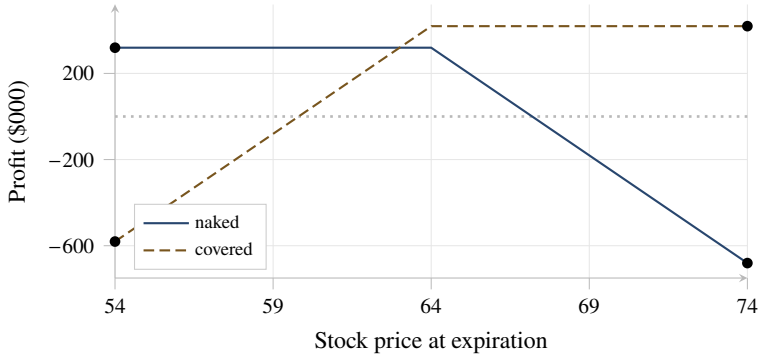


Figure 18.1: Profit at expiration on 100,000 written calls, $K = 64$, against the two unhedged responses. Marks are the table's two cases.

Switch between them: a stop-loss. The obvious repair is to run naked below \$64, buy the 100,000 shares when the stock rises through the strike, and sell them when it falls back. If every trade were done at exactly \$64, the dealer would hold the shares precisely when the option finishes in the money, the hedge would cost nothing, and the premium would be riskless profit. It cannot work. Purchases are made just above \$64 and sales just below, a stock near the strike crosses it again and again, and watching more closely makes each crossing cheaper but the crossings more numerous, so the hedge's error does not shrink. It is the naive hedge that delta hedging improves on.

What is needed is a position that holds up in *both* rows of the table, whichever way the stock moves, and Chapter 13 already found it: hold a quantity of stock that makes the portfolio insensitive to the stock's movement. That quantity is delta.

Delta is the first of several quantities the dealer has to control. Over a short interval Δt the value Π of the position changes because the stock moves by ΔS, because time passes, and because implied volatility moves by $\Delta\sigma$. Expanding that change in the things that move gives

$$\Delta\Pi \approx \Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma \tag{18.1}$$

where delta Δ, gamma Γ, theta Θ (per year, with Δt in years) and vega $\mathcal{V}$ are the position's partial derivatives with respect to S, S twice, t and σ. A change Δr in the interest rate adds a term $\rho\,\Delta r$ in the same way, and for equity options it is usually small because the rate barely moves over the interval. Each Greek is the coefficient of one term, and hedging is the choice of offsetting positions that cancel terms of this expansion. Sections 18.2 to 18.6 take the terms in turn.

18.2 DELTA

Delta, the coefficient of the first term of (18.1), is the rate of change of an option's value f with respect to the price S of the underlying:

$$\Delta = \frac{\partial f}{\partial S} \tag{18.2}$$

For a European call on a non-dividend stock, Section 15.4 identified it already:

$$\Delta_{\text{call}} = N(d_1), \qquad \Delta_{\text{put}} = N(d_1) - 1 \tag{18.3}$$

so a call's delta lies in $(0, 1)$ and a put's in $(-1, 0)$. With a dividend yield, both are multiplied by e^{-qT}.

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In the dealer's problem, $N(d_1) = 0.5199$. Holding 51,995 shares against the 100,000 written calls makes the portfolio delta-neutral: a small move in the stock changes the value of the shares and the options by equal and opposite amounts. Example 18.1 reads delta as a multiplier and converts it into the option's leverage.

Every Greek in this chapter is evaluated at the option's own implied volatility, the convention Section 15.8 set for pricing, so a book's delta adds Black–Scholes deltas computed at different volatilities. That is exact only if each option's implied volatility stays put as the stock moves; Section 19.11 shows what changes when the smile moves with the spot, and equation (19.4) corrects the delta for it.

Example 18.1 Delta as a multiplier, and an option's leverage

Delta as a multiplier. Delta converts a move in the stock into a move in the option. If the stock rises by \$1 to \$64, the call should gain about $0.5199 \times 1 = \$0.52$; repricing at \$64 gives 3.7385, a rise of 0.5442. The gap, 0.0242, is curvature: the gamma term of (18.1), $0.04886/2 = 0.0244$ with the gamma of Section 18.3. For a \$3 fall delta predicts $-3 \times 0.5199 = -1.5597$, but repricing at \$60 gives 1.8568, a fall of only 1.3376: the value curve stays above delta's tangent.

Leverage. Dividing by the option's value turns the multiplier into percentages:

$$\frac{\Delta f}{f} \approx \Omega \frac{\Delta S}{S}, \qquad \Omega = \frac{S\,\Delta}{f}$$

where Ω, the option's **elasticity**, is its leverage: the percentage change in its value for a 1% change in the stock. Here $\Omega = 0.5199 \times 63/3.19438 = 10.25$: a 1% rise to \$63.63 raises the call by 10.56% exactly, and the call's volatility is about $\Omega\sigma \approx 226\%$. At the same stock price Ω is 5.71 at a strike of 54 and 15.98 at 74. Out-of-the-money calls carry the most leverage, which is why they are the speculator's instrument; Ω is also the $N(d_1)V/E$ that Section 24.6 computes for equity as a call on the firm.

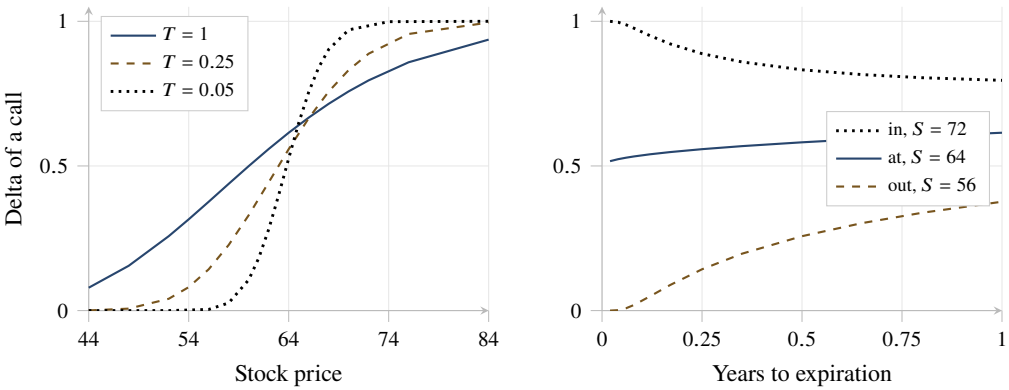


Figure 18.2: Call delta against the stock price and against remaining life. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.2 shows the feature that makes hedging difficult. As expiration approaches, delta stops being a smooth function and approaches a step: an option that will finish in the money has delta 1, one that will finish out has delta 0, and near the strike the transition becomes almost vertical. A hedge tracking a near-vertical function must buy or sell almost the entire position on a small move in the stock. The right panel says it along the other axis: hold the stock still and delta walks to 1, to 0, or to neither as expiration nears. This is the pin risk of Example 10.6 and the 0DTE mechanism of Section 10.12, both now visible as statements about $N(d_1)$.

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A hedging program, week by week

The dealer's position is best understood by running it. Tables 18.1 and 18.2 follow the same written position (100,000 calls, $K = 64$, eighteen weeks) along two simulated price paths, rebalancing weekly to restore delta neutrality. The two paths differ in one respect only: the first finishes in the money and the second finishes out.

The bookkeeping is the same in both. The dealer borrows to buy $\Delta \times 100{,}000$ shares at week 0, and each week borrows or repays to bring the holding back to the new delta. The cumulative cost column compounds the previous week's balance at 4% and then adds the week's trade, so it is the dealer's outstanding debt.

Table 18.1: Weekly rebalancing against 100,000 written calls, path finishing in the money. $S_0 = 63$, $K = 64$, $r = 4\%$, $\sigma = 22\%$, 18 weeks. Costs in \$000.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.12	0.421	−9,889	−604.4	2,673.8	2.1
2	62.51	0.488	6,651	415.8	3,091.6	2.4
3	60.86	0.394	−9,362	−569.7	2,524.2	1.9
4	58.46	0.261	−13,343	−780.0	1,746.1	1.3
5	61.57	0.418	15,788	972.1	2,719.5	2.1
6	62.02	0.438	1,915	118.8	2,840.4	2.2
7	62.03	0.431	−692	−43.0	2,799.6	2.2
8	61.55	0.391	−3,959	−243.7	2,558.1	2.0
9	63.48	0.513	12,183	773.4	3,333.5	2.6
10	61.94	0.396	−11,724	−726.2	2,609.8	2.0
11	64.41	0.574	17,822	1,147.9	3,759.8	2.9
12	64.71	0.597	2,360	152.7	3,915.4	3.0
13	63.86	0.523	−7,417	−473.7	3,444.7	2.6
14	63.25	0.455	−6,797	−429.9	3,017.5	2.3
15	66.43	0.781	32,561	2,163.0	5,182.9	4.0
16	70.36	0.988	20,695	1,456.1	6,642.9	5.1
17	69.67	0.998	975	67.9	6,716.0	5.2
18	66.04	1.000	238	15.7	6,736.8	
Shares delivered on exercise, proceeds $64 \times 100{,}000$					−6,400.0	
Cost of hedging the option					**336.8**	

Example 18.2 Reading the two hedging programs

The premium collected in Section 18.1 was \$319,438. Invested at 4% for eighteen weeks it would be worth \$323,900 at expiration, the number to compare the hedging cost against.

The hedge cost \$336,800 on the first path and \$309,300 on the second. One is 4.0% above the premium and the other 4.5% below. The dealer lost money on one path and made money on the other, having done nothing different in either case.

That is the whole content of the simulation below, seen transaction by transaction: discrete hedging does not reproduce the option payoff; it reproduces it *on average*, and the residual is the standard deviation Table 18.3 measures. Weekly rebalancing over eighteen weeks is eighteen adjustments, and eighteen is not many.

The trades themselves. In weeks 4 and 5 of Table 18.1 the stock fell to 58.46 and the dealer *sold* 13,343 shares, then rose to 61.57 and the dealer *bought* 15,788 back, \$3.11 higher, a loss of

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Table 18.2: The same position along a path finishing out of the money. Costs in $000.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.60	0.446	−7,443	−458.5	2,819.7	2.2
2	63.45	0.536	9,080	576.1	3,398.0	2.6
3	61.36	0.421	−11,549	−708.6	2,692.0	2.1
4	61.57	0.426	474	29.2	2,723.2	2.1
5	61.08	0.390	−3,530	−215.6	2,509.7	1.9
6	59.75	0.305	−8,524	−509.3	2,002.3	1.5
7	58.01	0.201	−10,372	−601.7	1,402.2	1.1
8	58.38	0.205	346	20.2	1,423.5	1.1
9	61.22	0.358	15,323	938.1	2,362.6	1.8
10	58.88	0.197	−16,084	−947.1	1,417.4	1.1
11	62.68	0.440	24,278	1,521.8	2,940.3	2.3
12	64.54	0.584	14,383	928.3	3,870.8	3.0
13	64.24	0.558	−2,599	−167.0	3,706.9	2.9
14	66.55	0.765	20,684	1,376.5	5,086.2	3.9
15	64.57	0.594	−17,061	−1,101.6	3,988.5	3.1
16	63.73	0.484	−11,027	−702.8	3,288.8	2.5
17	62.86	0.292	−19,212	−1,207.7	2,083.7	1.6
18	60.90	0.000	−29,162	−1,776.0	309.3	
Option expires worthless; no shares held					0.0	
Cost of hedging the option					**309.3**	

about $41,500 on the shares just let go. Nothing was mismanaged: as Example 13.5 showed, delta hedging sells low and buys high by construction, and the accumulated cost of doing so is the option premium.

Why the paths end where they do. In Table 18.1 delta walks up to 1.000 and stays there, so the dealer ends holding the full 100,000 shares and delivers them on exercise. In Table 18.2 delta walks down to zero and the last trade liquidates the remaining position. The program converges to the terminal payoff automatically (no decision is made at expiration) because $N(d_1)$ becomes the step function of Figure 18.2.

Where the cost came from. On the out-of-the-money path the option expired worthless and the hedge still cost $309,300. In weeks 13 to 15 the stock went 64.24, 66.55, 64.57, ending within a third of a dollar of where it started, while the dealer bought 20,684 shares at 66.55 and sold 17,061 of them back at 64.57: about $33,800, a tenth of the premium, lost on one whipsaw. That is the gamma cost of Section 18.3, arriving before the concept is defined: a delta hedge pays for movement, not direction, and most for movement near the strike, where gamma peaks.

Portfolio delta. Delta is additive. A book of options on the same underlying has

$$\Delta_{\text{portfolio}} = \sum_i w_i \Delta_i$$

where w_i is the number of options of type i held, negative for those written. A desk holding thousands of positions therefore computes one number and trades one quantity of stock.

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The price is the cost of the hedge

The two paths above showed the same program landing once above and once below the premium. Over many paths the claim behind every chapter since Chapter 13, verified on a two-step tree in Example 13.5, can be tested against the formula itself. Take the dealer's option of Section 18.1, worth \$3.1944, and run the program of Table 18.1 over 20,000 simulated paths, rebalancing n times and paying out $\max(S_T - K, 0)$ at expiration. The paths carry a *real-world* drift of $\mu = 10\%$, deliberately different from $r = 4\%$. Table 18.3 reports the cost in time-0 dollars.

Table 18.3: Cost of hedging 1 written call with no trading costs, in time-0 dollars. 20,000 paths; $\mu = 10\%$; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean − formula
5	3.2079	1.2050	+0.0135
10	3.1946	0.8560	+0.0002
20	3.1950	0.6265	+0.0006
50	3.1972	0.4011	+0.0028
100	3.1942	0.2873	−0.0002
250	3.1949	0.1830	+0.0005
Formula (15.2)	3.1944	—	—

The mean cost is the formula's value. Once rebalancing is frequent enough the two agree to within a few tenths of a cent, which is the simulation's own sampling error. The dealer who charges \$3.19 and hedges diligently breaks even on average; that is what the number means, and why the formula is a price rather than a forecast. And the drift did not matter: the paths were simulated at 10% while the formula uses 4%, which is Chapter 13's principle verified numerically.

Discrete hedging leaves risk, and it falls as $1/\sqrt{n}$. At five rebalances the standard deviation is \$1.21, more than a third of the option's value; at 250 it is \$0.18. The hedge is exact only in the continuous limit, and the residual is what Chapter 15's assumption of continuous trading conceals. Kamal and Derman (1999) give its standard deviation as approximately $\sqrt{\pi/4}\,\sigma\mathcal{V}/\sqrt{n}$, with $\mathcal{V}$ the option's vega (14.77 here; Section 18.5). That is 0.407 at $n = 50$ and 0.288 at $n = 100$, within 2% of the table from 50 rebalances on; at five, where the approximation is coarsest, it overstates by 7%. That residual is where Section 18.9 begins: the model says hedge continuously, the market charges for every trade, and Table 18.7 adds the charge to exactly this simulation.

18.3 GAMMA

Delta hedging is exact only for infinitesimal moves, because delta itself changes. **Gamma** measures how fast:

$$\Gamma = \frac{\partial^2 f}{\partial S^2} = \frac{N'(d_1)}{S_0\sigma\sqrt{T}} \tag{18.4}$$

where N' is the standard normal density, the slope of N:

$$N'(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}.$$

Gamma is positive for both purchased calls and purchased puts, and negative for written ones.

The practical meaning is direct. A delta-neutral portfolio has no first-order exposure, so its profit and loss $\Delta\Pi$ over a move ΔS, to second order, is what remains of (18.1) once the delta

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term is hedged away and time and volatility are held still:

$$\Delta\Pi \approx \frac{1}{2}\Gamma\,(\Delta S)^2 \tag{18.5}$$

Figure 18.3 is the same statement geometrically. The delta approximation is the tangent at the hedge point; the option's value curves away from it on both sides, so a written option hedged with stock loses whichever way the stock goes.

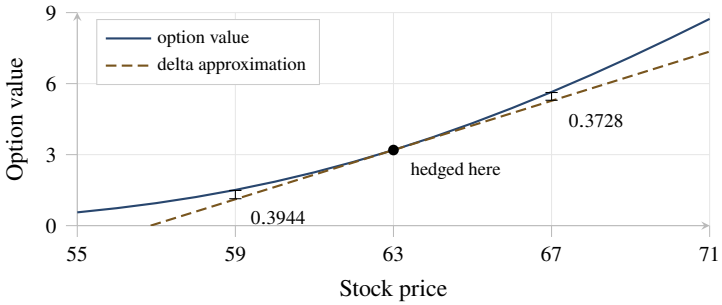


Figure 18.3: The dealer's option against its delta approximation at \$63. The marked gaps are Example 18.3's losses.

Example 18.3 What gamma costs a short option position

The dealer of Section 18.1 is delta-neutral at $S = 63$, with $\Gamma = 0.04886$ per option, so the book's gamma is $-0.04886 \times 100{,}000$, negative because the options are written.

Suppose the stock jumps before the hedge can be adjusted:

Stock	Actual P&L	$-\frac{1}{2}\Gamma(\Delta S)^2$
59	−0.3944	−0.3909
61	−0.0987	−0.0977
63	0	0
65	−0.0959	−0.0977
67	−0.3728	−0.3909

per option. The approximation is accurate to within two cents, and on 100,000 options a move to \$59 costs \$39,443 and a move to \$67 costs \$37,283.

Both columns are negative. A delta-neutral short-option position loses money whichever way the stock moves, and loses in proportion to the *square* of the move. Being short gamma means being short large moves regardless of direction, the same statement as Section 12.4's observation that a straddle buyer is betting on magnitude, seen from the other side of the trade.

What gamma is worth. Equation (18.5) puts a value on movement, and it depends on two things. The *sign* decides whether movement pays or costs: a book long gamma gains whichever way the stock goes, a book short gamma loses either way. The *size* decides how much, and because the relation is quadratic, doubling the move quadruples the amount. Section 18.4 sets that against what holding the position costs.

Figure 18.4 shows where the trouble concentrates. Gamma peaks at the money and rises sharply as expiration nears: at one year the peak is 0.031, at three months 0.058, at under three weeks 0.127, four times the one-year peak. At the money the contrast is sharper: with a week to run gamma is 0.20 against 0.027 for a one-year option (Figure 18.5), more than seven times. Gamma is the rate at which delta moves, so on the dealer's 100,000 options that means

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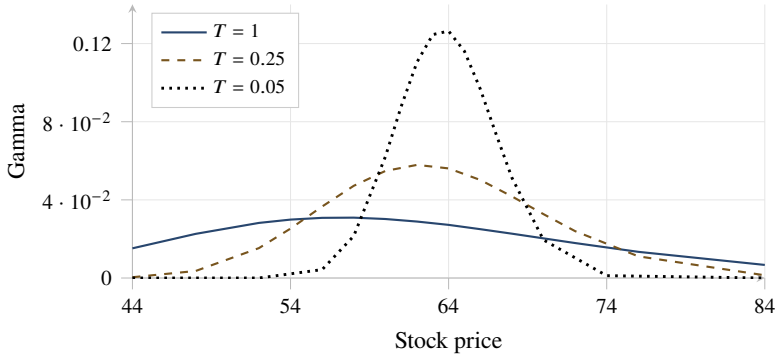


Figure 18.4: Gamma. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

rehedging some 20,000 shares for every dollar the stock travels, against 2,700 for the one-year option, which explains most of what Section 10.12 said about same-day options.

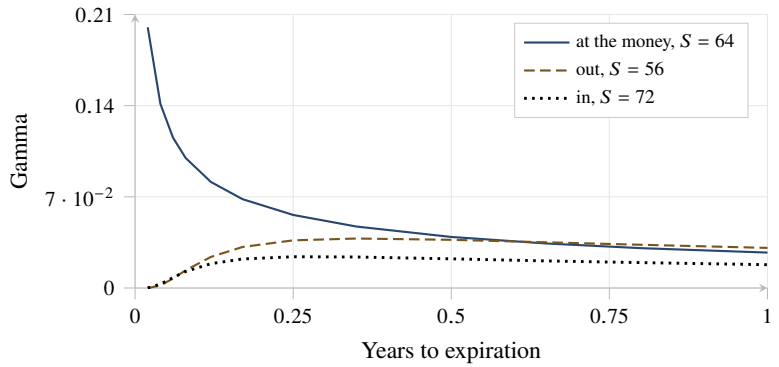


Figure 18.5: Gamma as expiration approaches. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.5 shows the other dimension. At the money gamma rises without limit as expiration approaches, while away from the money it collapses toward zero. An option that will finish decisively in or out of the money has nothing left to hedge; one that will finish near the strike has all of it left, and no time in which to do it.

Neutralizing gamma requires options. Delta can be hedged with the stock, because the stock's delta is 1. Gamma cannot: the stock's gamma is zero, so no quantity of it changes the book's gamma. Only another option can offset gamma, which is why dealers trade with each other and not only with clients, and why a desk short gamma across its whole book cannot reduce that exposure cheaply.

18.4 THETA, AND WHY IT IS GAMMA UNDER ANOTHER NAME

Theta, the coefficient of Δt in (18.1), is the rate at which an option loses value as time passes:

$$\Theta = \frac{\partial f}{\partial t} \tag{18.6}$$

For a European option on a non-dividend stock,

$$\Theta_{\text{call}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2), \qquad \Theta_{\text{put}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2) \tag{18.7}$$

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Both share the first term, the decay of time value, and differ in the second, the interest on the strike, negative for the call and positive for the put. Theta is usually quoted per calendar day, and for the dealer's option $\Theta = -5.8757$ per year, or about $-\$0.016$ a day.

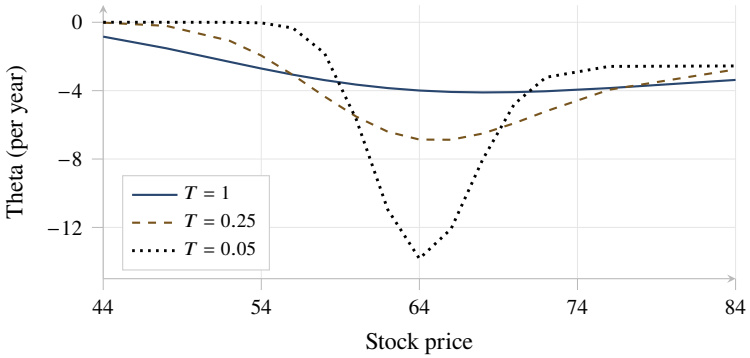


Figure 18.6: Call theta, per year. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.6 shows the shape. Set beside Figure 18.4 it is close to a mirror image (both concentrate near the money, and both sharpen as expiration nears), and the rest of this section shows that the resemblance is an identity rather than a coincidence.

Theta is not a risk in the sense delta and gamma are, because the passage of time is not uncertain; what matters is its tie to gamma.

Recall the Black–Scholes–Merton equation (15.1):

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rf$$

This holds for every derivative the model prices. For a *delta-neutral* portfolio, $\Delta = 0$ and the equation reduces to

$$\boxed{\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi} \tag{18.8}$$

Theta and gamma are two sides of one quantity. A delta-neutral book long gamma normally pays theta and one short gamma normally earns it, with the trade-off fixed by the model rather than chosen by the trader. The word is *normally* because, where gamma is nearly zero, the interest term in (18.8) is all that remains, which is how a deep in-the-money put carries positive gamma and positive theta at once. Example 18.4 checks the identity on the dealer's option.

Example 18.4 Verifying the identity

For the dealer's option: $\Theta = -5.8757$, $\Delta = 0.51995$, $\Gamma = 0.048862$, $S = 63$, $\sigma = 0.22$, $r = 0.04$, $f = 3.1944$.

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = -5.8757 + 0.04 \times 63 \times 0.51995 + \frac{1}{2} \times 0.0484 \times 63^2 \times 0.048862$$
$$= -5.8757 + 1.3103 + 4.6932 = 0.1278$$

and $rf = 0.04 \times 3.1944 = 0.1278$, exactly.

A trader long gamma is paid whenever the stock moves, by (18.5), and pays theta every day it does not. A trader short gamma collects theta daily and pays whenever the stock moves. Neither is a riskless profit; the model prices the exchange, and (18.8) is the price.

This also explains the standing warning of Section 12.11. A portfolio of butterflies, condors, and calendar spreads is not diversified: all are short gamma, so all collect theta and all lose together on a large move. Equation (18.8) says they are the same position.

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The break-even move. That exchange has a break-even. A delta-hedged position also carries its financing, the interest on the borrowing in Table 18.1's cumulative cost column, and that is the $r\Pi$ of (18.8). Net of it, the position's carry is $\Theta - r\Pi = -\frac{1}{2}\sigma^2 S^2 \Gamma$. Setting the gamma gain of (18.5) against that carry over one trading day, $\Delta t = 1/252$ on Section 15.8's clock, gives the move at which a delta-hedged position comes out level:

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\sigma^2 S^2 \Gamma \, \Delta t \qquad \Longrightarrow \qquad |\Delta S| = \sigma S \sqrt{\Delta t} \tag{18.9}$$

For the dealer's option that is $63 \times 0.22/\sqrt{252} = \$0.87$, or 1.39% of the \$63 spot. On a day the stock moves less than that the written position is ahead; on a day it moves more it is behind, and the whipsaw of Example 18.2 was a run of the second kind. Setting the gain against the option's theta alone, $|\Theta|/365$, would give \$0.81, which leaves out the financing and counts weekends as days on which the stock moves.

Gamma cancels, so the break-even is one standard deviation of the stock's daily move at the implied volatility, whatever the option. That is what it means for the option to have been fairly priced, and why a desk that has sold gamma is short realized volatility against the implied volatility it sold.

Which volatility to hedge at. Hedged at its implied volatility σ_i, a long option earns the gap between realized and implied variance day by day, weighted by gamma (Further Question 18.15), so its total depends on the path as well as on how much the stock moves. If the forecast is right, hedging at the volatility σ_r the trader expects to be realized locks in the total instead, at $f(\sigma_r) - f(\sigma_i)$, the option's value at that volatility less its value at the implied one; the position's marked value still swings until expiration (Ahmad and Wilmott, 2005). Most desks hedge at implied, the volatility the book is marked at.

18.5 VEGA

Vega measures sensitivity to volatility and is the coefficient of $\Delta\sigma$ in (18.1):

$$\mathcal{V} = \frac{\partial f}{\partial \sigma} = S_0 \sqrt{T}\, N'(d_1) \tag{18.10}$$

It is positive for all purchased options, calls and puts alike, as Section 11.1 established without a model.

For the dealer's option, $\mathcal{V} = 14.769$, meaning a one-percentage-point rise in volatility raises each option by \$0.148, or \$14,769 across 100,000 options. Set that beside the \$319,438 premium: a two-point move in implied volatility changes the position's value by a tenth of what the dealer was paid to take it on.

Two features matter in practice.

Vega is largest for long-dated options, scaling roughly with $\sqrt{T}$, which is the opposite of gamma. A book can therefore be short gamma and long vega at once, and desks routinely are.

Vega should not exist. The Black–Scholes–Merton model assumes σ is a known constant, so a sensitivity to changes in it is a sensitivity to something the model says cannot happen. That the market computes and hedges it anyway is the clearest admission that the model is being used as a quoting convention rather than as a description, which is Section 15.7's point, now visible in the risk report. Chapter 19 confronts the consequences.

Like gamma, vega peaks near the money (Figure 18.7); unlike gamma, it does not spike as expiration approaches but shrinks. Comparing with Figure 18.4 makes the point that governs how books are constructed: the short end carries the gamma and the long end carries the vega, so a desk's two largest risks live at opposite ends of the maturity curve and must be hedged with different instruments.

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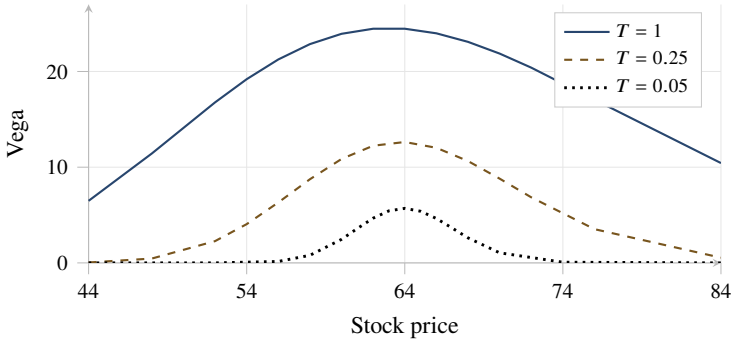


Figure 18.7: Vega. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

18.6 RHO AND THE REST

Rho measures sensitivity to the interest rate and supplies the $\rho\,\Delta r$ term that (18.1) sets aside:

$$\rho_{\text{call}} = KTe^{-rT}N(d_2), \qquad \rho_{\text{put}} = -KTe^{-rT}N(-d_2) \tag{18.11}$$

Only the discounted strike in Chapter 15's formula carries r, so rho is the time to expiration multiplied by the present value of what the holder expects to pay. For the dealer's option $\rho = 10.233$, so a one-point rise in rates is worth \$0.102 an option, or \$10,233 across the 100,000.

Set that beside vega: a one-point move in implied volatility is worth \$14,769 on the same position, the same order of magnitude. Rho is ignored on short-dated equity books not because the sensitivity is small but because the input barely moves: implied volatility travels several points in a week, policy rates in 25-basis-point steps a few times a year.

Maturity reverses that. Rho grows with T while an option's value grows roughly with $\sqrt{T}$, so the exposure per dollar of premium climbs with the term:

At-the-money call	Value	Rho	Per 100bp, % of value
18 weeks	3.739	11.29	3.0%
1 year	6.841	32.51	4.8%
5 years	18.008	147.71	8.2%
10 years	27.301	253.04	9.3%

So rho is watched wherever the term is long or the rate is the point: the ten-year employee options and warrants of Chapter 17, the convertible bonds of Chapter 25, the structured notes of Chapter 35; and the conversions, reversals and boxes of Chapter 11, which are option structures assembled so that the financing rate is the only exposure left, and which market makers use to lend and borrow. In Chapter 29's rate derivatives rho stops being a side-Greek altogether, because there the curve is the underlying.

In currency options rho is first-order. An option on an exchange rate has two rhos, one per currency, and Chapter 16's substitution gives both:

$$\rho = KTe^{-rT}N(d_2), \qquad \rho_f = -STe^{-r_fT}N(d_1)$$

where r is the domestic interest rate, r_f the foreign one, and S the exchange rate, as in Chapter 16. They almost cancel. Adding them gives exactly $-Tc$, so a parallel move in both rates does no more than discount the option, and what survives is the exposure to the *differential*, which is what sets the forward. Take Chapter 16's euro–dollar inputs: spot 1.0850, dollar rates 4%, euro rates 2.5%, volatility 8%. A one-year euro call struck at spot gains 14.4% of its premium when dollar rates rise 100 basis points and euro rates do not; the equity option of Section 18.1 gains 3.2% on the same shift, and even a three-month currency option gains 7.5%. That is why a

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central bank meeting is a scheduled volatility event in the currency market in a way it is not for a single stock.

Beyond the five, desks compute higher-order and cross sensitivities as the book requires: **vanna** ($\partial^2 f/\partial S\,\partial\sigma$, how delta moves with volatility) and **volga** ($\partial^2 f/\partial\sigma^2$, how vega moves with volatility) matter wherever the smile matters, which is Chapter 19's territory and essential for the barrier options of Chapter 34. **Charm** ($\partial^2 f/\partial S\,\partial t$, how delta moves with the passage of time) is a rounding error for a monthly option and the dominant hedging flow for the same-day options of Chapter 23, where a dealer's delta can fall by two-thirds over a session in which the index does not move.

Table 18.4 collects the five.

Table 18.4: Definitions, signs for a purchased option, and what each answers.

Greek	Derivative	Sign	Question it answers
Delta Δ	$\partial f/\partial S$	+ call, − put	How much stock to hold
Gamma Γ	$\partial^2 f/\partial S^2$	+	How often to rebalance, and what a jump costs
Theta Θ	$\partial f/\partial t$	−	What waiting costs per day
Vega $\mathcal{V}$	$\partial f/\partial\sigma$	+	What a change in implied volatility costs
Rho ρ	$\partial f/\partial r$	+ call, − put	What a rate move costs

18.7 MANAGING A BOOK

A desk does not hedge options one at a time. Every Greek is a derivative, so all are additive across positions on the same underlying, and a book of thousands of trades is five numbers.

Additivity means the book has a single expansion (18.1), whose coefficients are the summed Greeks, and hedging the book is canceling its terms in order. The order is set by which instrument reaches which term. Stock carries only the first term, and only options carry the second, so gamma is cleared with options before delta is cleared with stock; Example 18.6 does it in that order. Example 18.5 first checks the expansion against a full revaluation of the book in Table 18.5.

Table 18.5: Three positions on one stock. $S = 63$, $r = 4\%$, $\sigma = 22\%$.

Position	Quantity	Delta	Gamma	Vega
Calls K=64, 18 weeks	−100,000	−51,995	−4,886	−1,476,869
Puts K=58, 18 weeks	+50,000	−10,441	+1,761	+532,343
Calls K=70, 40 weeks	−40,000	−15,435	−1,259	−845,404
Book		**−77,870**	**−4,384**	**−1,789,929**

Desks report gamma in dollars rather than shares: the change in the position's dollar delta for a 1% move in the stock, $\Gamma S^2/100$. For this book that is $-4{,}384 \times 63^2/100 = -\$174{,}000$, the stock it must buy after every 1% rise. Vega is quoted per volatility point, one hundredth of the figures in the table, and theta per calendar day.

Example 18.5 The book's expansion against a full revaluation

One day passes ($\Delta t = 1/365$), the stock falls from \$63 to \$61 ($\Delta S = -2$), and implied volatility rises from 22% to 23% ($\Delta\sigma = 0.01$), with r unchanged at 4%. Table 18.5 gives the book's delta, gamma and vega. Its theta, from (18.7), is +601,812 per year, positive because the book is net short options. The terms of (18.1) are

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Term	Book Greek	Change	P&L ($)
$\Delta\,\Delta S$	−77,870	$\Delta S = -2$	+155,740
$\frac{1}{2}\Gamma(\Delta S)^2$	−4,384	$(\Delta S)^2 = 4$	−8,768
$\Theta\,\Delta t$	+601,812	$\Delta t = 1/365$	+1,649
$\mathcal{V}\,\Delta\sigma$	−1,789,929	$\Delta\sigma = 0.01$	−17,899
Sum of the four terms			+130,722
Full revaluation			+133,001
Residual			+2,279

The full revaluation prices each of the three positions by the Black–Scholes–Merton formula at $S = 61$, $\sigma = 23\%$ and one day less to expiration. The expansion accounts for all but $2,279 of the $133,001, or 1.7%. The delta term carries nearly everything. The day's theta of $1,649 is, by (18.8), largely the book's payment for being short 4,384 gamma, and by (18.9) its gamma and theta break even on a move of one daily standard deviation, $0.87; the $2 move costs more than five times the day's theta.

The residual is what the expansion leaves out: terms of third order in ΔS, and cross terms. About $1,850 of it is the cross term $(\partial^2\Pi/\partial S\,\partial\sigma)\,\Delta S\,\Delta\sigma$, the change in the book's vega as the stock moves, which the separate delta and vega terms cannot see. The expansion is local, and at this size what it misses is small; Section 18.10 takes up the moves where it is not.

Example 18.6 Neutralizing a book, in the right order

Table 18.5 gives a book that is short 77,870 delta, short 4,384 gamma, and short 1.79 million vega. Hedge it with one traded option, the 40-week call struck at 64:

	Delta	Gamma	Vega
The book	−77,870	−4,384	−1,789,929
40-week call, K=64, each	0.5692	0.03232	21.711

Gamma first, because it cannot be hedged with stock. Zeroing the book's gamma requires buying

$$\frac{4{,}384}{0.03232} = 135{,}644 \text{ options}$$

Then delta, with stock. Those 135,644 options carry $135{,}644 \times 0.5692 = 77{,}209$ of delta, which takes the book from −77,870 to −661. Buying 661 shares restores neutrality.

Vega is what remains. The hedging option carried $135{,}644 \times 21.711 = 2{,}944{,}967$ of vega, so the book moves from −1,789,929 to +1,155,038. Neutralizing gamma has flipped the sign of the vega exposure and left it large.

That is the general problem. There are three exposures and, on a single underlying, only two hedging instruments: the stock and one option. Neutralizing all three requires a *second* option of a different maturity, because gamma and vega have different term structures (Figures 18.4 and 18.7). Short-dated options are the gamma instrument and long-dated options are the vega instrument, and a desk needs both. Example 18.7 builds that hedge.

Example 18.7 Clearing gamma and vega together

Example 18.6 ended gamma-neutral but $1.16 million long vega. Two options of *different* maturities clear both at once. Keep the 40-week call and add an 8-week call at the same strike:

Instrument	Delta	Gamma	Vega	Gamma per unit vega
A: K=64, 8 weeks	0.4729	0.07321	9.835	0.00744
B: K=64, 40 weeks	0.5692	0.03232	21.711	0.00149

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The last column is the whole reason two maturities are enough. The short-dated option delivers five times as much gamma per unit of vega as the long-dated one, so the two are not redundant and any combination of gamma and vega can be reached.

Writing n_A and n_B for the quantities bought, neutrality is a 2×2 system:

$$0.07321\, n_A + 0.03232\, n_B = 4{,}384$$
$$9.835\, n_A + 21.711\, n_B = 1{,}789{,}929$$

with solution $n_A = 29{,}357$ and $n_B = 69{,}145$. Both are positive: the book is short both exposures, so both options are bought.

Then delta, with stock. The two positions carry $29{,}357 \times 0.4729 + 69{,}145 \times 0.5692 = 53{,}240$ of delta, taking the book from −77,870 to −24,630. Buying 24,630 shares closes it, and all three exposures are now zero.

Gamma and vega must be solved *together*, because every option moves both; delta is solved last, because the stock moves delta alone and so cannot disturb what has just been set. A hedge built in the other order has to be redone.

It is not free. The two option positions cost $421,218 to establish and carry a combined theta of about −$1,475 per day. That is Section 18.4 arriving as a cash flow: the desk has bought gamma, and buying gamma means paying theta every day the stock fails to move. Neutrality is a position with a running cost, not a state of rest.

Books across many underlyings

Table 18.5 aggregated positions on *one* stock. A single-stock desk may hold options on hundreds of stocks, and hedging each separately would multiply Table 18.7's transaction costs by the number of underlyings.

The standard response reuses Section 3.6's device. Convert each position's delta into an equivalent index exposure using the stock's beta,

$$\Delta_{\text{index-equivalent}} = \sum_i \beta_i\, \Delta_i\, S_i \tag{18.12}$$

where Δ_i is the book's delta in stock i, S_i is its price and β_i its beta against the index. Each term is therefore a dollar exposure rather than a share count, and the sum takes the place of βV_A in Section 3.6's hedge ratio, which divides it by the value of one futures contract to give the number to sell. One trade replaces hundreds.

The hedge removes the *market* component of the book's delta and leaves the idiosyncratic component, the part of each stock's movement unexplained by the index. That residual is small on any one day and is not diversified away, because the book's positions are not random: a desk accumulates whatever clients want to sell.

The same decomposition defines a trade. A book long single-stock options and short index options is long idiosyncratic volatility and short market volatility, which is the dispersion position Section 16.2 named. A desk that hedges by (18.12) without noticing has taken that position by accident, and it pays off when correlations fall and loses when they rise, which is when everything else is losing too.

Greeks for other underlyings

Chapter 16 extended the pricing formula to indices, currencies, and futures by the substitution $S_0 \to S_0 e^{-qT}$. The Greeks follow the same route, and Table 18.6 gives the results. The simplest case is a forward contract: its delta is e^{-qT}, or 1 without a yield, and its gamma is zero, so its hedge never has to chase the stock.

Two consequences follow. Delta is no longer bounded by 1 but by e^{-qT}: for an index option with $q = 3\%$ and $T = 2$, the factor $e^{-qT} = 0.942$ means a deep in-the-money call requires

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Table 18.6: Call Greeks for an underlying with yield q.

Greek	Formula
Delta	$e^{-qT}N(d_1)$
Gamma	$\dfrac{e^{-qT}N'(d_1)}{S_0\sigma\sqrt{T}}$
Theta	$-\dfrac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} + qS_0N(d_1)e^{-qT} - rKe^{-rT}N(d_2)$
Vega	$S_0\sqrt{T}\,e^{-qT}N'(d_1)$

only 0.94 units of the index, not 1. Currency desks often quote the premium-adjusted delta of Section 16.4 instead, so a quoted delta must be read with its convention. And for a *futures* option, $q = r$ makes every Greek a derivative with respect to the futures price F: the delta is $e^{-rT}N(d_1)$, the discounting of a payoff settled on F, and the hedge is held in the futures contract itself. The factor in Section 18.9's futures hedge (18.13) is a different adjustment, needed only when a spot-based option is hedged with futures.

18.8 WHAT DESKS ACTUALLY HEDGE

A textbook can neutralize every Greek. A desk cannot, and the choices it makes are informative.

Delta is hedged, continuously and almost automatically. It is cheap, since the underlying is the most liquid instrument available, and an unhedged delta swamps every other exposure.

Gamma is managed, not neutralized. Neutralizing it requires buying options, which costs theta by (18.8). Desks therefore run a gamma position deliberately, tightening it before events and letting it drift otherwise, and set limits rather than targets.

Vega is hedged by bucket. A single vega number is nearly useless because volatility does not move in parallel: the front of the curve moves more than the back, as Chapter 19 documents. Desks compute vega by maturity bucket and hedge each, which is the same bucketing logic Section 29.10 applies to interest rate deltas.

Rho is usually ignored in equities. It dominates in rates, where Chapter 29 takes it up.

18.9 HOW OFTEN TO REBALANCE

Table 18.3 showed hedging error falling as $1/\sqrt{n}$ with no transaction costs. Introducing costs changes the problem qualitatively, because they accumulate with every trade. Table 18.7 repeats that simulation, charging 0.10% of the value of each rebalancing trade, though not of the final settlement at expiration.

Table 18.7: Cost of hedging 1 written call, with a 0.10% proportional trading cost. 20,000 paths; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean + s.d.
5	3.2628	1.1924	4.4552
10	3.2926	0.8634	4.1560
20	3.3158	0.6212	3.9370
50	3.3685	0.4075	3.7760
100	3.4266	0.2976	**3.7242**
250	3.5457	0.2232	3.7689

The two columns move in opposite directions (Figure 18.8). Trading more often reduces the *uncertainty* of the outcome (the standard deviation falls from 1.19 to 0.22) while raising

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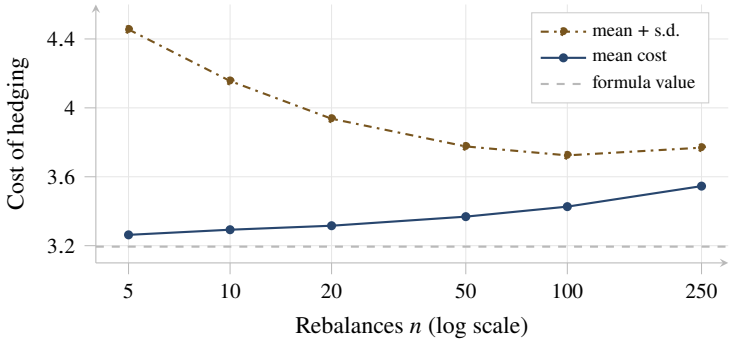


Figure 18.8: Hedging cost against rebalancing frequency, with a 0.10% trading cost.

its *expected* cost, from 3.26 to 3.55, as commissions accumulate. The total is minimized near $n = 100$.

Three conclusions follow.

There is a finite optimum. Once costs exist, continuous rebalancing is not merely impractical but wrong: the limit is strictly worse than a finite frequency.

Hedging is not free even at the optimum. At $n = 100$ the mean cost is 3.4266 against a formula value of 3.1944, so the dealer must charge about 7.3% above the model price to break even on trading costs. That margin is one component of the bid–ask spreads of Example 10.3, and it is why illiquid underlyings have wider option markets.

A clock is the wrong trigger. Table 18.7 rebalances at fixed intervals, which trades when nothing has happened and fails to trade when a great deal has. Desks instead rebalance when delta drifts outside a band, which saves trades in quiet markets and trades more often in volatile ones. Figure 18.4 shows what the band means for the stock: where gamma is high, each dollar the stock travels moves delta further, so the stock can travel less far before the band is crossed.

Hedging with futures instead

A desk hedging an index option uses futures rather than the basket, with Section 3.6's adjustment.

Because a futures price moves $e^{(r-q)(T^*-t)}$ times as far as the spot for a maturity T^*, the futures position required, in units of the underlying (dividing by the contract size gives contracts), is

$$N_{\text{futures}} = e^{-(r-q)(T^*-t)}\,\Delta_{\text{portfolio}} \tag{18.13}$$

Futures are more liquid and cheaper to trade than a basket, and they require margin rather than the full purchase price. The cost is basis risk (Section 3.3): the hedge tracks the futures price, and the option settles against the index. Where index futures trade persistently away from fair value, as China's index futures did after the 2015 trading restrictions, that basis is a running cost of a short futures hedge, charged at every roll.

18.10 BEYOND THE GREEKS

The Greeks are derivatives, so they describe behavior *near* the current state, where they are most reliable and least needed.

They fail for large moves. Equation (18.5) is a second-order approximation. Example 18.3 showed it accurate to within two cents for \$4 moves on a \$63 stock. For a 20% move it predicts a loss of \$3.88 an option either way, where repricing gives \$3.48 on a fall and \$3.07 on a rise, because gamma itself changes over so large a move. A risk report built entirely from Greeks

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describes a world in which nothing much happens, which is why every desk supplements them with **scenario analysis**: revalue the whole book at a grid of underlying prices and volatilities, and read the worst cell.

A typical grid moves the underlying by ±2%, ±5%, ±10%, ±20% and volatility by ±2, ±5, ±10 points, and reports the full revaluation rather than an approximation. The largest losses often sit in the corners, a big move *and* a volatility spike, where the cross term in $\Delta S\,\Delta\sigma$, which (18.1) omits, grows with the product of the two.

They assume you can trade. Every number in this chapter assumes the hedge can be adjusted at a quoted price. A gap denies it: a stock that closes at 63 and opens at 50 never traded at 56, so the delta is corrected after the loss rather than during it. Rebalancing more often does not help: Table 18.3's error falls as $1/\sqrt{n}$ only because the simulated path is continuous. On the dealer's position a 20% gap down costs $347,821 against the $319,438 collected, and a 30% one costs $663,836, the "roughly twice the premium" for which Section 18.1 rejected the naked position. The hedge is also financed and consumes liquidity: Table 18.1's cumulative cost column is borrowing, and Sections 18.11 and 18.12 describe what happens when every desk needs the same trade at the same moment.

They assume the inputs move independently. They do not: equity volatility rises when prices fall, which is the skew of Section 15.11 restated as a dynamic. The cross-derivative vanna of Section 18.6 captures this, and a book that is delta- and vega-neutral but vanna-exposed will discover the exposure on the day both move. Hull and White (2017), estimating the correction (19.4) from index and stock option data, find a minimum-variance delta below the Black–Scholes delta for calls and puts alike: fewer shares against a call, more against a put.

They assume the model. Every Greek in Table 18.4 is a derivative *of the Black–Scholes–Merton formula*. If the formula is the wrong map from prices to values, as Section 15.11 measured, its derivatives inherit the error. Where no closed form exists, the Greeks are themselves estimated, by bumping and revaluing, by pathwise or likelihood-ratio estimators, or by algorithmic differentiation, and Sections 20.10 and 32.13 show what each costs.

The Greeks remain indispensable, but they are a local linearization of an inexact model.

18.11 CREATING OPTIONS SYNTHETICALLY

Section 18.2 described a dealer hedging a written option. Running the same machinery in reverse manufactures an option that no one sold. That is what portfolio insurance did in 1987 (Section 40.2).

To create a synthetic long put on an index portfolio with yield q, hold the portfolio and maintain a short index position worth

$$e^{-qT}\,[1 - N(d_1)] \times \text{portfolio value}$$

taken in futures through (18.13) and adjusted continuously. As the market falls, $N(d_1)$ falls and the short grows; as it rises, the short shrinks. At expiration the combination has the payoff of portfolio-plus-put.

The attraction is that no premium is paid up front; Section 12.10 explained why that matters more to a committee than it should. The cost appears instead as the accumulated trading losses of Example 18.2. Two differences from buying a put are decisive.

The cost is uncertain rather than fixed. Buying a put fixes the price at inception. Replicating one pays the realized cost, which by Table 18.7 has a standard deviation that does not vanish. If

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realized volatility exceeds the volatility the program assumed, replication costs more than the put would have, exactly when protection is needed most.

The replication requires a liquid market, and it consumes liquidity. Selling futures into a decline is the mechanism, so the strategy needs a market able to absorb the sales at the moment everyone running the same program is selling. Buying a put transfers this problem to the writer; replicating one keeps it.

18.12 WHEN HEDGING MOVES THE MARKET

The Greeks describe how a dealer responds to the market. In aggregate those responses *are* part of the market, and the mechanism earlier chapters met without naming is gamma.

Consider the sign. A dealer who is **short gamma**, having sold options, must buy the underlying as it rises and sell as it falls, by (18.3). That is trading *with* the move, which amplifies it; a **long-gamma** dealer does the opposite, which damps it.

Five episodes in this book are now one phenomenon.

Pinning (Example 10.6). Near expiration, dealers long gamma at a strike buy below it and sell above it, holding the stock toward the strike. This is long-gamma damping, concentrated by the enormous gamma of Figure 18.4's shortest-dated curve.

Retail call buying (Section 10.12). Retail investors buy calls, leaving dealers short gamma, and dealer hedging amplifies moves: the opposite sign, and no contradiction, since which one prevails depends on which side of the options dealers hold.

Snowball knock-ins (Sections 12.12 and 35.7). Here the sign changes with the index. Issuers hold the barrier put the note's buyer has written, so above the barrier they are long gamma and buy as the index falls, damping the decline exactly as pinning does. At the barrier that reverses discontinuously: delta jumps, part of the hedge is sold, and, larger still, the bid that had been supporting the market stops.

Portfolio insurance (Section 40.2). This is the 1987 case, and the largest. Manufacturing a long put by trading means trading with the move, selling into declines: a short-gamma dealer's flow, executed by the option's would-be buyer rather than its writer.

Inverse volatility products (Section 22.8's case study). To keep a constant inverse exposure to VIX futures, an exchange-traded product must buy those futures when VIX rises, a short-gamma hedger's flow written into a prospectus; on February 5, 2018, that buying pushed the futures higher still, and the largest such product was terminated. Here the hedge was the product itself, so no dealer's risk report showed it.

Case Study 18.1 Reading dealer gamma

Because the sign of dealer gamma determines whether hedging damps or amplifies, estimates of aggregate dealer gamma have become a widely followed market indicator.

The estimates require assumptions that deserve skepticism. Open interest is observable, but who is long and who is short is not, and the usual convention (that dealers are long the calls, bought from investors writing calls against stock they hold, and short the puts, sold to institutions buying protection) is a guess that can be badly wrong. Retail call buying since 2020 has left dealers short many of the calls the convention counts as long; Chapter 23 shows how much of the answer the assumption decides.

The mechanism is nevertheless real: dealer hedging transmits option-market flow into the underlying, in a direction that depends on a position no one observes. That is enough to explain why volatility around expirations behaves differently and why large structured-product programs leave visible marks, but not to forecast a market.

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18.13 STATIC HEDGING

Everything so far has been **dynamic** hedging: hold a position, watch it drift, trade to restore it. Chapter 12 supplied the alternative, for when dynamic hedging breaks down.

Static hedging replicates a payoff with a portfolio of options bought once and held. Section 12.8 gave the recipe: any piecewise-linear payoff is a bond, a forward, and calls weighted by the change in slope at each kink. Such a hedge needs no rebalancing, incurs no gamma cost, and is indifferent to tomorrow's liquidity.

Static hedges use options, which are expensive and available only at traded strikes and maturities; dynamic hedges use the underlying, which is cheap and always available. One pays up front for certainty, the other as it goes for flexibility.

The choice is forced at discontinuities. A digital payoff is a step, so its gamma diverges near the strike at expiration. There, dynamic hedging becomes impractical, as it does near a barrier (Chapter 34); the usual answer is a static hedge, a tight call spread (Section 12.8), accepting a small mismatch for never having to trade. That is what went wrong in the snowball episodes of Sections 12.12 and 35.7: the issuers hedged dynamically where gamma diverged.

SUMMARY

A dealer who writes options and does nothing, or who buys the underlying outright, holds a directional bet in either case. Neither is a hedge. The Greeks are the coefficients that say how a book's value responds to each of the things that move it, and hedging is the business of controlling them.

Delta, $\partial f/\partial S$, is $N(d_1)$ for a call, and holding delta shares per option written makes a book insensitive to small moves. Scaled by S/f, delta becomes the option's elasticity, its leverage, which is highest out of the money. A delta hedge buys high and sells low by construction, and the accumulated cost of doing so is the option premium: that is what the premium pays for. Against a gap, though, there is no moment at which to trade.

Gamma measures how fast delta changes. A delta-neutral book gains or loses in proportion to the square of the move, so a short-gamma position loses on moves in either direction. Gamma peaks at the money near expiration and can be neutralized only with other options. Theta is gamma under another name: for a delta-neutral portfolio the Black–Scholes–Merton equation reduces to $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma = r\Pi$, so a book long gamma pays for waiting and is paid when the market moves, and a short-gamma book is the reverse. With its financing counted, a delta hedge comes out level when the stock moves by exactly the amount its implied volatility assigns.

Vega is largest for long-dated options, the opposite of gamma, so books are commonly short gamma and long vega at once, and that vega exists at all admits that the model is a quoting convention rather than a description of the world. Rho is ignored on short-dated equity books because rates barely move, but it is first-order in currency options, where the forward is a rate differential.

Desks hedge delta continuously, manage gamma within limits rather than neutralizing it, and hedge vega bucket by bucket because volatility does not move in parallel. Hedging error falls with the number of rebalances while transaction costs rise with it, so optimal rebalancing is finite and is usually triggered by a delta band rather than a clock. A book's gamma and vega are cleared together with options of two maturities, and delta last with stock. The Greeks are local: they understate large moves and say nothing about gaps, so desks revalue the book on a grid of scenarios.

Dealer hedging feeds back into the underlying, with the sign set by gamma. Pinning, retail-driven amplification, snowball knock-ins, 1987's portfolio insurance, and the inverse volatility products of February 2018 are one mechanism seen five times, differing only in who holds which side. Manufacturing a put by trading replaces a known premium with an uncertain

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cost; where gamma diverges, at a digital's strike or a barrier, a static hedge in options is the answer.

FURTHER READING

Taleb's *Dynamic Hedging* is the standard practitioner account and covers what textbooks smooth over, particularly the behavior of the Greeks near barriers and expirations. On transaction costs and optimal rebalancing, the Leland (1985) adjustment is the place to start and shows explicitly how the effective volatility at which a writer must price rises with trading costs. For the feedback effects of Section 18.12, the Brady Commission's 1988 report is the primary source on the mechanism operating at scale, and Grossman (1988) analyzed what dynamic hedging does to the market it trades in, which Section 18.12's argument follows.

PRACTICE QUESTIONS

18.1 A dealer writes 50,000 calls with delta 0.62. How many shares must be held to be delta-neutral, and what happens to that number as the stock rises?

18.2 Explain why neither a naked nor a covered position is a hedge, using the two outcomes in Section 18.1.

18.3 A delta-neutral portfolio has gamma $-4{,}000$. Estimate the profit or loss if the underlying moves by \$3, and state whether the direction of the move matters.

18.4 Using (18.8), find the theta of a delta-neutral portfolio worth \$500,000 with gamma $-2{,}000$, when $S = 80$, $\sigma = 25\%$, and $r = 4\%$.

18.5 Explain why gamma cannot be hedged with the underlying asset but delta can.

18.6 An option has vega 18. By how much does its value change if implied volatility rises from 22% to 24%? On a position of 20,000 options, what is the total?

18.7 Explain why a book can be short gamma and long vega simultaneously, and which maturities contribute each.

18.8 A trader holds a portfolio of butterflies and calendar spreads and describes it as diversified. Using (18.8), explain why it is not.

18.9 Using Table 18.7, state which rebalancing frequency minimizes the mean cost plus one standard deviation, and explain why the mean cost rises with the number of rebalances while the standard deviation falls. Explain also why the formula value of 3.1944 is not attainable at any frequency.

18.10 A pension fund holds an index portfolio and wants the payoff of portfolio-plus-put without paying a premium. State the futures position it must run and how that position changes as the market falls, and give two respects in which the result differs from buying the put outright.

18.11 A dealer who is short gamma must buy the underlying as it rises and sell as it falls. State whether that damps or amplifies the move, and explain why pinning near expiration (Example 10.6) and retail call buying (Section 10.12) push in opposite directions even though both are dealer delta-hedging.

18.12 An exchange-traded product offering inverse exposure to VIX futures must buy those futures when VIX rises. Explain why its rebalancing rule has that sign, why that rule trades the way a short-gamma hedger does, and what distinguishes this case from a dealer whose hedging incidentally moves the market.

18.13 For a European put with the same inputs as the dealer's call ($S_0 = 63$, $K = 64$, $T = 18/52$,

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$r = 4\%$, $\sigma = 22\%$), find its value by put–call parity, then its delta and the elasticity Ω of Example 18.1. Interpret the sign of each, and compare the put's leverage with the call's.

FURTHER QUESTIONS

18.14 Section 18.9 shows hedging error falling as $1/\sqrt{n}$ while expected transaction costs rise, roughly as $\sqrt{n}$ for a proportional cost. Write down the total expected cost as a function of n with a proportional trading cost k, find the n that minimizes it, and describe how the optimum moves with volatility and with k.

18.15 Example 18.4 verifies the Black–Scholes–Merton equation from which (18.8) follows. Show that a trader who is long gamma and delta-hedges continuously earns the gamma-weighted difference between realized and implied variance, $\int \frac{1}{2}\Gamma S^2(\sigma^2_{\text{realized}} - \sigma^2_{\text{implied}})\,dt$ over the option's life, and explain why the weighting matters to someone who buys options because they expect a specific event.

18.16 Section 18.12 argues that dealer hedging amplifies moves when dealers are short gamma and damps them when long. Construct the conditions under which the aggregate position flips sign during a single trading day, and explain why this makes gamma-based indicators unreliable precisely when they are most consulted.

18.17 A desk is short a large quantity of short-dated at-the-money options into an earnings announcement. Using Figures 18.2 and 18.4, describe what happens to the book's delta and gamma across the announcement, and state what the desk should have done beforehand and why it might not have.

18.18 Vega measures sensitivity to a parameter the model assumes constant. Construct the argument that computing vega is therefore incoherent, then the argument that it is nevertheless the right thing to do, and state which position a risk manager should take.

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Chapter 19
Volatility Smiles and Surfaces

Five chapters have deferred to this one. Section 11.3 derived conditions a whole option chain must satisfy and said they would be used here. Section 12.8 built a digital as the limit of a call spread and left the next derivative, the butterfly, which Section 13.3 met as a state price, to be read off the market here. Section 15.11 measured implied volatilities ranging from 56.9% to 33.2% across six strikes on a single stock on a single day, and observed that the model requires all six to be identical. Section 16.4 gave the FX market's quoting conventions, and Section 18.10 warned that the Greeks inherit the model's error.

The subject of this chapter is that error: the model requires one volatility, and the market's prices imply a different one at every strike. It is large, stable, well documented, and *informative*. The pattern of implied volatilities across strikes and maturities is not noise around a correct model. It is the market's statement about the distribution of the underlying, written in the only language the market has agreed on.

Sections 19.1 and 19.2 describe the two shapes the smile takes and what they say about the distribution. Section 19.3 introduces volatility indices and variance swaps, Section 19.4 the models that fit a smile, and Section 19.5 the constraints a surface must obey. Sections 19.6 to 19.10 cover the term structure, index against single-stock smiles, trading, the smile's causes and the volatility risk premium; Sections 19.11 and 19.12 cover how the smile moves and how the surface is used.

19.1 TWO SHAPES

An implied volatility is not observed. It is Section 15.7's construction: the σ that makes Chapter 15's formula reproduce a price the market actually traded at, so each one is a single traded option inverted. Plot those numbers against strike for a fixed maturity and one of two characteristic shapes appears. That curve is a **smile**, and the same option chain has one for every listed maturity. Taken together they define a function of two variables, one implied volatility for each strike K and maturity T, and that function is the **volatility surface**. It is what a desk quotes, marks and hedges from.

Put–call parity makes a call and a put at the same strike and maturity imply the same volatility (Section 15.11), so a surface carries one number per strike, read from the out-of-the-money side, where the bid–ask spread is tighter.

Equities: a skew. Figure 19.1 shows the shape that has characterized index and single-stock options since the crash of October 1987. Implied volatility falls as the strike rises, so that, measured in implied volatility, low-strike puts are expensive and high-strike calls cheap relative to at-the-money options; only at short maturities does the far call wing (the curve's high-strike end) turn up again. The pattern is called the **skew** or "smirk" rather than a smile because it is not symmetric.

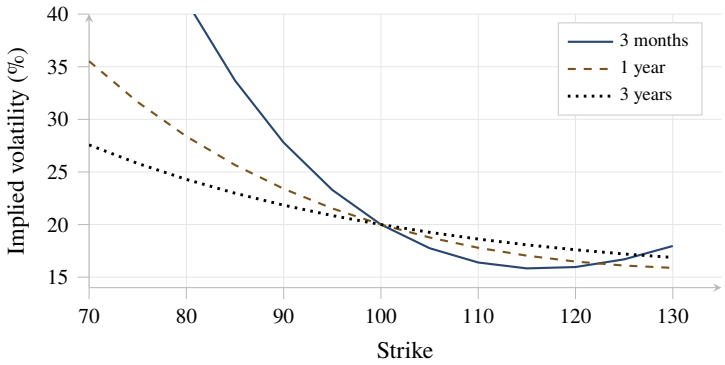


Figure 19.1: Equity implied volatility against strike, at three maturities. $S_0 = 100$; illustrative.

Currencies: a smile. Section 16.4 gave the FX shape: implied volatility is lowest near the money and rises in *both* wings, which is the symmetric curve the word "smile" describes. It is tilted (the risk reversal quoted there measures by how much), but the tilt changes sign as sentiment moves, whereas equity skew does not; only in pairs with a known crash direction, an emerging-market currency or a carry trade's funding currency, does it keep one sign for years.

The difference is structural, and Section 19.9 explains it. An exchange rate is a ratio of two currencies, and a crash in one is a spike in the other, so the distribution has no preferred direction. An equity is a levered claim on a firm: it can be wiped out in a crash, but it has no matching way to multiply in value just as suddenly.

The term structure. Figure 19.1 also shows that the skew flattens as maturity lengthens. Measure the skew by its slope, the fall in implied volatility, in volatility points, for each 10 points of strike between strikes of 90 and 110: the slope falls from 5.7 at three months to 2.8 at one year to 1.6 at three years.

Most of that is the unit of measurement: the same 20 points of strike are about two standard deviations at three months and little more than half of one at three years, as the next subsection measures. The central limit theorem of Section 13.9 would flatten it faster than the change of unit alone, since independent shocks (jumps above all) average out over a long horizon, which is why jump models generate skew mainly at the short end (Sections 21.5 and 21.7). Volatility shocks are persistent and do not average out so quickly, and they keep the long-dated skew alive.

Measuring moneyness

The axis of Figure 19.1 is the strike, which is what the contract names and the worst of the available coordinates for comparing one option with another. Six are in use, set out in Table 19.1, and they are not interchangeable.

They are one axis in different units, and three things follow.

The coordinate decides what "the same option" means from one day to the next. Section 19.11's two rules for how the smile moves with the spot, sticky strike and sticky delta, are a disagreement about which coordinate it stands still in, not about today's prices, and (19.4) is its cost to a hedger.

A fixed strike is a different option at every maturity. At 20% volatility the 90 strike of Figure 19.1 is 1.05 standard deviations below the money at three months, 0.53 at one year and 0.30 at three, because $\sigma\sqrt{T}$ shrinks as expiration approaches. Measured per unit of standardized moneyness rather than per point of strike, the figure's three curves have slopes of 5.68, 5.59 and 5.56 volatility points per standard deviation: they are one curve in that coordinate. Real

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Table 19.1: The coordinates a surface is quoted in.

Measure	Definition	What it is for
Strike	K	The contract term, and how an exchange lists the option
Spot moneyness	K/S	Section 19.11's sticky-delta rule; the columns of Table 19.3
Forward moneyness	K/F	Removes carry, so at the money means $K = F$
Log-moneyness	$\ln(K/F)$	Scale-free: an underlying at 40 and one at 4,000 report the same skew for the same shape, which is why Chapter 21 measures in it
Standardized moneyness	$\ln(K/F)/(\sigma\sqrt{T})$	Distance from the money in standard deviations; comparable across maturities, as delta is
Delta	$N(d_1)$	The FX quoting convention of Section 16.4 (spot or forward delta, premium-adjusted in some pairs), already rescaled by volatility and time

surfaces flatten more slowly than that: Section 21.7 measures the decay as $T^{-0.4}$ against the $T^{-1/2}$ a curve stationary in standardized moneyness would give, so in that coordinate the skew slowly *steepens* with maturity.

Delta is the only one a trader can act on directly, since it is the hedge ratio, already scaled by volatility and time. That is why FX quotes in it, and it is also the circular one: delta depends on the volatility being quoted, so a 25-delta option's strike and its implied volatility depend on each other and have to be solved for together.

Shapes across asset classes

Table 19.2 sets out the surface's shape in each market; the differences are systematic.

Table 19.2: Characteristic surface shapes and quoting conventions.

Market	Shape	Why
Equity index	Steep skew	Leverage, crash risk, one-sided protection demand
Single stock	Milder skew	Same forces, plus takeover possibility lifting the right wing
Currency	Smile, tilt varies	Ratio of two economies; no preferred direction
Commodity	Often *reverse* skew	Supply shocks push prices up, not down
Interest rates	Skew, quoted in normal volatility	Rates can be negative, so lognormal fails outright

Commodities frequently skew the other way. A grain, natural-gas or power price is cushioned on the way down by production cuts and unbounded above when supply fails, so the fat tail is on the *upside*; crude oil and base metals, driven more by demand, often carry a put skew instead. Section 5.9's negative oil settlement does not undercut the cushioning argument: it required storage to run out.

Rates are quoted in normal volatility: the standard deviation of the rate's change measured in basis points, rather than of its percentage change. Chapter 29 explains why: once policy rates went negative, a lognormal model assigning zero probability to $r < 0$ became untenable, and the market moved to quoting basis-point volatility.

19.2 WHAT THE SMILE SAYS ABOUT THE DISTRIBUTION

Section 12.8 built a digital as the limit of a tight call spread, priced at $-\partial c/\partial K$. Section 13.3 pushed the same idea one derivative further, to a state price. An infinitesimal butterfly centered at K (the call spread with strikes $K - \varepsilon$ and K bought against the one with strikes K and $K + \varepsilon$

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sold, scaled by $1/\varepsilon^2$) pays only if the underlying finishes at K, so its price is the state price for that outcome, and

$$\boxed{g(K) = e^{rT}\,\frac{\partial^2 c}{\partial K^2}} \tag{19.1}$$

where g is the risk-neutral density of S_T and c is the market call price as a function of strike, so the derivatives run along the smile. This is the Breeden–Litzenberger result, and it means a complete set of option prices contains the market's entire probability distribution for the underlying. [TN19] derives it in two differentiations, shows why nonnegativity of the density is the convexity condition of Section 11.3, and sets out what changes when the chain is finite.

Applied to a smile, not a single volatility, it makes the chapter's central point visible.

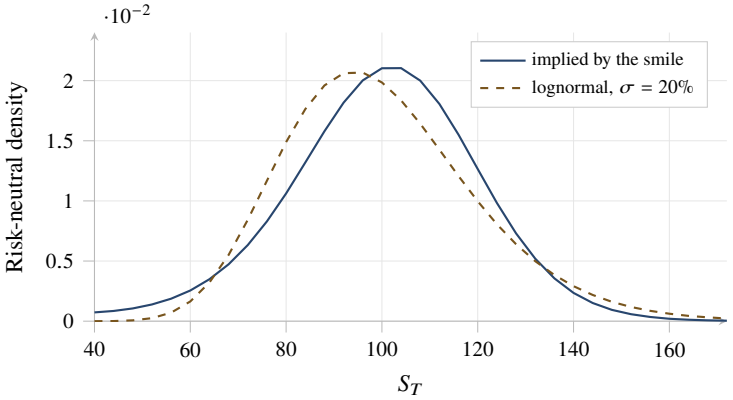


Figure 19.2: Density implied by a one-year equity skew, against the lognormal it would be if the model held. $S_0 = 100$, $r = 0$.

Example 19.1 Reading the tails

Figure 19.2 plots the density obtained from (19.1) applied to the one-year skew

$$\sigma(K) = 0.20 - 0.15\ln(K/100) + 0.15\ln^2(K/100)$$

the shape of Figure 19.1 with a milder left wing, against the lognormal density that a constant 20% volatility would produce.

Each probability is read off as a digital, $\Pr(S_T < K) = 1 + e^{rT}\,dc/dK$ with $r = 0$, where dc/dK is the call price's slope along the smile; this is the density integrated from zero:

	Implied by smile	Lognormal
$\Pr(S_T < 60)$	3.0%	0.7%
$\Pr(S_T < 70)$	6.9%	4.6%
$\Pr(S_T < 80)$	14.8%	15.5%
$\Pr(S_T > 130)$	6.0%	7.9%

The market assigns more than four times as much probability as the lognormal does to a 40% decline and 1.5 times as much to a 30% decline, and less probability to a 30% rise. Between the tails the implied density runs below the lognormal from about 65 to 97 and above it near the center, which is why the two columns nearly agree at 80: the distribution is peaked, with mass drawn from the left shoulder into the center and the far tail.

Expensive low-strike puts and cheap high-strike calls are not a pricing anomaly: they are what a fat left tail and a thin right tail look like when translated into Black–Scholes volatilities.

One caveat matters for practice. A tail probability at a quoted strike is the digital $1 + e^{rT}\,dc/dK$, so it depends only on the smile's level and slope there. Below the lowest traded strike the smile

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must be extrapolated, but the probability of finishing below that strike is still fixed: it is a digital put, which the put spread between the two lowest traded strikes approximates. What the extension decides is how that probability is spread over lower prices, and so anything integrated over the tail, and it must be free of arbitrage to do even that. Extended as a quadratic, this smile is not: the density turns negative below a strike of about 6, and integrated upward from that point it totals 1.06 rather than 1.00, a 6% error created entirely outside the observable range.

The same calculation on real quotes

Figure 19.2 used a smooth parameterized smile. The point of (19.1) is that it needs no such thing: six quoted prices and arithmetic are enough, as Example 19.2 shows.

Example 19.2 Extracting a distribution from six quotes

Return to the option chain of Table 10.2: a stock at \$48.30, 45 days to expiration, six strikes \$2.50 apart. Section 15.11 solved each call's midquote (the bid–ask midpoint) for implied volatility:

Strike	Call mid	Implied vol	$C_{i-1} - 2C_i + C_{i+1}$	Density $g(K)$
42.50	6.650	56.9%	—	—
45.00	4.425	46.0%	0.4700	0.0757
47.50	2.670	39.7%	0.5600	0.0902
50.00	1.475	36.7%	0.4300	0.0692
52.50	0.710	34.5%	0.3600	0.0580
55.00	0.305	33.2%	—	—

The fourth column is the price of a butterfly with wings \$2.50 wide. Every entry is positive, so the chain passes Section 11.3's convexity test and admits no butterfly arbitrage. The fifth column divides by the wing width squared and compounds forward, which is (19.1) in difference form: $g(K) = e^{rT}(C_{i-1} - 2C_i + C_{i+1})/h^2$ with $h = 2.50$, $r = 5\%$ and $T = 45/365$.

Multiplying each density by \$2.50 gives, approximately, the probability the market assigns to each bucket, the \$2.50 band around a strike: 18.9%, 22.5%, 17.3%, 14.5%. The four together cover \$43.75 to \$53.75 and carry 73.3% of the probability.

The tails come from the outer call spreads. A call spread's price, compounded forward and divided by its width, approximates the digital at the midpoint of its strikes, so $\Pr(S_T > 43.75) \approx e^{rT}(6.650 - 4.425)/2.50 = 89.6\%$ and $\Pr(S_T > 53.75) \approx e^{rT}(0.710 - 0.305)/2.50 = 16.3\%$. The left tail is the complement of the first: $\Pr(S_T < 43.75) = 10.4\%$.

Why the total is exactly 100%. 10.4% + 73.3% + 16.3% = 100.0%, but the total is an *identity*, not a test: the middle differences telescope, leaving only the two outer call spreads, which are precisely the tail digitals, so *any* six numbers would sum to 100%.

A lognormal at the at-the-money implied volatility of 39.7%, centered on Example 11.4's forward of \$47.60, puts only 53.0% of its mass in that same central band. The lognormal that *does* put 73.3% there has $\sigma = 25.6\%$.

So the market is pricing the middle of the distribution as if volatility were about 26%, while the strike 12% below spot is quoted near 57%: the distribution is peaked in the center, and the little mass below \$43.75 must lie far enough out to justify that price. More mass in the center and in the far tails than a normal distribution carries is called **leptokurtosis**, and this is what it looks like when you measure it rather than assume it.

Two limits apply. The calls are American and the stock pays a dividend before expiration, so the deep-in-the-money \$42.50 quote, the one the left tail is read from, carries early-exercise premium a European call would not. Every \$0.10 of it lowers the measured $\Pr(S_T < 43.75)$ by about 4 points, so the 10.4% understates the European tail and the 57% overstates the wing. And the whole left tail rests on that one call spread: it fixes the 10.4%, but six strikes spanning −12% to +14% of spot cannot say how that 10.4% is distributed below them, as Example 19.1 warned. A chain tells you a tail's mass, and through the lowest put its mean, but never its shape.

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19.3 VOLATILITY INDICES AND VARIANCE SWAPS

Chapter 1 introduced VIX and Section 18.12 used it; this section defines it. **VIX** is Cboe's volatility index for the S&P 500: the annualized volatility, in percentage points, that the index's listed options imply over the next 30 days. It is a construction from the surface, and the construction is Chapter 12's replication result applied once more.

The static portfolio that pays variance. Suppose one wants a payoff equal to the *realized variance* of the underlying over $[0, T]$ (a traded contract defines it as a discrete sum of squared daily log returns, Section 22.2; the continuous-time definition used here is for pricing and hedging). Section 12.8 showed that any payoff can be built from options; the specific answer here is that realized variance is replicated by a static portfolio of out-of-the-money options weighted by $1/K^2$, together with a dynamic position in the underlying. The fair value of that portfolio gives the **fair variance swap rate**:

$$\sigma_{\text{var}}^2 = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{19.2}$$

where F is the forward price of the underlying to expiration, and P and C are put and call prices at strike K. VIX is 100 times the square root of (19.2) at 30 days, interpolated in total variance $\sigma^2 T$ between the two listed expirations that bracket 30 days. Cboe computes the same index at other horizons, from nine days (VIX9D) to three months (VIX3M) and, since April 2023, one day (VIX1D), the reference for Chapter 23's same-day options.

The $1/K^2$ weighting is not a modeling choice; it is what makes the portfolio's payoff proportional to variance rather than to something else. Section 22.3 derives this strip of options, and [TN22] takes up the correction the exchange applies to it in practice. The weighting has a consequence that decides how the index behaves: low strikes receive far more weight than high ones. Because the skew makes those low strikes expensive, a volatility index built this way systematically exceeds the at-the-money implied volatility, by more than four volatility points on Figure 19.1's one-year skew, as Section 22.6 measures. And because real quotes do not run to zero and infinity, Cboe truncates the strip by a mechanical rule, so the published index depends on how far out options happen to be quoted that day.

The instrument behind the formula is the **variance swap**, which pays the difference between realized variance and the fixed rate σ_{var}^2. It became the standard volatility instrument because its payoff is the realized variance whatever price level the moves occurred at, and because its replicating options are held *statically*. The direction of the logic matters here: VIX is not an input to the surface; it is an output of it.

19.4 MODELS THAT FIT THE SMILE

The Black–Scholes–Merton model assumes one volatility, and the market quotes many. Three families of model resolve the contradiction, and each buys something at a price.

Local volatility. Let volatility be a deterministic function of the price and time, $\sigma(S, t)$. Dupire showed that for any arbitrage-free surface, given at every strike and maturity, there is exactly one such function reproducing it. Local volatility therefore fits the smile *perfectly*, by construction, and it preserves completeness: the market has one source of risk, so Section 13.4's replication argument survives and every option is hedgeable.

The cost is its dynamics. A local-volatility model that matches today's surface generally predicts that tomorrow's smile will flatten and shift in ways the market does not display, which matters for anything whose value depends on how the surface moves, such as Section 19.11's delta and the exotic options of Chapter 34.

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Stochastic volatility. Let volatility be a second random process, correlated with the price. Heston's model is the standard example, and the negative correlation between price and volatility generates the equity skew directly, the leverage explanation of Section 19.9 as a model.

The cost is completeness. There are now two sources of risk and one hedging instrument, so by Section 13.4 the market is incomplete, no unique price exists, and options must be used to hedge options. That is not a defect of the model; it is the model reporting a fact about the world that the Black–Scholes–Merton model concealed.

Jumps. Allow the price to move discontinuously. Jumps produce fat tails at short maturities without requiring an implausibly high volatility for the continuous moves, which is exactly where Figure 19.1 shows the steepest skew. They also break completeness, and for a sharper reason: a jump of unknown size cannot be hedged by any position in the underlying taken before it.

SABR. In interest rate markets the standard is SABR (stochastic alpha, beta, rho), a stochastic-volatility model with a closed-form implied-volatility approximation, which lets a desk fit a smile with four intuitive parameters rather than a numerical procedure. Chapter 29 uses it for swaptions, and Section 29.9 fits one to market data.

No model dominates: each is used where its virtue is the requirement, and exotics desks mostly run a hybrid of the first two, the local-stochastic volatility of Section 21.9, and compare it against the others. Section 19.12 explains why the disagreement between them is itself the number to look at.

19.5 CONSTRAINTS ON A SURFACE

Because the implied density must be a density, a volatility surface cannot take just any shape. Section 11.3 derived the constraints without a model; they now become tests for a surface.

Nonnegative density. From (19.1), $\partial^2 c/\partial K^2 \geq 0$, which is exactly the convexity condition (11.3). A surface that makes call prices nonconvex in the strike implies negative probabilities and admits the butterfly arbitrage of Example 11.2.

Monotonicity. $c(K)$ must be decreasing in K, though by no more than the present value of the strike gap (Section 11.3).

Calendar consistency. For a fixed strike (strictly, a fixed forward moneyness K/F, which is the same thing when there is no carry), total variance $\sigma^2(K,T)\,T$ must be nondecreasing in T. If it were not, a calendar spread would have a negative price for a nonnegative payoff.

These are the conditions any interpolation must respect, and respecting them is harder than it sounds. Fitting a smooth curve through quoted volatilities is not the same as fitting an arbitrage-free surface, and a naive spline through market data routinely produces negative densities somewhere between the quotes. Dealers' pricing systems parameterize the surface instead, most often with Gatheral's stochastic-volatility-inspired (SVI) parameterization, which fits each expiration's total variance with five parameters and is linear in log-strike in the wings; its surface extension, surface SVI (SSVI; Gatheral and Jacquier 2014), comes with explicit conditions under which the whole surface passes the tests above. The linear wings are required: Lee (2004) showed that without arbitrage total implied variance can grow at most linearly in log-strike, $\sigma^2 T$ at most about $2\,|\ln(K/F)|$ far out in either wing. A quadratic smile like Example 19.1's grows faster and must eventually imply an arbitrage; that one does so near a strike of 6, where its density turns negative.

In a rising call wing the monotonicity condition is the one to watch, and Example 19.3 turns it into a number a trader can carry.

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Example 19.3 How fast a wing may steepen

Take spot at 100, six months to expiration and a 3% rate. The strike enters the price twice, directly and through the implied volatility the smile assigns to it, so differentiating along the smile is a total derivative:

$$\frac{dc}{dK} = \frac{\partial c}{\partial K} + \frac{\partial c}{\partial \sigma}\frac{d\sigma}{dK} = -e^{-rT}N(d_2) + \mathcal{V}\frac{d\sigma}{dK}$$

The first partial is minus the digital a flat volatility would give (Section 12.8), negative because a higher strike is worth less; the second is vega, positive. A rising wing pushes against the first term, and the call price stops falling in the strike (the call spread turns negative) once

$$\frac{d\sigma}{dK} > \frac{e^{-rT}N(d_2)}{\mathcal{V}}$$

The right-hand side is a hard ceiling on the wing's slope, and it is not constant:

Strike	σ	Vega	Maximum steepening per 5 of strike
110	22.5%	25.77	5.34 volatility points
115	24.0%	22.83	4.45 volatility points
120	26.0%	20.19	3.86 volatility points
125	29.0%	18.60	3.49 volatility points

The ceiling tightens as the strike moves out, because $N(d_2)$ falls faster than vega does, the opposite of an extrapolation that steepens the wing as it goes.

Now breach it. Take a chain quoting 24.0%, 23.2%, and 22.8% at 100, 105, and 110, then an aggressive wing at 25.0% and 31.0% for 115 and 120:

Strike	σ	Call	Call spread
100	24.0%	7.4824	
105	23.2%	5.0715	+2.4109
110	22.8%	3.3237	+1.7478
115	25.0%	2.6563	+0.6673
120	31.0%	3.0442	**−0.3878**

The 115/120 call spread has a payoff that is never negative and a price of −\$0.3878: you are paid \$0.39 to receive something worth between \$0 and \$5.

The chain tolerates $\sigma(120)$ up to 29.27% before the spread turns, so the quoted 31.0% was 1.7 volatility points past the boundary, far too small a discrepancy to see by eye on a plotted smile, and an arbitrage nonetheless. That is why the check is arithmetic rather than visual, and why it is run on the quotes before any curve is fitted through them.

19.6 THE TERM STRUCTURE AND ITS DYNAMICS

The surface has a second dimension, and it moves differently from the first.

Figure 19.3 shows the characteristic pattern. In calm markets the term structure slopes *upward*: short-dated volatility is low and long-dated volatility is higher. In stressed markets it *inverts*, sometimes violently, with one-month volatility far above two-year.

The mechanism is mean reversion. Volatility is not a random walk. It clusters, as Section 14.12 noted and Section 38.10 models, but it also returns toward a long-run level. A short-dated option prices volatility over a period in which today's level dominates; a long-dated option prices an average over a period in which mean reversion has had time to act. The slope simply reports which side of the long-run level the market is currently on.

Two consequences follow for the desk.

Vega buckets are not interchangeable. Section 18.8 said desks compute vega by maturity bucket rather than as one number, and Figure 19.3 is the reason: short maturities move several times as much as long ones. Hedging total vega hedges a parallel shift that does not occur.

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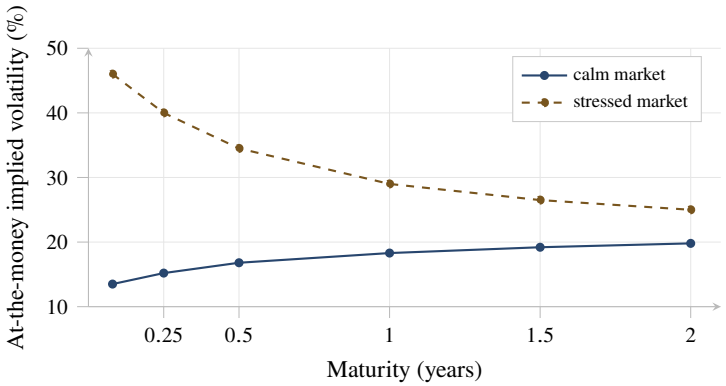


Figure 19.3: At-the-money implied volatility against maturity, in two regimes. Illustrative.

Calendar spreads are a term-structure trade. Section 12.5 described the calendar spread as a bet on time decay and on the term structure. Now the second half is precise: selling near-dated and buying far-dated volatility is a bet that short-dated volatility stays low or, when it is high, falls back toward its long-run level, and it loses in exactly the stress episodes that invert the curve.

A scheduled event bends both dimensions. When a single large move is expected on a known date (an earnings release, a court ruling), the distribution of the price just after it can be bimodal; near-the-money options are then worth more relative to the wings than a lognormal implies, and the smile becomes a frown. Only expirations after the date contain the event. If the one just before trades at 25% and one fifteen trading days out at 35%, and the fourteen ordinary days carry 25%, subtracting their variance from the later expiration's total leaves the event day's: $(0.35^2 \times 15 - 0.25^2 \times 14)/252 = 0.0038$. Its square root, 6.2%, is the standard deviation the market prices for the announcement-day move, against $0.25/\sqrt{252} = 1.6\%$ on an ordinary day.

19.7 THE INDEX SMILE AND THE SINGLE-STOCK SMILE

Table 19.2 recorded that an index skews more steeply than its own constituents. That is not an observation about two unrelated markets: the index *is* the constituents, so the two surfaces are linked, and what links them is correlation.

For a portfolio with weights w_i and component volatilities σ_i,

$$\sigma_I^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \, \rho_{ij} \, \sigma_i \sigma_j \tag{19.3}$$

Everything on the right except the correlations is observable from single-stock option prices, and the left-hand side is observable from index option prices. If all pairs are assumed to share one correlation, that one unknown can be solved for, and the answer is called the **implied correlation**, which Example 19.4 computes.

Example 19.4 Implied correlation and what dispersion trades

Take an index of 50 equally weighted stocks, each with a one-year implied volatility of 30%, and a one-year index implied volatility of 18%. With a common correlation ρ, (19.3) collapses to

$$\sigma_I^2 = \sigma^2 \left[\frac{1}{n} + \rho \left(1 - \frac{1}{n} \right) \right]$$

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so $(0.18/0.30)^2 = 0.36 = 0.02 + 0.98\rho$ and $\rho = 34.7\%$.

Where the trade is. Suppose realized correlation over the past year has been 25%. At that correlation the index volatility would be $0.30\sqrt{0.02 + 0.25 \times 0.98} = 15.4\%$, against the 18% the index options actually cost. Selling index volatility and buying the components, the **dispersion trade** of Sections 16.2 and 18.7, collects that gap of 2.6 volatility points, a difference in volatility rather than a 2.6% return, if correlation stays at 25% and the components realize their 30%.

Why the gap persists. It is Section 19.10 again, in a specific form. Index puts are the instrument institutions buy for portfolio protection, and protection is a demand for correlated downside, so the premium concentrates in the index rather than in the constituents. Implied correlation exceeding realized correlation is the price of that demand.

And why the trade is not free. In a crash component volatilities and correlation rise together, so index volatility rises proportionally more than single-stock volatility, and the long single-stock leg covers only part of the loss. The dispersion seller is short exactly the thing that moves most, which is why the strategy earns steadily and loses in large, infrequent amounts (Example 22.8).

The same mechanism explains the shape difference the table recorded. Correlation is not a constant: it rises when markets fall and falls when they rise. On the downside the index gets hit twice, by higher component volatility and by higher correlation together; on the upside dispersion increases and damps the index move. So the index inherits its constituents' skew and then adds a correlation skew on top of it, and the sum is steeper than any single stock's.

19.8 TRADING THE SMILE

A single surface has three dimensions of shape (level, skew, and curvature), and Section 16.4 already observed that the FX market quotes precisely those three because they are what it trades. The instruments are Chapter 12's, and correlation, which spans two surfaces, is traded through the dispersion trade of Section 19.7.

Level is traded with straddles or, more cleanly, with the variance swaps of Section 19.3.

Skew is traded with the risk reversal of Section 12.6: long an out-of-the-money call, short an out-of-the-money put, or the reverse. A trader who thinks puts are too expensive relative to calls buys the risk reversal, selling the expensive put and buying the cheap call, and must include Section 19.11's delta correction in the hedge.

Curvature is traded with the vega-weighted butterfly, the position the foreign-exchange market's butterfly quote of Section 16.4 prices: a strangle bought against an at-the-money straddle, sized so the two vegas cancel. Buying the wings against the at-the-money straddle is a bet that the market will come to price more extreme outcomes in both directions.

Who is on each side. Most of the participants are the ones Section 16.2 named, and their positions in surface terms are now legible. Protection buyers are structurally long the put wing. Covered-call writers are structurally short the call wing. Structured-product issuers hold the other side of what their products embed: long the barrier puts that buyers of Section 12.12's reverse convertibles and autocallables write, which sit low in the strike dimension, and short the calls inside principal-protected notes. Each of these leaves a persistent mark on a specific region of the surface, and dealers hold whatever the end users leave unmatched.

A region expensive relative to its neighbors usually reflects who has been buying it rather than a superior forecast, and Section 19.10 shows that, on average, the buyers overpay.

19.9 WHY THE SMILE EXISTS

Four explanations are offered; all four contribute, and their relative weights are contested.

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The distribution really is not lognormal. Section 14.12 gave the reasons: prices jump, and volatility is neither constant nor independent of the price level. A model with jumps or with stochastic volatility produces fatter tails than the lognormal, and fitting the Black–Scholes–Merton model to options priced under such a model generates a smile as an artifact. This explanation is certainly part of the answer, and it explains the FX smile well.

Leverage. A share is a residual claim on a levered balance sheet. When the equity value falls, leverage rises, so the equity becomes more volatile: volatility and price are negatively correlated by construction. This produces exactly the asymmetry seen in equities and *not* in currencies, where no such mechanism exists.

Crash risk. The equity skew dates from October 1987, as this section's case study shows. Before that day the market had not priced a 20% one-day decline; afterward it did. On this reading the skew is the price of a risk the lognormal model assigns essentially zero probability to. Rubinstein (1994), who documented the change, called it "crash-o-phobia."

Supply and demand. One-directional demand for low strikes and one-directional supply of high strikes (Section 19.8) will produce the observed shape whatever the distribution.

Case Study 19.1 The day the skew appeared

The four explanations above are hard to separate on cross-sectional data. History separates them rather sharply.

Studies of index option prices before October 1987 find implied volatilities close to *flat* across strikes, a smile so mild it was regarded as a minor pricing puzzle rather than a feature. The Black–Scholes–Merton model, on this evidence, described the market it was written for tolerably well for its first fourteen years.

After October 19, 1987, the skew appears, and it does not go away. Nearly four decades later the pattern of Figure 19.1 is a permanent fixture of every equity index option market in the world.

That timing is difficult to reconcile with the leverage explanation, which describes a mechanism that was equally true in 1986. It is difficult to reconcile with a change in the true distribution, since nothing about the economy changed that Monday. What changed was the market's belief about the probability of a very large decline, revised by a single observation and never revised back.

The supply-and-demand explanation gets support from the same event, which Section 40.2 tells in full: the failure of portfolio insurance destroyed the belief that crash protection could be manufactured by dynamic trading, which turned investors who had made their own puts into buyers of listed ones. Both readings point the same way, and both say the skew is priced crash risk rather than a measured distribution.

Because each predicts a similar shape, the best evidence is the next section's: a skew that merely described the distribution would be fairly priced on average, and it is not.

19.10 THE VOLATILITY RISK PREMIUM

Sections 10.12 and 12.4 asserted a result this chapter owes: implied volatility exceeds subsequently realized volatility, on average, across most markets and most periods. That average gap is the volatility risk premium that Section 15.7 named.

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The magnitude is typically two to four volatility points on equity indices (an implied volatility of 18% against a realized 15%, say), and the gap is larger for low strikes than at the money, so the skew is steeper than the realized distribution justifies.

Two readings are offered, and both are partly right.

It is compensation. Selling options means accepting a payoff that is worst precisely when markets fall and investors most need money. It is Chapter 18's short-gamma position, in economic terms. A risk that pays badly in bad states commands a premium, exactly as equities do relative to bills.

It is supply and demand. The buyers of protection described in Section 16.2 are price-insensitive relative to the sellers, who are constrained by capital. The premium is then a rent earned by whoever can bear the risk.

Section 12.4's straddle buyer pays this premium, so buying options is a bet that realized will exceed implied, not that the stock will move. Section 12.12's covered-call funds collect it, so their option income earns a genuine expected return, though part of each distribution is the investor's own capital handed back. And selling options earns a risk premium with a short-gamma position, profitable on average and, as Section 18.10's gap example showed, capable of losing more than the whole premium in a single move.

This also settles the question Section 15.7 raised. When an implied volatility of 29.85% meets a historical estimate of 20.4%, there are two readings: the market anticipates something the history does not contain, or buyers are paying the premium. Within a single implied volatility the two components cannot be told apart; averaged over enough observations, the forecast-error component tends to zero, and what remains is the premium.

19.11 HOW THE SMILE MOVES

For a desk, what matters more than the smile at one instant is what happens to it when the underlying moves, and Section 18.10 warned that the answer changes the Greeks.

Two reference conventions describe it, and a third completes Derman's (1999) taxonomy.

Sticky strike holds that the implied volatility of a *given strike* stays put as the spot moves. The smile is fixed in strike space, and Black–Scholes deltas are correct as computed.

Sticky delta (or sticky moneyness) holds that the implied volatility of a given *moneyness* stays put, so the whole smile translates with the spot. A fixed strike then acquires a different implied volatility as the market moves.

Sticky implied tree, named for the tree form of local volatility, holds that the local-volatility function of Section 19.4 stays put, so a rise in the spot *lowers* the implied volatility at a fixed strike and the delta falls below the Black–Scholes delta. Derman associated sticky strike with a range-bound index, sticky delta with a trending one, and the implied tree with a market that fears a sharp fall.

Figure 19.4 shows what each convention does to the one-year smile when the spot rises from 100 to 105.

Under sticky strike nothing happens to the curve, and the 100-strike option keeps its 20.00% volatility. Under sticky delta the curve slides to the right in proportion to the spot, and the 100-strike option, now 5% in the money rather than at the money, picks up the 21.45% that used to belong to the 95.2 strike. The same option at the same instant carries volatilities 1.45 volatility points apart under the two conventions, and Example 19.5 shows what that does to the hedge.

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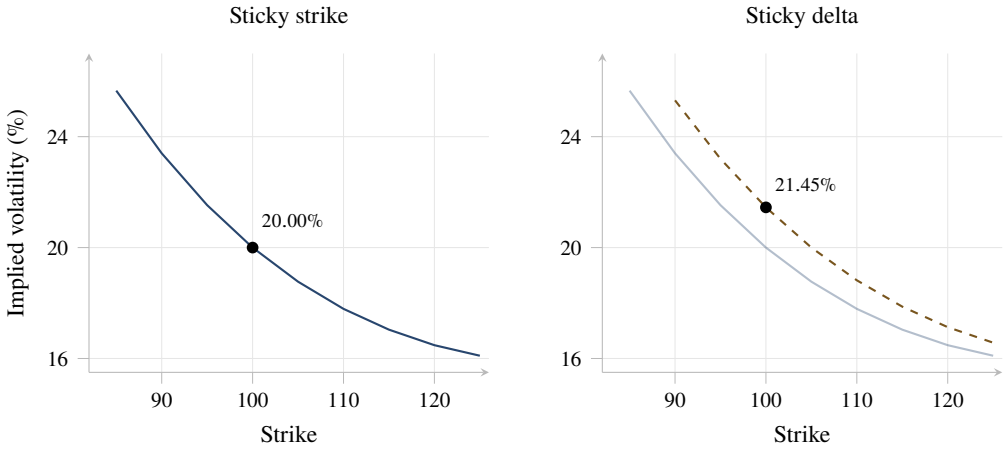


Figure 19.4: The one-year smile after the spot rises from 100 to 105. The marked point is the 100-strike option. Faint curve: the smile before the move.

Example 19.5 The delta correction

Under sticky delta the spot enters the price twice, directly and through the implied volatility the smile hands the option as it slides, so the total sensitivity to the spot is a total derivative:

$$\Delta_{\text{total}} = \frac{dc}{dS} = \frac{\partial c}{\partial S} + \frac{\partial c}{\partial \sigma}\frac{\partial \sigma}{\partial S} = \Delta_{\text{BS}} + \mathcal{V}\frac{\partial \sigma}{\partial S} \tag{19.4}$$

The first partial is the usual delta, computed with the smile held still; the second is vega times the rate at which the sliding smile assigns the option a new volatility. In Chapter 18's expansion, equation (18.1), the delta and vega terms were treated as independent; a sliding smile makes $\Delta\sigma$ move with ΔS, so part of the vega term is first order in the stock and belongs in the delta hedge.

For the one-year skew of Figure 19.1 with $S = 100$ and $r = 0$:

Strike	Type	σ	Δ_{BS}	Correction	Δ_{total}
90	put	0.2340	−0.2853	+0.1256	−0.1596
100	call	0.2000	+0.5398	+0.1092	+0.6490
110	call	0.1779	+0.3275	+0.0683	+0.3958

The corrections run from 0.07 to 0.13 in delta, a 20% error on the at-the-money hedge, or 10,920 shares on a position of 100,000 options. A desk hedging with Black–Scholes deltas while the market behaves in a sticky-delta way has every delta too low: too few shares against a call, too many against a put.

Under an equity skew, volatility falls as strike rises. Sticky delta means the smile moves with spot, so a rise in S lowers the option's moneyness K/S and therefore *raises* its implied volatility. Since vega is positive, the option's value rises by more than delta alone predicts, so every delta in the table moves up: a call's hedge grows and a put's shrinks.

Which convention holds is an empirical question, and for equity indices the measured answer lies outside the first two: when the index falls, at-the-money volatility rises by more than sticky strike implies, though by less than local volatility predicts (Bergomi 2009). The correction for an index therefore has the opposite sign to this table's, and the minimum-variance delta lies below the Black–Scholes delta, as Section 18.10 reported from Hull and White (2017). FX is closer to sticky delta, which is why Section 16.4 quotes in delta terms. Desks measure the relationship rather than assuming it, and the measured slope enters (19.4) directly.

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19.12 THE SURFACE IN PRACTICE

A desk maintains a volatility surface per underlying, and it is the single most important piece of market data an option business owns.

What one actually looks like. Table 19.3 is an S&P 500 surface in the form a desk reads it: implied volatility in percent, strikes across as a percentage of spot, maturities down the side. Its wings are milder than Figure 19.1's.

Table 19.3: Implied volatility (%) by strike, as a percentage of spot, and by maturity. A calm market; illustrative, not market data.

	Strike (% of spot)						
Maturity	**80**	**90**	**95**	**100**	**105**	**110**	**120**
1 month	30.0	22.4	19.6	17.5	15.8	14.9	15.6
3 months	27.5	21.9	19.9	18.5	17.1	16.0	14.8
6 months	26.0	21.7	20.3	19.3	18.2	17.2	15.6
1 year	25.0	21.9	20.8	20.0	19.3	18.5	17.2
2 years	24.2	22.0	21.3	20.8	20.2	19.7	18.7

Every row falls from left to right through the money, which is the skew of Section 19.1. A one-month put struck 20% below spot is marked at 30.0% against 17.5% at the money; the same strike two years out is 24.2% against 20.8%. Measured between the 90 and 110 strikes the slope runs 3.75 volatility points per 10% of spot at one month, 1.70 at one year and 1.15 at two, so two-thirds of the skew is gone by the two-year point. Measured in Section 19.1's standardized coordinate instead, the same rows give 1.89 volatility points per standard deviation at one month against 3.37 at two years: the skew *steepens*.

Read down the at-the-money column instead and the second dimension appears: 17.5% rising to 20.8%, the calm-market term structure of Section 19.6.

Two features are worth pointing out because they look like errors and are not. Total variance rises down every column: at the 110 strike $\sigma^2 T$ runs from 0.0019 at one month to 0.0776 at two years. That is Section 19.5's calendar condition holding, and a column in which it failed anywhere would be a calendar-spread arbitrage rather than a steep surface. And the top right corner turns back up (the one-month 120 strike at 15.6% against the 110's 14.9%), because no one sells a far out-of-the-money one-month call at the price a straight continuation of the skew would imply.

It is built from quotes, not from a model. Each expiration's forward is read first from put–call parity near the money (Section 11.6), and time runs on Section 15.8's trading-day clock, refined to Chapter 23's business time, which weights each day by its expected variance; both matter most at the short end, where Section 23.9 prices the same-day smile this way. Liquid strikes and maturities are then marked to the market; everything else is interpolated subject to the constraints of Section 19.5.

It prices the illiquid from the liquid. Section 15.8's practical rule was to use each option's own implied volatility. The surface is what makes that possible for an option that does not trade: locate it in strike–maturity space, interpolate, price. *This is what the Black–Scholes–Merton formula is actually for.* Example 19.6 carries it through on Table 19.3.

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Example 19.6 Pricing an option that does not trade

A client asks for a nine-month put struck at 4,600 on an index trading at 5,000. No line of Table 19.3 is that option: the strike is 92% of spot and the maturity falls between two quoted rows. It sits inside the rectangle formed by the 90 and 95 strikes and the six-month and one-year maturities, and those four corners are what price it.

Interpolate across the strikes first, linearly, at each of the two maturities:

$$\sigma(92, 0.5) = 21.7 + 0.4 \times (20.3 - 21.7) = 21.14\%$$
$$\sigma(92, 1) = 21.9 + 0.4 \times (20.8 - 21.9) = 21.46\%$$

Then move down the maturities, but not in volatility. Interpolate in total variance $\sigma^2 T$, the quantity Section 19.5 requires to be nondecreasing:

$$w(0.5) = 0.2114^2 \times 0.5 = 0.02234, \qquad w(1) = 0.2146^2 = 0.04605$$

At nine months $w = 0.03420$, so $\sigma = \sqrt{0.03420/0.75} = 21.35\%$. Interpolating the two volatilities directly would have given 21.30%. Five hundredths of a point is nothing here, but only the variance rule always respects the calendar bounds the two quoted maturities impose.

With $r = 4\%$ and a dividend yield of 1.5%, the Black–Scholes–Merton formula at 21.35% gives a put worth 157.9 index points.

Interpolated down the 100 column the same way, the nine-month at-the-money volatility is 19.77%; priced at that single volatility, the same put is worth 136.2. A desk doing so would have sold the option 13.8% below its market value, and the missing 21.7 index points are the skew: the option's own strike was never allowed to enter its price.

It generates the risk report. Chapter 18's vega is computed bucket by bucket against this surface, and the buckets are chosen where the surface actually moves independently.

Its failures are exotic risk. A vanilla option is priced by reading the surface. An exotic is not: Chapter 34 shows that barrier and path-dependent payoffs depend on how the surface *evolves*, which no static surface specifies. Two models that fit today's surface perfectly can price the same barrier option far apart. Section 21.11 measures the gap at nearly half the price for a reverse knock-out (a call knocked out by a barrier above its strike, so it dies where it is in the money), and at about one percent for a down-and-out call struck at the money (knocked out by a fall through a barrier below). That gap, not the vanilla price, is the model risk that Section 13.4's completeness discussion predicted, and its size depends on where the payoff sits relative to the region the surface leaves unconstrained.

SUMMARY

Implied volatility is not constant across strikes. Equities show a skew (low strikes expensive, high strikes cheap) that has persisted since October 1987; currencies show a roughly symmetric smile whose tilt changes sign with sentiment. The skew flattens with maturity, mostly because a fixed strike gap spans fewer standard deviations at longer horizons; persistent volatility keeps the long-dated skew from vanishing as independent shocks alone would let it. A skew number means nothing without the moneyness convention it was measured in.

By Breeden–Litzenberger, $g(K) = e^{rT}\partial^2 c/\partial K^2$, so a complete set of option prices contains the market's entire risk-neutral distribution. Applied to a skew it gives a fat left tail and a thin right one, with the market assigning a large decline several times the probability the lognormal does; run on a quoted chain rather than a fitted curve, butterflies give the density and call spreads give the tails. Beyond the lowest traded strike, extrapolation decides the tail's shape.

Because the density must be nonnegative, Section 11.3's convexity condition becomes a

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constraint on any surface, alongside monotonicity in strike and nondecreasing total variance in maturity. Fitting a smooth curve is not the same as fitting an arbitrage-free surface: a wing can steepen only so far across a range of strikes before the quotes are themselves an arbitrage.

A volatility index is an output of the surface: a $1/K^2$-weighted strip of out-of-the-money options prices variance, and the skew lifts it above at-the-money volatility. Models that fit the smile trade one virtue against another: local volatility fits today's surface exactly and keeps the market complete but mispredicts how it moves, while stochastic volatility and jumps capture the dynamics and short-dated tails at the cost of completeness.

Four explanations contribute: a distribution that is not lognormal, leverage, crash risk priced since 1987, and one-directional demand. Leverage and crash risk give equities a direction that currencies lack; the timing of 1987 and the index's steeper skew point to priced crash risk and one-directional demand.

The term structure moves on its own logic: at-the-money volatility slopes upward in calm markets and inverts in stressed ones, which is why vega is bucketed by maturity. Component and index volatilities together imply a correlation, and selling it against realized correlation is the dispersion trade. Implied volatility on equity indices also exceeds subsequently realized volatility by a few points on average, so buying a straddle is a bet that realized will exceed implied rather than a bet that the market will move.

How the smile moves determines the hedge. The total delta is $\Delta_{\mathrm{BS}} + \mathcal{V}\,\partial\sigma/\partial S$, a correction too large to ignore; sticky delta makes it positive, while on equity indices, where fixed-strike volatility falls as the index rises, it is negative, so desks measure it rather than assume it. The surface is built from quotes subject to those constraints, prices an option that does not trade consistently with ones that do, and generates Chapter 18's vega buckets. Its limitation is that it says nothing about its own evolution, which is where exotic model risk lives.

FURTHER READING

Breeden and Litzenberger (1978) is the original statement of (19.1) and is short. Derman (1999) sets out the sticky-strike, sticky-delta and sticky-implied-tree regimes of Section 19.11, and Bergomi (2009) measures how index smiles actually move against them. Daglish, Hull and Suo (2007) derive the no-arbitrage condition a moving surface must satisfy and test the sticky rules against it and against S&P 500 data; a skewed surface is consistent with neither rule in its basic form. Gatheral's *The Volatility Surface* (2006) is the standard reference for arbitrage-free parameterizations and the constraints of Section 19.5; Gatheral and Jacquier (2014) give SSVI and Lee (2004) the wing bound. On the volatility risk premium, Carr and Wu (2009) measure it from synthetic variance swaps across indices and single stocks, and Bollerslev, Tauchen and Zhou (2009) relate it to expected stock returns; Chapters 21 and 22 return to it when volatility becomes a traded asset.

PRACTICE QUESTIONS

19.1 Explain why equity options show a skew while currency options show a roughly symmetric smile, giving the structural reason in each case.

19.2 Using the three-month and six-month rows of Table 19.3, compute the slope between strikes 90 and 110 per 10% of spot and per standard deviation of standardized moneyness (use each row's at-the-money volatility), and explain why the two disagree about whether the skew flattens.

19.3 State the Breeden–Litzenberger result and explain, without algebra, why the second derivative of the call price with respect to strike is a probability density.

19.4 Using the table in Example 19.1, state by what factor the market's probability of a 30%

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decline exceeds the lognormal's, and explain what feature of the smile produces it.

19.5 Using Example 19.3, state the largest rise in implied volatility the 115/120 spread can carry, describe the arbitrage that appears beyond it, and explain why the permitted slope shrinks as strikes move further out of the money.

19.6 An at-the-money option has vega 40 and Black–Scholes delta 0.54. Under sticky delta with $\partial\sigma/\partial S = 0.0015$, compute the corrected delta and the error in shares on a 50,000-option position, and state which convention gives the larger hedge for a call under an equity skew.

19.7 Explain why the volatility risk premium implies that a straddle buyer is not simply betting the market will move.

19.8 In Example 19.2 the bucket probabilities and the two tail probabilities sum to exactly 100%. Explain why this had to happen and what it would mean if it did not; then replace the \$50 call's midquote of 1.475 with 1.050, and say which butterfly turns negative and what arbitrage it offers.

19.9 An index of 100 equally weighted stocks has component implied volatilities of 25% and an index implied volatility of 16%. Compute the implied correlation, and explain why index skew is steeper than single-stock skew.

19.10 Explain why the volatility term structure inverts in a crisis, and why this makes total vega a poor summary of an option book's exposure.

FURTHER QUESTIONS

19.11 Example 19.1 notes that its quadratic smile, extended below the observable range, implies a density that turns negative below a strike of about 6 and totals 1.06 when integrated upward from there. Explain why this is not a numerical error to be corrected, describe two different extrapolation choices and how each would distribute the probability below the lowest observable strike (a total the chapter shows is fixed), and state what a risk manager should conclude about tail estimates derived this way.

19.12 Section 19.9 gives four explanations for the skew. Design an empirical test that would distinguish the leverage explanation from the supply-and-demand explanation, stating what data you would need and what each hypothesis predicts.

19.13 Section 19.10 states that the volatility risk premium is larger at low strikes than at the money. Explain what this implies about the profitability of selling puts versus selling straddles, and why the strategy with the higher average return is not obviously the better one.

19.14 Show that under sticky delta the at-the-money-forward implied volatility does not change when the spot moves, even though the correction in (19.4) for the option struck there today does not vanish (Example 19.5), and explain why this makes the at-the-money-forward straddle of Section 11.4 a natural reference point for a desk.

19.15 Section 19.12 claims two models fitting today's surface perfectly can price the same barrier option far apart, and that the size of the gap depends on the payoff. Explain what information a barrier price requires that a vanilla price does not, connect this to Section 13.4's completeness argument, and explain why a reverse knock-out should be far more exposed to the modeling choice than a down-and-out struck at the money.

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Chapter 20
Numerical Procedures

Chapter 13 opened by saying that its tree was a teaching device rather than a computational method, and that how to build trees efficiently, how many steps they need, and how to make them converge belonged here. Chapter 15 said the same of the free boundary in an American option, Chapter 14 of the choice between two ways to discretize a process, and Chapter 17 of the simulation needed to value a relative-performance award; the smile-fitting models of Section 19.4, local volatility above all, have no formula and are computed the same way. This chapter answers those questions.

The subject is narrow and the stakes are not. Every price in Chapters 13 through 19 that does not come from a closed-form formula is produced by one of the three methods in this chapter, and the difference between a method that converges and one that merely appears to is the difference between a book of trades that is valued correctly and one that is not.

Section 20.1 says what numerical methods are for. Sections 20.2 to 20.5 are trees in practice: the control variate, trinomial trees, and dividends. Sections 20.6 to 20.10 are Monte Carlo: discretizing the process, variance reduction, several assets, and Greeks from a simulation. Section 20.11 is American Monte Carlo and Section 20.12 finite differences; Section 20.13 chooses among the three methods, and Section 20.14 records where they came from.

20.1 WHAT NUMERICAL METHODS ARE FOR

A closed-form price exists only when three things happen to be true at once: the payoff is simple, the process is simple, and there is no early exercise. Break any one and a number must be computed.

Table 20.1: Why a problem needs a numerical method, and which one suits it.

Feature	Example	Natural method
Early exercise	American put, Bermudan swaption	Tree or finite difference
Path dependence	Asian, lookback, barrier	Monte Carlo
Many state variables	Basket, quanto basket, CVA	Monte Carlo
Both at once	Callable structured note	American Monte Carlo
Greeks across a grid	A whole book, revalued	Finite difference

Table 20.1 is the chapter in one page, and the pattern in it has a reason. [TN15] showed that the pricing partial differential equation and the risk-neutral expectation are the same statement (the Feynman–Kac relation), so a price can be obtained either by solving the equation backward or by simulating the expectation forward. The two directions have opposite strengths, and everything in Table 20.1 follows from that:

Backward methods (trees, finite differences) sweep from expiration to today and therefore know, at every node, the value of continuing. That is exactly what an early-exercise test needs. They also produce the value at every state, so the Greeks come out of the same computation.

Their weakness is dimension: a grid with m points along each of d state variables has m^d nodes (a million at $m = 100$ and $d = 3$), which is fatal beyond about three.

Forward methods (Monte Carlo) generate whole paths and average. **Path dependence**, a payoff that depends on the route and not only the destination, therefore costs no more to evaluate than a terminal payoff does: the whole path has already been generated, and an Asian or a lookback payoff simply reads it. The cost also grows linearly rather than exponentially in the number of state variables. Their weakness is the mirror image: at any intermediate date a simulated path does not know its own continuation value, which is why early exercise is hard and why Section 20.11 exists.

Transform methods are a fourth route, and a narrow one. Where a model's characteristic function (the Fourier transform of its terminal distribution) is known in closed form (Heston, Bates and the jump models of Chapter 21), a European price is a single integral over it, which the fast Fourier transform of Carr and Madan (1999) or the cosine expansion of Fang and Oosterlee (2008) evaluates fast enough for Chapter 21 to calibrate Heston to a whole surface inside an optimizer. Having no early exercise and no path dependence, the route supplements the other three rather than replacing any.

Convergence is the whole subject. Every method here produces an answer for any settings at all; the question is always whether that answer is close, and how one would know. The chapter's recurring warning is that a number that looks stable can be wrong, and that the cheapest defense is to compute something whose answer is already known.

20.2 BINOMIAL TREES IN PRACTICE

Chapter 13 built the tree and Section 13.9 observed that its value settles to a limit at a rate of about $1/n$, oscillating between odd and even n. Here is that statement made precise, because both halves of it matter.

Take a European put with $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, and six months to expiration. The Black–Scholes–Merton value is 3.582934.

Table 20.2: Cox–Ross–Rubinstein tree against the Black–Scholes–Merton value.

n	Tree	Error	n	Tree	Error
5	3.789185	+0.206251	100	3.572449	−0.010485
10	3.479418	−0.103516	101	3.592980	+0.010046
11	3.675951	+0.093017	200	3.577688	−0.005246
20	3.530789	−0.052145	201	3.587979	+0.005045
21	3.631453	+0.048519	500	3.580835	−0.002099
50	3.561990	−0.020943	1000	3.581884	−0.001050

Two features of Table 20.2 deserve attention.

The error halves when n doubles. From 100 to 200 to 500 to 1000 the error runs −0.0105, −0.0052, −0.0021, −0.0011. This is first-order convergence, and it is bad news: to gain one decimal place you must take ten times as many steps, and the work at n steps is of order n^2 nodes, written $O(n^2)$, so a factor of ten in accuracy costs a factor of a hundred in time.

The sign alternates with the parity of n. Even n undershoots, odd n overshoots, and they do so by almost the same amount. The reason is geometric: with $S_0 = K$, an even number of steps places a terminal node exactly at the strike while an odd number straddles it, so the discretized distribution sits differently against the payoff's kink in the two cases.

The alternation is a property of this contract, not of trees. What sets the error is where the strike falls between two terminal nodes, and a node lands on it exactly when $\kappa + n$ is an even

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integer, for $\kappa = \ln(K/S_0)/(\sigma\sqrt{\Delta t})$. At the money $\kappa = 0$ and that reads "n even." Otherwise κ grows like $\sqrt{n}$, the strike drifts through the lattice as steps are added, and the signs stop alternating: at $K = 55$ the errors at $n = 26, 27, 28$ run −0.006, +0.026, +0.003. Section 20.3 turns this into a rule for choosing n at any strike.

How many steps. There is no universal answer, but the shape of Table 20.2 gives the discipline. Never report a tree value without computing it at two step counts. If 200 and 400 steps disagree in the third decimal, the third decimal is not real. Dealers' pricing systems typically run 200 to 1000 steps for a vanilla American option and use the correction of the next section to do better with fewer.

20.3 THE CONTROL VARIATE

This section treats one technique, the main repair for the slow, oscillating convergence Section 13.9 deferred: it reaches far beyond trees, and the rest of the chapter keeps returning to it.

The idea is general. Suppose you want A, which you can only estimate, and there is a related quantity E that you can estimate *by the same method* and whose true value E^{true} you already know. Then

$$\boxed{\hat{A}_{\text{corrected}} = \hat{A} + \left(E^{\text{true}} - \hat{E}\right)} \tag{20.1}$$

where $\hat{A}$ is the method's estimate of the target, $\hat{E}$ its estimate of the companion (the same method at the same settings, on the same tree or the same paths), E^{true} the companion's known exact value, and $\hat{A}_{\text{corrected}}$ the number reported. The correction is the method's own error on a problem where the error is visible.

For an American option the natural companion is the European option with the same strike and expiration: price both on the same tree, and correct the American value by the tree's error on the European, which the Black–Scholes–Merton formula gives exactly.

Example 20.1 The control variate on an American put

Take the same contract, now American: $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, six months. Its value is 3.69702, computed two ways that share no arithmetic (corrected trees at thousands of steps, and the finite-difference grid of Section 20.12), against the European 3.582934, an early-exercise premium of 0.114.

n	Plain tree	Error	Corrected	Error
50	3.68552	−0.01150	3.70646	+0.00944
51	3.71691	+0.01989	3.69700	−0.00002
100	3.69135	−0.00567	3.70183	+0.00481
101	3.70717	+0.01015	3.69712	+0.00010
200	3.69419	−0.00283	3.69944	+0.00242
201	3.70213	+0.00511	3.69709	+0.00007

On odd step counts the correction is extraordinary: at $n = 201$ the error falls from +0.00511 to +0.00007, a factor of about 75, for roughly twice the work. On even step counts it changes the sign of the error and shrinks it by less than a fifth.

The asymmetry says exactly what the control variate does. With $S_0 = K$ and odd n, no node sits at the strike, and the American and the European tree carry the same large error from straddling the payoff kink—a *shared* error, which (20.1) removes completely. With even n a node sits exactly on the strike and the two errors part company: the European's is as large as before, but the American's is only about half of it, because the American value also carries the error in locating

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the early-exercise boundary, which the European option does not have and therefore cannot correct.

A control variate removes what the two problems share and nothing else. The practical corollary is to choose a companion problem as close to the target as possible.

The step count is the second lever, and it works at any strike. Nothing in that explanation mentions parity: what decides the matter is whether a node sits on the strike, which by Section 20.2 happens when $\kappa + n$ is even and is furthest from happening when $\kappa + n$ is odd. At the money $\kappa = 0$ and that reads "take n odd," which is Section 20.2's alternation turned into an instruction. Away from the money it reads "choose n so that $\kappa + n$ is close to an odd integer," whatever the parity of n.

At $K = 55$, step counts that leave the strike near a midpoint are about sixteen times more accurate for the same work.

Notice also, in Figure 20.1, what the correction does to the shape of the sequence. The plain tree oscillates, so extrapolating across it is dangerous; the corrected sequence oscillates too, but its odd branch already sits on zero and its even branch decays smoothly toward it, so extrapolating along a branch is safe. Following the correction with **Richardson extrapolation** (taking $2V(2n) - V(n)$, which cancels an error of the form C/n provided both step counts carry the same C) is the standard recipe in practice, and it reaches four-decimal accuracy at a few hundred steps. The proviso is not automatic, because C depends on the strike's **offset**, its position between the two terminal nodes that bracket it: the two step counts must share an offset, which at the money means both even.

Other parameterizations remove the oscillation at its source. The equal-probability tree fixes $p = 1/2$ and puts the drift in u and d; Leisen and Reimer (1996) choose u, d and p by inverting the normal distribution so that the tree is centered on the strike, which makes convergence monotone and close to second order with no extrapolation. It is a common choice for vanilla American equity options.

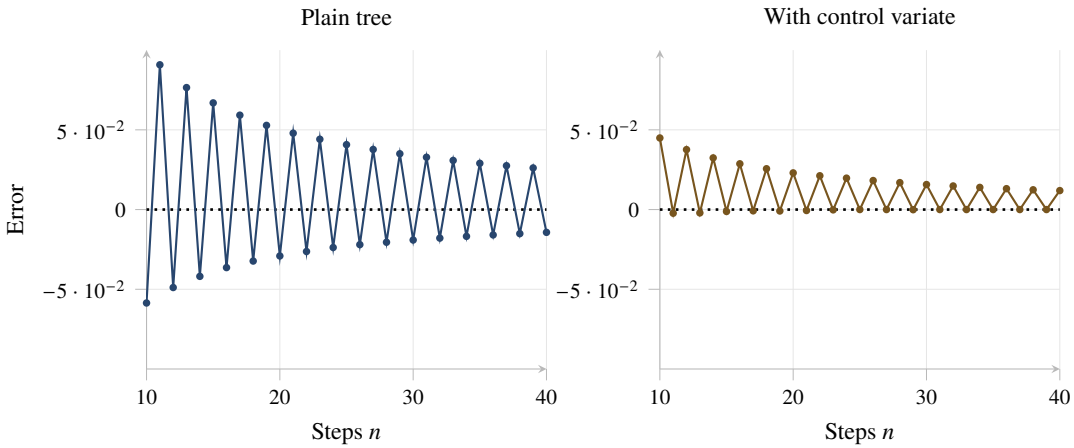


Figure 20.1: American put error against step count, same vertical scale.

20.4 TRINOMIAL TREES

Let the price move up, down, *or stay*. With the log-price $x = \ln S$ spaced Δx apart and drift $\nu = r - q - \sigma^2/2$,

$$p_u = \frac{1}{2}\left(\frac{\sigma^2\Delta t + \nu^2\Delta t^2}{\Delta x^2} + \frac{\nu\Delta t}{\Delta x}\right), \quad p_d = \frac{1}{2}\left(\frac{\sigma^2\Delta t + \nu^2\Delta t^2}{\Delta x^2} - \frac{\nu\Delta t}{\Delta x}\right), \quad p_m = 1 - p_u - p_d \tag{20.2}$$

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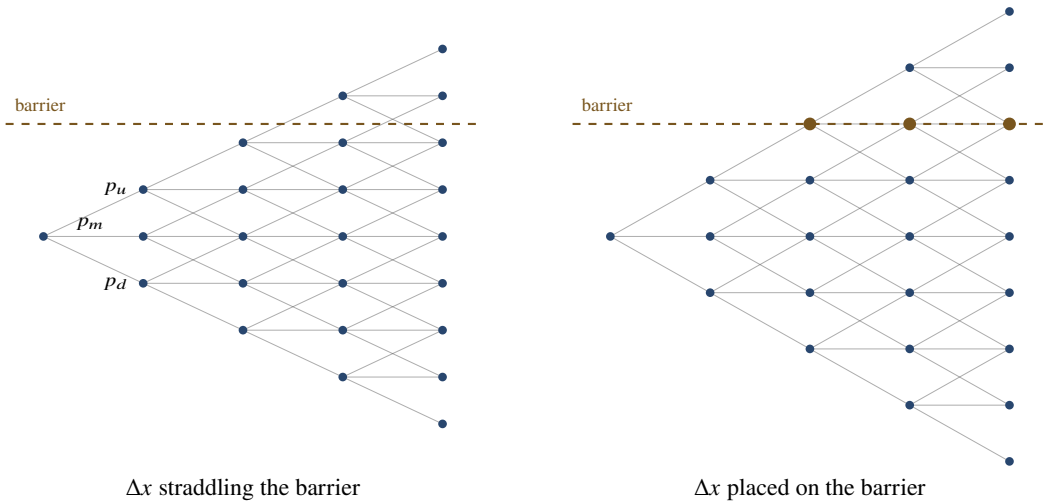
These are not a definition but a consequence. Over one step the change δx in the true log-price has

$$E[\delta x] = \nu\,\Delta t, \qquad E[\delta x^2] = \sigma^2\Delta t + \nu^2\Delta t^2$$

while the tree moves x by $+\Delta x$, 0 or $-\Delta x$ and so delivers $(p_u - p_d)\,\Delta x$ and $(p_u + p_d)\,\Delta x^2$. Setting each pair equal, together with $p_u + p_m + p_d = 1$, gives three equations in three unknowns, and (20.2) is their solution. The tree is built to reproduce the first two moments of the process it stands for, which is what Chapter 13's binomial tree does with two branches rather than three.

That leaves Δx free, and $\sigma\sqrt{3\Delta t}$ is the common choice: it puts $p_u + p_d$ at about $1/3$, so the middle branch carries roughly $2/3$ of the probability and the two outer ones stay safely positive.

Figure 20.2 shows the geometry, and the recombination is the property that matters: a step adds two nodes rather than tripling them, so the work grows with the square of the number of steps instead of with 3^n.



Δx straddling the barrier Δx placed on the barrier

Figure 20.2: Four steps of a trinomial lattice, drawn against the same barrier at two choices of Δx.

The extra branch buys two things and does *not* buy a third.

It does not converge faster. On the European put above, the trinomial errors at $n = 50, 100, 200$ are -0.0207, -0.0103, -0.0052, against binomial errors of -0.0209, -0.0105, -0.0052. Both are $O(1/n)$ and they are numerically almost identical. Anyone who reaches for a trinomial tree expecting accuracy for free will not get it.

It buys a free parameter. In (20.2), Δx and Δt are chosen independently, whereas the binomial tree ties them by $u = e^{\sigma\sqrt{\Delta t}}$. That freedom is what lets a node be placed exactly on a barrier, as the right panel of Figure 20.2 does and the left one fails to. Chapter 34 shows that a barrier option priced on a tree whose nodes straddle the barrier converges badly, because the barrier is effectively moved to the nearest node.

It buys room for a varying volatility. A local-volatility model in the sense of Section 19.4 needs a different σ at each node; in a binomial tree that changes u from node to node, and the paths stop meeting. The trinomial keeps Δx fixed and absorbs σ in the probabilities.

It is also a finite difference scheme. Section 20.12 shows that the explicit finite difference method *is* a trinomial tree, with the three probabilities appearing as the coefficients of the discretized derivatives. The two subjects are one subject.

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20.5 DIVIDENDS AND THE NON-RECOMBINING TREE

Section 13.10 flagged this as the first place the tree's convenience breaks. Two of the three ways a stock can pay a dividend cost the tree nothing. A *continuous yield* is handled in the drift: replace r by $r - q$ and the geometry is untouched. A *discrete proportional* dividend (a known fraction δ of whatever the price turns out to be at the ex-date t_d) is free for a different reason: every node at that date is multiplied by the same $1 - \delta$, so the lattice is rescaled rather than distorted and recombines exactly as before. A known cash dividend is not free. If the price drops by \$2 on the ex-date, an up-then-down path and a down-then-up path no longer meet, because \$2 is subtracted at different price levels: subtracting a constant moves nodes by different proportions, where multiplying by one does not. Every node standing at the ex-date then starts a tree of its own, so each dividend multiplies the node count by the number of nodes at its date, and with several ex-dates the count grows geometrically until the tree is unusable. Figure 20.3 shows the difficulty.

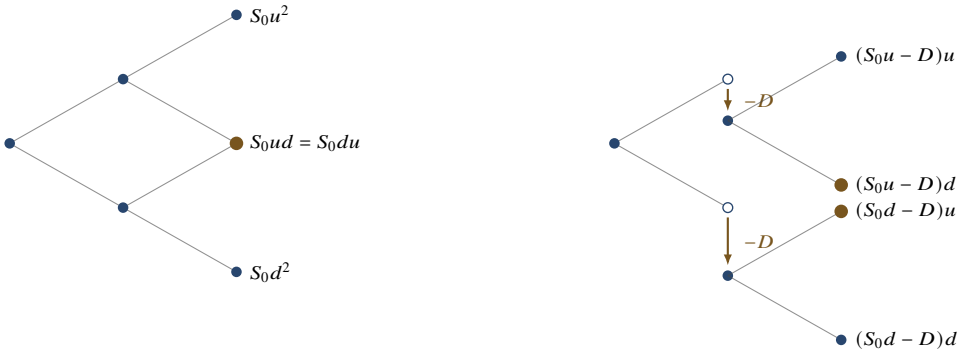


Figure 20.3: Two steps around an ex-date, without and with a cash dividend D.

The standard remedy is the **displaced tree**, which splits the stock price into a part that behaves like a lognormal and a part that does not.

Write $S = S^* + De^{-r(t_d - t)}$ for $t < t_d$, where D is the dividend and t_d its date. The second term is a known, riskless amount, the present value of a certain payment. Build the tree on S^*, which has no dividend and therefore recombines, then add the present value of the dividend back at every node when computing the exercise decision and the payoff. One caveat goes with it: the σ on the tree is the volatility of S^*, so a volatility quoted on S must be scaled up to roughly $\sigma S/S^*$. Example 20.2 takes its 30% as the volatility of S^*.

Figure 20.4 draws the construction rather than its result. The dashed lattice is S^*: it is the tree that is actually built, it recombines, and it runs straight through the ex-date without noticing it. The solid nodes above it are what the exercise test compares against, S^* with the dividend's present value added back, and past the ex-date that value is zero and the two are one lattice. The recombination lives in S^* and the economics in S, which is the trick.

Example 20.2 A cash dividend done three ways

Give the American put of Example 20.1 a \$2.00 dividend at $t = 0.25$.

The displaced spot is $S^* = 50 - 2e^{-0.05\times 0.25} = 48.0248$, and a tree built on it gives 4.6063, 4.6096 and 4.6100 at 100, 400 and 1,600 steps. That sequence looks settled and is not: the converged American put value is 4.6098, which the tree reaches only after tens of thousands of steps and the finite-difference method of Section 20.12 confirms. The corresponding European

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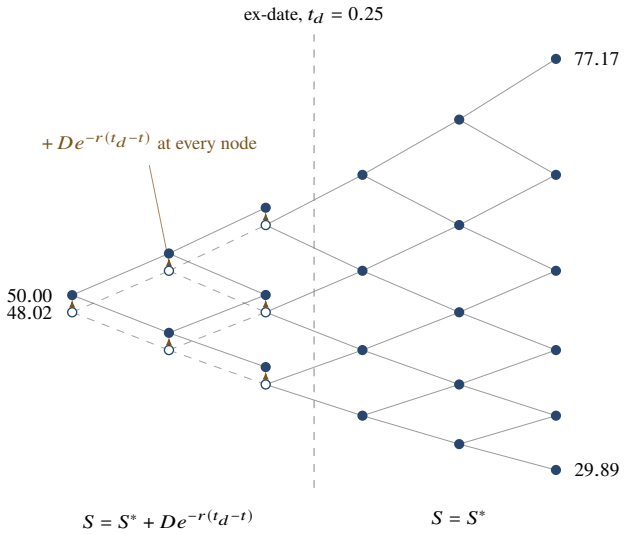


Figure 20.4: Five steps on the displaced tree of Example 20.2: dashed, S^*; solid, S^* plus the dividend's present value. $S_0 = 50$, $D = 2$ at $t_d = 0.25$, $\sigma = 30\%$, $r = 5\%$.

value is 4.4689 from the Black–Scholes–Merton formula on S^*.

Compare that with two shortcuts:

Treatment	Value	Error
Displaced tree (correct)	4.6098	—
Dividend ignored	3.6970	−19.8%
Spot reduced by the full \$2.00	4.6385	+0.6%

Ignoring the dividend is catastrophic and obviously so. Subtracting the undiscounted dividend instead of its present value is a small error here (\$0.03, or 0.6%), but it is the kind that scales: with a larger dividend, a longer time to the ex-date, or a higher rate, the discounting matters more, and the error is one-directional rather than random.

The early-exercise premium also moves. Without the dividend it was 0.114; with it, 4.6098 − 4.4689 = 0.141. The rise comes from the lower price, which leaves the put deeper in the money, and not from the dividend itself: on a non-paying stock at the same spot of 48.0248 the premium is 0.157. Before the ex-date a dividend makes early exercise of a *put* less attractive, not more: the put holder wants the stock price low, and the dividend does that for them without their having to exercise. That runs opposite to the call case of Section 11.10.

Holding the dividend *amount* fixed shows what its form and timing are worth. A 4% proportional dividend at t_d has a present value of exactly $\delta S_0 = \$2.00$, since $e^{-rt}S_t$ is a martingale, against \$1.9752 for the \$2.00 of cash, and the put is worth correspondingly more: 4.6223 against 4.6098. Spread the same 4% smoothly over the six months instead ($q = -\ln(0.96)/0.5 = 8.16\%$) and the terminal distribution is unchanged, $0.96S_0e^{(r-\sigma^2/2)T+\sigma W_T}$ either way, so the European value is 4.4810 under both; but the American falls to 4.4812: the early-exercise premium drops from 0.1414 to 0.0002, because a continuous yield lowers the drift to $r - q = -3.16\%$ everywhere, and against a stock expected to fall a put is worth holding rather than exercising. The form of the payment moves the value by 0.3%, its timing by more than 3%.

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20.6 MONTE CARLO SIMULATION

Simulate the risk-neutral process forward, compute the payoff on each path, average, and discount:

$$\hat{f} = e^{-rT}\,\frac{1}{M}\sum_{i=1}^{M}\text{payoff}\big(\text{path}_i\big) \tag{20.3}$$

This is the definition of a risk-neutral price from Section 13.2, evaluated by brute force. Nothing about it is subtle, and its two properties follow immediately from the law of large numbers and the central limit theorem.

It is unbiased. Each sampled payoff has the price as its discounted expectation, and their average converges to it regardless of the payoff's shape, which is why a path-dependent payoff needs no special handling.

Its error falls like $1/\sqrt{M}$. Write $f_i = e^{-rT}\,\text{payoff}(\text{path}_i)$ for the discounted payoff on path i. The standard error of (20.3) is

$$\text{s.e.}(\hat{f}) = \frac{s}{\sqrt{M}}, \qquad s^2 = \frac{1}{M-1}\sum_{i=1}^{M}\left(f_i - \hat{f}\right)^2 \tag{20.4}$$

where s is the sample standard deviation of the M discounted payoffs and $\hat{f}$ is their average, which is (20.3) itself. Both come out of one pass over the paths, so a simulation reports its own accuracy, which nothing else in this chapter does. To gain one decimal place you must run a hundred times as many paths, and Example 20.3 shows what that costs.

Example 20.3 The cost of a decimal place

Simulate the European put of Table 20.2, whose exact value is 3.582934:

Paths	Estimate	Standard error	Actual error
1,000	3.71346	0.16249	+0.13053
10,000	3.63958	0.05062	+0.05664
100,000	3.58181	0.01595	−0.00113
1,000,000	3.58916	0.00504	+0.00623

The standard error falls by $\sqrt{10} \approx 3.16$ for every tenfold increase in paths, which is the $1/\sqrt{M}$ rule holding to within 2%. The actual errors bounce around inside it (at a million paths the estimate happens to be worse than at a hundred thousand), which is the point of a standard error: it is a statement about the distribution of the answer, not about this answer.

A million paths bought a standard error of half a cent on an option worth \$3.58. A 200-step binomial tree, costing a small fraction of the time, was already inside a cent. For a vanilla European option, Monte Carlo is the wrong tool, and reporting a simulated value where a formula exists is a sign that something has gone wrong in the design rather than in the code.

The standard error also has a use beyond reporting: it is a diagnostic. A simulation whose standard error is not falling like $1/\sqrt{M}$ has a bug, a badly behaved payoff, or a random number generator with structure. Always compute it, and always compare it with the observed error on a case where the answer is known. Example 20.4 does both, on a calculation Chapter 9 needed.

Example 20.4 The exposure profile of Chapter 9

Chapter 9 valued counterparty risk on a five-year swap using an expected exposure profile it did not simulate. The calculation is a good one to meet first because the answer can be checked in closed form.

A dealer holds a five-year payer swap on \$100 million, struck at the market so that it is worth

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zero today. Take the swap rate to diffuse as a normal random walk, so that at a horizon t it stands at $S_t = S_0 + \sigma_r \sqrt{t}\, Z$ for a standard normal Z, with $\sigma_r = 1.456\%$ a year an absolute volatility rather than a proportional one. That is the normal-volatility convention of Section 19.1, which rates markets adopted because it permits a negative rate. It also makes the closed form below available: $V(t)$ is linear in the rate and so inherits its distribution. By equation (7.6) the swap is then worth

$$V(t) = L\,(S_t - K)\,A(t)$$

where L is the \$100 million notional, $K = S_0$ is the fixed rate, and $A(t)$ is the annuity of the $5 - t$ years still to run, which we take as $5 - t$ years of unit payments.

The simulation. Draw Z, form $V(t)$, keep $\max(V(t), 0)$, and average. Ten thousand paths at $t = 2.5$ give \$2.29 million, with a standard error of about \$0.03 million.

The closed form. What the simulation just estimated is the expected exposure $\mathrm{EE}(t) = E[\max(V(t), 0)]$: the average over states of what would be lost if the counterparty failed at t, with the states in which the dealer owes money contributing nothing. Because $V(t)$ is normal with mean zero, equation (9.3) evaluates that same expectation directly, with $\sigma_V(t)$ the standard deviation of the swap's value at that horizon:

$$\mathrm{EE}(t) = 0.3989\,\sigma_V(t) = 0.3989\,L\,\sigma_r\sqrt{t}\,A(t)$$

At mid-year points this gives, in millions, \$1.848, \$2.490, \$2.296, \$1.630 and \$0.616, the column Example 9.2 used; that example and Figure 9.1 explain the shape of the profile.

The closed form exists only because the swap's value is linear in a normal rate. Add an option, a barrier, or a second currency, and the expectation stops being available in closed form, and that is when the three-stage simulation of Section 9.1 stops being an illustration and becomes the only method.

20.7 DISCRETIZING THE PROCESS

Section 14.4 set out two ways to step a geometric Brownian motion forward and said the choice would return here.

The **Euler** scheme discretizes the stochastic differential equation directly:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \varepsilon \sqrt{\Delta t} \tag{20.5}$$

The **exact** scheme uses the solution rather than the equation:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\varepsilon\sqrt{\Delta t}\right] \tag{20.6}$$

They are not equivalent, and the difference matters in two ways.

Equation (20.6) has no discretization error at all. It is the exact distribution of $S_{t+\Delta t}$ given S_t, so for geometric Brownian motion a single step from 0 to T is as accurate as a thousand. Any intermediate steps are needed only because the *payoff* looks at them: an Asian option needs the dates its average is taken on, a barrier option needs a fine grid.

Equation (20.5) can go negative. A large negative ε drives S below zero, which is impossible for the process it claims to simulate. With $\sigma\sqrt{\Delta t}$ small this is rare; with a large step or a high volatility it is not, and a simulation that produces negative prices will produce nonsense payoffs without necessarily producing an error message.

So use (20.6) whenever it exists, which for lognormal processes it always does. The reason Euler schemes are ubiquitous is that for most other processes (stochastic volatility, mean-reverting rates, jump models) no exact solution is available, and then the discretization error is real and must be managed.

A **predictor–corrector** step addresses one of its two parts, the error in the drift. Where the drift depends on the state being simulated, (20.5) freezes that drift at the value it has when the interval opens, and the step is biased whenever the state moves appreciably before the interval

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closes. Write the process as

$$dX = a(X,t)\,dt + b(X,t)\,dW$$

Take the Euler step first, as a prediction,

$$\hat{X} = X_t + a(X_t,t)\,\Delta t + b(X_t,t)\,\varepsilon\sqrt{\Delta t} \tag{20.7}$$

then step again from X_t, averaging the drift at the two ends of the interval,

$$X_{t+\Delta t} = X_t + \frac{1}{2}\big[a(X_t,t) + a(\hat{X},t+\Delta t)\big]\Delta t + b(X_t,t)\,\varepsilon\sqrt{\Delta t} \tag{20.8}$$

The same ε stands in both, the draw of (20.7): drawing again would add a second independent source of noise where the intention was to remove a bias. The cost is one extra drift evaluation per step and no extra random numbers, and Example 20.5 measures what it buys.

Example 20.5 What a predictor–corrector corrects

Geometric Brownian motion lets the drift error be priced rather than simulated. Taking expectations through (20.5) multiplies S by $1 + \mu\Delta t$ over a step, the first two terms of $e^{\mu\Delta t}$, and through (20.8) by $1 + \mu\Delta t + \frac{1}{2}(\mu\Delta t)^2$, the first three. Each scheme therefore arrives at a different forward, and the Black–Scholes–Merton formula prices each of those forwards exactly.

Take $S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$ and $T = 1$. The exact forward is 110.5171 and the call is worth 13.2697; at twelve steps Euler reaches a forward of 110.4713 and the corrector 110.5170. The premiums those imply:

Steps	Euler	Error	Pred.–corr.	Error	Ratio
1	12.931946	0.337730	13.258455	0.011222	30
4	13.180623	0.089053	13.268935	0.000742	120
12	13.239625	0.030052	13.269593	0.000083	360
52	13.262708	0.006968	13.269672	0.0000045	1560

Halving the step halves the Euler error and quarters the corrector's: the drift is discretized to first order in Δt by (20.5) and to second order by (20.8).

One error is absent from the table. Pricing each scheme's forward under Black–Scholes isolates the drift, and an Euler simulation carries a second error on top: its terminal distribution is a product of normals rather than a lognormal, which is the same defect that lets (20.5) go negative. A predictor–corrector repairs the first and leaves the second alone.

The limit of the method is the distributional error just named. The diffusion term of (20.8) is the Euler one unchanged, so a quantity that depends on the fine structure of the path rather than on its expectation gains nothing from correcting the drift. Nor does the correction help a continuously monitored barrier, whose error comes from crossings missed between dates: that needs, on each step, the probability that the path crossed the barrier between the two simulated points, or Chapter 34's continuity correction.

That leaves a rule with two branches. Use the exact scheme where one exists. Use a predictor–corrector where the drift depends on the state and is not small beside the diffusion. Chapter 32's market model is where the predictor–corrector is normally applied, because there the drift of each rate is a function of all the others, and Example 32.4 shows it earning almost nothing, for the reason this section supplies: at a 3% rate the drift is about 1% of the diffusion, so correcting it cannot move the answer.

20.8 VARIANCE REDUCTION

Since the error falls like $1/\sqrt{M}$, buying accuracy with paths is expensive. Buying it with structure is cheap. Four techniques follow.

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Antithetic variates. For every path generated with shocks $\varepsilon_1, \ldots, \varepsilon_n$, generate its mirror with $-\varepsilon_1, \ldots, -\varepsilon_n$ and average the two payoffs. Each path has the right distribution, and for a payoff monotone in the shocks the two are negatively correlated, so the variance of the average is lower. On the European put, the standard error at 100,000 paths (50,000 antithetic pairs) falls from 0.0159 to 0.0112, a factor of 1.4. It costs almost nothing and is worth doing for monotone payoffs, but it is not transformative, and for a payoff symmetric in the shocks, such as a straddle at the money, it is useless.

Control variates. Equation (20.1) returns in a more powerful form. Estimate a quantity Y whose true value is known alongside the target X, and report

$$\hat{X}_{\text{corrected}} = \hat{X} - \beta\left(\hat{Y} - Y^{\text{true}}\right), \qquad \beta = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(Y)} \tag{20.9}$$

With β chosen as in (20.9), the variance is reduced by a factor $1/(1-\rho^2)$, where ρ is the correlation between X and Y. Everything depends on ρ, and the difference between a good control variate and a mediocre one is enormous.

Example 20.6 A control variate that is worth 58 million paths

Price an arithmetic-average Asian call: $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, one year, twelve monthly fixings. There is no formula.

There is a formula for the *geometric*-average version, because the geometric average of lognormals is lognormal. It gives 4.012352 exactly. And the geometric and arithmetic averages of the same twelve prices are nearly the same number, so their payoffs are nearly perfectly correlated.

At 100,000 paths:

	Estimate	Standard error
Plain Monte Carlo	4.23399	0.02020
Geometric control variate	4.23671	0.00084

The measured correlation is $\rho = 0.99914$ and $\beta = 1.0485$. The variance falls by a factor of 583: to reach the same standard error, plain Monte Carlo would need 58 million paths.

The pattern that produced it is the one that gave the tree its accuracy in Example 20.1: find a problem as close as possible to the target and solvable in closed form, let the method price both, and correct the target by the error on the companion.

Importance and stratified sampling. Both put the paths where the answer is decided. A deep out-of-the-money option pays on a few paths in a thousand; **importance sampling** shifts the drift of the simulation toward the payoff region and reweights each path by how much likelier it was under the true drift than under the shifted one, which keeps the estimate unbiased, and it is the standard tool for deep out-of-the-money prices and for the rare tail losses of credit portfolios. **Stratified sampling** draws a fixed number of paths from each of several equal-probability bins of the normal draw that drives the path, so the sample cannot overweight a region by chance; moment matching, which rescales the draws to a sample mean of exactly 0 and variance of exactly 1, is a cruder relative, at the cost of a small bias.

Quasi-Monte Carlo. Replace pseudo-random numbers with a low-discrepancy sequence (Sobol or Halton) designed to fill the space more evenly than randomness does. Convergence can approach $1/M$ rather than $1/\sqrt{M}$ when a few of the draws decide most of the answer. The cost is that the usual standard error is no longer valid, since the points are not random; estimating the error requires several randomized repetitions. It works best when a few of the

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random numbers drawn for each path decide most of its payoff, and paths are often built in a special order to make that so.

20.9 SIMULATING SEVERAL ASSETS

Monte Carlo's advantage over trees is dimension, and correlated assets are the reason to use it.

To simulate d assets with correlation matrix $\mathbf{R}$, factor $\mathbf{R} = \mathbf{L}\mathbf{L}^\top$ by Cholesky decomposition and set $\boldsymbol{\varepsilon} = \mathbf{L}\mathbf{z}$, where $\mathbf{z}$ is a vector of d independent standard normals. Here $\mathbf{L}$ is lower triangular (zero above the diagonal), and that shape is not incidental: it is what makes the construction sequential, with ε_1 built from z_1 alone, ε_2 from z_1 and z_2, and each shock in turn from the draws already used plus one new one. For two assets it reduces to

$$\varepsilon_1 = z_1, \qquad \varepsilon_2 = \rho z_1 + \sqrt{1-\rho^2}\, z_2$$

which is all that most problems need.

It is the correlation matrix that is factored, not the covariance matrix. Each asset's volatility enters separately, in its own price update, so the ε_i must come out with unit variance, which the two-asset form shows, since $\rho^2 + (1-\rho^2) = 1$. Factoring the covariance matrix instead returns shocks that already carry σ_i, and applying σ_i to them a second time is an error that produces plausible prices at the wrong volatility.

The factorization also needs an internally consistent matrix (technically positive semidefinite: one that gives no portfolio a negative variance), and one assembled pair by pair, from estimates over different windows or from traders' manual adjustments, often is not; Cholesky then fails partway through. Risk systems replace such a matrix with the nearest consistent one, by Higham's (2002) method.

Example 20.7 Valuing a relative-TSR performance award

Section 17.7 said that a performance award with a market condition is valued by simulation at grant and never trued up, and pointed here for the machinery.

The award: for each target unit, the executive receives shares at vesting in three years, in a quantity that scales linearly from 0% if the company's total shareholder return trails the index by 30% or more, through 100% if the two are equal, to a cap of 200% if the company beats the index by 30% or more. Settlement is in shares.

Take $S_0 = 50$, company volatility 35%, index volatility 20%, correlation 0.60, $r = 5\%$, three years. Simulating 400,000 correlated pairs gives:

Expected units earned, $E[u]$	0.8361
Expected share price, $E[S_T]$	58.1795
Naive product, $e^{-rT}E[u]\,E[S_T]$	41.8704
Correct value, $e^{-rT}E[u\,S_T]$	60.2468

Before reading anything off that table, apply the chapter's own rule to it. The exact $E[S_T]$ is $50e^{0.15} = 58.0917$, so the run came in 0.15% high, about one and a half standard errors, which is what 400,000 paths should deliver. The award value carries a standard error of roughly 0.13.

Two results follow.

The expected number of units is below one, at 0.836, even though the payout schedule is symmetric around a 100% payout. The company's higher volatility drags its median return below the index's (the $-\sigma^2/2$ term of Section 14.9), so the relative performance is more often slightly negative than slightly positive, while the 200% cap truncates the compensating upside.

And yet the award is worth 60.25 per unit, against a share price of 50. Expected units times expected share price gives 41.87, which understates the award by 30%; equivalently, the correct value is 44% higher. The gap is entirely the covariance: units are earned precisely on the paths where the share price is high, so $E[u\,S_T] \neq E[u]E[S_T]$.

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Section 17.7's asymmetry now has a number attached: that 60.25 is expensed in full whether or not the award ever pays. Simulation is the natural method because the payoff, a capped and floored function of two correlated prices multiplied by one of them, is trivial to write inside a loop and awkward to integrate in closed form.

20.10 GREEKS FROM A SIMULATION

A simulated price is one number; a desk needs its sensitivities, because they are the coefficients of Chapter 18's expansion, equation (18.1), whose terms its hedges are built to cancel. The obvious approach, to revalue with S_0 bumped by a small h, difference the two prices, and divide by h, has a trap in it.

The trap is independent draws. If the bumped and unbumped simulations use independent random numbers, the difference of two estimates each carrying a standard error of, say, 0.016 has a standard error of 0.023, while the delta being measured might move the price by 0.05. The noise swamps the signal, and no number of paths fixes it efficiently. With common random numbers (the same draws reused in both runs) the two runs differ only through the bump, the noise largely cancels, and the estimator is usable.

This is the same principle as the control variate: the two runs share an error, and sharing it lets it cancel.

None of this difficulty arises on a lattice or a grid, which produce the value at every state and hand over delta, gamma and theta as differences of numbers already computed; Section 20.12 gives the formulas. Simulation is the method that has to work for them.

Two refinements remove the bump instead of taming it. **Pathwise differentiation** differentiates the payoff along each path, holding the draw fixed, which for a call is one application of the chain rule:

$$\frac{\partial}{\partial S_0}\, e^{-rT}(S_T - K)^+ = e^{-rT}\,\mathbf{1}\{S_T > K\}\,\frac{\partial S_T}{\partial S_0} = e^{-rT}\,\mathbf{1}\{S_T > K\}\,\frac{S_T}{S_0}$$

The second step is that $S_T = S_0 \exp[(r - \sigma^2/2)T + \sigma\sqrt{T}\,\varepsilon]$ is proportional to S_0 once ε is fixed, so $\partial S_T/\partial S_0 = S_T/S_0$. Average the right-hand side over the paths and the delta is in hand, from the paths that already produced the price, with no second simulation. The average is exactly unbiased here, because $e^{-rT}\mathbb{E}[S_T\mathbf{1}\{S_T > K\}] = S_0 N(d_1)$ makes the estimator's mean $N(d_1)$, the Black–Scholes delta itself. At $S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$ and $T = 1$ that delta is 0.7257, and 400,000 paths give 0.7262.

What this buys is not lower variance. Against a bump run on common random numbers the two variances agree (the ratio is 0.976 at $h = 1$ and 1.000 at $h = 0.01$), because as h falls the difference quotient becomes the derivative it approximates. What it buys is that there is no h to choose: the $h = 1$ bump reports 0.7344 against the exact 0.7257, an error of 0.0086 where the standard error is 0.0008. That error is bias. For a continuous payoff a small bump, or one taken on both sides of S_0, removes it at no cost in noise; the choice of h bites for a discontinuous payoff, where the bumped difference is nonzero on a vanishing fraction of paths and its variance grows like $1/h$.

It does require a differentiable payoff, and the failure is worse than it sounds. For a cash-or-nothing digital the payoff *is* the indicator, its derivative is zero wherever it exists, and the estimator returns zero on every path against a true delta of 0.0167: not a noisy answer but a confidently wrong one.

The **likelihood-ratio method** differentiates the density instead, which leaves the payoff untouched. Writing the price as $e^{-rT}\int \Pi(s)\,p(s;S_0)\,ds$, with Π the payoff and p the density

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of S_T, and differentiating under the integral,

$$\frac{\partial}{\partial S_0}\, e^{-rT}\,\mathbb{E}[\Pi] = e^{-rT}\,\mathbb{E}\left[\Pi(S_T)\,\frac{\partial \ln p}{\partial S_0}\right] = e^{-rT}\,\mathbb{E}\left[\Pi(S_T)\,\frac{\varepsilon}{S_0\sigma\sqrt{T}}\right]$$

where the last step is a short calculation: $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, so $\partial \ln p/\partial S_0$ reduces to $(\ln S_T - \text{mean})/(\sigma^2 T S_0)$, and that is the draw ε divided by $S_0\sigma\sqrt{T}$. The estimator is therefore the discounted payoff times the draw, scaled: one multiplication on top of the price, for any payoff whatever. On the digital it returns 0.0166 against the exact 0.0167, with a standard error of 0.00004.

Nothing is free. Because the payoff is left whole and the derivative is pushed onto the draw, the estimator inherits the payoff's full variability: on the call it carries 8.6 times the variance of the pathwise estimator, which is 2.9 times the standard error and so about nine times the paths for the same precision. Dealers' pricing systems use pathwise where they can and likelihood ratio where they must. Applied to every input at once, pathwise differentiation becomes adjoint algorithmic differentiation (AAD). A simulated price is a chain of elementary steps (build the exponent, exponentiate, take the payoff, discount, average), and the chain rule can run along that chain in either direction. Run forward, it carries the derivative with respect to one input through every step, so n inputs cost n passes, no better than bumping. Run backward, it starts from the price and asks at each step how much the price moves per unit change in that step's result, a number called the step's **adjoint**; one backward pass then delivers the derivative with respect to every input at once. AAD software records the steps during the pricing run and replays them in reverse, so all the sensitivities together cost a fixed multiple of one price, typically three to five times, however many inputs there are; Section 32.13 shows what that does to the cost of a whole portfolio's sensitivities. The cost is memory, since every intermediate value must be kept for the backward pass, and the limit is pathwise differentiation's: a digital has no derivative to pass back. Example 20.8 runs it on one path.

Example 20.8 One path, run backward

Take the call above ($S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$, $T = 1$) and a single draw $\varepsilon = 0.5$. The forward pass computes and stores

$$x = \left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,\varepsilon = 0.18, \qquad S_T = S_0 e^{x} = 119.72, \qquad V = e^{-rT}(S_T - K)^+ = 17.84$$

The backward pass, writing $\bar{y}$ for the adjoint of a quantity y, starts from $\bar{V} = 1$ and multiplies by one local derivative per step,

$$\bar{S}_T = e^{-rT}\,\mathbf{1}\{S_T > K\} = 0.9048, \qquad \bar{x} = \bar{S}_T\, S_T = 108.33$$

and each input reads its sensitivity off the step where it entered:

$$\frac{\partial V}{\partial S_0} = \bar{S}_T\,\frac{S_T}{S_0} = 1.083, \quad \frac{\partial V}{\partial \sigma} = \bar{x}\left(\sqrt{T}\,\varepsilon - \sigma T\right) = 32.50, \quad \frac{\partial V}{\partial r} = \bar{x}\,T - T\,V = 90.48, \quad \frac{\partial V}{\partial K} = -0.905$$

One backward pass has produced four sensitivities, each the pathwise estimate on this path; averaged over the paths they are delta, vega, rho and the sensitivity to the strike. Bump-and-revalue would have needed four more simulations.

20.11 AMERICAN MONTE CARLO

American Monte Carlo is the one genuinely difficult technique in the chapter. The problem, as Section 20.1 noted, is structural: a simulated path standing at an intermediate date does not know its own future, and an early-exercise decision requires exactly that.

The Longstaff–Schwartz method resolves it by *estimating* the continuation value with a

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regression across paths.

Carry two numbers per path. Let C_i be the cash flow path i receives under the rule built so far, and τ_i the date it is received; initialize C_i to the payoff at expiration and τ_i to T. Now work backward through the exercise dates. At each date t_k:

1. Select the paths that are in the money at t_k. The decision is trivial elsewhere, and confining the fit to the region where it is close is what keeps the regression useful.

2. For each selected path form $Y_i = C_i e^{-r(\tau_i - t_k)}$: the value at t_k of the cash flow that path is currently scheduled to receive. Regress Y on a few functions of the current price, the regression's **basis functions** (1, S, S^2 will do), across those paths only.

3. Take the fitted value at each path's own S_{t_k} as its continuation value, and exercise where the immediate payoff exceeds it. On those paths overwrite C_i with the immediate payoff and τ_i with t_k, discarding the later cash flow: a path can be exercised once.

4. Step back to the previous date.

There is no regression at time zero. The estimate is the average of the discounted cash flows, $\frac{1}{M}\sum_i C_i e^{-r\tau_i}$, compared with the immediate payoff if the option may be exercised at once.

The regression is doing something specific and limited: it is estimating a conditional expectation from the cross-section of paths, using the fact that many paths pass near the same state. Example 20.9 runs the algorithm by hand.

Example 20.9 Longstaff–Schwartz on ten paths

Take the American put of Example 20.1 ($S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, six months), with three exercise dates a sixth of a year apart and ten paths. Ten is far too few to price anything; it is enough to run the algorithm by hand.

Path	$t = 1/6$	$t = 2/6$	$t = 3/6$
1	58.83	55.93	62.73
2	47.62	56.43	57.46
3	43.85	54.54	46.88
4	42.21	43.86	45.49
5	51.71	47.49	48.36
6	48.92	47.20	46.39
7	64.47	71.06	70.94
8	40.90	41.89	49.10
9	55.01	49.11	61.99
10	42.38	39.52	38.54

At expiration $C_i = \max(50 - S_T, 0)$, which is nonzero on paths 3–6, 8 and 10.

At $t = 2/6$ six paths are in the money. Each carries $Y_i = C_i e^{-0.05/6}$, and the fitted regression is $\hat{Y} = 306.71 - 12.925\,S + 0.13674\,S^2$:

Path	S	Y	Continue	Exercise	Decision
4	43.86	4.4726	2.8718	6.1400	exercise
5	47.49	1.6264	1.2975	2.5100	exercise
6	47.20	3.5800	1.2909	2.8000	exercise
8	41.89	0.8925	5.2346	8.1100	exercise
9	49.11	0.0000	1.7579	0.8900	continue
10	39.52	11.3649	9.4837	10.4800	exercise

Path 9 is the instructive row: its own future cash flow is zero, so $Y_9 = 0$, yet the regression

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assigns it a continuation value of 1.76 from what happened to the *other* paths near $S = 49$, and since that exceeds the 0.89 on offer, the rule continues. That is the idea: the continuation value is estimated from the cross-section, not read off the path's own future.

At $t = 1/6$ the cash flows just written are discounted back one more step, and the fit is $\hat{Y} = 375.53 - 15.604\,S + 0.16274\,S^2$:

Path	S	Y	Continue	Exercise	Decision
2	47.62	0.0000	1.5224	2.3800	exercise
3	43.85	3.0684	4.2281	6.1500	exercise
4	42.21	6.0890	6.8491	7.7900	exercise
6	48.92	2.7768	1.6620	1.0800	continue
8	40.90	8.0427	9.5716	9.1000	continue
10	42.38	10.3930	6.5367	7.6200	exercise

Path 10 shows what the method is and is not. It was scheduled to collect 10.48 at $t = 2/6$, worth 10.39 here, but the regression puts its continuation value at 6.54 and the immediate payoff is 7.62, so the rule exercises and the path collects 7.62. The rule is estimated, not clairvoyant, and on this path it gives up value that waiting would have earned.

Discounting each path's surviving cash flow from its own exercise date and averaging over all ten gives 3.6940. The converged value is 3.6970, and the agreement is luck: the European option priced on the same ten paths comes out at 2.46 against its true 3.5829, which is the honest measure of what ten paths are worth.

Three details decide whether an implementation works.

The regression needs far more paths than basis functions. With three basis functions and only three in-the-money paths the fit is exact, each fitted continuation value equals that path's own realized cash flow, and the exercise rule silently acquires perfect foresight, reporting a confident answer that is far too high. Keep the in-the-money count comfortably above the number of basis functions, or use fewer of them.

Scale the basis. At $S \approx 50$ the three columns hold values near 1, 50 and 2,500, and least squares on columns differing that much in scale loses precision. Replacing S by S/K throughout the basis makes the columns 1, S/K and $(S/K)^2$, all of order one near the strike, which removes the problem.

Two biases run in opposite directions. Fitting the regression and valuing the option on the same paths leaks a little foresight, which pushes the answer up. The suboptimal exercise rule pushes it down, and by more. Example 20.10 shows the net at path counts that matter. The low estimate also has a partner: Rogers (2002) and Haugh and Kogan (2004) showed independently that any exercise rule can also be turned into an *upper* bound on the value (the dual method), and Andersen and Broadie (2004) made that bound computable, so a least-squares price can be bracketed from both sides and the width of the bracket measures how good the rule is.

Example 20.10 Longstaff–Schwartz against the tree

The target is the American put of Example 20.1, worth 3.69702. With 25 or 50 exercise dates the method prices a **Bermudan** option, one exercisable only on a fixed set of dates, so that is its benchmark: on a tree that may exercise only on the same dates it is worth 3.6907 and 3.6938, below the American by 0.0063 and 0.0032. Using antithetic paths, with each pair counted once in the standard error, and a quadratic regression:

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Paths	Dates	Estimate	Std. error	Bermudan	Error
20,000	25	3.69681	0.0175	3.6907	+0.0061
100,000	25	3.66563	0.0079	3.6907	−0.0251
100,000	50	3.68495	0.0077	3.6938	−0.0089
400,000	50	3.68705	0.0038	3.6938	−0.0068

In one run the rows scatter about their Bermudans by a standard error or more, and the first lands above; what is structural is the direction of the error on average, not its size in any row, because the regression produces an imperfect exercise rule and any rule other than the optimal one earns less. Adding paths shrinks the standard error but not the bias; a better regression basis reduces the bias, and more exercise dates move the target closer to the American.

A 200-step binomial tree gave this value to within 0.003 in a fraction of the time, so for a one-dimensional American option Longstaff–Schwartz is strictly the wrong choice. Its value is that it is the only method that works when the tree cannot be built: a callable note on a basket of five assets, a Bermudan swaption in a multi-factor model, or the exposure profiles of Chapter 9, where the early-exercise decision sits inside a simulation that must run anyway.

20.12 FINITE DIFFERENCE METHODS †

This section develops the third route to a price: solving the Black–Scholes–Merton partial differential equation numerically.

Write $V(S,t)$ for the price of the derivative being valued. Chapter 15 called it f; V is the usual symbol in this literature and is used throughout the section. Section 15.9 derived the form that carries a dividend yield,

$$\frac{\partial V}{\partial t} + (r-q)S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV$$

Lay a grid over (S,t), replace each derivative by a difference quotient, and solve backward from the known terminal condition.

Three edges of the grid are filled in before the calculation can work backward from expiration, and they are what the scheme is solved from. The terminal column comes from the payoff. The two price boundaries come from what the option is worth when the underlying is worthless or very large: for an American put, $V(0,t) = K$, since it would be exercised at once, and $V(S_{\max},t) = 0$. A European put takes the discounted strike, $V(0,t) = Ke^{-r(T-t)}$, since it cannot be exercised before expiration. $S_{\max}$ has to sit far enough out that the option really is worthless there, and the scale to measure it against is $\sigma\sqrt{T}$, the standard deviation of $\ln S_T$: on the contract used below, an $S_{\max}$ three of these above $\ln S_0$ already makes the error from cutting the grid off less than 10^{-6}, and the examples' 200 (four times the strike) is more than six above it. Reaching far enough out is easy; the trap is that $S_{\max}$ also fixes $\Delta S = S_{\max}/M$, where M is now the number of price steps in the grid (not the path count of Section 20.6), and unless $S_0/\Delta S$ is a whole number the grid has no node at today's price at all. With $S_{\max} = 150$ and $M = 400$ the nearest node sits 0.125 below S_0, and the put comes back 0.053 too high, which is that 0.125 times the delta of −0.43 that Example 20.13 reports, and nothing to do with the scheme. The *coarser* $\Delta S = 0.5$, which does land on S_0, is out by 0.002. Choose M together with $S_{\max}$ so the grid has a node at S_0, or interpolate between the two that straddle it. Figure 20.5 shows the arrangement.

In the log variable $x = \ln S$ the coefficients stop depending on the state,

$$\frac{\partial V}{\partial t} + \nu\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 V}{\partial x^2} = rV, \qquad \nu = r - q - \frac{1}{2}\sigma^2$$

and that ν is the drift of (20.2). The examples below set $q = 0$, matching the trees they are

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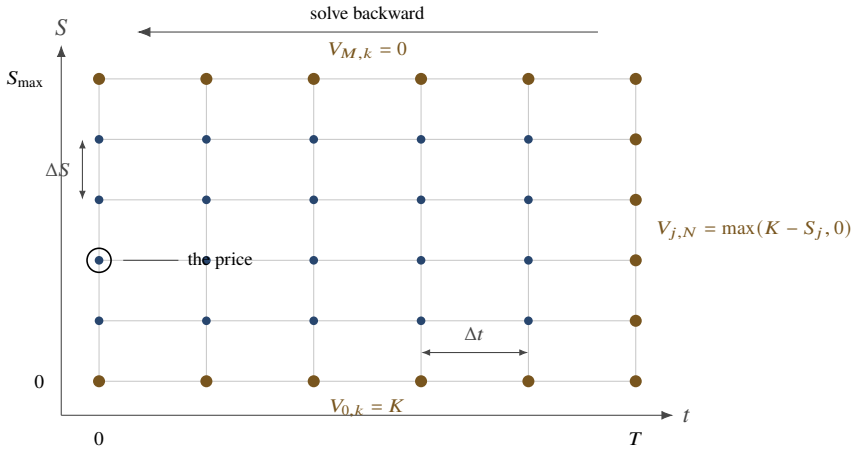


Figure 20.5: The grid for an American put. Larger (amber) nodes are boundary values; the rest are computed backward to $t = 0$.

measured against.

Three schemes differ in *when* the derivatives in S are evaluated. Write $V_{j,k}$ for the option's value at the grid point $S_j = j\,\Delta S$ and the date $t_k = k\,\Delta t$ (the first index counts price steps, the second counts dates), and $\mathbf{V}_k$ for the whole column of values at one date. The calculation runs backward, from the terminal column $\mathbf{V}_N$ to $\mathbf{V}_0$. Write $\mathcal{L}$ for the operator the equation puts against the time derivative,

$$\mathcal{L}V = rV - (r-q)S\frac{\partial V}{\partial S} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}, \qquad \frac{\partial V}{\partial t} = \mathcal{L}V$$

Each derivative is then replaced by a difference over neighboring grid points:

$$\frac{\partial V}{\partial S} \approx \frac{V_{j+1,k} - V_{j-1,k}}{2\,\Delta S}, \qquad \frac{\partial^2 V}{\partial S^2} \approx \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{\Delta S^2}, \qquad \frac{\partial V}{\partial t} \approx \frac{V_{j,k+1} - V_{j,k}}{\Delta t}$$

The two space differences are centered on node j and carry an error of order ΔS^2; the time difference is one-sided and carries an error of order Δt. Substituting them, and using $S_j = j\,\Delta S$, makes every ΔS cancel, leaving three coefficients that depend on the node index alone:

$$(\mathcal{L}\mathbf{V}_k)_j = A_j V_{j-1,k} + B_j V_{j,k} + C_j V_{j+1,k}$$

$$A_j = \frac{1}{2}\left[j(r-q) - \sigma^2 j^2\right], \qquad B_j = r + \sigma^2 j^2, \qquad C_j = -\frac{1}{2}\left[j(r-q) + \sigma^2 j^2\right]$$

That the grid spacing has vanished is why the stability condition below is a statement about how many nodes there are rather than about how wide they are. With θ a weight between 0 and 1, the share of the operator taken at the earlier date, the scheme is

$$\frac{\mathbf{V}_{k+1} - \mathbf{V}_k}{\Delta t} = \theta\,\mathcal{L}\mathbf{V}_k + (1-\theta)\,\mathcal{L}\mathbf{V}_{k+1}$$

and gathering the unknown column on the left turns it into one linear system per step:

$$(\mathbf{I} + \theta\,\Delta t\,\mathcal{L})\,\mathbf{V}_k = (\mathbf{I} - (1-\theta)\,\Delta t\,\mathcal{L})\,\mathbf{V}_{k+1}$$

That equation is the method. Its right-hand side is known, its left-hand side is not, and each step delivers the column one date earlier. The three schemes are three readings of it, and Figure 20.6 draws what each costs.

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Explicit, $\theta = 0$. The left side is $\mathbf{V}_k$ by itself, so nothing is solved and every interior node is read off directly:

$$V_{j,k} = a_j V_{j-1,k+1} + b_j V_{j,k+1} + c_j V_{j+1,k+1}, \qquad j = 1, \ldots, M-1$$

with $a_j = -\Delta t A_j$, $b_j = 1 - \Delta t B_j$ and $c_j = -\Delta t C_j$. Those three weights are the arrows in the figure's left panel, and on the log grid they approximate the probabilities of (20.2).

Implicit, $\theta = 1$. Now $(\mathbf{I} + \Delta t\, \mathcal{L})\mathbf{V}_k = \mathbf{V}_{k+1}$, which one row at a time reads

$$\Delta t A_j V_{j-1,k} + \left(1 + \Delta t B_j\right) V_{j,k} + \Delta t C_j V_{j+1,k} = V_{j,k+1}$$

for $j = 1, \ldots, M-1$. The first and last rows reach past that range, to $V_{0,k}$ and $V_{M,k}$, which are the boundary values of Figure 20.5 and therefore known, so they move across:

$$\begin{pmatrix} 1+\Delta t B_1 & \Delta t C_1 & & \\ \Delta t A_2 & 1+\Delta t B_2 & \ddots & \\ & \ddots & \ddots & \Delta t C_{M-2} \\ & & \Delta t A_{M-1} & 1+\Delta t B_{M-1} \end{pmatrix} \begin{pmatrix} V_{1,k} \\ V_{2,k} \\ \vdots \\ V_{M-1,k} \end{pmatrix} = \begin{pmatrix} V_{1,k+1} - \Delta t A_1 V_{0,k} \\ V_{2,k+1} \\ \vdots \\ V_{M-1,k+1} - \Delta t C_{M-1} V_{M,k} \end{pmatrix}$$

The matrix is tridiagonal, so it is never inverted. Write the row for $V_{j,k}$ as $\alpha_j V_{j-1,k} + \beta_j V_{j,k} + \gamma_j V_{j+1,k} = d_j$, with $\alpha_j = \Delta t A_j$, $\beta_j = 1 + \Delta t B_j$, $\gamma_j = \Delta t C_j$, and d_j the right-hand side above. The **Thomas algorithm** is Gaussian elimination that uses the three bands and touches nothing else: a forward pass clears the subdiagonal and a back substitution reads the column off. Starting from $\gamma'_0 = d'_0 = 0$, the forward pass runs for $j = 1, \ldots, M-1$:

$$\gamma'_j = \frac{\gamma_j}{\beta_j - \alpha_j \gamma'_{j-1}}, \qquad d'_j = \frac{d_j - \alpha_j d'_{j-1}}{\beta_j - \alpha_j \gamma'_{j-1}}$$

and the back substitution, starting from $V_{M-1,k} = d'_{M-1}$, runs back down for $j = M-2, \ldots, 1$:

$$V_{j,k} = d'_j - \gamma'_j V_{j+1,k}$$

Each pass is a single loop over M points, so a column costs $O(M)$ operations, the same order as an explicit step. And since the matrix is the same at every step, the γ'_j depend on nothing that changes: compute them once before the time loop begins, and each column costs only the d'_j recursion and the back substitution.

Crank–Nicolson, $\theta = 1/2$. The same system with half the operator on each side,

$$\left(\mathbf{I} + \frac{\Delta t}{2}\,\mathcal{L}\right)\mathbf{V}_k = \left(\mathbf{I} - \frac{\Delta t}{2}\,\mathcal{L}\right)\mathbf{V}_{k+1}$$

so a step costs the implicit scheme's solve plus one tridiagonal multiplication to form the right-hand side.

And, for an American option, one more line. Once a column has been computed, the early-exercise condition is imposed across it:

$$V_{j,k} \leftarrow \max\left(V_{j,k},\ K - S_j\right)$$

This is the free boundary Chapter 15 deferred, handled by projection. The boundary is never located; it is enforced wherever it binds, and where it does the node holds the payoff instead of the solved value. For the explicit scheme the projection is exact; after an implicit or Crank–Nicolson solve it carries an error of order Δt, which costs Crank–Nicolson its second

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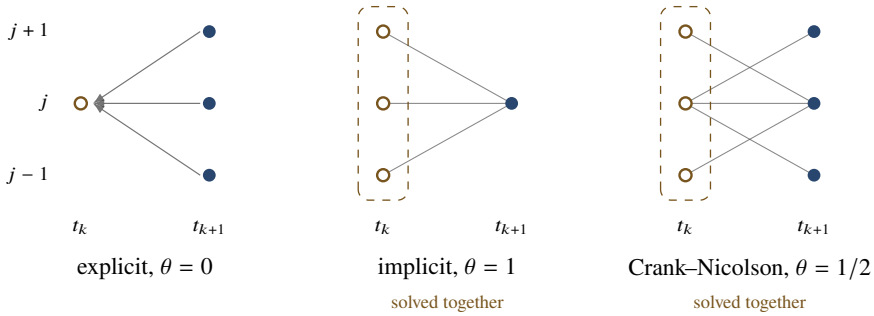


Figure 20.6: The nodes entering one equation under each weight. Filled nodes are known, hollow ones are being solved for, and t_k is the earlier date.

order. Imposing the condition inside the solve restores it, by Brennan and Schwartz's ordering of the elimination for a put or by projected iteration in general.

The weight is what sets the order in time. At $\theta = 1/2$ the right-hand side is the average of the operator at the two dates, which is its value at the midpoint between them to within $O(\Delta t^2)$, so the one-sided time difference above becomes a centered one about that midpoint. That is why Crank–Nicolson is second-order in time for a smooth problem where the other two are first-order, and Example 20.12 measures what survives of it on an American put.

The explicit scheme is a trinomial tree. At $\theta = 0$ each new value is an explicit weighted average of three neighbors, and on the log grid those weights are (20.2) to first order in Δt. The tree carries a $\nu^2 \Delta t^2$ term the scheme drops, and discounts by $e^{-r\Delta t}$ where the scheme multiplies by $1 - r\Delta t$. Both differences fall by a factor of ten for every factor of ten in Δt, so Section 20.4's remark holds in the limit: the two methods are the same computation described in two vocabularies.

And it is only conditionally stable. The weights are probabilities only if Δt is small relative to $1/(\sigma^2 M^2)$, where M is the number of space steps. Violate that and the scheme does not merely lose accuracy: it explodes.

Example 20.11 What instability looks like

Solve the American put of Example 20.1 explicitly on a grid running to $S_{\max} = 200$:

Space steps	Time steps	Value
100	1,000	3.6766
200	1,000	beyond 10^{40}
200	4,000	3.6918

Refining the space grid while leaving the time grid alone destroyed the calculation. The middle row is not slightly wrong; it is dozens of orders of magnitude wrong, because the scheme amplifies rounding error geometrically once the stability condition fails, and its exact size depends on nothing more than the order of the arithmetic.

Doubling the space resolution therefore requires four times as many time steps. That is the practical objection to the explicit scheme, and it is why the implicit and Crank–Nicolson schemes exist: both are unconditionally stable, at the cost of solving a tridiagonal system at each time step, which the Thomas algorithm does in $O(M)$ operations.

There is one consolation. An unstable explicit scheme fails loudly, and a number beyond 10^{40} cannot be mistaken for a price. The failures worth fearing are the quiet ones, and Crank–Nicolson

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has one: it damps error that flips sign from one node to the next only slowly, so the payoff's kink leaves oscillations near the strike that persist for many steps and show in gamma before the price. Rannacher's repair is to take the first few steps fully implicit.

Figure 20.7 draws the boundary that the three rows sit on either side of. It is a parabola, so the time steps a finer space grid needs rise with the square of its resolution: the middle row was not unlucky, only on the wrong side of the curve.

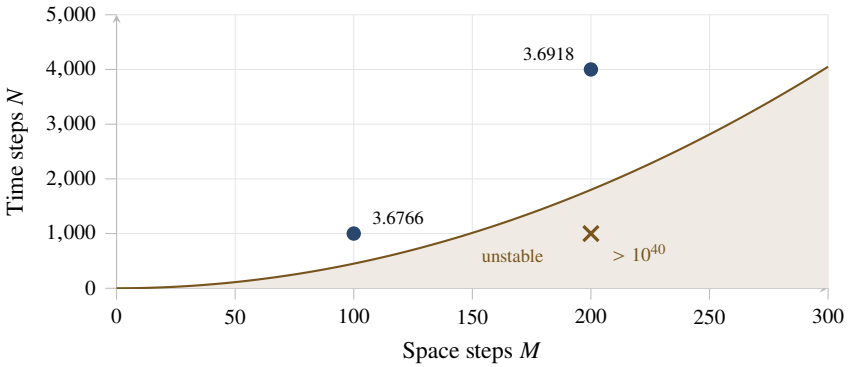


Figure 20.7: Stability of the explicit finite-difference scheme on the American put of Example 20.1. Below the curve $N = T\sigma^2 M^2$ the scheme diverges; the marks are the three grids of Example 20.11.

Example 20.12 Implicit against Crank–Nicolson

The same American put, benchmark 3.69702, gives:

Space	Time	Implicit	Error	Crank–Nicolson	Error
100	50	3.65776	−0.03926	3.67363	−0.02339
200	100	3.68196	−0.01506	3.69033	−0.00669
400	200	3.69056	−0.00646	3.69494	−0.00208
800	400	3.69403	−0.00299	3.69631	−0.00071

The implicit errors fall by roughly a factor of two per refinement, which is first order. The Crank–Nicolson errors fall by about three and a half, then three, then less: the symmetric time-averaging delivers second order on the smooth part of the problem, and the projection's first-order error shows through as the grid is refined. For nearly the same work Crank–Nicolson is more accurate at every grid size, and its lead grows with refinement, from under twice as accurate on the coarsest grid to four times on the finest, which is why it is the default.

Both converge from below, as the European put does on the same grids: the sign is the discretization's, not the exercise constraint's.

Why a desk uses finite differences at all. The method produces the value at every grid point, so the quantities a desk needs are already differences of numbers it holds. Writing j_0 for the node nearest the spot,

$$\Delta \approx \frac{V_{j_0+1,0} - V_{j_0-1,0}}{2\,\Delta S}, \qquad \Gamma \approx \frac{V_{j_0+1,0} - 2V_{j_0,0} + V_{j_0-1,0}}{\Delta S^2}, \qquad \Theta \approx \frac{V_{j_0,1} - V_{j_0,0}}{\Delta t}$$

which are the same three quotients the scheme is built from. A tree gives them as cheaply, from the nodes at steps one and two. For a book that must be revalued, and its risk recomputed, across a range of spot prices every evening, that is decisive.

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Vega and rho do not come free: volatility and the rate are parameters of the equation rather than state variables, so each costs a second solve at a bumped input, which is why a desk needing a full set of Greeks reruns twice rather than five times.

Example 20.13 Greeks from the tree and from the grid

Ask each method for the Greeks of the American put of Example 20.1 as well as its price, at no extra cost in either case:

	Value	Δ	Γ	Θ
Tree, 6,000 steps	3.6969	−0.4300	0.03953	−3.187
Crank–Nicolson, 800×400	3.6963	−0.4299	0.03953	−3.190

Two methods sharing no arithmetic agree to three significant figures on all four numbers: Section 20.13's second habit applied to the Greeks, and the cheapest check available on an implementation.

The European option's closed-form delta and gamma are −0.4114 and 0.03668, against the American values above. The early-exercise feature shows more strongly in the Greeks than in the price: the value differs by 3%, the delta by 5%, and the gamma by 8%. A desk hedging an American book off European Greeks is wrong by more than the pricing error would suggest.

With a second solve, raising σ by one volatility point moves the price by 0.1373, and raising r by a basis point moves it by −0.0009.

20.13 CHOOSING A METHOD

The chapter reduces to a short list of decisions.

One state variable and early exercise → tree or finite difference. Use a control variate against the European, then extrapolate between step counts that share an offset. A few hundred steps suffice.

Path dependence → Monte Carlo, with a control variate. Look hard for a closely related closed form before accepting plain simulation; Example 20.6 is what that search is worth.

More than three state variables → Monte Carlo, necessarily. Grid methods are out of reach.

Both early exercise and high dimension → Longstaff–Schwartz, accepting the downward bias and reporting it.

Greeks across a grid of spot prices → finite difference, which delivers them as a by-product.

Desks divide along the same lines. Market makers in listed American equity options price on trees, fast enough to update quotes across every strike and expiration continuously; rates desks run one-factor callable and Bermudan books on trees or grids (Chapter 31) and simulate with regression only when the model has several factors (Chapter 32); XVA desks simulate a netting set's many state variables and take their sensitivities by AAD. Stochastic-volatility models are calibrated by transform pricing, and a market-condition award is valued for the accounts by simulation, as in Example 20.7.

Two habits matter more than the choice.

Always price something you already know. Run the European option through the same machinery and compare with the Black–Scholes–Merton formula. It costs almost nothing, it catches most implementation errors, and by (20.1) the discrepancy is not merely a diagnostic but a correction.

Always compute the answer twice. Two step counts, two path counts, two grid resolutions. A number computed once is not a number; it is a hypothesis. Between them the two habits catch most failures in this chapter: the instability of Example 20.11 falls to the second immediately, while the bias of Example 20.10, which a second path count cannot reveal, shows only against an independent answer, the Bermudan tree there or the dual upper bound where no tree exists.

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20.14 WHERE THIS CAME FROM

Boyle (1977) introduced Monte Carlo to option pricing, but early exercise resisted simulation for two decades. Methods for it appeared through the 1990s, among them Broadie and Glasserman's (1997), and it was Longstaff and Schwartz's (2001) regression of the continuation value on the paths themselves that made American Monte Carlo routine, and that lets the exposure simulations of Chapter 9 price Bermudan options at all. The finite-difference route is older: Brennan and Schwartz (1977) priced the American put that way, and in 1978 showed the explicit scheme to be a lattice approximation of the diffusion, the identity Section 20.12 uses.

On lattices the pathologies came before the fixes. Boyle and Lau (1994) explained why a barrier priced on a tree converges badly and how node placement repairs it; Broadie, Glasserman and Kou (1997) gave the continuity correction for discretely monitored barriers that Chapter 34 checks against simulation. Barone-Adesi and Whaley (1987) supplied a fast analytic approximation to the American premium, which lattices and later methods now match in accuracy at little cost.

SUMMARY

A closed form requires a simple payoff, a simple process, and no early exercise; break one and a number must be computed. The pricing equation and the risk-neutral expectation are the same statement, so a price may be obtained backward by solving the equation or forward by simulating the expectation, and the two directions have opposite strengths: backward methods handle early exercise but fail in high dimension, forward methods handle path dependence and dimension but not early exercise.

Binomial trees converge at $O(1/n)$, with an error set by where the strike falls between two terminal nodes; at the money that is the parity of n. The control variate exploits it, correcting the target by the method's own error on a companion whose answer is known and cutting the American put's error by orders of magnitude; it removes what the two problems share and nothing else, so the companion should sit as close to the target as possible.

Trinomial trees do not converge faster. What they buy is an independent choice of Δx, which lets a node be placed on a barrier, and the explicit finite difference scheme is a trinomial tree in another notation. A known cash dividend destroys recombination (a proportional one, discrete or continuous, does not), and the remedy is to build the tree on the price net of the dividend's present value and add it back at each node.

Monte Carlo is unbiased with a standard error falling like $1/\sqrt{M}$, so a decimal place costs a hundredfold increase in paths, and structure beats brute force: a geometric-average control variate for an arithmetic Asian option removes so much variance that matching it with plain paths would take an impractical number of them. Lognormal paths are simulated exactly, and an Euler step, which can go negative, is a fallback for processes without a solution. Greeks from a simulation need common random numbers if bumped; the pathwise estimator needs a differentiable payoff and the likelihood-ratio estimator does not, and adjoint differentiation returns all sensitivities at once for the cost of a few prices. Correlated assets go through a Cholesky factor, and the reason to bother shows in a relative total shareholder return award, where multiplying expected units by expected price understates the value badly, because units are earned on precisely the paths where the share price is high. Longstaff–Schwartz regresses the continuation value on the cross-section of in-the-money paths and is biased downward, since a suboptimal exercise rule earns less; a dual calculation supplies the matching upper bound, and the method is the right choice only where a tree cannot be built.

Finite difference schemes differ in when the derivatives in S are evaluated. The explicit scheme is only conditionally stable, while implicit and Crank–Nicolson are unconditionally stable, and Crank–Nicolson is second-order in time because its weight centers the time

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difference, an order that early exercise imposed after the solve gives back. A grid returns delta, gamma and theta as differences of values it has already computed; vega and rho need a second solve. Two habits govern everything: price something whose answer is already known, and compute every answer twice.

FURTHER READING

Glasserman's *Monte Carlo Methods in Financial Engineering* is the standard reference, and the one that explains why variance reduction works rather than only how to implement it. Longstaff and Schwartz (2001) is the original least-squares Monte Carlo paper and is short; the subsequent literature on the method's bias and on choosing regression bases is large and worth sampling before implementing it. Wilmott, Howison and Dewynne treat the partial differential equation route with the numerical analysis intact, which is the place to go for stability conditions derived rather than quoted. Capriotti (2011) is the practitioner account of adjoint differentiation for Greeks. On trees, Broadie and Detemple (1996) smooth the terminal payoff by putting the Black–Scholes–Merton value at the last step, and Boyle and Lau (1994) show how node placement interacts with a barrier; between them they explain a family of convergence pathologies that Chapter 34 meets again. For low-discrepancy sequences, the practical literature diverges sharply from the theoretical bounds, and any implementation should be tested against a known answer at the dimensions actually used.

PRACTICE QUESTIONS

20.1 A binomial tree gives 3.5724 at 100 steps and 3.5777 at 200. Estimate the exact value by Richardson extrapolation, state what you must check before trusting it, and explain why the binomial error alternates in sign with the parity of n when $S_0 = K$.

20.2 A tree gives an American put value of 3.6914 and a European put value of 3.5724 at the same step count, and the Black–Scholes–Merton formula gives 3.5829 for the European. Compute the control-variate corrected American value, and explain why the correction helps enormously at odd step counts and only slightly at even ones.

20.3 State two things a trinomial tree buys over a binomial tree and one thing it does not, and name the finite difference scheme that is a trinomial tree in another notation.

20.4 Explain why a known cash dividend destroys the recombination of a binomial tree, and describe the standard remedy.

20.5 A Monte Carlo estimate has a standard error of 0.04 at 10,000 paths. How many paths are needed for a standard error of 0.005?

20.6 Explain why the exact lognormal simulation step is preferable to an Euler step, giving both reasons.

20.7 A control variate has correlation 0.95 with the target. By what factor is the variance reduced? Compare with the 0.99914 achieved by the geometric Asian control variate and explain why the improvement is so much larger than the correlations differ.

20.8 Explain why bumping the spot price and revaluing gives a poor delta if the two simulations use independent random numbers.

20.9 State the direction of the Longstaff–Schwartz bias, explain why it goes that way, and state when the method is nonetheless the right choice.

20.10 Explain why the explicit finite difference scheme becomes unstable when the space grid is refined, and state how the time grid must respond.

20.11 Explain why a relative total shareholder return award cannot be valued as expected units

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times expected share price, and state the direction and rough size of the error.

FURTHER QUESTIONS

20.12 Richardson extrapolation can return an answer worse than either of the values it is built from. Derive the condition on the error sequence under which extrapolation improves the estimate, explain how the offset of Section 20.3 decides whether two step counts satisfy it, and propose a diagnostic a pricing system could run automatically to detect a violation before reporting a price.

20.13 Example 20.1 argues that a control variate removes the error the two problems share and nothing else. Formalize this: write the tree's American and European errors as a shared component plus an idiosyncratic one, show what (20.1) does to each, and use the result to predict how the correction's effectiveness changes as the option moves deep into or out of the money.

20.14 The Asian control variate of Example 20.6 achieves a variance reduction of 583 through a correlation of 0.99914. Explain why the geometric and arithmetic averages are so highly correlated, determine how the correlation and hence the reduction would change if the number of fixings rose from 12 to 250, and discuss whether the closed form for the geometric case remains available.

20.15 Example 20.7 finds that the naive product of expected units and expected share price understates the award's value by 30%, so that the correct value is 44% higher. Decompose the correct value into the naive product plus a covariance term, explain the economic content of that term, and state what it implies about a company that grants relative total shareholder return awards and simultaneously reports their value using an expected-payout methodology.

20.16 Section 20.11 states that Longstaff–Schwartz is biased low and that adding paths does not remove the bias. Explain why the bias is not a small-sample effect, describe how one could construct an *upper*-biased estimate to bracket the true value, and discuss what a risk manager should report when only the lower estimate is available.

20.17 Example 20.11 shows an explicit scheme returning a number beyond 10^{40}. Derive the stability condition for the scheme, show that it is equivalent to requiring the three trinomial weights of (20.2) to be nonnegative, and explain why this equivalence means the instability is a statement about probabilities rather than about arithmetic.

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